



Children's ENCYCLOPEDIA

The book that explains everything





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Contents

How this book works	10	Aztecs	35
Africa	12	Bicycles	36
Aircraft	13	Big Bang	37
American West	14	Biology	38
Amphibians	15	Birds	39
Ancient China	16	Black holes	40
Ancient Egypt	17	Body cells	41
Ancient Greece	18	The story of storytelling	42
Ancient India	19	Books	44
Ancient Rome	20	Brain	45
Animal families	21	Bridges	46
Animal groups	22	Bronze Age	47
Animal homes	23	Buildings	48
Antarctica	24	Carbon cycle	49
Arctic	25	The story of the changing world	50
The story of color	26	Cars	52
Art	28	Castles	53
Asia	29	Cats	54
Asteroids	30	Caves	55
Astronauts	31	Cells	56
Astronomy	32	Changing states	57
Atmosphere	33	Chemistry	58
Atoms	34	Circuits	59
		Climate change	60
		Clocks	61
		The story of clothing	62
		Clouds	64





Coding	65	Europe	94
The story of codes	66	Evolution	95
Comets	68	Explorers	96
Communication	69	Factories	97
Compass	70	Farming	98
Computers	71	Feelings	99
Conservation	72	Film	100
Constellations	73	Fish	101
Coral reefs	74	Flags	102
Crafts	75	Flowers	103
Dance	76	The story of eating	104
Day and night	77	Food	106
Deserts	78	Food chains	107
Digestion	79	Forces	108
Dinosaurs	80	Forests	109
Dogs	81	Fossil fuels	110
Early humans	82	Fossils	111
Earth	83	Fractions	112
Earth's surface	84	French Revolution	113
Earthquakes	85	Friction	114
Eggs	86	Fruit and seeds	115
Electricity	87	Galaxies	116
The story of energy	88	Gases	117
Elements	90	Gems	118
Engineering	91	Genes	119
Engines	92	The story of water	120
Erosion	93	Glaciers	122





Governments	123	Magnets	151
Grasslands	124	The story of pets	152
Gravity	125	Mammals	154
Habitats	126	Maps	155
Hearing	127	Mars	156
Heart	128	Materials	157
Hibernation	129	Maya	158
Human body	130	Measuring	159
Imperial Japan	131	Medicine	160
Incas	132	Mercury	161
Industrial Revolution	133	Metals	162
Insects	134	Metamorphosis	163
Inside Earth	135	Meteorites	164
The story of inventions	136	Microscopic life	165
Internet	138	Migration	166
Invertebrates	139	Milky Way	167
Iron Age	140	Mixtures	168
Jupiter	141	Money	169
Knights	142	Monkeys and apes	170
Lakes	143	Moon	171
Language	144	Mountains	172
Law	145	Muscles	173
Life cycle	146	Mushrooms	174
Light	147	Musical instruments	175
Liquids	148	The story of music	176
Lungs	149	Myths and legends	178
Machines	150	Native Americans	179





The story of exploration	180	Reptiles	210
Navigation	182	Rivers	211
Neptune	183	Robots	212
North America	184	Rock cycle	213
Numbers	185	Rocks and minerals	214
Oceania	186	Satellites	215
Oceans and seas	187	Saturn	216
Orchestra	188	Science	217
Philosophy	189	The story of the sciences	218
Photography	190	Seashore	220
Photosynthesis	191	Seasons	221
Physics	192	Shapes	222
Pirates	193	Sharks	223
Plants	194	Ships	224
Plastic	195	Sickness	225
Pluto	196	Sight	226
Polar habitats	197	Sinking and floating	227
Pollution	198	Skeleton	228
Precious metals	199	Skin	229
The story of gold	200	Slavery	230
Prehistoric life	202	Sleep	231
Radio	203	Smell	232
Rain forests	204	Solar system	233
Recycling	205	Solids	234
The story of festivals	206	Sound	235
Religion	208	South America	236
Renaissance	209	Space travel	237





Spiders	238	Volcanoes	268
Sports	239	Volume	269
The story of games	240	Water cycle	270
Stars	242	Weather	271
Stone Age	243	The story of school	272
The story of homes	244	Work	274
Storms	246	World	275
Sun	247	World War I	276
Symmetry	248	World War II	277
Taste	249	The story of war	278
Telephone	250	Writing	280
Television	251	Zoo	281
Temperature	252		
Theater	253	Reference	282
Tides	254	Glossary	294
Time zones	255	Index	298
Touch	256	Acknowledgments	304
Trade	257		
The story of transportation	258		
Trains	260		
Trees	261		
Turkish Empire	262		
Universe	263		
Uranus	264		
Venus	265		
Vertebrates	266		
Vikings	267		



How this book works

Ever wonder what a planet is? Do you know the difference between frogs and toads? Every page in this encyclopedia is packed with fun facts and amazing photographs. Dive right in or jump to a page that catches your eye to discover all about our wonderful world. The helpful tips here will guide you through using this book.

Alphabetical pages

The book's pages are arranged alphabetically. You can look up topics using the contents list, which starts on page four, or flip through the pages in whichever order you like.

See also

If you enjoy reading a page, you can use the "see also" box to find similar pages in the book and see where they are. This lets you make links across topics and subject areas and create your own journey through the encyclopedia.

Amphibians

Amphibians are animals that spend their lives in and around water. They hatch in water from eggs. As they grow they develop lungs that let them breathe on land. Amphibians skin must always be wet, so they stay close to water. There are three main groups of amphibians.

Reptiles and amphibians
Reptiles and amphibians are cold-blooded animals. Reptiles have scales and lay eggs. Amphibians have smooth skin and lay eggs in water.

Frogs and toads
Frogs and toads are amphibians. They have long legs and can jump. Frogs have long, thin bodies and can swim. Toads have short, stout bodies and can crawl.

Snakes and lizards
Snakes and lizards are reptiles. They have long, thin bodies and can crawl. Snakes have no legs and can move in many ways. Lizards have four legs and can jump.

Snails
Snails are mollusks. They have a hard shell and can move slowly.

Butterflies
Butterflies are insects. They have four wings and can fly.

Beetles
Beetles are insects. They have six legs and can crawl.

Ants
Ants are insects. They have six legs and can crawl.

Worms
Worms are invertebrates. They have no legs and can crawl.

Plants
Plants are living organisms. They have roots, stems, and leaves.

Animals
Animals are living organisms. They have heads, bodies, and tails.

Humans
Humans are animals. They have heads, bodies, and tails.

Follow the suggestions in the "see also" box to get to related topics elsewhere in the book.

Metamorphosis

Metamorphosis is a process where an animal changes its body structure. This process happens in many animals, including insects, amphibians, and reptiles.

Butterflies
Butterflies go through four stages of metamorphosis: egg, larva, pupa, and adult. The larva stage is when the butterfly is a caterpillar. The pupa stage is when the butterfly is a chrysalis. The adult stage is when the butterfly is a fully formed butterfly.

Frogs
Frogs go through four stages of metamorphosis: egg, tadpole, pupa, and adult. The tadpole stage is when the frog is a small, aquatic animal. The pupa stage is when the frog is a small, aquatic animal. The adult stage is when the frog is a fully formed frog.

Snakes
Snakes go through four stages of metamorphosis: egg, larva, pupa, and adult. The larva stage is when the snake is a small, aquatic animal. The pupa stage is when the snake is a small, aquatic animal. The adult stage is when the snake is a fully formed snake.

Ants
Ants go through four stages of metamorphosis: egg, larva, pupa, and adult. The larva stage is when the ant is a small, aquatic animal. The pupa stage is when the ant is a small, aquatic animal. The adult stage is when the ant is a fully formed ant.

Worms
Worms go through four stages of metamorphosis: egg, larva, pupa, and adult. The larva stage is when the worm is a small, aquatic animal. The pupa stage is when the worm is a small, aquatic animal. The adult stage is when the worm is a fully formed worm.

Plants
Plants go through four stages of metamorphosis: seed, seedling, juvenile, and adult. The seedling stage is when the plant is a small, aquatic animal. The juvenile stage is when the plant is a small, aquatic animal. The adult stage is when the plant is a fully formed plant.

Animals
Animals go through four stages of metamorphosis: egg, larva, pupa, and adult. The larva stage is when the animal is a small, aquatic animal. The pupa stage is when the animal is a small, aquatic animal. The adult stage is when the animal is a fully formed animal.

Humans
Humans go through four stages of metamorphosis: egg, larva, pupa, and adult. The larva stage is when the human is a small, aquatic animal. The pupa stage is when the human is a small, aquatic animal. The adult stage is when the human is a fully formed human.

Reptiles

Reptiles are cold-blooded animals. They have scales and lay eggs. Reptiles are found in many habitats, including deserts, forests, and mountains.

Snakes
Snakes are reptiles. They have long, thin bodies and can crawl. Snakes have no legs and can move in many ways.

Lizards
Lizards are reptiles. They have four legs and can jump. Lizards have long, thin bodies and can move in many ways.

Turtles
Turtles are reptiles. They have a hard shell and can crawl. Turtles have four legs and can move in many ways.

Alligators
Alligators are reptiles. They have long, thin bodies and can crawl. Alligators have four legs and can move in many ways.

Crocodiles
Crocodiles are reptiles. They have long, thin bodies and can crawl. Crocodiles have four legs and can move in many ways.

Follow the suggestions in the "see also" box to get to related topics elsewhere in the book.

The color at the top and bottom of the page shows which type of subject the page is about. This turquoise color shows that the amphibians page is about nature (see below).

Subject areas

The encyclopedia covers nine different subject areas. Each one has its own color, which is shown at the top and bottom of the page.



The story of...

Pets

Pets have become an important part of our lives. Many animals are kept as companions, for work, or to help people with disabilities. Here are some of the most common pets.



The story of...

These pages bring together information from the different subjects, to get you thinking about things from lots of different angles. These pages are not arranged alphabetically.



Reference

The reference section contains useful lists, diagrams, and tables. For example, there is a map of the world, and there are lists of great artists, scientists, and writers.

Glossary

The glossary is a list of some of the more difficult words used in the book. If you would like an explanation of what a word means, you can look for it in the glossary.

Glossary

Abacus A short, thick column that supports a horizontal structure, such as a roof or a balcony.	Abandon To leave something or someone without caring about them.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.
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Index

Abacus A short, thick column that supports a horizontal structure, such as a roof or a balcony.	Abandon To leave something or someone without caring about them.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.	Abandonment The state of being abandoned or left alone.
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Index

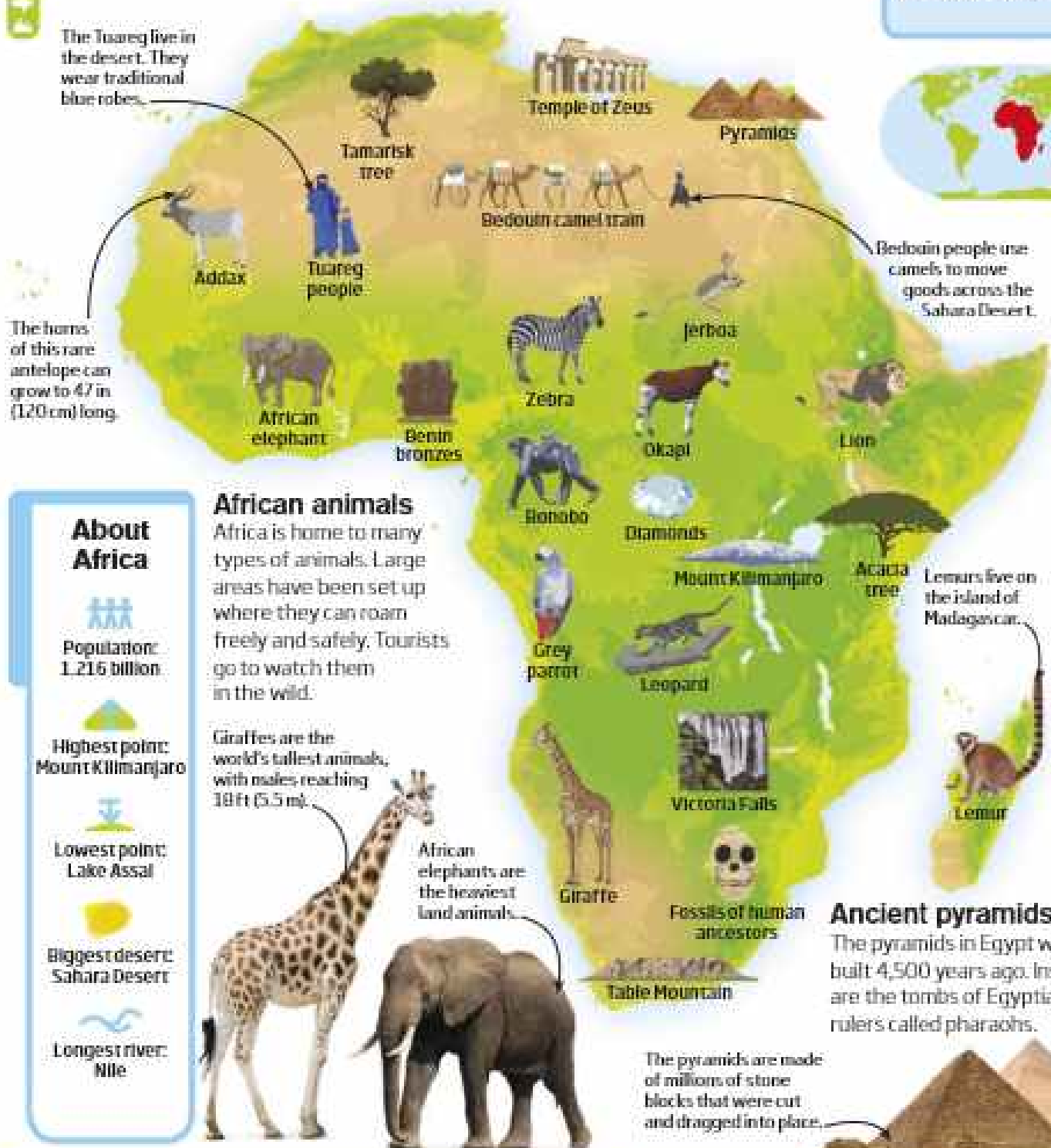
The index lists alphabetically everything covered in the book, along with its page number. If there is anything you want to know about, you can look it up here.

Africa

Africa is a very hot continent, and a lot of the landscape is made up of deserts and dry plains. Its central region is covered in rain forest. This continent was home to the first people on Earth millions of years ago.

SEE ALSO

- ▶ Ancient Egypt p.17
- ▶ Conservation p.72
- ▶ Deserts p.78
- ▶ Early humans p.82
- ▶ Language p.144
- ▶ World p.275



About Africa



Population:
1.216 billion



Highest point:
Mount Kilimanjaro



Lowest point:
Lake Assal



Biggest desert:
Sahara Desert



Longest river:
Nile

Ancient pyramids

The pyramids in Egypt were built 4,500 years ago. Inside are the tombs of Egyptian rulers called pharaohs.

The pyramids are made of millions of stone blocks that were cut and dragged into place.



Aircraft

Aircraft are types of transportation that let us travel through air. They take people on vacations and are also used by fire crews, doctors, and farmers. Types of aircraft include planes, helicopters, and hot air balloons.

SEE ALSO

- Atmosphere p.33
- Birds p.39
- Forces p.108
- Gravity p.125
- Transportation pp.258-259

Plane

The Airbus A380 is the world's largest passenger plane. It can carry more than 800 people on two decks and fly nonstop from the US to Australia.

Ailerons are flaps on each wing that move up and down to turn the aircraft.

This 79 ft (24 m) high tail section has a flap called a rudder that steers the aircraft.

This part of the tail helps to keep the aircraft flying straight.

The cockpit is where the pilot and copilot fly the aircraft.

The A380 is powered by four huge jet engines, each as long as a family car.

Helicopter

A helicopter uses fast-spinning blades, called rotors, to fly through the air. These rotor blades raise the helicopter and move it forward. Smaller rotors on the tail keep the craft straight.



How aircraft fly

The green arrows show the four pushes and pulls, or forces, acting on an aircraft during flight. The weight of gravity pulls it down, lift raises it up, thrust moves it forward, and drag pulls it back. The pilot uses the controls to manage these forces to take off, fly, and land safely.



American West

Between the years of 1840 and 1900, many people living in the eastern United States moved west to find adventure and start new lives. Some, called settlers, set up farms and cattle ranches. Others dug mines in search of gold.

Wagon train

Settlers traveled west in groups of wagons called wagon trains. They took with them everything they needed to make their new homes.

Wagons were pulled by horses or oxen. They carried people, supplies, and farming equipment.

A wagon train could have up to 250 wagons.



SEE ALSO

- ▶ Gold pp.200-201
- ▶ Native Americans p.179
- ▶ North America p.184
- ▶ Transportation pp.258-259
- ▶ Trains p.260

Native American wars

Wars were fought between the settlers and the local Native American tribes. Despite winning some battles, the tribes ultimately lost the wars and their land.



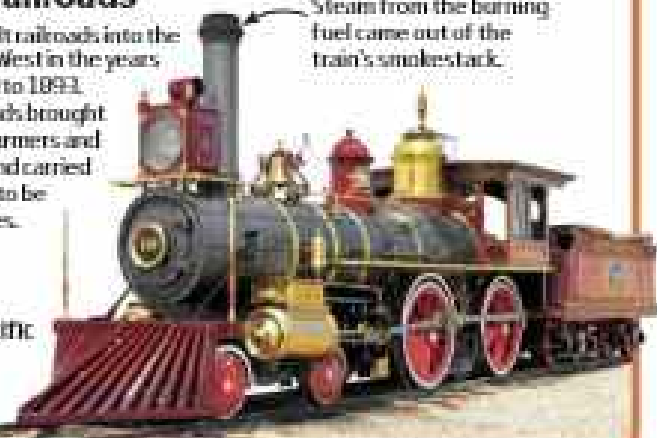
Painting of Custer's Last Stand, a Native American victory

Early railroads

People built railroads into the American West in the years from 1869 to 1893. The railroads brought in lots of farmers and settlers, and carried out goods to be sold in cities.

Steam from the burning fuel came out of the train's smokestack.

Union Pacific Railroad No.119



Amphibians

Amphibians are animals that spend their lives in and around water. They hatch in water from eggs. As they grow they develop lungs that let them breathe on land. Amphibian skin must always be wet, so they stay close to water. There are three main groups of amphibian.

SEE ALSO

- ▶ Eggs p.86
- ▶ Water pp.120-121
- ▶ Invertebrates p.139
- ▶ Metamorphosis p.163
- ▶ Reptiles p.210
- ▶ Skin p.229

Newts and salamanders

These long-tailed amphibians can replace injured body parts by growing new ones. An eye, leg, or tail takes just weeks to replace.

Bright yellow markings warn attackers that the fire salamander is poisonous.



Fire salamander

Caecilians

Though they look like worms, caecilians (see-sill-yens) are amphibians. They live underwater or in underground burrows on land.



Congo caecilian

Forever young

Axolotls are unusual, because they stay in water all their lives. They keep their tadpole-like fins and feathery gills even as an adult.



Axolotl

Frogs and toads

The most common group of amphibians includes frogs and toads. Frogs are usually wetter, smaller, and smoother than toads.

Toads have dry, bumpy skin, unlike frogs that have wet, smooth skin.



Oriental fire-bellied toad

Large, bulging eyes help toads and frogs to see in all directions.

Amphibians can breathe through their skin.



Green tree frog

Webbed feet on the back legs help frogs and toads to swim in water.



Ancient China

Chinese culture is thousands of years old. By 200 BCE, China was united under a family of emperors called the Han Dynasty from the city of Hanzhong. China produced many new inventions that spread to the rest of the world.

SEE ALSO

- ▶ Eating pp.104-105
- ▶ Inventions pp.136-137
- ▶ Exploration pp.180-181
- ▶ Trade p.257
- ▶ Writing p.280



This silk costume was worn by a woman in a Chinese opera.

Inventions

The Chinese first made silk clothes about 5,500 years ago. They also invented paper, gunpowder, printing, mechanical clocks, the compass, porcelain, and umbrellas.

The Great Wall

Chinese emperors built strong walls to keep out northern tribes. The Great Wall of China is 5,500 miles (8,850 km) long and 500 years old.

The wall has 25,000 watchtowers. Soldiers used them to look out for the enemy.

Hilltop walls were easier to defend.

The top of the wall is wide enough for soldiers to march along.

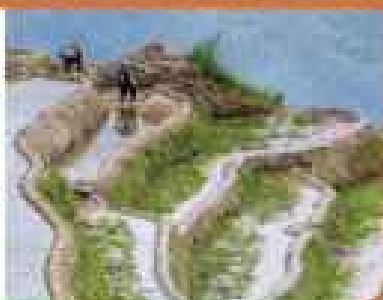


Writing

Chinese writing dates to about 1400 BCE. Each of the 4,000 symbols represents a word or part of a word. The text is read from top to bottom and right to left.

Rice-growing

Rice was first grown in China about 10,000 years ago. It is still farmed today. Rice grows in flooded fields, such as these terraces.



Ancient Egypt

Thousands of years ago, between 7000 BCE and 395 CE, Egypt was led by powerful rulers called pharaohs. The Egyptian people farmed the land next to the Nile river and built amazing monuments for their pharaoh and the gods.

Nile River

The Nile was very important in Egyptian life. Farmers grew food along the river's banks and people used it to travel up and down the whole country.

SEE ALSO

- Buildings p.48
- Governments p.123
- Life cycle p.146
- Rivers p.211
- Ships p.224
- Weather p.271



Pyramids

The pyramids were tombs built to protect the pharaoh when he died. They were filled with treasure for the pharaoh to use in the afterlife. The largest pyramid is nearly 450 ft (140 m) tall.

The mummy was put in a painted wooden mummy case.



Words were written with picture signs, called hieroglyphs.

Living forever

The Egyptians turned their bodies into mummies when they died. The body was dried out using salt, then wrapped in bandages. This way, they hoped they would live forever.

Ancient Greece

The ancient Greeks were among the most creative people in history. They were great builders and artists who invented theater, politics, history, writing, science, and sports. Some of the words they created are still used today. Greek civilization was at its best between 510 and 323 BCE.

SEE ALSO

- ▶ Ancient Rome p.20
- ▶ Buildings p.48
- ▶ Crafts p.75
- ▶ Religion p.208
- ▶ Games pp.240-241
- ▶ Sports p.239

Parthenon

The most famous Greek temple is the Parthenon in Athens. It was built for the goddess Athena, protector of the city.

The Parthenon stands on the Acropolis, a hill overlooking Athens.

The Parthenon was built with white marble.

There were 46 main columns.

Greek pottery

Greek vases were often painted with scenes from myths. This vase shows one of 12 labors, or tasks, carried out by the mythical hero Hercules.



Vases like this were all-purpose containers, for oil, wine, honey, or other foods.

Olympics

The Greeks held the first athletic competitions, such as the Olympic Games. This lifelike statue shows someone throwing a round weight, called a discus.

Gods and goddesses

The Greeks worshipped dozens of gods. Here are six of the most important ones, who are all members of the same family.



Zeus, king of the gods



Aphrodite, goddess of love



Apollo, god of music



Poseidon, god of the sea



Artemis, goddess of hunting



Hades, god of the Underworld



Ancient India

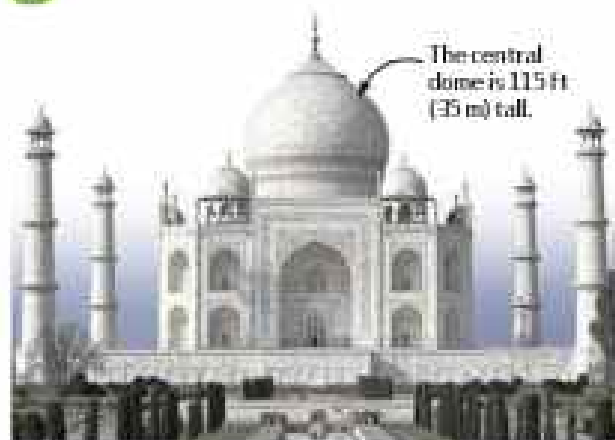
Great cities were built in India 5,000 years ago. This marked the start of a unique civilization. In the years 1526-1857, India was united as the Mughal Empire. The Mughals made scientific breakthroughs and beautiful works of art.

SEE ALSO

- ▶ Art p.28
- ▶ Asia p.29
- ▶ Astronomy p.32
- ▶ Buildings p.48
- ▶ Religion p.208
- ▶ War pp.278-279

Mughal Empire

The Mughal Empire was founded by Babur, a ruler from Central Asia who conquered northern India in 1526. The Empire was ruled by Babur's family for more than 300 years.



The central dome is 115 ft (35 m) tall.

Taj Mahal

Built by the Mughal ruler Shah Jahan in 1632, the Taj Mahal is the tomb of his wife, Mumtaz Mahal. The huge white marble building took 20,000 men more than ten years to finish.

This globe is marked with the positions of the stars.

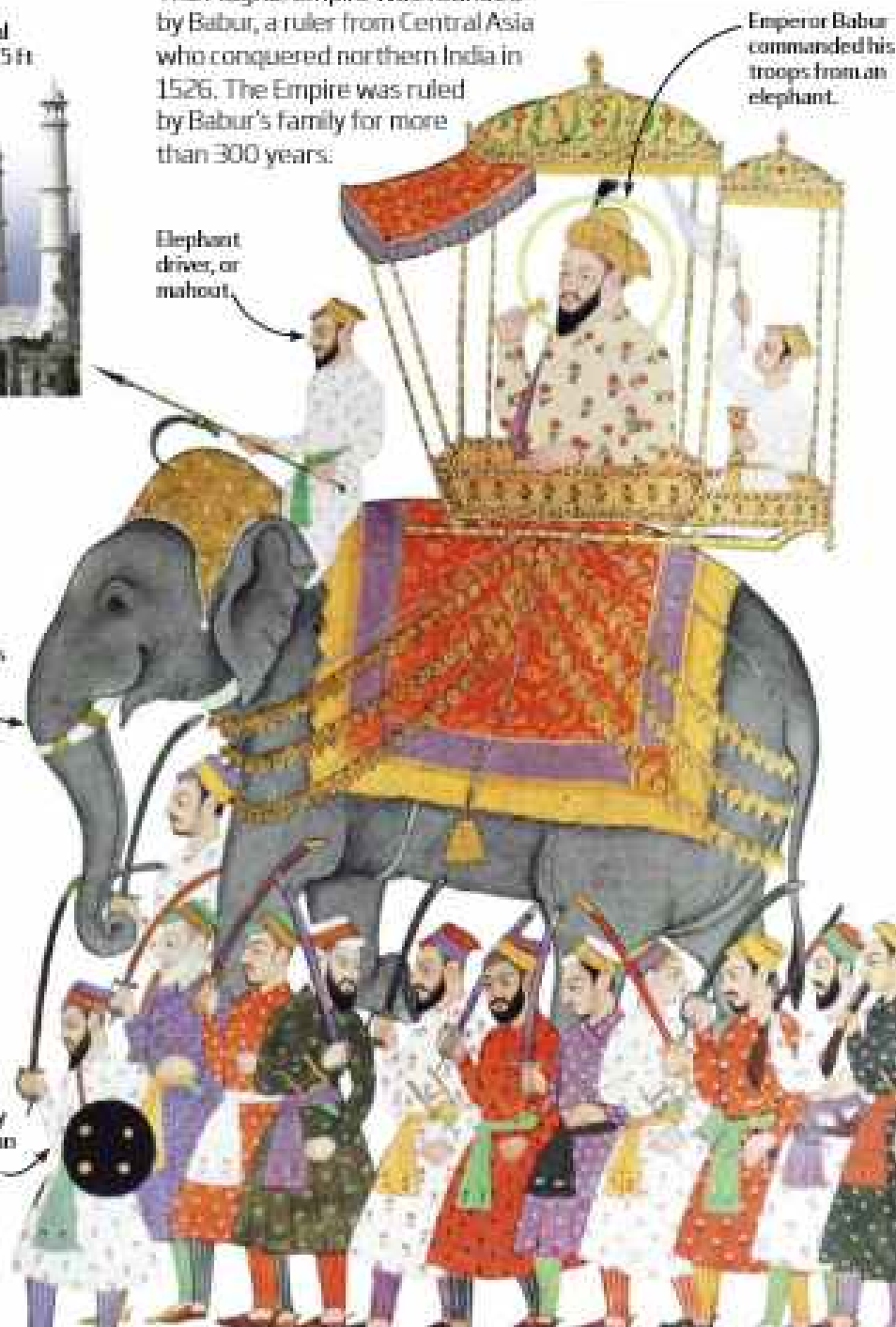


Science

The Mughals studied the stars and made brass globes to map their positions. Scientists invented shampoo and new ways to work with metal.

War elephants used their trunks and tusks to kill the enemy.

Babur's army had more than 80,000 men.



Emperor Babur commanded his troops from an elephant.

Elephant driver, or mahout.

Ancient Rome

About 2,000 years ago, the ancient Romans ruled a great empire, including all the lands around the Mediterranean Sea. The Roman Empire was well organized, and it lasted for hundreds of years.

SEE ALSO

- Buildings p.48
- Europe p.94
- Governments p.123
- Law p.145
- Maps p.155
- Slavery p.230
- War pp.278-279

Roman society

Within the empire, there were different groups of people, with different rights. Citizens had more rights than non-citizens, and slaves had no rights at all.

Emperor
The ruler of the empire, who was all-powerful.

Citizen
Only citizens were able to vote and become government officials.

Freed person
A former slave, freed by their owner.

Slave
A person owned as property.



Roman Empire

The Roman Empire started as just one city, Rome, in what is now Italy. Over time it grew across Europe.

Roman soldiers

The Romans conquered their empire using well-trained armies, called Legions. This is a centurion, an officer in charge of 80 legionnaires (foot soldiers).

Centurions wore a helmet crest.

Armor was made from metal links called chain mail.

Shin guards



Only Roman citizens and above were allowed to wear togas.

Roman buildings

The Romans were expert builders, and many of their buildings still stand. This is the Pont du Gard, an aqueduct that carried water to Nîmes, a Roman city in France.



Animal families

Animals live in many different types of family groups. Some animals live in big groups, called colonies. They work together to bring up their babies. Other animals form male and female pairs. Family groups help animals to survive.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Animal homes p.23
- ▶ Birds p.39
- ▶ Insects p.134
- ▶ Mammals p.154
- ▶ Homes pp.244–245

Pair

After mating, a pair of emperor penguins take turns looking after the egg and then feeding the baby. They live with up to 5,000 other penguins.



Herd

Zebras move around in large groups, called herds. When babies are born, the large numbers help protect them from other animals' attacks.



Colony

Ant families can be very large. The queen is the leader of the colony and lays the eggs. The other ants work to protect and feed the colony.



Small family

After giving birth, a female river otter cares for her pups for two to three years until they are ready to hunt and look after themselves.



Animal groups

Animals can be divided into groups, depending on their body features. Animals that look and act in similar ways are grouped together. This is called classification.

SEE ALSO

- ▶ Animal families p.21
- ▶ Fish p.101
- ▶ Insects p.134
- ▶ Invertebrates p.139
- ▶ Spiders p.238
- ▶ Vertebrates p.266

The green band shows which animal groups are vertebrates.

Amphibians
These animals have wet skin. They live in or near water. The babies hatch from eggs, and change-body shape to become adults.

Birds
Birds have feathers, which keep them warm and help most to fly. They have beaks to catch or pick up food.

Animal groups

Animals are divided into two main groups. Vertebrates have spines, invertebrates don't. There are more groups inside these two main groups.

Invertebrates
There are many groups of invertebrates, including insects, slugs, spiders, and shellfish.

Fish
Fish live in water. They are covered in bony plates, called scales, and have special organs called gills for breathing.

Reptiles
Reptiles have scaly skin. They are cold-blooded, which means they must warm up in the sun before they can move.

Mammals
Mammals have fur or hair on their bodies. They feed their babies on milk made by the mothers.

Animal homes

Animals need homes for shelter and to keep their young safe. Animal homes are built in many different places and in all shapes and sizes. Some animals work together to build large structures. Others move every night, making new homes as they go.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Birds p.39
- ▶ Insects p.134
- ▶ Mammals p.154
- ▶ Homes pp.244-245
- ▶ Work p.274

Weaver bird nest

Male weaver birds loop leaves and grass together to build their nests. The entrance is at the bottom to stop other animals from getting in.



Termite mound

Termites work together to build large mounds. The chimney shape of the mound helps to keep the termites inside cool.

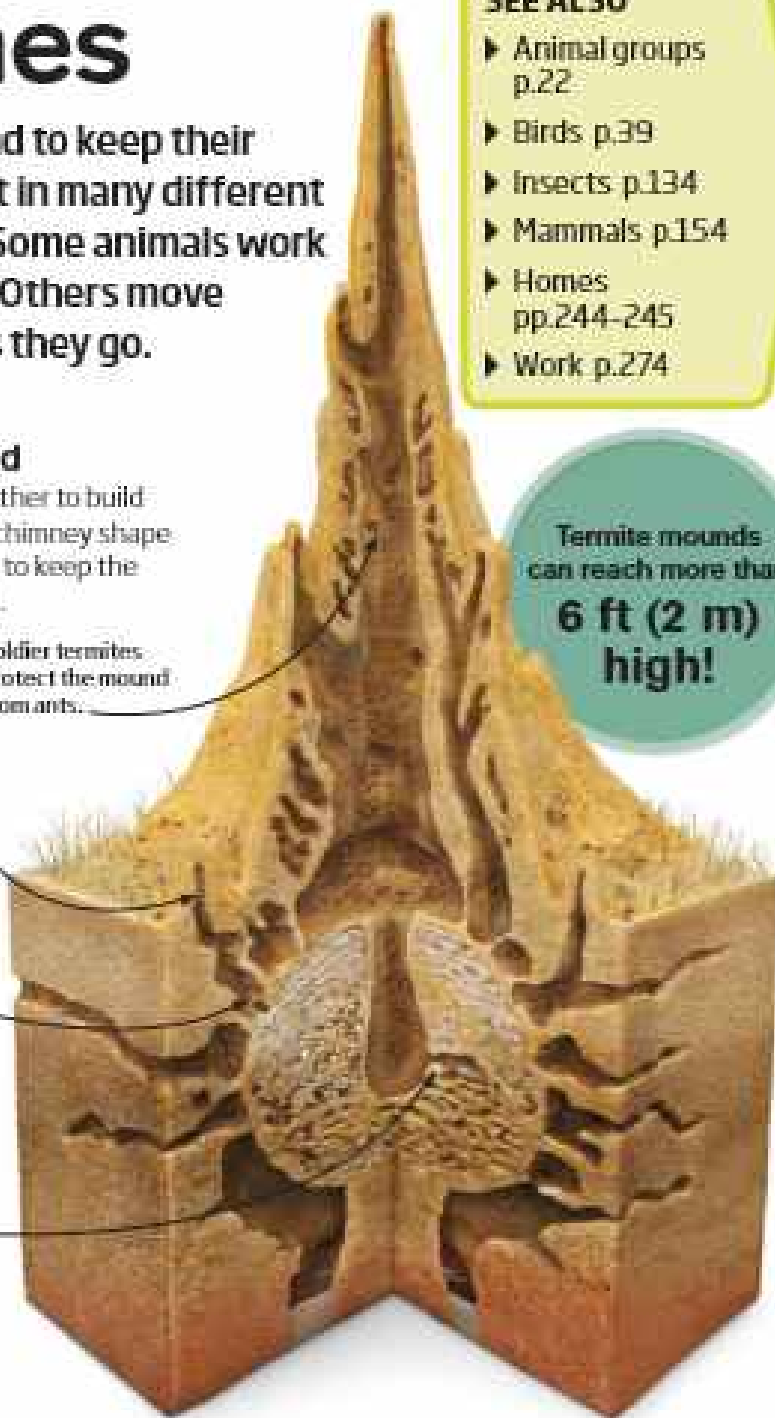
Soldier termites protect the mound from ants.

Turrets are built for different entrances and exits.

The termite workers store grass in the outer areas.

The queen termite lives in the center of the colony.

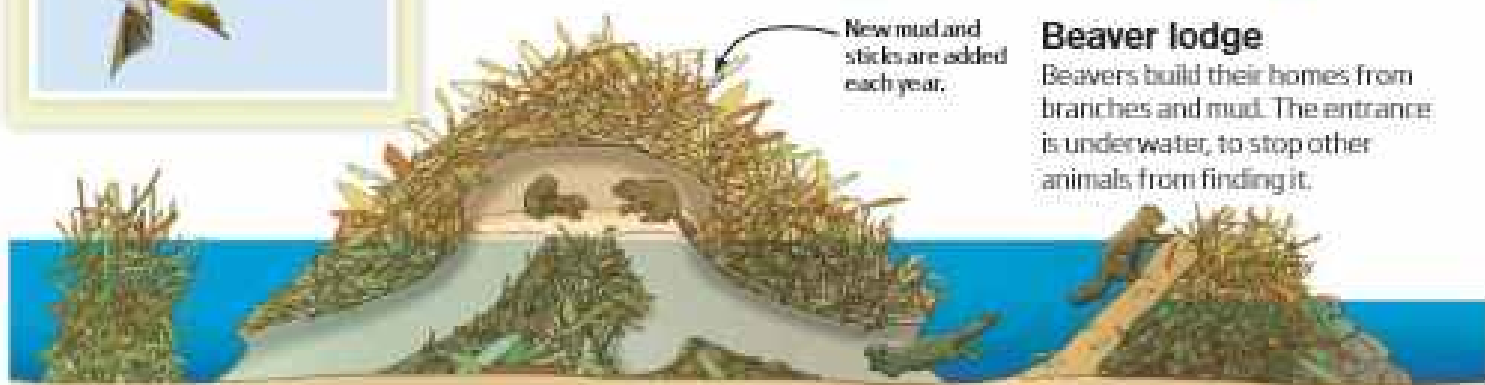
Termite mounds can reach more than **6 ft (2 m)** high!



New mud and sticks are added each year.

Beaver lodge

Beavers build their homes from branches and mud. The entrance is underwater, to stop other animals from finding it.

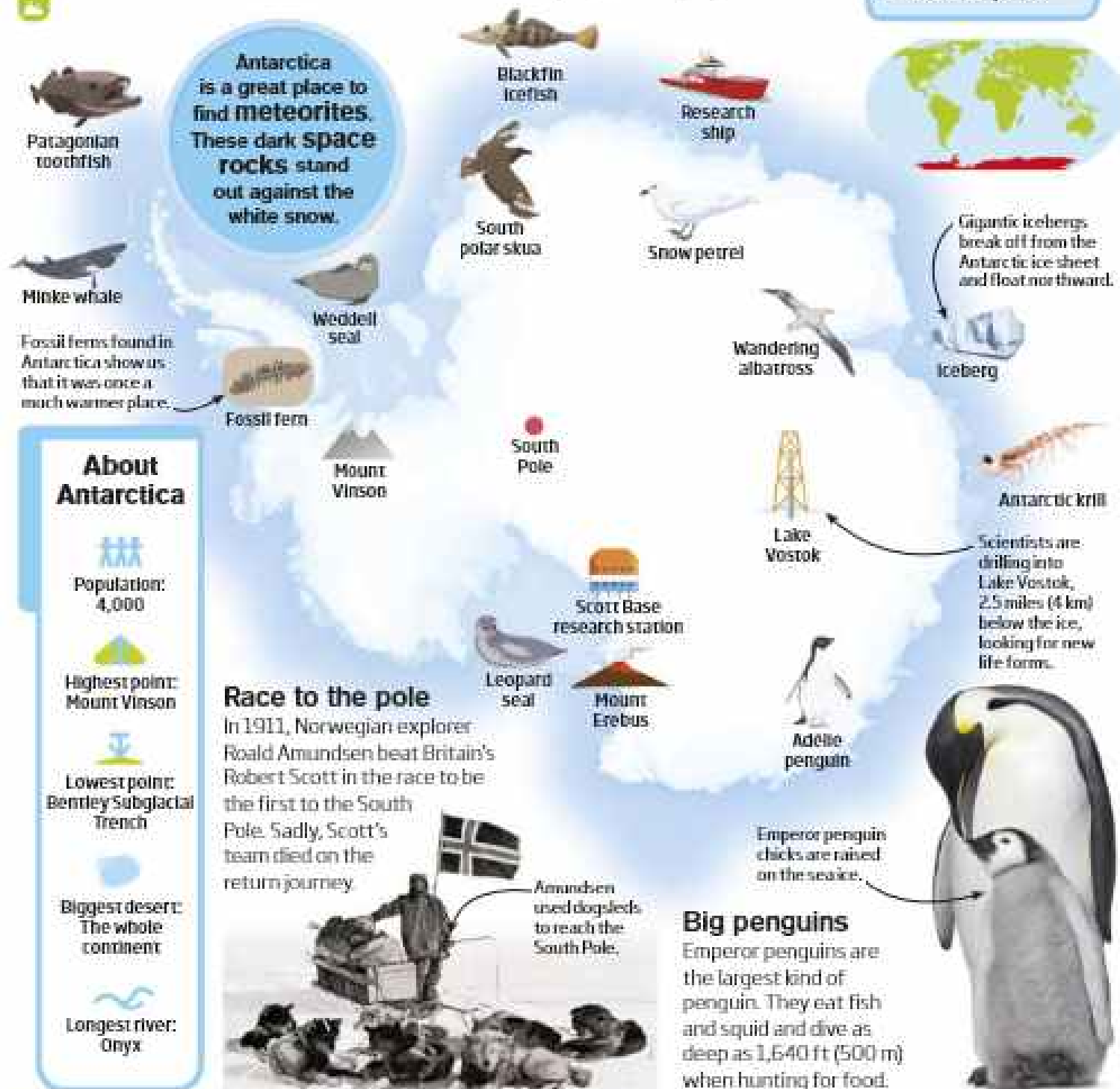


Antarctica

Antarctica is the fifth largest and most southern continent. It is very cold and windy. Most of the land is buried under huge ice sheets, which stretch far out into the sea. In winter, it can be as cold as -130°F (-90°C), and windspeeds during storms can reach an incredible 200 mph (320 kph).

SEE ALSO

- ▶ Arctic p.25
- ▶ Birds p.39
- ▶ Changing world pp.50-51
- ▶ Climate change p.60
- ▶ Explorers p.96
- ▶ Glaciers p.122



About Antarctica



Population:
4,000



Highest point:
Mount Vinson



Lowest point:
Bentley Subglacial Trench



Biggest desert:
The whole continent



Longest river:
Onyx

Race to the pole

In 1911, Norwegian explorer Roald Amundsen beat Britain's Robert Scott in the race to be the first to the South Pole. Sadly, Scott's team died on the return journey.



Amundsen used dog sleds to reach the South Pole.

Big penguins

Emperor penguins are the largest kind of penguin. They eat fish and squid and dive as deep as 1,640 ft (500 m) when hunting for food.



Emperor penguin chicks are raised on the seacoast.

Arctic

The Arctic is a cold region that surrounds the North Pole. It is mostly sea, which is frozen for much of the year, but it also includes most of Greenland and the northernmost parts of North America, Europe, and Asia.

SEE ALSO

- ▶ Antarctica p.24
- ▶ Climate change p.60
- ▶ Oceans and seas p.187
- ▶ Polar habitats p.197
- ▶ World p.275

Chilly circle

The Arctic Circle is about twice the size of the US. The animals that live there need to be able to survive in the cold. Land animals have thick fur or fluffy feathers, and sea animals have a thick layer of fat.

About the Arctic



Population:
At least 500,000 in summer, but fewer in winter



Highest point:
Gunnbjørn Fjeld, Greenland



Lowest point:
Arctic Ocean



Biggest desert:
Arctic Polar Desert



Lemmings are small rodents. They eat plants and live in burrows.

The Nenets people of Arctic Russia are reindeer herders who live in tents made of reindeer skin.

55 million years ago, the Arctic was so hot that there was no ice, and alligators lived in the sea.

All change

Our planet is getting warmer, and the Arctic sea ice is melting. This means that ships can now travel across the Arctic between the Atlantic and Pacific oceans.

Polar bears

Polar bears live in the Arctic and roam the sea ice, hunting seals. The bears are at risk from climate change because the sea ice is melting. This makes it harder for the bears to find seals to eat.



The **story** of...

Color

Our world is full of beautiful colors. The colors we see around us are actually different types of light bouncing off objects and into our eyes. Colors also have meanings. For example, a red light means “stop” in traffic, and a white flag means “surrender” in war.

Paint

Artists make paint by mixing something called a pigment with water. A pigment is a material that changes the color of reflected light by absorbing some colors and reflecting other colors.

Artists mix paint on a palette.

Rainbow colors

White light is made from all the colors of the rainbow—red, orange, yellow, green, blue, indigo, and violet. When white light from the sun passes through rain, it can refract (split) to make a rainbow.

The male peacock is brightly colored compared to the female.

Animal colors

Male birds are often colorful to attract female birds. Some animals use bright colors as a warning, and some can change color. Others use their coloring to blend in with their surroundings.

Female peacock

Male peacock



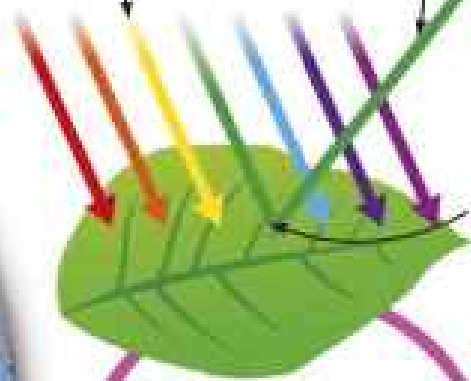
All the colors of light except for green are absorbed by the leaf.

Green light reflects into our eyes.



Eye

The green light is not absorbed and bounces off the leaf.

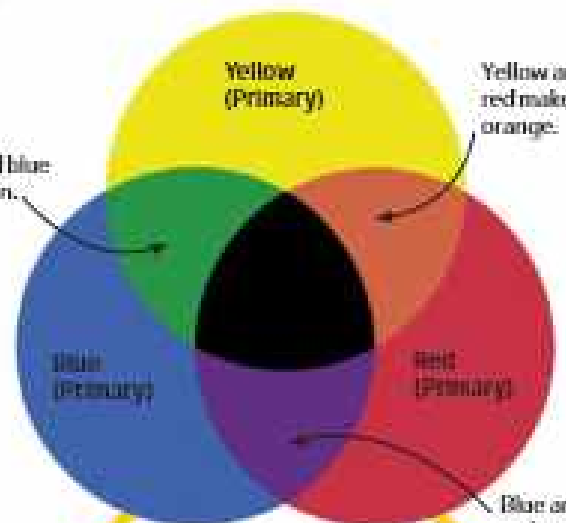


Reflecting colors

A plant looks green because green light reflects off it into our eyes. The other colors of light from the sun are absorbed by the plant's leaves.

Yellow and blue make green.

Yellow and red make orange.



Blue and red make purple.

Mixing colors

Primary paint colors can be mixed to make secondary colors. The primary colors are red, yellow, and blue. Secondary colors can also be mixed to make new colors—for example, mixing orange and green makes brown.

Fashion

All around the world, different styles and colors are used in clothing to help people express themselves. Fashion changes over time. Clothes are very different now from those worn 250 years ago.

Silk dress from 1750



Art

When you create a picture or sculpture, you are making art. Art can show something from real life or the imagination, and is made of different materials. Throughout history, people have drawn or painted pictures of the world around them. Art helps us show our feelings about the world and creates beautiful objects to look at.

SEE ALSO

- ▶ Ancient Rome p.20
- ▶ Color pp.26-27
- ▶ Buildings p.48
- ▶ Crafts p.75
- ▶ Photography p.190
- ▶ Stone Age p.243

Painting

Painters use an object such as a brush loaded with colored paint to create an image on paper, board, or canvas. Paintings can be detailed or simply show lines and shapes.

Stone Age handprints at Cueva de las Manos



Cave painting

The first paintings were made on cave walls around 40,000 years ago. They showed handprints, people, and animals.

Bedroom in Arles by Vincent van Gogh, 1888



Lifelike painting

Many paintings show lifelike images from the world. This could be an indoor or an outdoor view, a person, or an object.

Landscape at Ceret by Juan Gris, 1911



Abstract painting

Abstract art uses colors and shapes to show amazing pictures that aren't lifelike, but could show something real.

Sculpture

Sculptors work in clay, wood, stone, metal, or other materials to create objects. These might show people or abstract shapes.

Dancer Adjusting her Shoe by Degas, c.1890



Drawing

Artists use pencils, colored crayons, chalk, charcoal, and ink to draw beautiful images on paper. Drawing is quicker than painting, so it is a good way of recording real-life scenes.

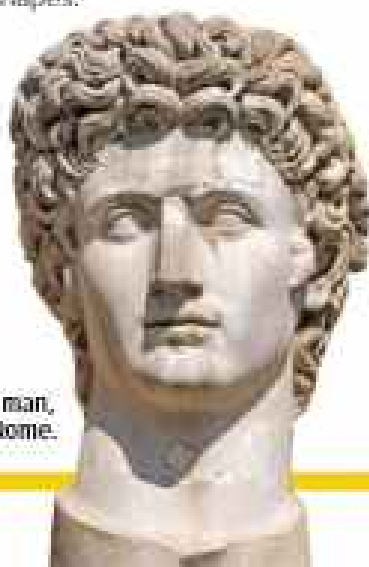
The Great Wave by Hokusai, c.1831



Printing

Pictures can be cut out of a material such as wood and covered in ink or paint. The ink or paint on the cutout picture is then transferred onto paper, to create a print.

Sculpture of a man, from ancient Rome.



Asia

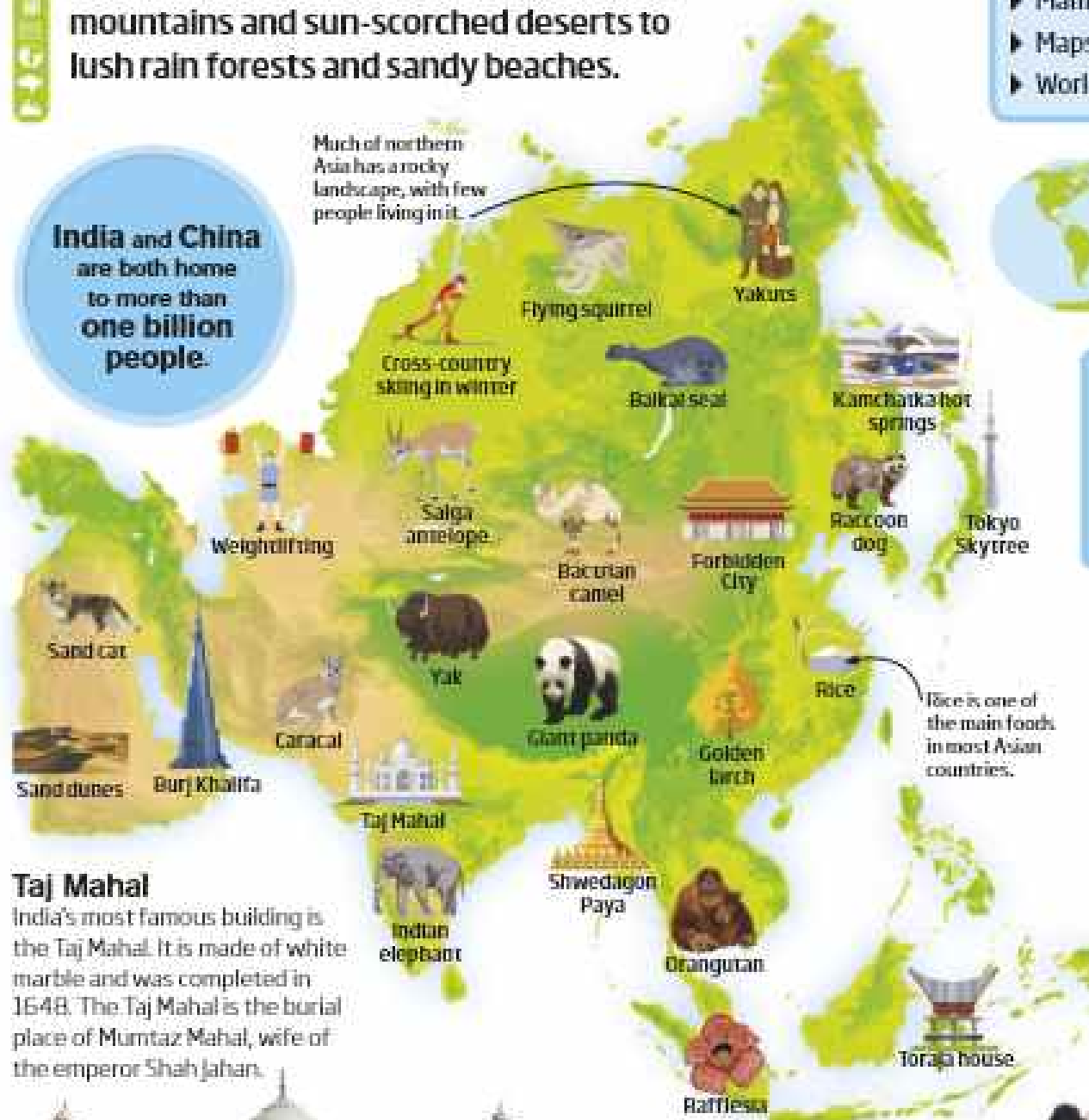
The biggest continent on Earth is Asia. More than 60 percent of the world's population live here. Asia has many landscapes, from snow-capped mountains and sun-scorched deserts to lush rain forests and sandy beaches.

SEE ALSO

- ▶ Ancient China p.16
- ▶ Ancient India p.19
- ▶ Buildings p.48
- ▶ Mammals p.154
- ▶ Maps p.155
- ▶ World p.275

India and China are both home to more than one billion people.

Much of northern Asia has a rocky landscape, with few people living in it.



Taj Mahal

India's most famous building is the Taj Mahal. It is made of white marble and was completed in 1648. The Taj Mahal is the burial place of Mumtaz Mahal, wife of the emperor Shah Jahan.



About Asia



Population: 4.427 billion



Highest point: Mount Everest



Lowest point: Dead Sea, Israel



Biggest desert: Arabian Desert



Longest river: Yangtze, China

Pandas

These rare black-and-white bears live in the mountains of China. They spend most of their time eating bamboo. Pandas are a symbol of peace and friendship in China.



Asteroids

Asteroids are rocky or metallic objects that travel around the sun. They formed at the same time as the planets. Most asteroids are covered in craters, or dents, from where they have smashed into each other.

SEE ALSO

- Gravity p.125
- Metals p.162
- Meteorites p.164
- Rocks and minerals p.214
- Solar system p.233

Asteroid shapes

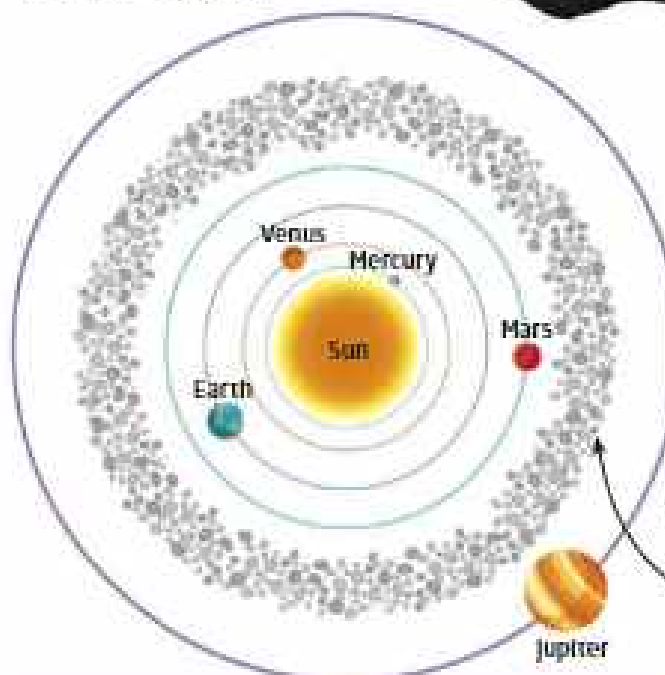
Most asteroids have uneven shapes, but the biggest asteroids are round. The biggest asteroids are also called dwarf planets.

The craters on an asteroid's surface are made by bumping into smaller asteroids.

The asteroid Toutatis is 3 miles (5 km) long.

Scientists think Toutatis is made of two separate pieces held together by gravity.

It takes Toutatis four years to travel around the sun.



The main belt

Most asteroids travel around the sun in the "main belt" between Mars and Jupiter. There are millions of asteroids in this belt, but they are very far apart.

The asteroid belt

Asteroid mining

Scientists think that in the future people will mine asteroids for their metals, minerals, and water. Spaceships may stop off on asteroids as they travel around the solar system.



Astronauts

Astronauts are people who are specially trained to take part in missions in space. They help us learn more about the universe we live in. Fewer than 600 people have been into space, and only 12 have walked on the moon.

The helmet visor has a special filter that shields sunlight, as it can be extremely bright in space.

A camera records what the astronaut is seeing.

The backpack holds oxygen supplies for breathing.

Tools can be attached to the front of the spacesuit.

The suit is made of many layers of fabric that keep the astronaut safe and warm.

In 1961, Russian space explorer Yuri Gagarin became the first human to go into space.

Astronaut suit

Space can be both very hot and very cold. To keep themselves safe, astronauts wear special suits with helmets, gloves, boots, and an air supply.

SEE ALSO

- Explorers p.96
- Moon p.171
- Exploration pp.180-181
- Solar system p.233
- Space travel p.237
- Universe p.263



The International Space Station

The International Space Station is a permanent base for astronauts. The station is 250 miles (400 km) above the Earth. It can house six astronauts at a time.

Becoming an astronaut

It takes many years of work to become an astronaut. Astronauts have to learn many new skills. They also train hard to make sure they are fit and healthy.



Astronauts train underwater because they float weightlessly, like in space.

Astronomy

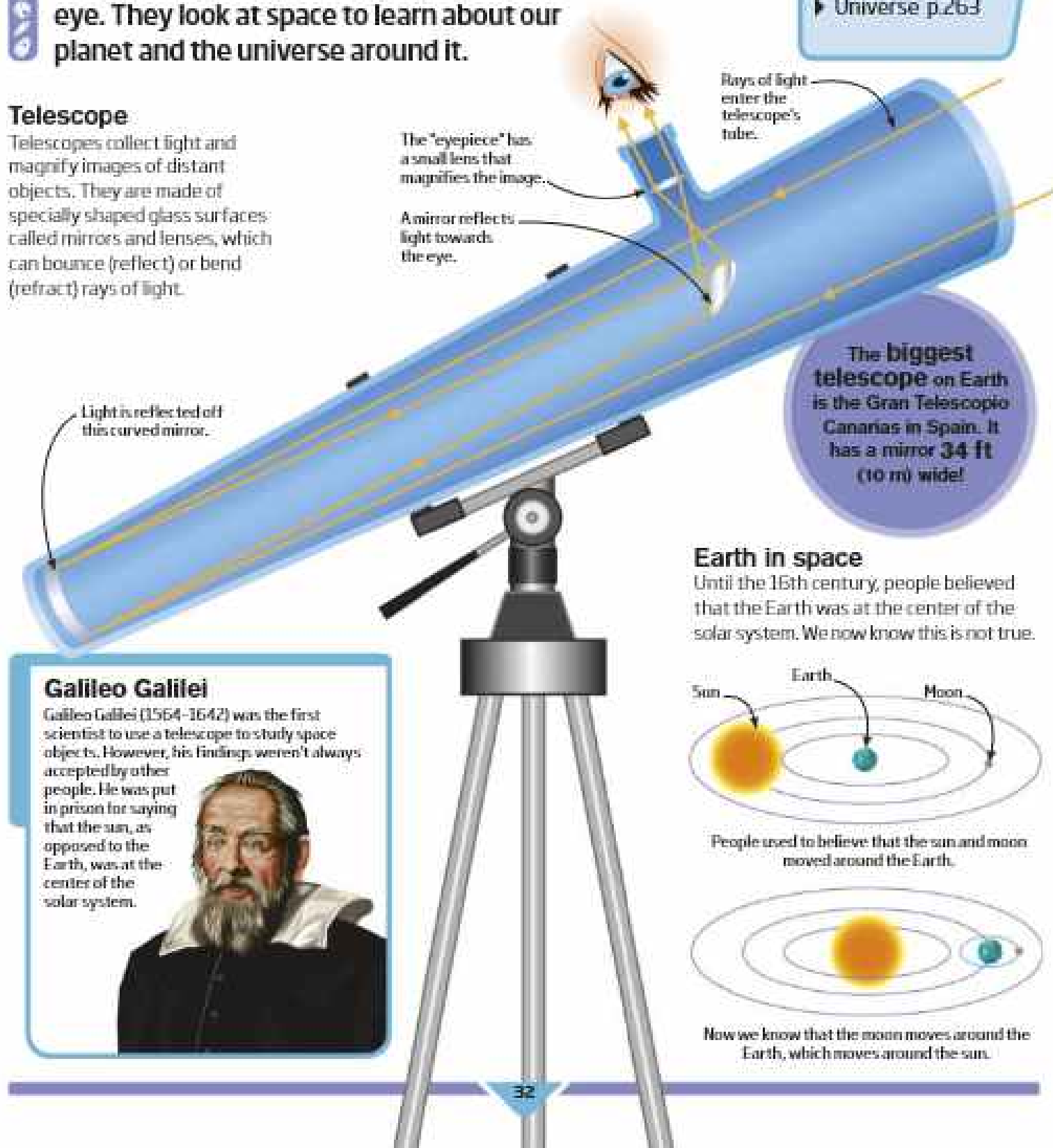
Astronomy is the science of studying space. The first astronomers looked at the night sky with just their eyes. Modern astronomers use binoculars and telescopes to look at things too far away to be seen with the naked eye. They look at space to learn about our planet and the universe around it.

SEE ALSO

- ▶ Earth p.83
- ▶ Light p.147
- ▶ Science p.217
- ▶ The sciences pp.218-219
- ▶ Sun p.247
- ▶ Universe p.263

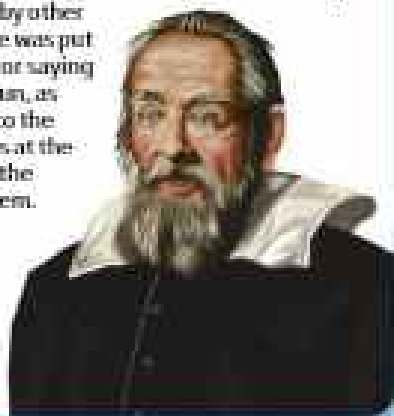
Telescope

Telescopes collect light and magnify images of distant objects. They are made of specially shaped glass surfaces called mirrors and lenses, which can bounce (reflect) or bend (refract) rays of light.



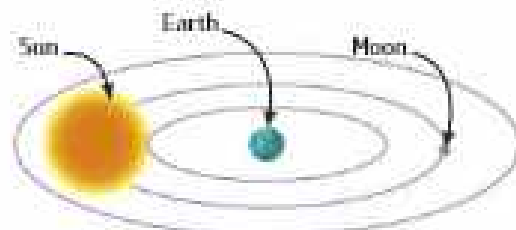
Galileo Galilei

Galileo Galilei (1564-1642) was the first scientist to use a telescope to study space objects. However, his findings weren't always accepted by other people. He was put in prison for saying that the sun, as opposed to the Earth, was at the center of the solar system.

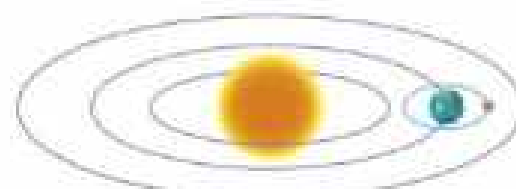


Earth in space

Until the 16th century, people believed that the Earth was at the center of the solar system. We now know this is not true.



People used to believe that the sun and moon moved around the Earth.



Now we know that the moon moves around the Earth, which moves around the sun.

Atmosphere

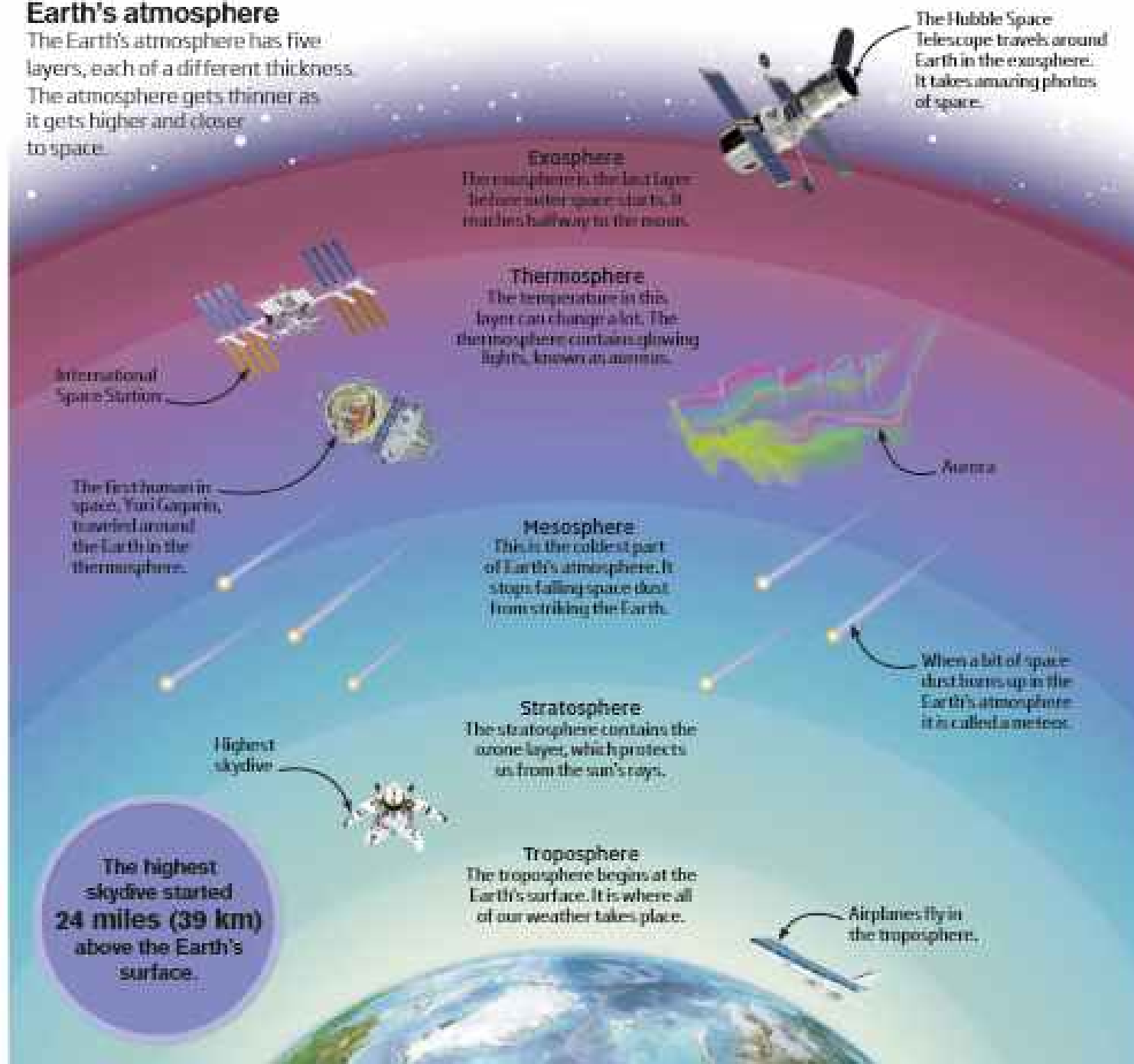
An atmosphere is a blanket layer of gases that surrounds a planet or moon. The Earth's atmosphere gives us the air we breathe. It also keeps our planet warm, blocks harmful rays from the sun, and stops space rocks from hitting us.

SEE ALSO

- ▶ Atoms p.34
- ▶ Earth p.83
- ▶ Gases p.117
- ▶ Meteorites p.164
- ▶ Sun p.247
- ▶ Temperature p.252
- ▶ Weather p.271

Earth's atmosphere

The Earth's atmosphere has five layers, each of a different thickness. The atmosphere gets thinner as it gets higher and closer to space.





Atoms

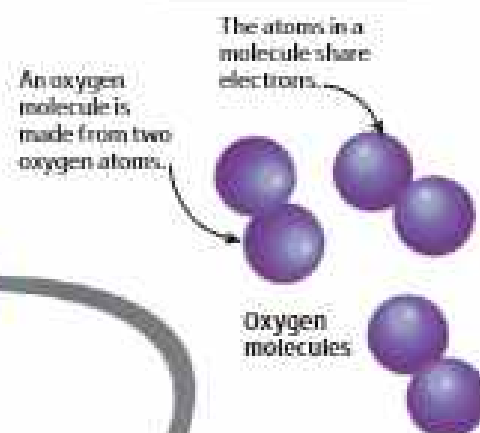
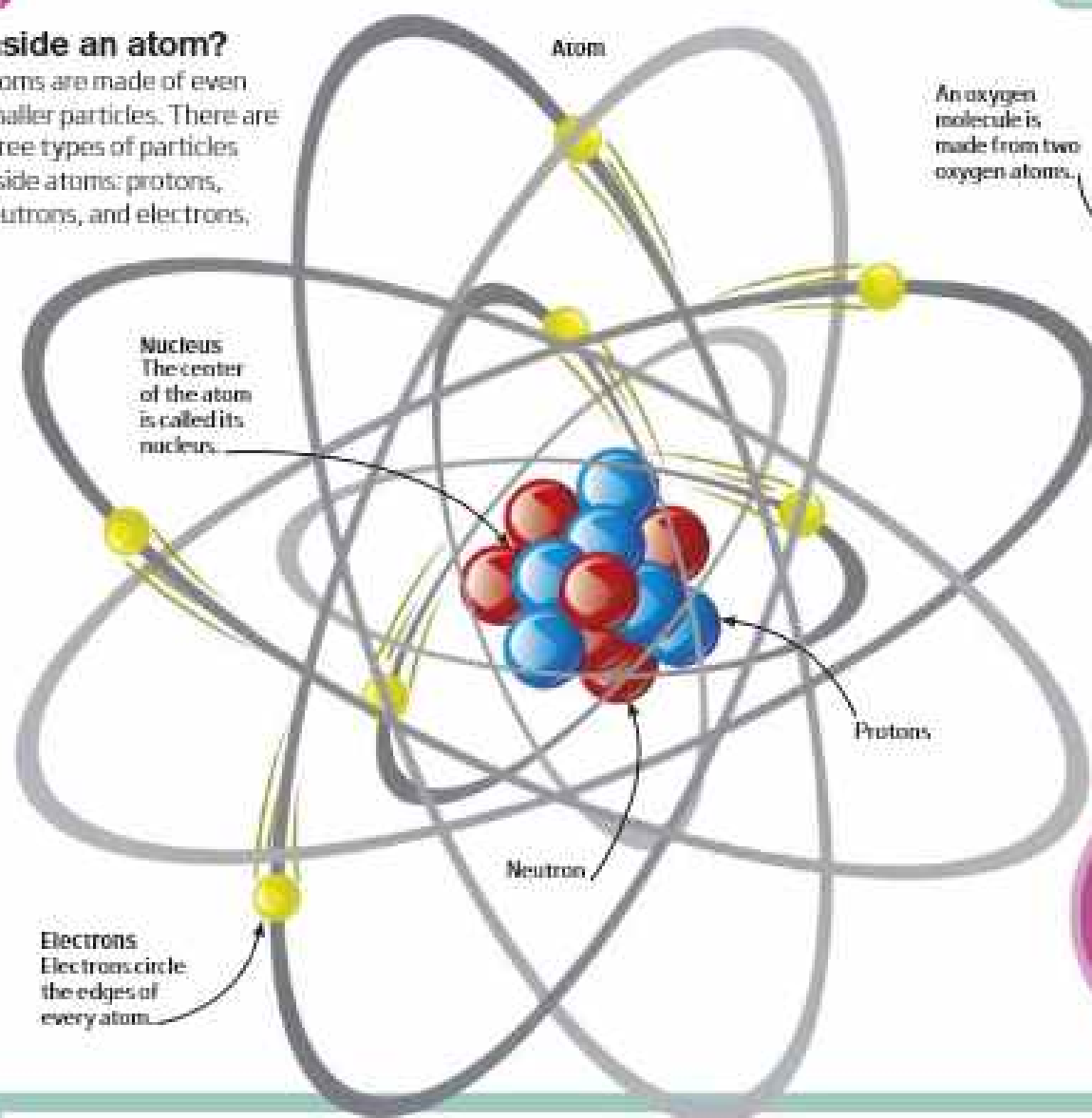
Atoms are tiny, round building blocks that build everything in the universe. They are so small we can't see them and are mostly made of empty space. Humans, cars, stars, and everything else around us are all made from tiny atoms.

SEE ALSO

- ▶ Carbon cycle p.49
- ▶ Changing states p.57
- ▶ Chemistry p.58
- ▶ Elements p.90
- ▶ Solar system p.233

Inside an atom?

Atoms are made of even smaller particles. There are three types of particles inside atoms: protons, neutrons, and electrons.



Molecules

Two or more atoms stuck together are called a molecule. Molecules can be long chains of atoms, or arranged in shapes.

There are
7 billion
atoms in the period
at the end of this
sentence—more than
there are people
on Earth.

Carbon atoms

Carbon joins with other atoms in molecules to form all living things. Carbon atoms can be arranged in different ways to make different materials.



Diamond



Pencil



Aztecs

The Aztecs were a people who lived in Central America. They had a huge empire between 1400 and 1519 CE, during which time they built great stone cities. Farmers produced corn and avocados, and bred turkeys.

Aztec city

At the center of each city stood a group of temples. Most temples were set on top of enormous stepped pyramids that could be more than 197 ft (60 m) tall.

SEE ALSO

- Buildings p.48
- Crafts p.75
- Farming p.98
- Incas p.132
- Maya p.158
- Religion p.208

The turquoise stone symbolized the breath of the gods.



Mask

The Aztecs used masks in religious ceremonies or to display in temples. This wooden mask is covered with pieces of a precious blue-green stone called turquoise.

Priests sacrificed animals and humans using knives of flint.

Many pyramids had twin temples on top.

Only priests and those who were to be sacrificed climbed the pyramid.

Pyramids and palaces were made of stone.

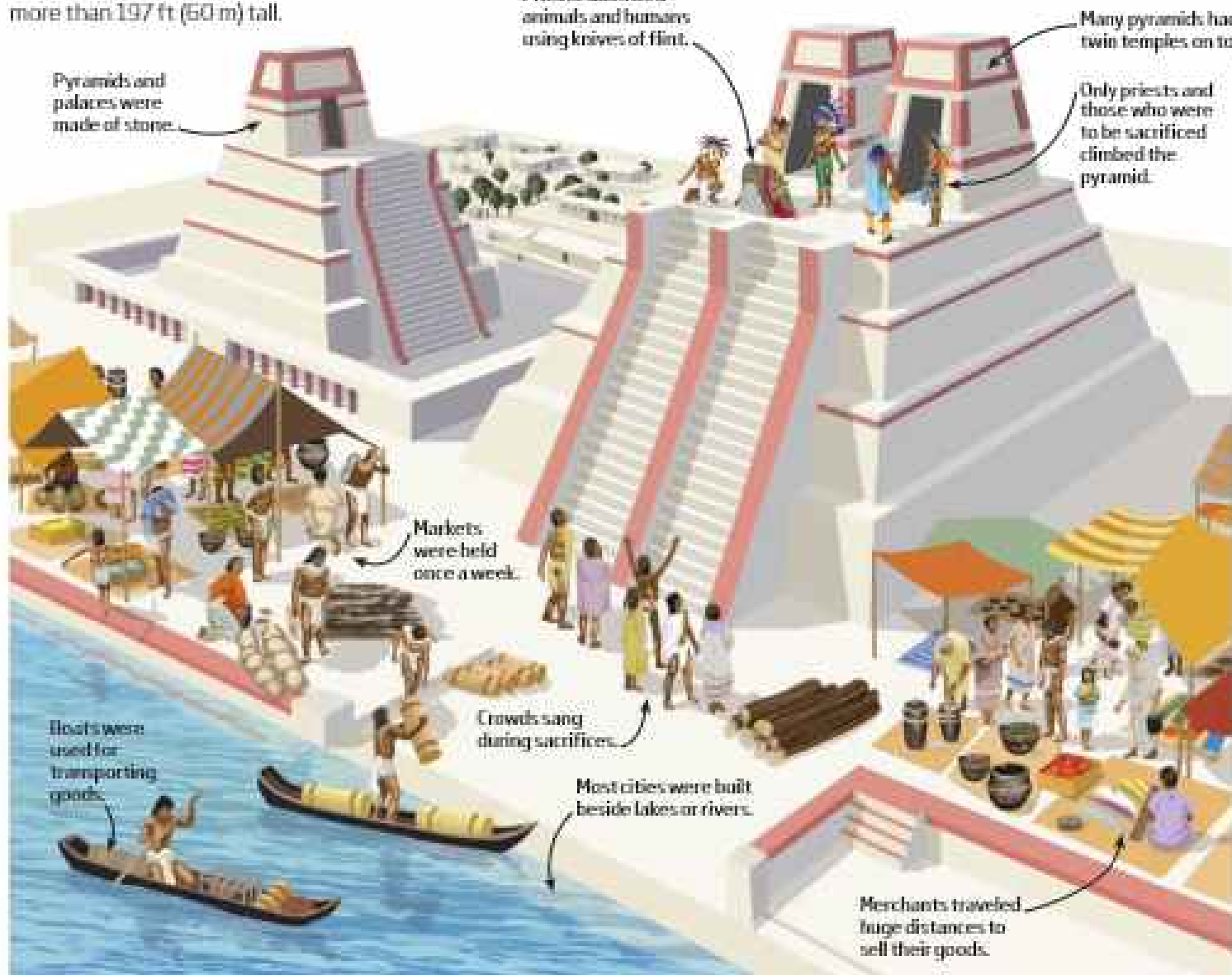
Markets were held once a week.

Crowds sang during sacrifices.

Most cities were built beside lakes or rivers.

Merchants traveled huge distances to sell their goods.

Boats were used for transporting goods.



Bicycles

A bicycle is a two-wheeled vehicle. There are many types of bicycles. Some are for cycling on the road, others for riding up and down mountains, and others for racing around a track. Cyclists wear helmets to protect their heads.

SEE ALSO

- Inventions pp.136-137
- Metals p.162
- Sports p.239
- Games pp.240-241
- Transportation pp.258-259

How a bicycle works

To move a bicycle forward, a cyclist pushes down on the pedals. These move a chain connected to the rear wheel. The bike is steered with a set of handlebars, which have brake levers to help the rider to slow down.



The hard outer covering and spongy inner material of bicycle helmets protect riders' heads if they fall.

The right lever applies the front brake, and the left lever applies the rear brake.

The curved handlebars on this bike allow the rider to crouch down for an easier ride.

This racing bike has very thin tires for extra speed.

The derailleur is a part of the bike that moves the chain to a different cog for going up or down hills.



Traditional spoked wheel

Carbon-fiber racing wheel

Bicycle wheels

Most bicycle wheels have traditional spokes (wire rods) that join the hub at the center of the wheel to the rim at the edge. Racing wheels have just a few large spokes.

Cycle racing

In the fastest type of bike racing, riders compete on bicycles without brakes or gears. They ride on sloped tracks in arenas called velodromes. Other types of races take place on mountain tracks and on roads. The most famous of these races is the 2,175 mile (3,500 km) Tour de France.



Big Bang

Scientists believe that the universe began in a dramatic event called the Big Bang, 13.8 billion years ago. The universe was tiny at the start, but it has been expanding ever since.

SEE ALSO

- Atoms p.34
- Elements p.90
- Gases p.117
- Light p.147
- Solar system p.233
- Universe p.263
- Volume p.269

1. The Big Bang

The universe was tiny and very hot. It started to grow.

2. Atoms

After 380,000 years, tiny particles formed, called atoms.

3. First stars and galaxies

The first stars and clusters of stars, called galaxies, appeared 100 to 200 million years after the Big Bang.

Most scientists think our universe will keep expanding forever.

4. Our solar system

The sun and the rest of the solar system formed nine billion years after the Big Bang.

5. Today's universe

The universe is still expanding. Scientists study this stretching of space by measuring how fast other galaxies are moving away from us.

Creation

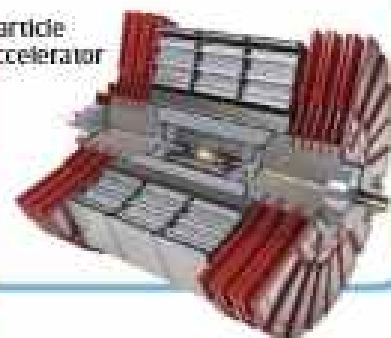
The universe started as a tiny point of pure energy. It then began to quickly expand and cool down. Over time it got bigger and bigger.

Scientists think the first stars were probably very big and very bright.

Re-creating the Big Bang

Scientists try to create the conditions of the Big Bang in huge machines called accelerators, which smash tiny particles together.

Particle accelerator



Light from distant galaxies travels for billions of years before it reaches Earth.

The space between galaxies keeps expanding.

The universe may keep expanding forever.



Biology

Biology is the science of living things. It studies how plants and animals interact with each other and their surroundings. It includes grouping and labeling living things and investigating how they live.

SEE ALSO

- ▶ Cells p.56
- ▶ Evolution p.95
- ▶ Food p.106
- ▶ Food chains p.107
- ▶ Habitats p.126
- ▶ Human body p.130
- ▶ Plants p.194

Each part of a plant has a name. This is the petal.

The human body is made up of many connecting parts.

Human biology
This investigates how the human body works—how it's put together, and what it needs to stay healthy.

Botany
This is the study of plants, from tiny mosses and algae all the way to massive trees.

Ecology
The study of how plants and animals depend on each other and their environment, to survive.

Biology
Biology covers lots of different areas. It can be broken down into smaller sections that often overlap.

Zoology
Zoology is the study of animals—how their bodies work and develop and how they behave.

Microbiology
This is the study of tiny living things like bacteria, viruses, and fungi.

Living things are made of tiny parts called cells.

Birds

Birds are animals that have feathers and beaks. They lay hard eggs, which their chicks hatch out of. Most birds can fly, and they are found all over the world.

SEE ALSO

- ▶ Aircraft p.13
- ▶ Animal groups p.22
- ▶ Dinosaurs p.80
- ▶ Eggs p.86
- ▶ Fruit and seeds p.115
- ▶ Rain forest p.204



Parrots

These tropical birds are colorful and often very noisy. They are strong fliers and eat fruit, nuts, and seeds.

Macaw

Short beak to pick up nuts and seeds



Yellow warbler

Songbirds

Most birds in the world are part of a huge family called songbirds. Each type of songbird has its own special song.



Bald eagle

Birds of prey

Birds of prey hunt for their food. They have sharp beaks, fly fast, and grab food, such as fish, with their feet.

There are almost **10,000** different kinds of birds.

Long, curved bill used to find food.

Wading birds

These long-legged birds wade around in the mud. They search for small animals, such as crabs, in the water to eat.



Scarlet ibis

Swimming birds

Not all birds can fly. Penguins swim underwater instead. Their feathers are waterproof, and they use their wings to steer.



Emperor penguin

Black holes

Black holes are the universe's most mysterious objects. They form when a star much more massive than the sun runs out of fuel. It explodes as a "supernova," then collapses under its own gravity, creating a black hole.

SEE ALSO

- ▶ Galaxies p.116
- ▶ Gravity p.125
- ▶ Light p.147
- ▶ Physics p.192
- ▶ Stars p.242
- ▶ Sun p.247
- ▶ Volume p.269

Invisible monster

We can't see black holes, as even light gets trapped by their gravity. However, many are surrounded by hot discs of gas and dust, which give off high-energy radiation that can be seen using special telescopes.

Black holes can collide with other black holes and get larger.

Supermassive black hole

The most massive black holes are found at the center of galaxies such as the Milky Way. They may form when massive clouds of gas collapse.



The enormous gravity of a black hole distorts space and time.

The edge of a black hole is called an event horizon.

The center of a black hole is called a singularity.

Spaghettification

Things that fall into black holes are stretched out, or "spaghettified." An imaginary astronaut would feel a stronger pull on one end of their body than the other, stretching them apart.



Body cells

Body parts are made up of tiny things called cells. Cells have important jobs to do, like sending messages, turning food into energy, and fighting off germs. Every cell has a particular job to do to keep the body healthy.

SEE ALSO

- ▶ Cells p.56
- ▶ Genes p.119
- ▶ Human body p.130
- ▶ Microscopic life p.165
- ▶ Sickness p.225
- ▶ Skin p.229

Inside a cell

All our body cells have an outer shell with a liquid inside. In the very center of the cell is a core, called the nucleus.

Mitochondria

These tiny structures release energy to power the cell.

Cytoplasm

Cytoplasm is a liquid inside the cell where chemicals mix together to bring the cell to life.

Nucleus

The nucleus is the control center of the cell. It contains instructions called genes.

Cell membrane

The membrane is the edge of the cell. It allows things to move in and out.

There are approximately
37.2 trillion
cells in the
human body!

Types of cell

Cells come in many different sizes and shapes. Each one is perfectly suited for the job it does in the body. Cells can divide to make more of themselves.



Red blood cells pick up oxygen from the lungs and transport it around the body.



White blood cells change shape and squeeze in between other cells to kill germs.



Nerve cells have long stems. They carry electrical messages to the brain.



Fat cells store and release energy. They protect the body from bumps and knocks.



Intestine cells are frilly. The frills absorb useful nutrients from food.

The **story** of...

Storytelling

Storytelling is the activity of telling or writing stories. They can be real or made up, can be of any length, and can be about any subject. People have always told stories to entertain each other or to inform people about, and make sense of, their world.

Oral history

Because early people couldn't read or write, they passed on stories by telling them to each other. The storyteller often acted out bits of the story. People still tell stories in this way today.



Storytelling in a library

In ancient India, people recited all

10,600

verses of the sacred hymn book, *Rigveda*, from memory.

Animal stories

Many stories feature animals as their main characters. These animals speak and dress like real people, although they often live in animal homes. Br'er Rabbit is a character from the South.

Br'er Rabbit wears human clothes.



Br'er Rabbit

What is a story?

A story is a fictional (made-up) account of imaginary or real events and people. A story has a beginning, a middle, and an end. The Chinese story of Pangu explains the creation of the Earth.



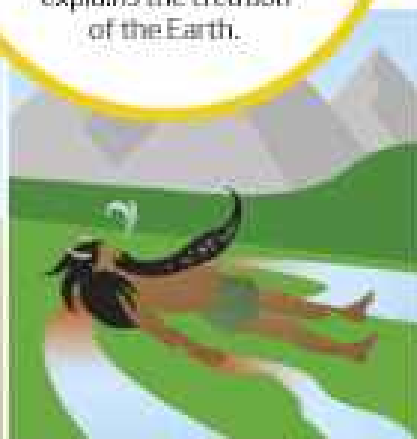
Beginning

In the beginning, there was chaos. Out of chaos came an egg, and out of this egg hatched Pangu, the first creature.



Middle

As Pangu grew up, he created the Earth and the sky, and stood between them to force them slowly apart.



End

When Pangu died, his breath became the wind, his voice became thunder, and his bones became valuable minerals.





Poetry

Poetry is a type of literature. It often has short, rhyming lines joined together in verses. A poem uses words very carefully to suggest different meanings or ideas. It can be any length and about any subject.

Beowulf is a very long poem about an ancient hero.

Fairy tales

A fairy tale is a story that contains magic and characters such as fairies, witches, goblins, or giants. Fairy tales tell the story of good versus evil, and usually have a happy ending. *Sleeping Beauty*, *Aladdin*, and *The Boy Who Cried Wolf* are all examples of fairy tales.

In the fairy tale *Aladdin*, a genie magically appears from a lamp.



Novels

A novel is a long story about people and their lives. Novels can be set in imaginary or real worlds, and can take place at any point in time. There are many different types of novel. For example, a historical novel would be set in the past and teach about history.

With more than **8,000 pages**, the world's longest novel is *Men of Goodwill* by French author Jules Romains.



Filming *Romeo and Juliet*

Films

Films are a visual form of stories. Actors perform the story in a real-life setting. The actors speak the lines of the story and try to make the film as realistic and lifelike as possible.

Children's stories from around the world

Books

Books are collections of written words put together to tell stories or give you information. Before books were invented, people shared stories by remembering and telling them to each other. Millions of books have been created since paper was invented, and many are now read on electronic devices such as tablets.

SEE ALSO

- ▶ Storytelling pp.42-43
- ▶ Inventions pp.136-137
- ▶ Language p.144
- ▶ Materials p.157
- ▶ Writing p.280

Early books

Books have been written and decorated by hand for thousands of years. Pages were often made from parchment, which is the thinned skin of an animal. These books took a long time to make.

J.K. Rowling's first Harry Potter book has sold more than **100 million** copies since 1997.

Printing

In around 1440 in Germany, Johannes Gutenberg invented the printing press. He carved single letters onto metal blocks, then put them together as words, and printed them onto pages.

1900s printing press



Medieval prayer book

Golden book

In Europe, monks wrote the first books in Latin. They decorated them with real gold.



Fiction

A book of fiction is a story in which the author writes about imaginary people and events. Fiction books can also be called novels.

Non-fiction

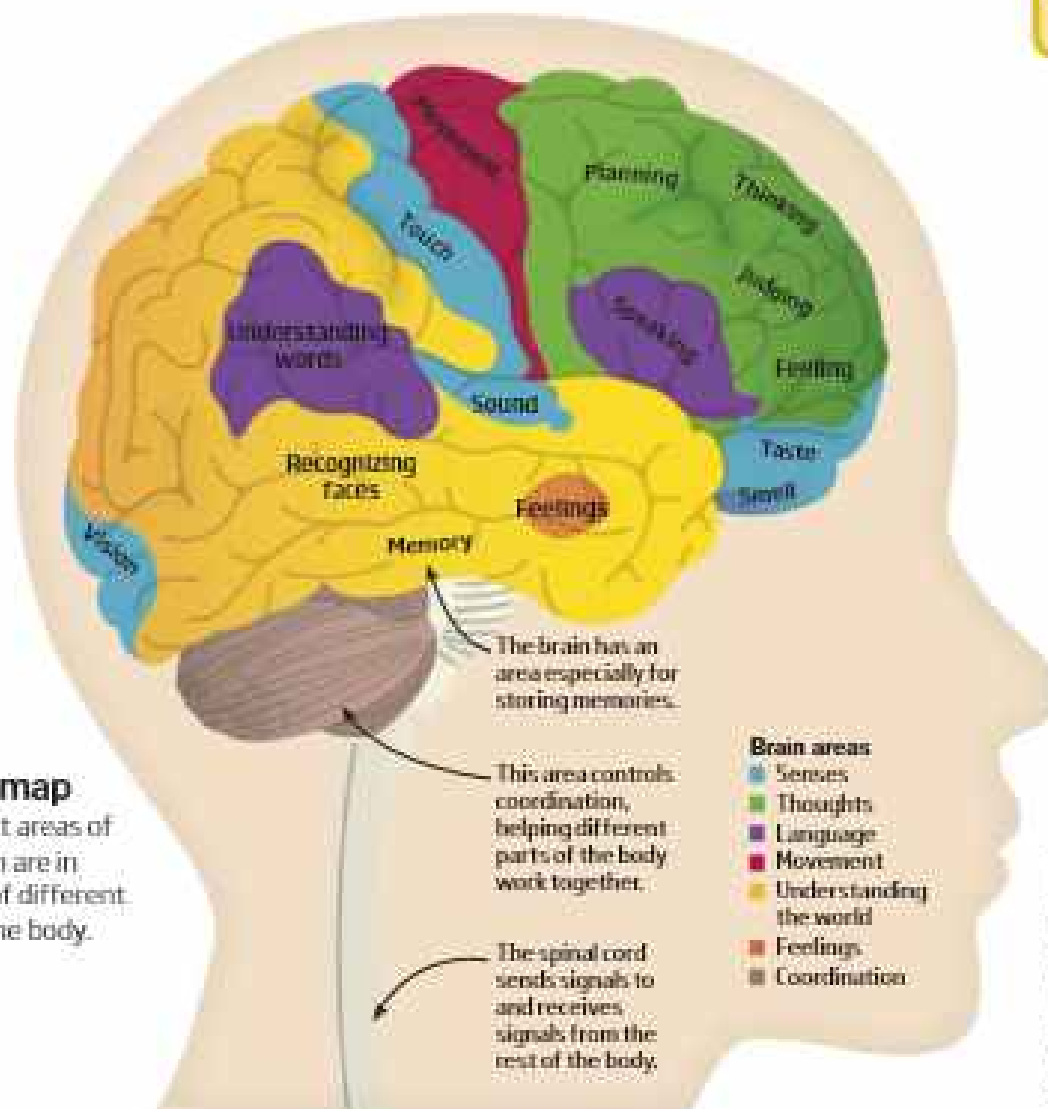
A non-fiction book contains facts about the real world. Dictionaries, atlases, cookbooks, and books about history and animals are all examples of non-fiction books.

Brain

The brain controls the whole body. Every time we think, feel, or move, the brain is at work. It even keeps working when we're asleep. The brain is the most complicated organ in the living world.

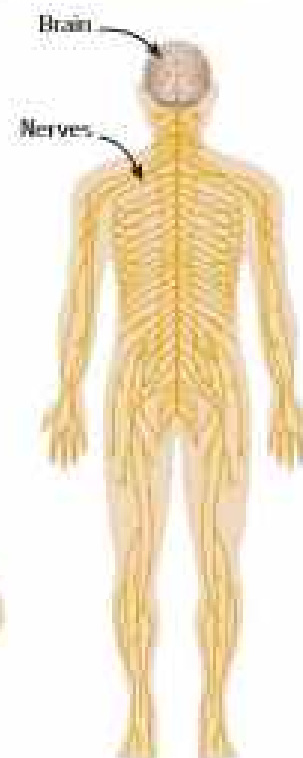
SEE ALSO

- Body cells p.41
- Feelings p.99
- Human body p.130
- Robots p.212
- Sight p.226
- Sleep p.231



Brain map

Different areas of the brain are in charge of different jobs in the body.

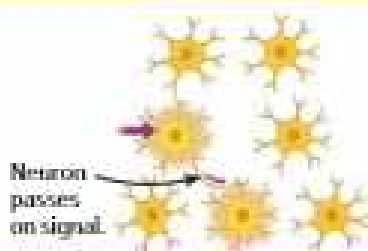


Nervous system

Nerves connect the brain to the rest of the body in a network called the nervous system. Nerves carry information between different body parts.

Thinking

The brain is made up of tiny cells called nerves, or neurons. They look like little trees. When we think, tiny electrical and chemical signals move quickly through the cells.



Acting alone

The brain does some things without us having to think. For example, it keeps our hearts pumping blood around our bodies at the right speed and controls our breathing.



Bridges

Bridges are structures that carry people and vehicles over obstacles. They are usually built over rivers, valleys, and roads. Bridges are designed to hold heavy loads and survive bad weather.

SEE ALSO

- Cars p.52
- Engineering p.91
- Materials p.157
- Rivers p.211
- Storms p.243
- Transportation pp.258-259

Bridge types

Engineers design different types of bridges depending on the size of the gap to be crossed, the type of land around it, and the weight of the crossing traffic.

Suspension bridges

A suspension bridge can carry heavy loads. Steel cables are anchored to strong supportive towers, spreading the weight.

Cables provide support from above.

Golden Gate Bridge, San Francisco, CA

The wires used for the cables on the **Golden Gate Bridge** could go around the Earth three times.

Tall towers at each end of the bridge are anchored deep in the ground.

Road or walkway made from strong concrete.



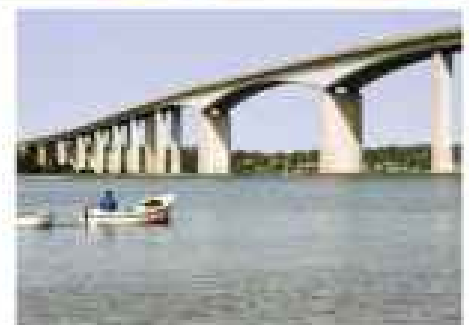
Arch

Arch bridges are usually made of stones cut into exactly the right wedge shapes to form an arch.



Truss

A truss bridge is built from triangles called trusses. Triangles are the strongest shapes, so this bridge can carry heavy weights.



Beam

A beam bridge is the most simple design. All the weight is placed directly on top. It is built to be stiff and not bend.

Bronze Age

The Bronze Age is a period of history after the Stone Age and before the Iron Age. It began about 5,500 years ago when societies first learned to make a metal called bronze. Bronze is a mix of the metals tin and copper. Combined, these metals are stronger than they would be alone.

SEE ALSO

- Buildings p.48
- Crafts p.75
- Iron Age p.140
- Metals p.162
- Stone Age p.243
- Trade p.257
- Writing p.280



Bronze Age bracelets

Bronze weapons were used by the first armies during the Bronze Age.

Bronze Age spear tip

Bronze

Using bronze tools allowed societies to clear more land for farming, and to grow, store, and trade more food and goods. Bronze was also used to make weapons and jewelry.

People traded more with other societies during the Bronze Age. Some traders used their money to buy bronze jewelry.

Early writing

The first written language was invented in the Bronze Age. It was called cuneiform. Writers used pointed reeds called styluses to make marks in soft clay tablets, which then hardened.



Bronze Age cuneiform tablet from Iraq

Settlements

During the Bronze Age, people lived together in large groups for the first time. Settlements were bigger than earlier villages had been. There were even towns and cities.

These buildings in Germany are copies of Bronze Age houses.

The houses have wooden frames and are built on stilts.



Buildings

A building is a solid structure fixed in one place. It has walls and a roof to shelter us from the weather. The shape of the building depends on its purpose. Buildings can be many different things, including hospitals, schools, and houses.

SEE ALSO

- ▶ Ancient Greece p.18
- ▶ Ancient India p.19
- ▶ Bridges p.46
- ▶ Castles p.53
- ▶ Engineering p.91
- ▶ Factories p.97

Types of buildings

Many towns and cities have a mix of buildings from across history. These buildings have different designs and are made from a variety of materials.



Construction

Machines are used to put up large buildings. Foundations are dug deep into the ground to stop the building from falling down. Cranes lift heavy building materials such as steel beams and panes of glass for windows.



Carbon cycle

Without the element called carbon, our world would be frozen and lifeless. Carbon constantly moves between living things, the atmosphere, the oceans, and the Earth below us. This movement is known as the carbon cycle.

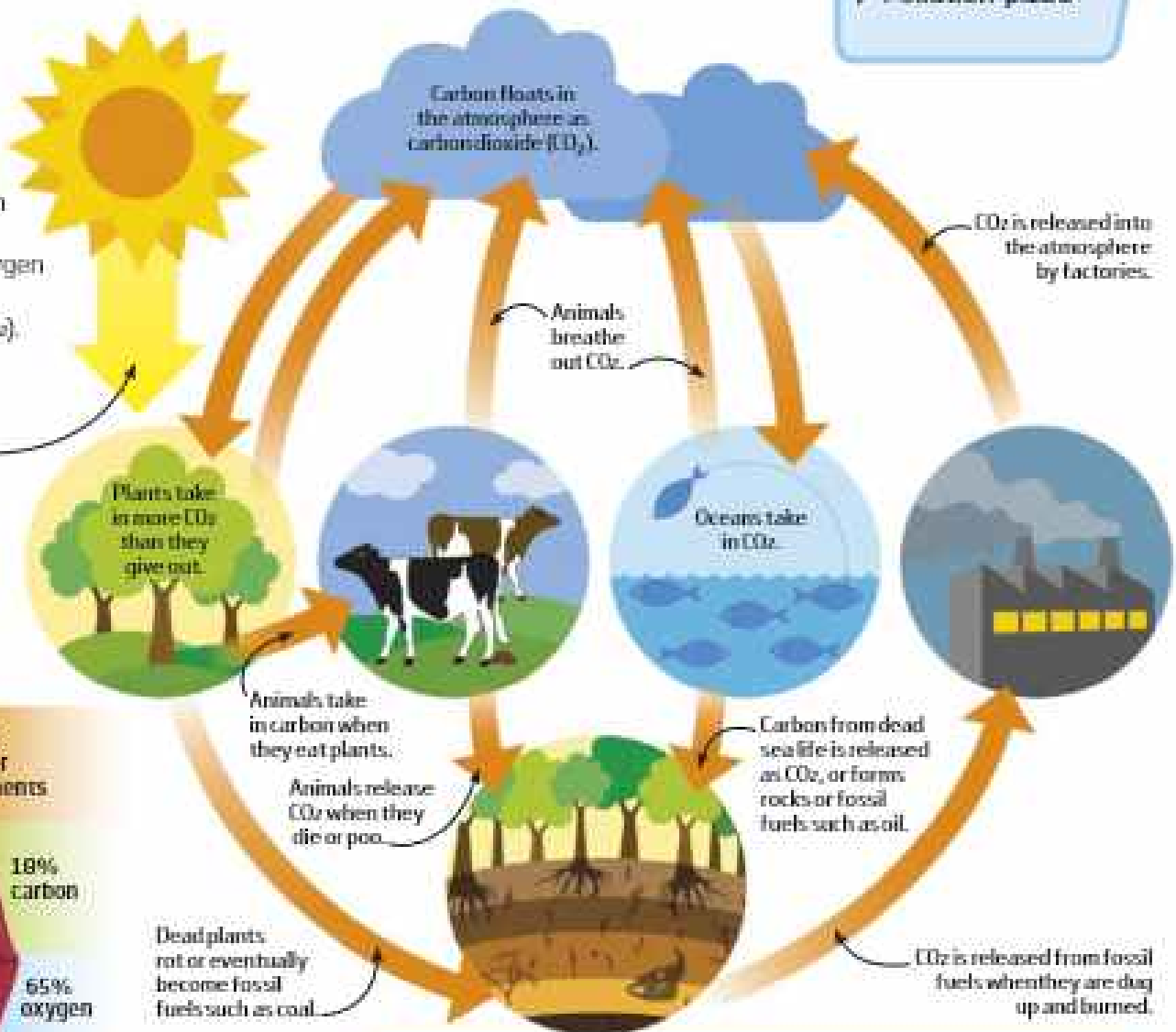
SEE ALSO

- Atmosphere p.33
- Climate change p.60
- Elements p.90
- Fossil fuels p.110
- Gases p.117
- Pollution p.198

Carbon movement

Carbon is always on the move. When it is in the air, it is combined with oxygen to make the gas carbon dioxide (CO_2).

Plants use sunlight and CO_2 to make their own food.

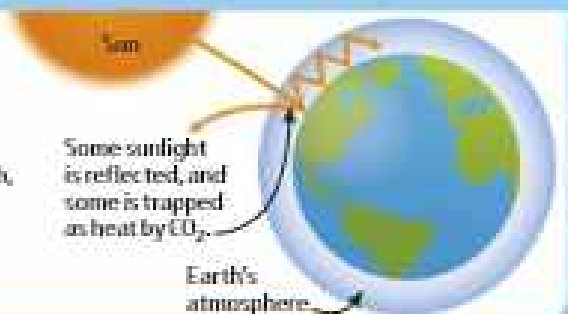


Carbon in the human body

Almost one-fifth of the human body is made of carbon. After we die, this is returned to the Earth and naturally recycled.

Keeping the Earth warm

Carbon dioxide in the atmosphere acts like a blanket around the Earth, trapping the sun's heat. Without it, the Earth would be very cold.



The **story** of...

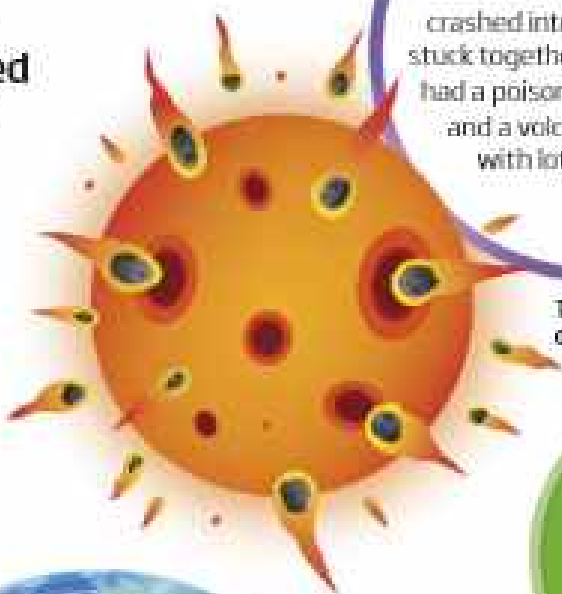
Changing world

Since it formed, the Earth has been covered with molten lava, bombarded by rocks from space, and blanketed in ice. In 4.5 billion years, the Earth has changed from a hot, lifeless planet into a watery home, full of life.

Earth is born

Our planet was formed when rocks orbiting the sun crashed into each other and stuck together. The early Earth had a poisonous atmosphere and a volcanic landscape with lots of craters.

The formation of the Earth



On the move

Earth's land is split up into huge chunks called continents. Throughout Earth's history, the continents have shifted around.

Today, there are seven continents, but they have not always looked the way they do today.

The continents were grouped together in a "supercontinent" called Pangea.

250 million years ago



Mountains are created when continents **move** and hit each other.

Marrella, a life form from 540 million years ago.



Life begins

At first, nothing lived on Earth. The earliest life forms appeared 4,600 million years ago. Over time, more species developed. Now there are millions of different types of life, and over 7.5 billion people.

120 million years ago

Pangea split in two, forming Laurasia in the north and Gondwana in the south.





Changing surface

The Earth's surface changes constantly. As continents crash together or move apart, mountain ranges are forced up, then worn down. Rainforests become icy wastelands; oceans expand and shrink, and glaciers turn into deserts.

The Andes mountains formed 45 million years ago.

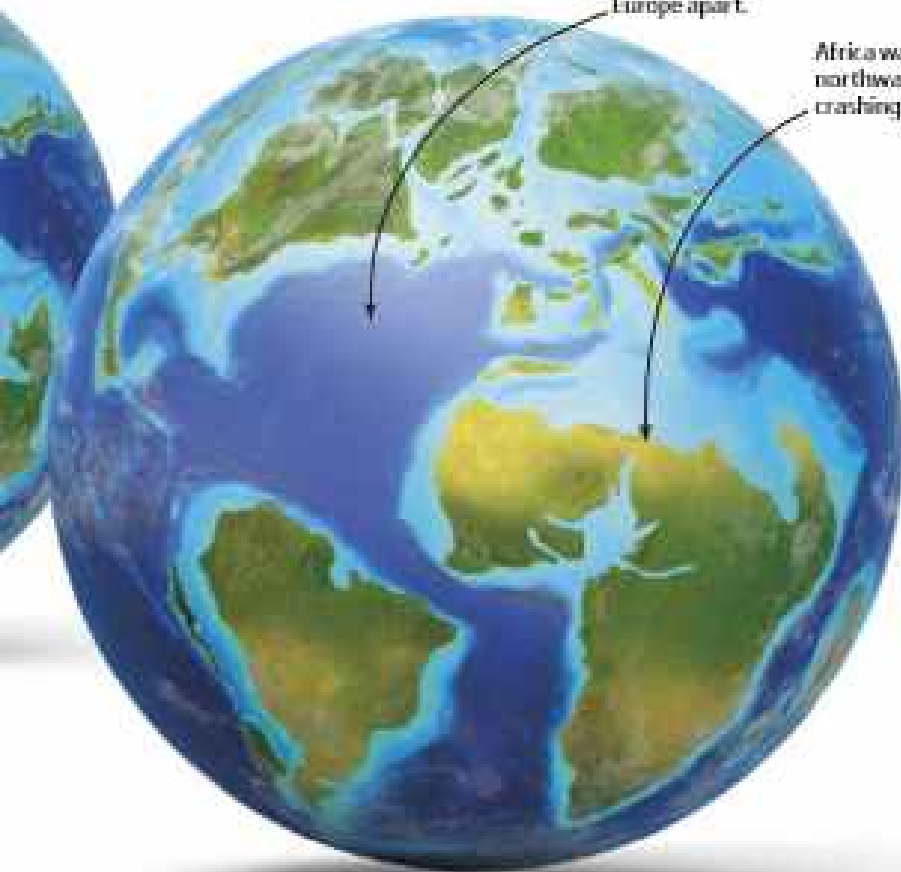


Mammoths lived during the last Ice age.

The Atlantic Ocean was opening, pushing North America and Europe apart.

Africa was moving northward, on its way to crashing into Europe.

If the sea level today was as high as it was 500 million years ago, London, New York, and Sydney would all be underwater.



80 million years ago

Climate change

Humans have a direct impact on Earth's climate. We burn fossil fuels such as coal, oil, and gas to make power. This releases dangerous gases into the air, making Earth hotter.



Burning fossil fuels

Cars

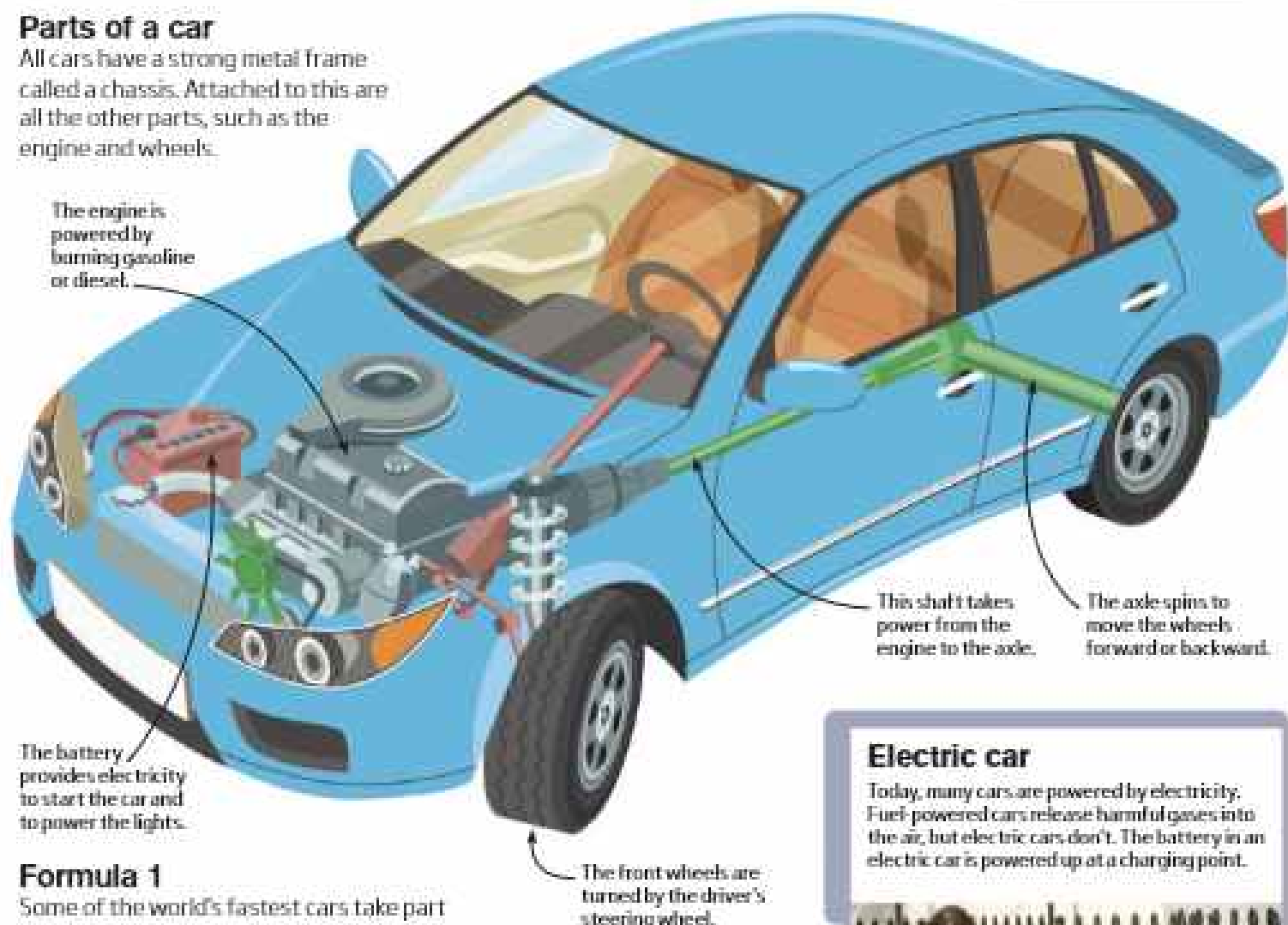
Cars are vehicles that travel on roads. The shape and size of a car depends on its use, such as for a family or for racing. Cars are powered by engines. Engines can be powered by fuel such as gasoline or diesel, or by using electricity.

SEE ALSO

- ▶ Electricity p.87
- ▶ Engines p.92
- ▶ Fossil fuels p.110
- ▶ Inventions pp.136-137
- ▶ Sports p.239
- ▶ Transportation pp.258-259

Parts of a car

All cars have a strong metal frame called a chassis. Attached to this are all the other parts, such as the engine and wheels.



Formula 1

Some of the world's fastest cars take part in a racing competition called Formula 1. They compete on special tracks and around some city streets. The cars have long, low shapes to help them go super fast.



1998 Ferrari F300 F1

Electric car

Today, many cars are powered by electricity. Fuel-powered cars release harmful gases into the air, but electric cars don't. The battery in an electric car is powered up at a charging point.



Castles

Most castles were built between 1000 and 1500 CE. They had homes and workshops inside their walls. Kings and wealthy people built castles as protection from enemy attack. Once cannons were invented, they were no longer safe. Many fell apart over time, but some are still standing.

SEE ALSO

- Buildings p.48
- Engineering p.91
- Europe p.94
- Imperial Japan p.131
- Knights p.142
- Homes pp.244-245

Stone castles

Later castles in Europe had stone walls and towers. The biggest ones could hold hundreds of soldiers. Smaller stone castles were home to just a single family and its servants.

The bailey was an open space inside the castle with areas for growing food.

The largest tower in the castle was called the keep.

Stone walls were difficult for attackers to climb.

Sheep and other livestock were kept safe in the castle.

The drawbridge could be lifted to keep the enemy out of the castle.

Early castles

Hundreds of "motte and bailey" castles were built from 1020–1200. They had two parts—a mound of earth (motte) topped by a tower, and a courtyard (bailey).

People kept watch for the enemy from tall towers.

Some castles had a water-filled moat around them for extra protection.

Many castles were built on hilltops so they had a clear view of everything around them.

Himeji Castle in south Japan is called "the white heron" as its peaks look like wings flapping.

Japanese fortress

Castles in Japan were designed with towers built of both wood and stone. The central tower was used as the main hideout if the castle was attacked.



Cats

The cat family are all meat-eaters. They have sharp teeth that let them slice up their food. Cats are very fast, with strong bodies that let them run, jump, and even swim.

Small wild cats

Most wild cats are much smaller than lions. The colors of their fur help them to be camouflaged, or hidden, in their habitat.

Cat tongues are covered in sharp hairs. They are used to clean meat off a bone and to wash their fur.

Cats have good hearing.

Whiskers help cats sense their surroundings.



Chartreux

Pet cats

Cats were one of the first animals that humans tamed from the wild to become pets, around 12,000 years ago.

Big cats

The largest types of cat are known as "big cats." They include lions, tigers, leopards, and jaguars. These are the only cats that are able to roar.

Male lions have longer hair over their shoulders, called manes.



Lion

Night vision

Cats hunt at dawn and dusk. Their eyes are good at seeing when there is not much light. Their eyesight is six times better than a human's.



Leopard

SEE ALSO

- Dogs p.81
- Food chains p.107
- Habitats p.126
- Pets pp.152-153
- Sight p.226
- Vertebrates p.266

Lynx

Caves

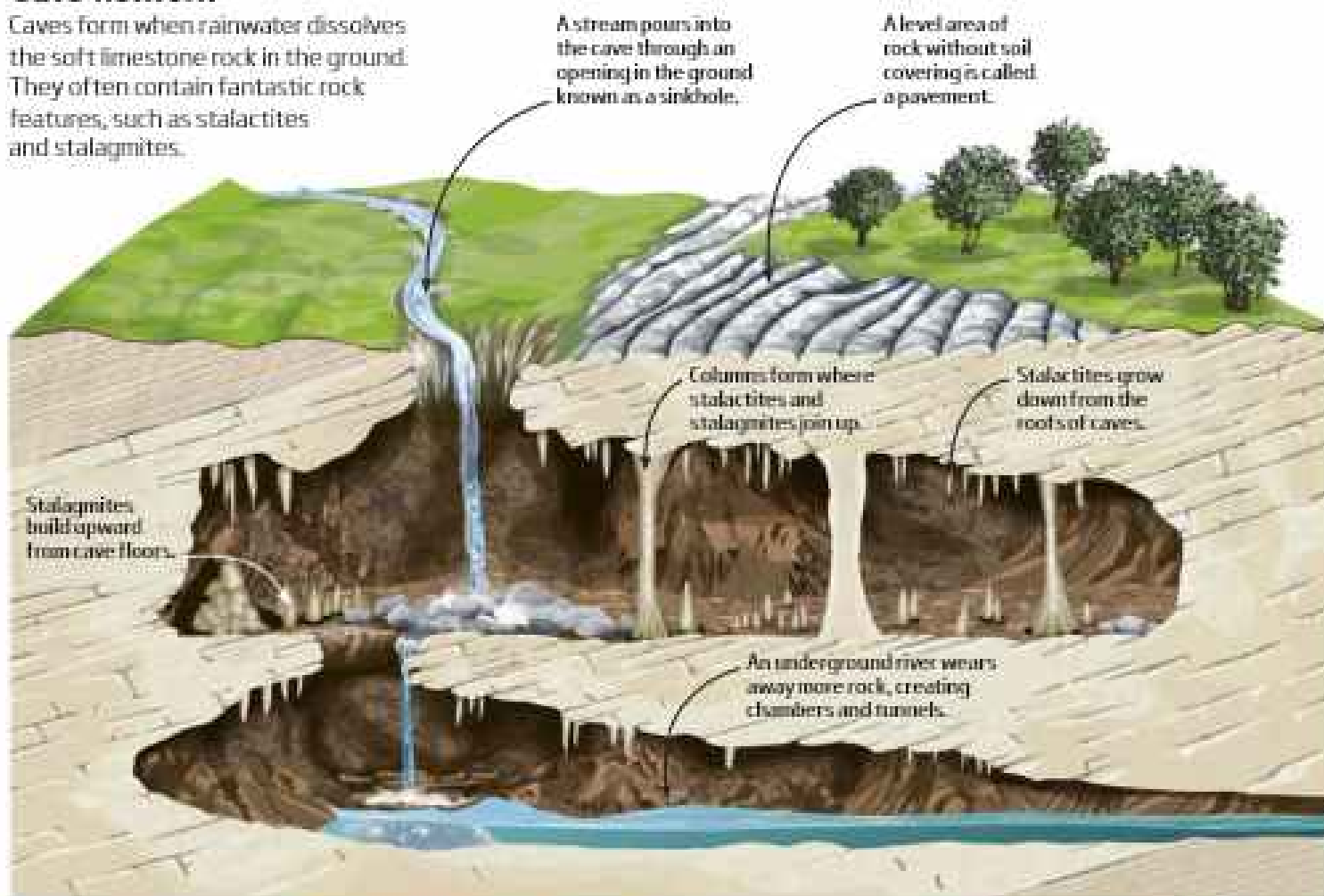
Caves are large natural holes in the ground. They are usually formed when rock is hollowed out by running water over millions of years. Prehistoric humans used caves for shelter, and some people still live in caves today. They are also home to lots of animals, including bats.

SEE ALSO

- ▶ Animal homes p.23
- ▶ Erosion p.93
- ▶ Glaciers p.122
- ▶ Rocks and minerals p.214
- ▶ Homes pp.244-245

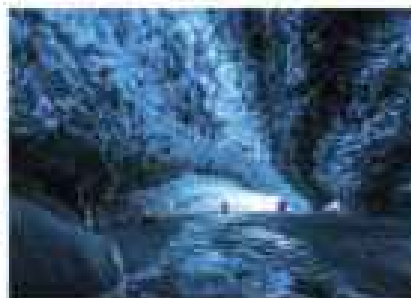
Cave network

Caves form when rainwater dissolves the soft limestone rock in the ground. They often contain fantastic rock features, such as stalactites and stalagmites.



Ice caves

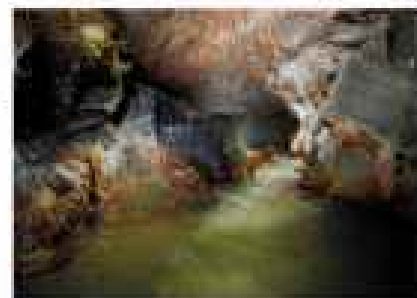
Glaciers are rivers of ice that move very slowly. Some glaciers have caves inside them. They are made by streams of water tunneling through the glacier.



This ice cave is inside a glacier in Iceland.

Biggest cave

The 40-story-high Hang Son Doong cave in Vietnam is the world's biggest cave. Inside, it has a river, a forest, and even its own clouds!



A river flows through part of the huge cave.

Cells

Living things are made from tiny parts that stick together, called cells. Cells come in different shapes and sizes depending on the job they do. They can divide and make copies of themselves.

SEE ALSO

- ▶ Body cells p.41
- ▶ Genes p.119
- ▶ Heart p.128
- ▶ Photosynthesis p.191
- ▶ Plants p.194
- ▶ Sickness p.225

Plant cell

Plant cells can collect air and make their own food from sunlight. They have a strong cell wall, which gives the plant strong stems and leaves.

Cell wall

The strong outer wall gives the cell and plant their shape.

Cytoplasm

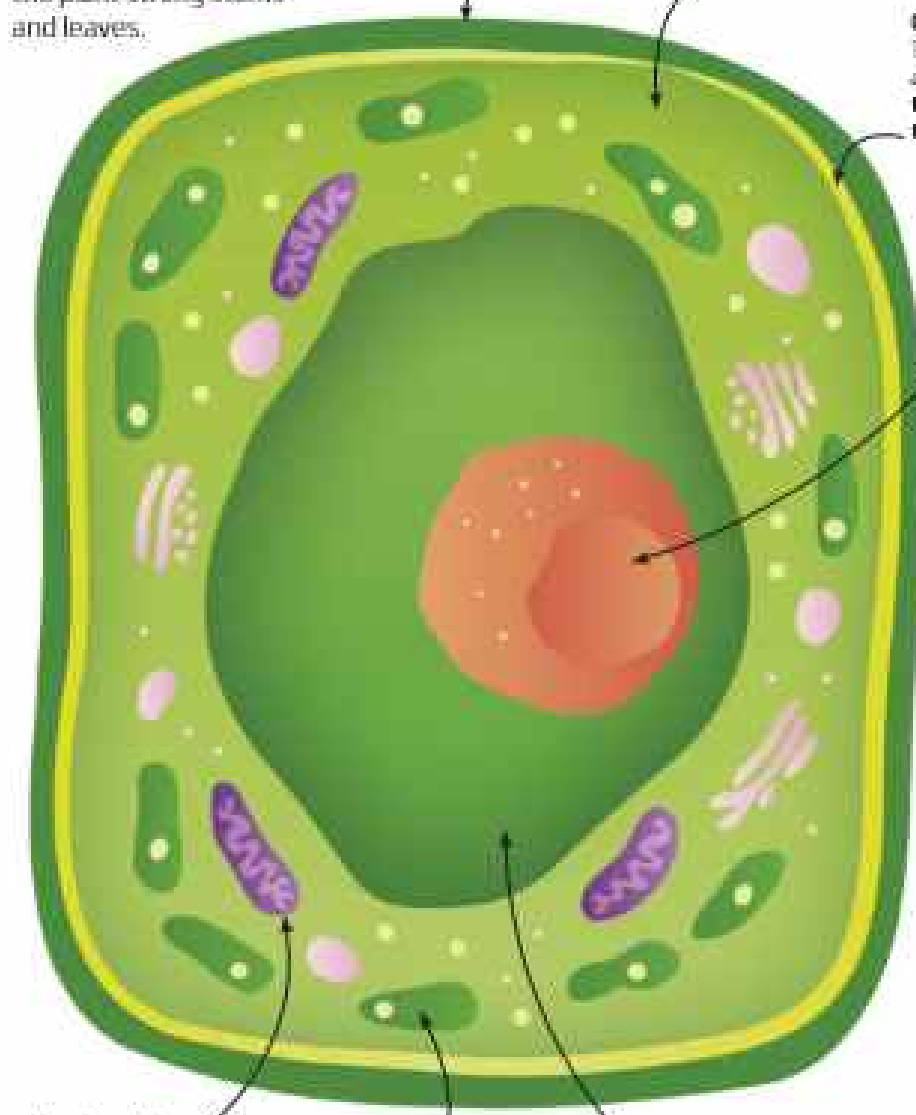
This is the liquid inside the cell that everything else floats in. Chemicals mix together here.

Cell membrane

The cell membrane is a barrier that keeps the cytoplasm inside the cell.

Nucleus

The nucleus controls the cell and contains instructions called genes.



Mitochondrion

This is where energy from sugar is released to power the different parts of the cell.

Chloroplast

Chloroplasts collect sunlight and use air and water to make food.

Vacuole

The vacuole is a storage bubble filled with liquid food, water, or waste.

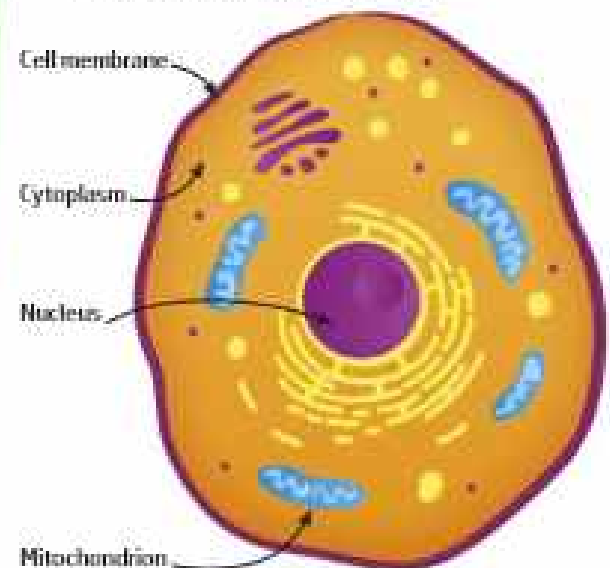
Bacteria

Bacteria are living things made from just one cell. They can divide to make copies of themselves. This is how germs spread and make us sick.



Animal cell

Animal cells use oxygen to break down sugar and make energy. Animals get sugar from the food they eat, while oxygen gets to the cells through the blood.





Changing states

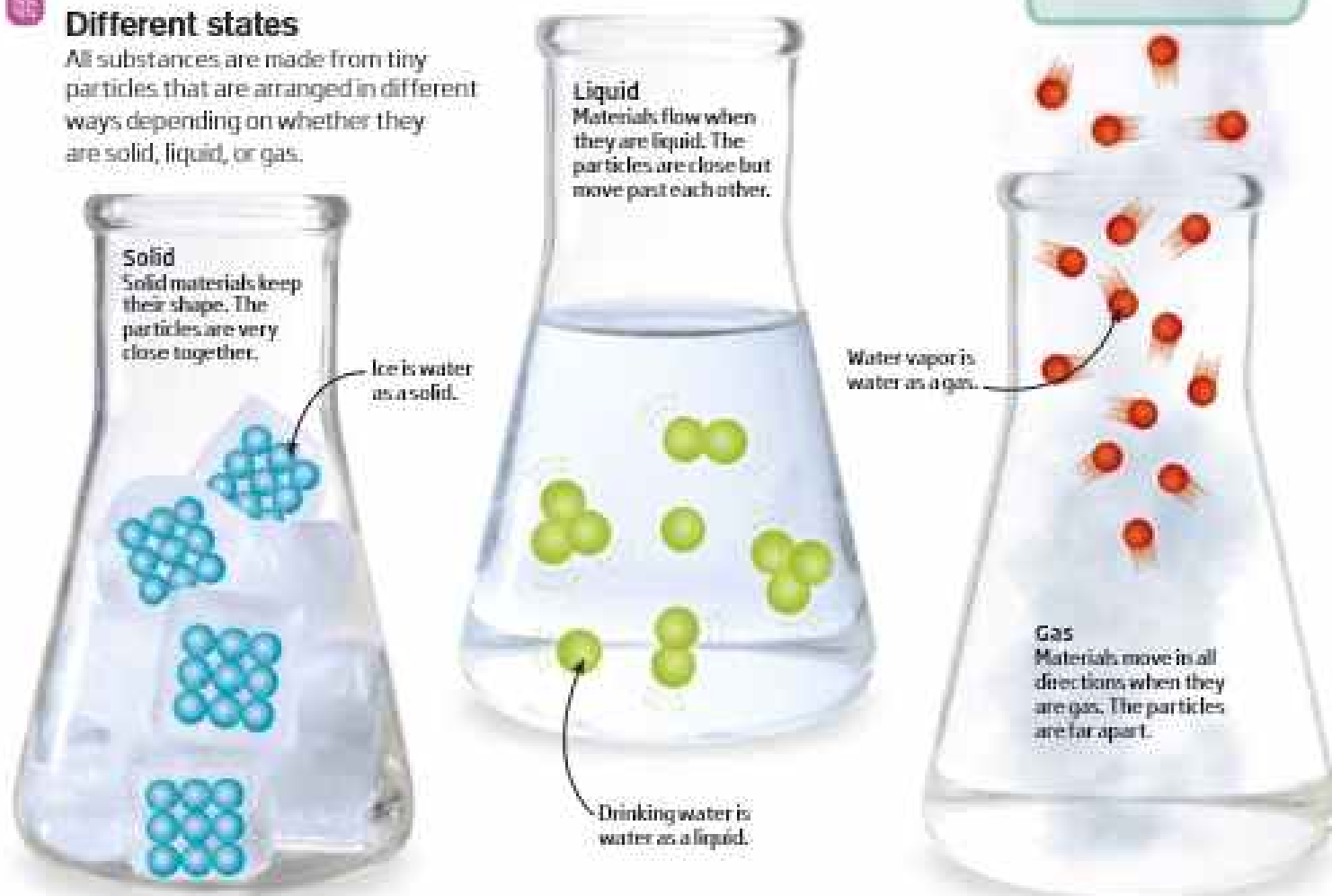
Most materials can change between being solid, liquid, or gas, depending on their temperature and how tightly packed they are. Substances such as water can change from one state to another and then back again.

SEE ALSO

- ▶ Atoms p.34
- ▶ Gases p.117
- ▶ Water p.120-121
- ▶ Liquids p.148
- ▶ Solids p.234
- ▶ Temperature p.252

Different states

All substances are made from tiny particles that are arranged in different ways depending on whether they are solid, liquid, or gas.



Shifting states

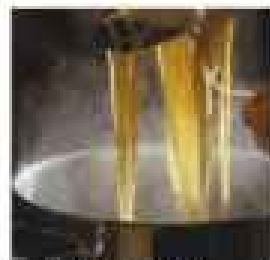
The same material can change from being solid to liquid and from liquid to gas when it is heated, and back again as it cools. These changes are called melting, solidifying, evaporating, and condensing.



Melting When we heat a solid, it melts and changes to a liquid.



Solidifying When a liquid like this lava cools, it becomes solid.



Evaporating When a liquid changes into a gas, it evaporates as steam.



Condensing When water turns from a gas back into a liquid form, this is condensing.



Chemistry

Chemistry is a science that looks at the smallest ingredients of everything, called elements. It studies how elements react when we mix them together, and how tiny particles in materials are arranged and can be rearranged.

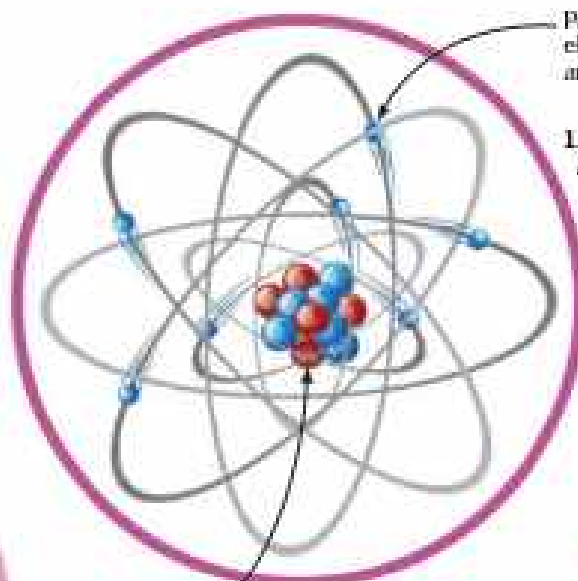
SEE ALSO

- ▶ Atoms p.34
- ▶ Body cells p.41
- ▶ Elements p.90
- ▶ Engineering p.91
- ▶ Changing states p.57
- ▶ Materials p.157

Building blocks

Everything around us is made from tiny parts called atoms. Atoms join to each other and other types of atoms to make arrangements called molecules. These basic building blocks are what chemistry investigates.

An atom is mostly empty space. If an atom was the size of a football stadium, the nucleus would be the size of a marble.



Particles called electrons move around the outside.

The center is called the nucleus. It contains particles called protons and neutrons.

1. Atoms

Atoms are so small we can't see them and they're mostly empty space. They have even smaller particles inside. The type of atom depends on the number of particles inside it.

2. Elements

Materials that have only one type of atom are known as elements. Elements have just one ingredient - themselves. Antimony, plutonium, and gold are all elements.

Gold

Antimony

Plutonium

Chemical reaction

When two or more elements come together to make a new compound, it is called a chemical reaction. Reactions can fizz, burn, or even explode when new compounds are made.



Iron and oxygen react to form rust.



3. Compounds

Elements that have joined together are called compounds. For example, water is a compound made from the elements oxygen and hydrogen.

Oxygen and hydrogen combined make the compound water.



Circuits

A circuit is when an object that uses electricity is connected with wires to a power source. Circuits in our homes connect lights and appliances, like the fridge or television, to electricity.

SEE ALSO

- ▶ Computers p.71
- ▶ Electricity p.87
- ▶ Light p.147
- ▶ Measuring p.159
- ▶ Television p.251

How circuits work

When a circuit is a complete loop with no gaps, electricity can flow around it. We use symbols to represent the different parts of the circuit.

Electricity travels through metal wires.



Battery "cell"

Batteries are power sources that hold electricity. Electricity can flow around the circuit when we connect it to both ends of a battery.

Wires are covered in plastic to stop electricity escaping.

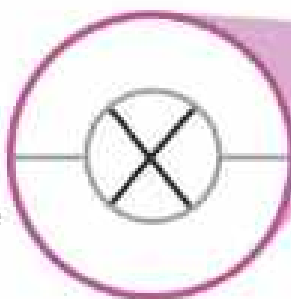
Switch

Electricity only flows around the circuit when the switch closes and is switched on. Once the switch is opened, the circuit breaks.



Light bulbs

Electricity makes the light bulb switch on and light up. Objects in a circuit that need electricity to work are called components.



Crocodile clips connect circuit wires to objects.

The flow of electricity around the circuit is called the **current**.

Circuit boards

Computers have tiny boards in them called circuit boards. Circuit boards have lots of wires connected up to tiny components to make the computer work.



Climate change

Climate is the average weather for an area. The way we live is changing the Earth's climate. It is becoming warmer. This change is causing extreme weather conditions, such as long periods of little rainfall, called droughts, and huge storms. Many countries are now trying to stop climate change.

SEE ALSO

- ▶ Arctic p.25
- ▶ Carbon cycle p.49
- ▶ Fossil fuels p.110
- ▶ Industrial Revolution p.133
- ▶ Pollution p.198
- ▶ Storms p.246

More than **39 billion tons** (35 billion metric tons) of **CO₂** is released every year.

What is the cause?

Power stations, factories, and cars pump gigantic amounts of the gas carbon dioxide (CO₂) into the atmosphere. This gas acts like a blanket, trapping the sun's heat and warming our planet.

Exhaust fumes from road traffic are one of the biggest causes of climate change.

How is the climate changing?

Climate change is making our summers hotter. Floods, droughts, and powerful storms are becoming more common. The ice in cold parts of the world is melting, which is making sea levels rise.



Flooding in New Orleans, Louisiana

What can we do?

We need to reduce the amount of fuels we burn that release CO₂. We can do this by using sources of energy that do not produce CO₂, such as sunlight, wind, and water.

Solar panels make energy from sunlight and do not release harmful gases.



Clocks

A clock is a device used to keep track of time. Ancient civilizations measured time by falling sand, flowing water, or the position of the sun. Modern clocks are either digital or have clockwork mechanisms.

SEE ALSO

- Machines p.150
- Measuring p.159
- Numbers p.185
- The sciences pp.218-219
- Sun p.247
- Time zones p.255

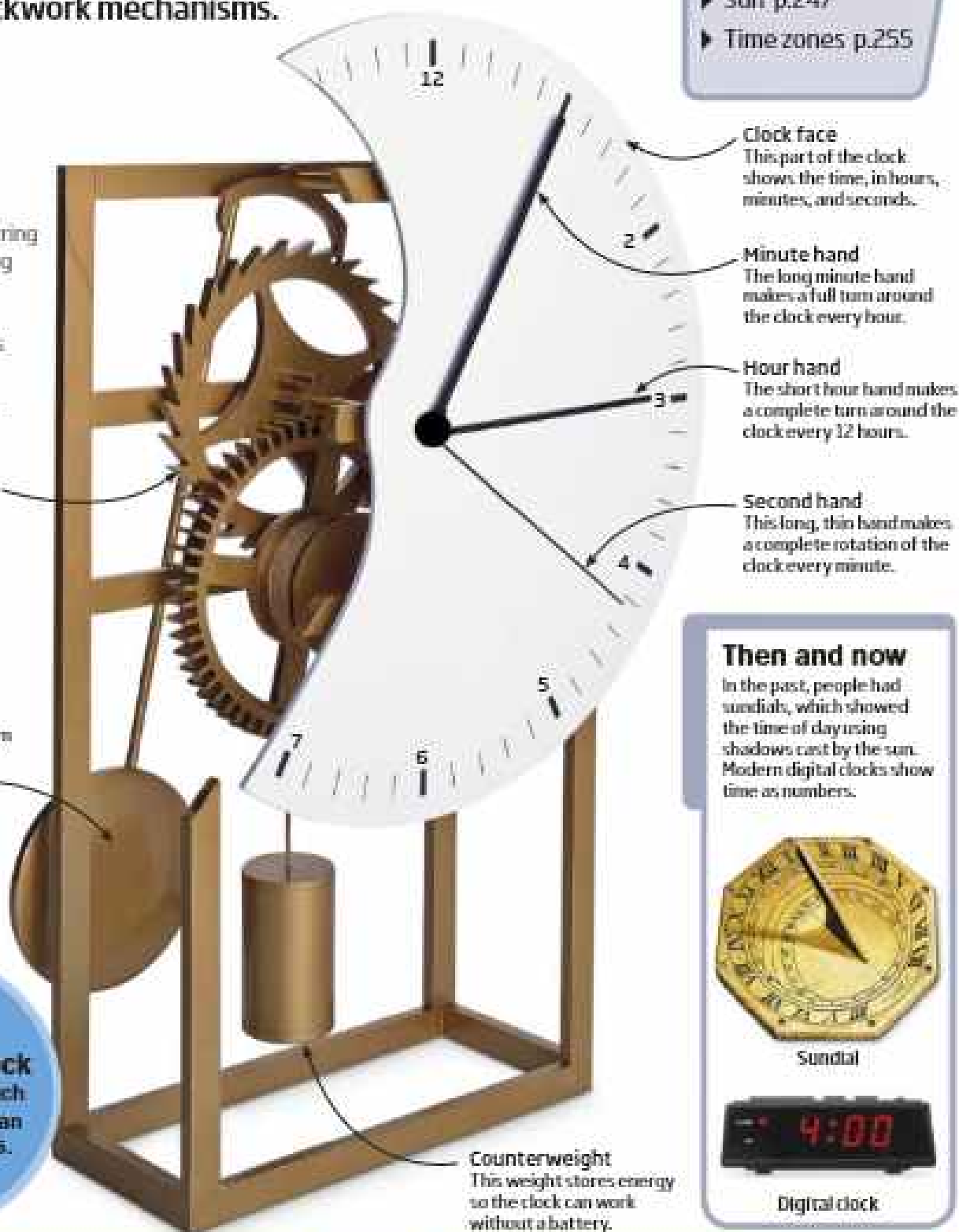
How does a clock work?

Clocks make regular movements for measuring time. They do this using machinery known as clockwork. The clock face has moving hands that show the hours, minutes, and seconds.

Gears
Gears are wheels that lock together. In a clock, they make the hands turn at different speeds.

Pendulum
Each swing of the pendulum turns the gears one click, usually once a second.

The first pendulum clock was made by Dutch scientist Christiaan Huygens in 1656.



Clock face
This part of the clock shows the time, in hours, minutes, and seconds.

Minute hand
The long minute hand makes a full turn around the clock every hour.

Hour hand
The short hour hand makes a complete turn around the clock every 12 hours.

Second hand
This long, thin hand makes a complete rotation of the clock every minute.

Counterweight
This weight stores energy so the clock can work without a battery.

Then and now

In the past, people had sundials, which showed the time of day using shadows cast by the sun. Modern digital clocks show time as numbers.



Sundial



Digital clock

Clothing

Over time, the things we wear have changed a lot. The clothes that people wear often reflect where they live, what jobs they do, and how much money they have. Clothes are generally made from woven materials such as cotton, wool, or silk, with special designs. They are often worn for show, but can also be practical or even worn for fun.



First clothes

The first people wore clothes made of animal skins to keep out the cold and wet. They later discovered how to clip the wool off a sheep's back and spin it into a thread that could be woven to make material.



The Roman Emperor Nero wore a purple toga and punished anyone else who wore purple with death.

Roman clothes

The main item of clothing for ancient Romans was a simple tunic. For special occasions, men wore a long piece of cloth called a toga on top of their tunic. Women wore a woollen shawl called a palla.



This long, embroidered dress is made of silk.

A toga wraps around the body and is thrown over the left shoulder.

Court finery

In the royal courts of Europe in the 1500s and 1600s, men and women wore special, expensive clothing. The women wore long, embroidered dresses, while the men wore padded jackets with short trousers and silk stockings.



A padded jacket, called a doublet.

A long cotton or silk sari is wrapped around the waist.

A sari can be up to **27 ft** (8 m) long.



Kimono

Traditional costume

Around the world, people wear clothes that are unique to their country. For traditional events, Indian women wear saris, while Japanese women wear long, decorated kimonos with wide sleeves. A kimono is tied at the back with an *obi*, or sash.



Hats were designed to match the rest of the outfit.

A New Look skirt was full and long, with a small waist.

Weaving

The cotton, silk, wool, or linen material used to make clothes is woven on a loom. Different colored threads can be used to make patterns, such as checks or stripes.



Loom

A suit is often worn with a shirt and tie.



Traditionally, the bottom button of a suit jacket should be left undone.

The New Look

During World War II (1939–45), material for new clothes was in short supply. In reaction to this, fashion designer Christian Dior introduced in Paris in 1947 a “New Look” for women. The full skirts were shorter and became fashionable around the world.

Smart suit

All around the world, businessmen and women wear a suit of a jacket and matching pants. The suit was developed in Europe during the 1800s and is a practical uniform for work and other formal occasions.

Clouds

Clouds are made of tiny droplets of water or ice. They form when air that contains water rises and cools. As well as making rain, snow, and hail, clouds help control our planet's temperature.

SEE ALSO

- Gases p.117
- Water pp.120-121
- Storms p.246
- Temperature p.252
- Water cycle p.270
- Weather p.271

Types of cloud

There are many different types of cloud. Some float very high up, while others hug the ground. Clouds may look white and fluffy or dark and stormy.

High-level clouds

Cirrus
These wispy clouds form very high up in the air.

Cirrocumulus
Cirrocumulus clouds sometimes break up to form these little clouds.

Altostratus
These clouds form a thin sheet across the sky.

Medium-level clouds

Altostratus
These clouds are broken into segments, or "cloudlets."

Cirrostratus
These thin clouds are made of tiny ice crystals.

Low-level clouds

Stratus
These flat sheets of cloud can be white or gray.

Stratocumulus
These big clouds form a lumpy layer in the sky.

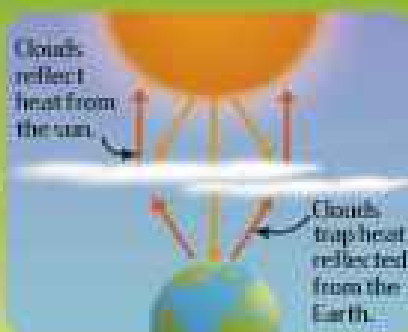
Cumulonimbus
Towering and huge, cumulonimbus clouds are often seen during thunderstorms.

Nimbostratus
Tall and gray, these clouds bring hours of rain or snow.

Cumulus
These fluffy heaps of cloud are often seen on sunny, breezy days.

Temperature control

Clouds reflect the sun's heat, stopping the Earth from getting too hot. They also trap heat below them, so a cloudy night is warmer than a clear one.



Cloud or UFO?

Clouds that do not move can form in the sheltered air behind areas of high ground, such as mountains. They are shaped like saucers or lenses and have been mistaken for unidentified flying objects (UFOs).





Coding

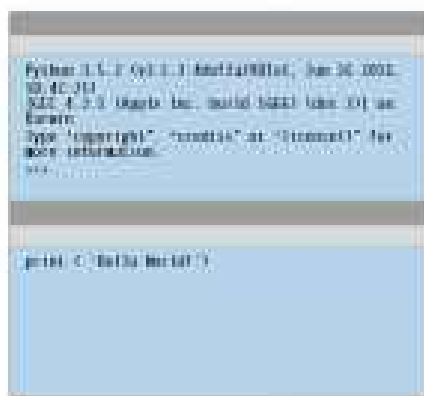
Computers follow instructions from special programs. These instructions are known as code, and writing them is called coding. Code can be written in many different coding languages.

SEE ALSO

- ▶ Codes p.66–67
- ▶ Communication p.69
- ▶ Computers p.71
- ▶ Internet p.138
- ▶ Language p.144
- ▶ School pp.272–273

Computer languages

Programming languages tell computers what to do. This example shows a text-based language called Python.



Input
Instructions are typed into a text window. These instructions tell the computer to show, "Hello World!"



Output
When the program is run, it follows the typed instructions. Here, the computer screen shows, "Hello World!"

Learning to code

Some computer languages are easier to learn than others. Scratch uses colorful blocks of code that you can arrange to make your own games.



Output
The blocks control the actions of the characters here, on the "Stage."

Input
In Scratch, coding is done by putting together instruction blocks.

Codes

Codes are words, letters, and numbers that are used to represent other words, letters, and numbers. People used codes to communicate with each other, or to keep things secret. Others, such as DNA, are just instructions.



The ancient Egyptians used **hieroglyphics** as a **writing system** to record their history.

Writing with pictures

The ancient Egyptians used drawings to communicate, instead of written words. These symbols are called hieroglyphics. Hieroglyphics were not understood by modern people until a stone was discovered with a translation of hieroglyphics into Greek, allowing the "code" to be worked out.

Programming

Computers need instructions to operate. These are created by programmers, who use combinations of symbols and words. Programming is often called coding.

1000101110
0101010
011010110
010010
10111000
0 1011
0101010
0110101110001
010010 001010 10110
101 10001011 01011010101

Morse Code

Morse Code represents letters and numbers as dots and dashes. It was used to send messages through wires before telephones were invented.



Morse Code tapping machine

您好

Nice-How, Mandarin

Bonjour

Boh-zho, French

Hello

English

こんにちは

Konnichiwa, Japanese

Languages

The different languages humans speak are types of code. Until you learn a foreign language, hearing someone using it to speak or reading their writing will make little sense to you.

Merhaba

Mehr-hah-bah, Turkish

Holá

Oh-lah, Spanish

здравствуйте

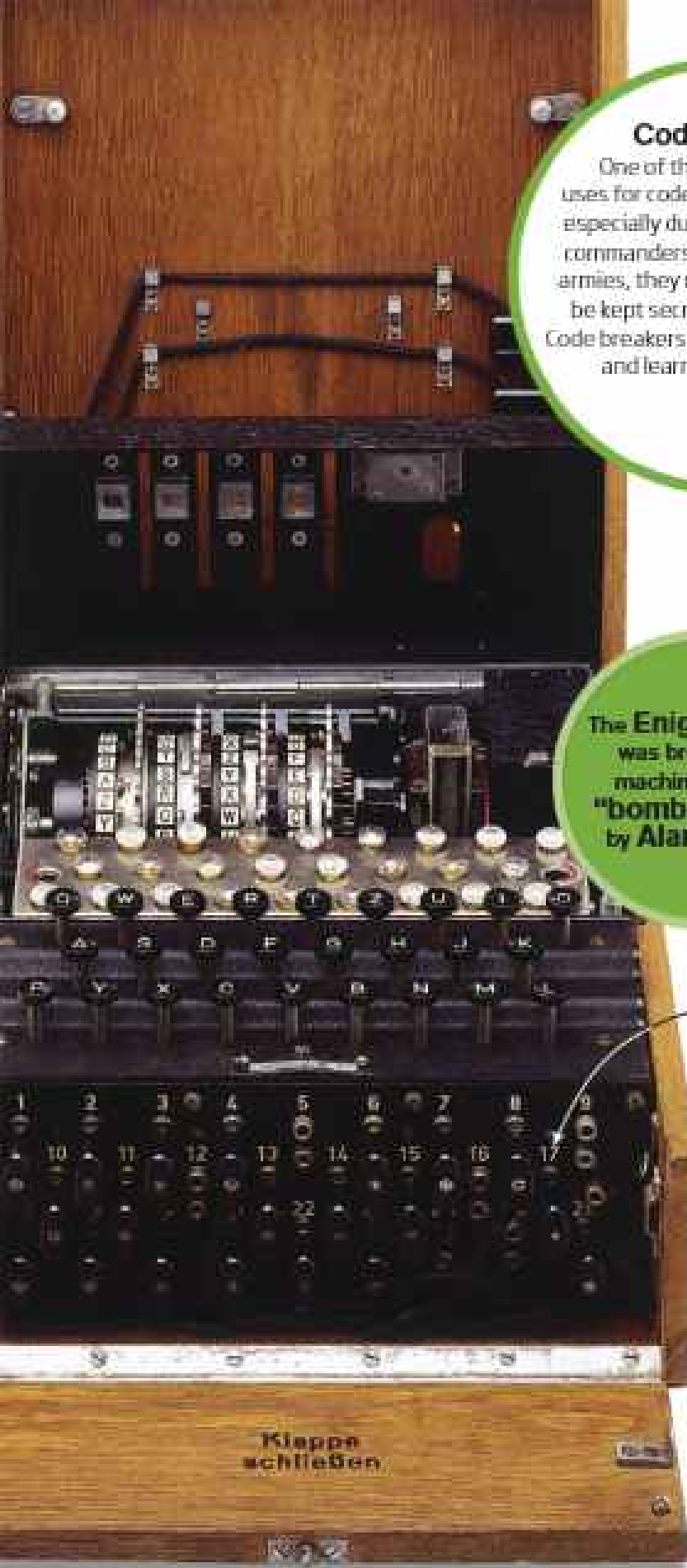
Zdrast-wul-tyeh, Russian

Jambo

Ja-m-boh, Swahili

নমস্কার

Nomostkar, Bengali



Codes in war

One of the most common uses for codes is to keep secrets, especially during wartime. When commanders give orders to their armies, they need those orders to be kept secret from the enemy. Code breakers try to crack the codes and learn enemy secrets.

This system of writing from ancient Greece is still not understood.

Linear A tablet



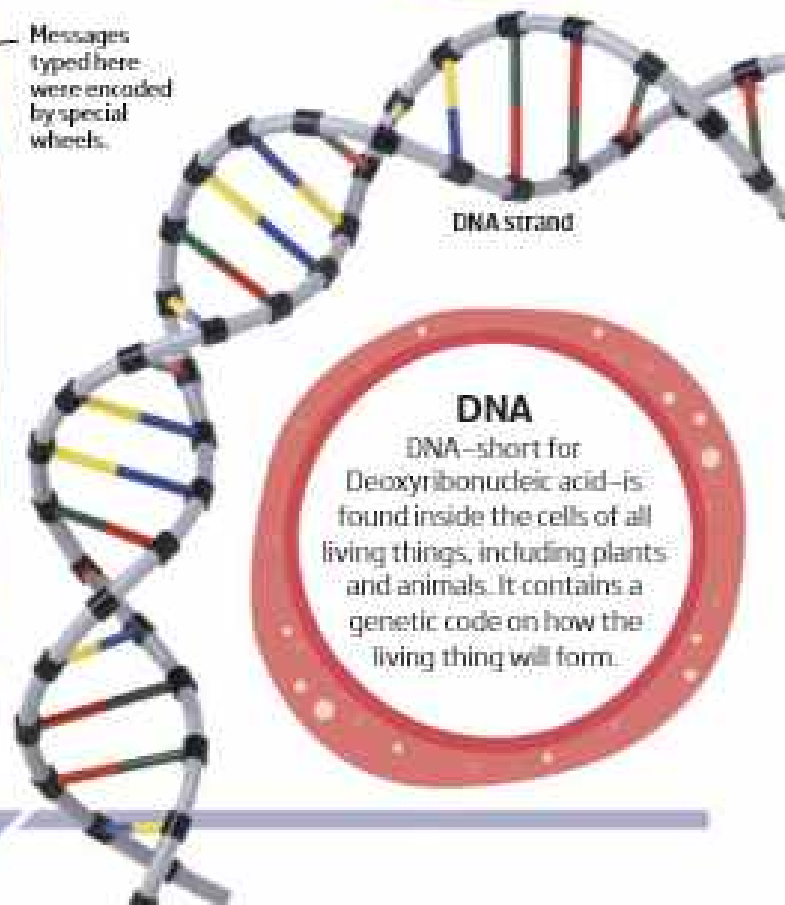
The Enigma Code was broken by a machine called a "bombe," created by Alan Turing.

Unbroken codes

Some languages remain a mystery—their writings have been discovered but never translated. It is likely that we will never know what they mean.

Messages typed here were encoded by special wheels.

DNA strand



DNA

DNA—short for Deoxyribonucleic acid—is found inside the cells of all living things, including plants and animals. It contains a genetic code on how the living thing will form.

The Enigma machine was used by Germany during World War II.

Comets

Comets are objects in the solar system made of ice, dust, and rock. They have a hard core and long tails of gas and dust. Every now and again they appear within sight of Earth, before disappearing into deep space.

SEE ALSO

- ▶ Asteroids p.30
- ▶ Gases p.117
- ▶ Gravity p.125
- ▶ Meteorites p.164
- ▶ Solar system p.233
- ▶ Sun p.247

Dust tail

Gas tail

Two tails

When a comet gets close to the sun, its ice melts and two tails form: one made of gas and one made of dust.

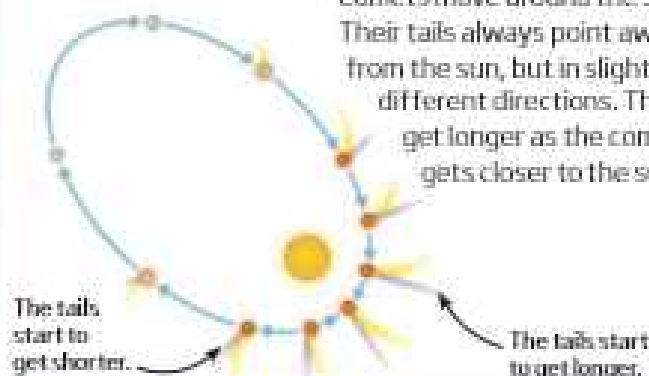
Halley's Comet

Halley's Comet makes one full circle around the sun every 75 years, and it has been recorded by historians for more than 2,000 years. The Bayeux Tapestry, which tells the story of the Battle of Hastings, shows it moving across the sky in the year 1066.



Circling the sun

Comets move around the sun. Their tails always point away from the sun, but in slightly different directions. The tails get longer as the comet gets closer to the sun.



Communication

The different ways people keep in touch are all forms of communication. People have many ways of communicating, such as talking to each other or writing letters. Most modern communication other than talking is done using mobile phones.

SEE ALSO

- ▶ Codes pp.66–67
- ▶ Computers p.71
- ▶ Internet p.138
- ▶ Satellites p.215
- ▶ Games pp.240–241
- ▶ Telephones p.250

Mobile phones

Mobile phones are very useful, as they let people communicate in many different ways.

Email

Email lets us send written messages. It is much faster than sending a letter through the mail.



Photo messaging

Photos can be taken and then shared with other people.



Phone

Using a phone lets you talk to others around the world.



Video calling

This type of call lets you see the person you are speaking to.



Gaming

People can chat and talk to one another while they play games.



Internet

The internet lets people find out information very quickly.



Texting

Texts are short written messages. Texting is a popular way to communicate.

Past and future

People have always found clever ways to communicate. The first humans drew pictures for each other on cave walls. Modern communication is becoming more high tech.



Telegraph

In the early 1800s, messages were sent in a series of dots and dashes (called Morse Code) over a thin wire.



HoloLens

This is a computer you can wear. It displays 3-D objects that the wearer can interact with.

Compass

A compass is a simple instrument that helps people find their way around by showing directions. It is usually round and contains a freely rotating, magnetic needle that always points north-south. This lets you work out other directions.

SEE ALSO

- ▶ Ancient China p.16
- ▶ Inside Earth p.135
- ▶ Magnets p.151
- ▶ Maps p.155
- ▶ Navigation p.182

How to use a compass

When using a compass, lay it flat and then turn it until the north end of the needle is above north on its base. Now you know which direction is north, you can find the other directions.

Direction hand

This hand can be turned to mark the direction that you want to travel toward.

Compass rose

The base of the compass shows all the different directions and is known as the compass rose.

Directions

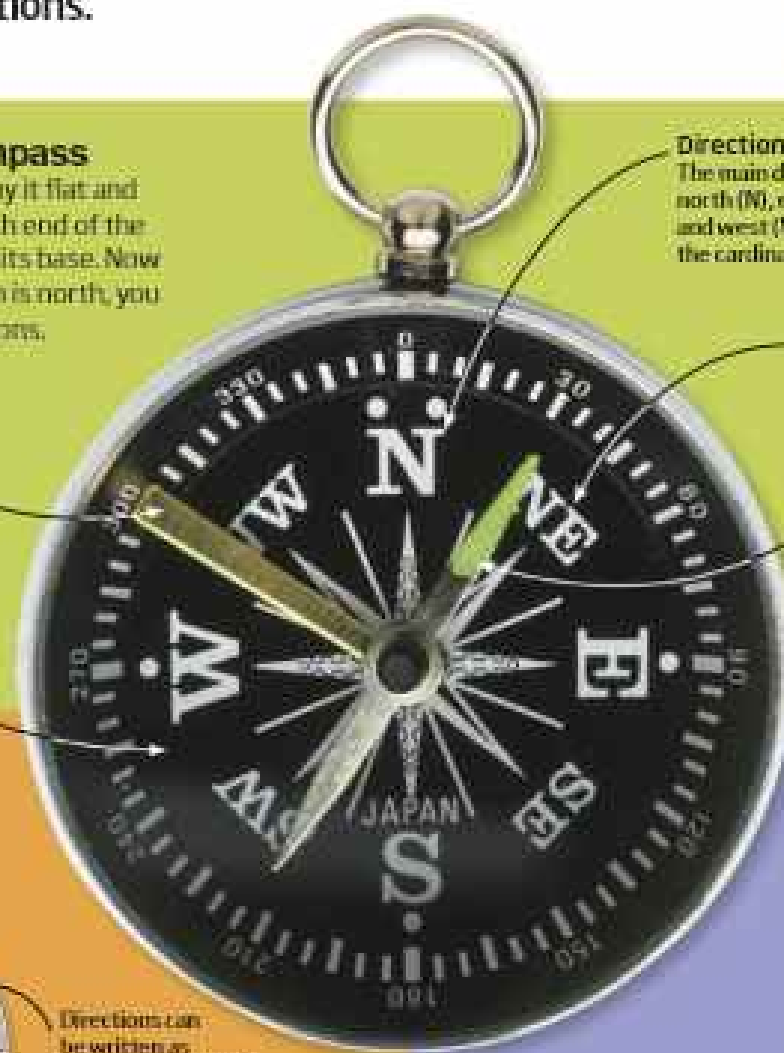
The main directions are north (N), east (E), south (S), and west (W). They are called the cardinal directions.

Finer directions

Between the four cardinal directions are more precise ones, such as northeast (NE) and southwest (SW).

Needle

The magnetic needle detects Earth's magnetic field and lines up with north-south. The end of the needle pointing to north is usually colored or marked.



Directions can be written as angles. For example, southwest is 225° .

Walking compass

A walking compass has a see-through back so that it can be used on top of a map. This lets you work out where you are and which direction you want to go in.

A compass on a phone shows which direction the phone is pointing in.



21st-century compass

Many of today's phones contain a device called a magnetometer. It detects the Earth's magnetic field and lets you use your phone as a compass.

Computers

Computers are machines that store information and can be programmed to perform tasks. Many show information on a screen, such as on a mobile phone or a laptop. Others are hidden inside objects to make them work.

SEE ALSO

- ▶ Coding p.65
- ▶ Codes pp.66–67
- ▶ Communication p.69
- ▶ Internet p.138
- ▶ Machines p.150
- ▶ Robots p.212

How computers work

Computers are programmed in code to perform different tasks. This is called software. The programs are stored and run by the parts, or hardware, of the computer.



Screen
Words and images are displayed on the screen.

USB port
Work can be saved and moved between devices on a tiny hard drive, called a USB. It is plugged in here.

Keyboard
Words typed on the keyboard appear on the screen.

Battery
Computers need electricity to work. The battery provides and stores the electricity.

Processor
This part performs the calculations that make the computer work.

Motherboard
This connects everything, so that different parts of the computer can communicate.

RAM
This part stores information, but only while the computer is switched on.

Hard drive
This stores information even when the computer is switched off.

How we use computers

Computers are found inside many everyday objects. Computers let us program these machines to perform specific tasks.



Traffic lights
Computers control when the lights change color.



Video games
Game consoles are computers that let you play games on a television screen.



Industrial robot
Computers tell big machines what to do—often the same thing over and over again.

Conservation

Conservation is the protection of habitats and the plants and animals that live in them. This is important because human activity, such as cutting down trees and throwing away garbage, destroys animal homes.

SEE ALSO

- Climate change p.60
- Farming p.98
- Forests p.109
- Habitats p.126
- Pollution p.198
- Zoo p.281

Habitats

Animals live in areas we call habitats. When people damage habitats, they threaten many animals and plants. Wildlife reserves protect habitats and the animals that live in them.

Forests

People cut down trees for wood and to make space for farms. Some wood is used to make paper. Recycling paper helps save trees.

In danger

Many animals are in danger of dying out. When a type of animal has completely died out, we say it is extinct. National parks and laws try to stop animals from becoming extinct.

Under threat

Here are some of the ways humans have affected animals and their habitats, or homes. We can help protect animals by making small changes in our everyday lives.

Pollution

Pollution happens when people release dirty or harmful substances into the world around them. It can kill wildlife. Reusing and recycling things reduces the amount we throw away.

Fishing

People have caught so many fish from the sea that many species are becoming rare. We can now choose to eat farmed fish instead, leaving wild fish alone.

Constellations

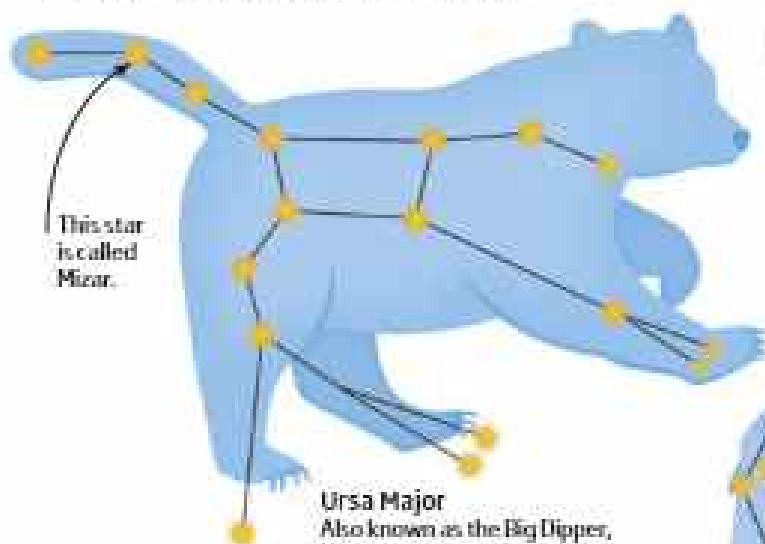
Ancient civilizations looked at the night sky and grouped stars into patterns or constellations to represent heroes, creatures, and objects. As the Earth moves, the constellations appear to move across the sky. The stars are all at different distances from the Earth.

SEE ALSO

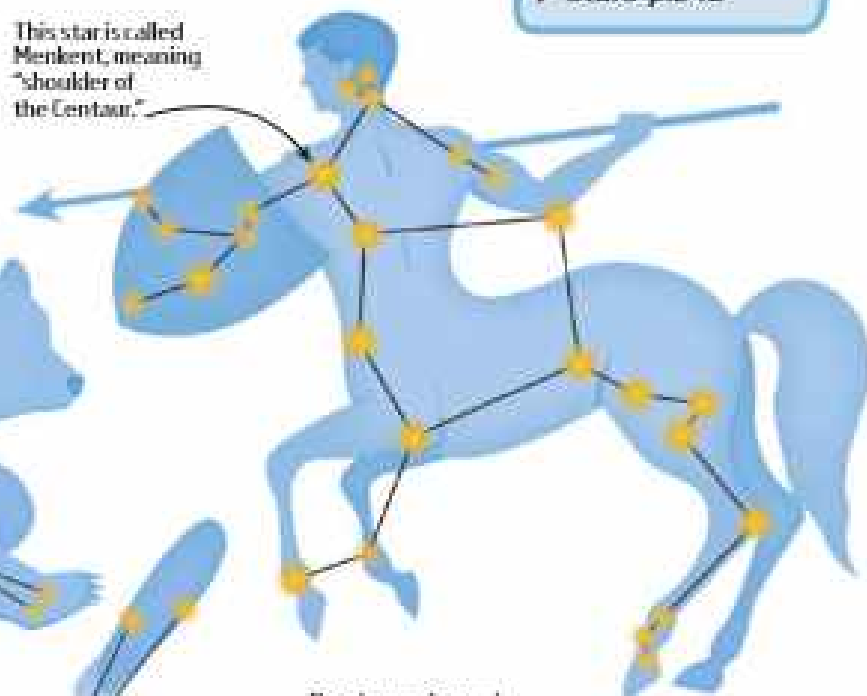
- ▶ Ancient Greece p.18
- ▶ Galaxies p.116
- ▶ Myths and legends p.178
- ▶ Navigation p.182
- ▶ Seasons p.221
- ▶ Stars p.242

Modern constellations

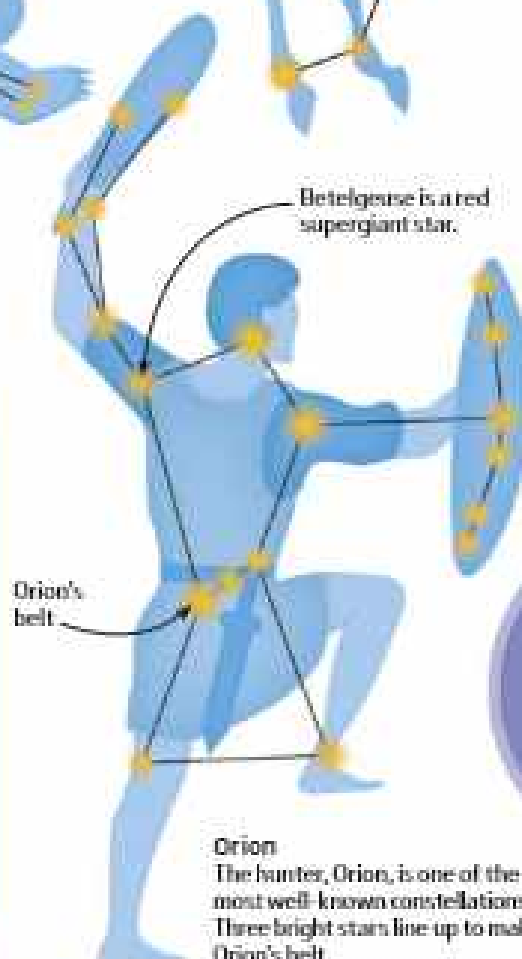
Astronomers today look at 88 constellations. Some can be seen from both the north and south parts of the world, others from only one or the other.



Ursa Major
Also known as the Big Dipper, Ursa Major ("great bear") can only be seen from the northern half of Earth.



Centaurus
This constellation represents a half-man, half-horse creature from Greek myth. It is only visible from the southern half of Earth.



Orion
The hunter, Orion, is one of the most well-known constellations. Three bright stars line up to make Orion's belt.

Navigation

Ancient sailors used the constellations to find where they were. By looking at the pattern of stars they could work out where they were on Earth. One key signpost was Polaris, the North Star.



Over thousands of years, stars shift and constellations change their shape.

Coral reefs

Coral reefs are underwater structures where many plants and animals live. They are made by tiny animals called corals that grow hard shells. When they die, the shells remain and new corals grow on top of them. Some coral reefs can grow very big.

SEE ALSO

- ▶ Animal homes p.23
- ▶ Fish p.101
- ▶ Invertebrates p.139
- ▶ Oceans and seas p.187
- ▶ Vertebrates p.266

Coral reef

Coral reefs cover only a tiny part of the ocean, but are home to nearly a quarter of all ocean life. The reefs are full of food for sea creatures.

Trigger fish eat coral, using strong teeth that crush the corals' shells.

Green sea turtles have sharp beaks to break off seagrasses to eat.

Great Barrier Reef

This reef is home to more than 1,500 different types of fish. It is the longest reef in the world and is found off the east coast of Australia.

Tiny clown fish live among the tentacles of anemones.

This red coral forms a branchlike shell.

The organ pipe coral has leathery tentacles that trap food.

Anemones have swaying tentacles that sting fish if they get too close.

These hawkfish hide among the coral, swimming out to grab small fish and shrimp.

The shells of these corals are covered in small black spines.

Blind coral shrimp are small animals that feed by cleaning the skin of fish.

Blue corals grow together to form columnlike groups.

If threatened by other animals, reef crabs pretend to be dead.

Fan corals bend and sway in the water, catching food.

Many small animals use the seagrass as hiding places.

Crafts

A craft is something done by hand, often with a lot of skill. People have always used natural and artificial materials, such as clay or glass, to make things. Craft objects can be for everyday use, such as plates for eating, or for decoration, such as jewelry.

SEE ALSO

- ▶ Ancient Rome p.20
- ▶ Art p.28
- ▶ Books p.44
- ▶ Clothing pp.62-63
- ▶ Inventions pp.136-137

Pottery

Potters use clay from the ground to make things such as plates, bowls, cups, and vases. They shape the clay, then put it in a special oven called a kiln to heat it up and make it hard.



African wooden figure

Woodcarving

Woodworkers shape wood into different things. They can make furniture, bowls, and other useful or decorative items.



Red terra cotta clay

Ancient Egyptian vase

Weaving

Weavers bring together wool, silk, or cotton threads to make material. This can be used for many things, such as clothes, rugs, and wall decorations.



Native American woven rug



Glass Roman jug

Glasswork

When sand is heated to a very high temperature, it becomes liquid glass. This can be shaped and cooled into solid objects such as jugs.



Beads are made in different sizes and shapes.

Beading

Materials such as glass can be made into beads. Beads can be threaded onto string to make jewelry, or sewn onto clothing.

Ancient Middle Eastern beads

Dance

Moving your body in time to a beat is called dancing. People dance to music to enjoy themselves, to be close to their friends, and to show their skills. Dances can be very formal with set movements to follow, or much more casual and relaxed.

SEE ALSO

- ▶ Clothing pp.62-63
- ▶ Music pp.176-177
- ▶ Festivals pp.206-207
- ▶ Religion p.208
- ▶ Sports p.239
- ▶ Theater p.253



Fans used in dance

Traditional dance
Many countries or regions have their own dances, called traditional dances. Korean fan dancing involves making shapes with fans.



Tribal dance
Many African tribal dances follow drum beats and have strong rhythms. The historical dances of tribes can include whole crowds.



Arms above the head

Feet lift high off the ground

Religious dance
Some people in the Islamic religion dance by spinning around in circles. This is called Sema whirling and helps them feel closer to God.



Elegant arm positions

Ballet
Ballet is a formal dance style with graceful and strong moves. Ballet dancers use a series of precise steps, leaps, and lifts.

Pointed toes



Bollywood dance
Bollywood films from India are famous for their dance routines. Often, the whole cast perform exciting dances with precise arm movements and footwork.

Precise hand shapes



Acrobatic moves

Street dance
Street dance often involves dancers making up their own moves to hip-hop music. They usually do flips and spins.



Complicated footwork

Latin dance
Latin dance began in Latin America. Dances such as the tango involve two people dancing close together, as if they are in love.

Day and night

Day and night are times of light and darkness that occur because the Earth spins, or rotates. One full day-and-night cycle is called a “day” for short. The half of the spinning Earth that faces the sun has daylight, and the half facing away from the sun is in darkness.

SEE ALSO

- ▶ Earth p.83
- ▶ Light p.147
- ▶ Moon p.171
- ▶ Seasons p.221
- ▶ Solar system p.233
- ▶ Sun p.247
- ▶ Tides p.254

What makes day and night?

As the Earth spins round, parts of it move in and out of the sun's light. The light parts are in day and the dark parts are in night.

It takes the Earth 24 hours to make one rotation.

Half of the Earth is in darkness.

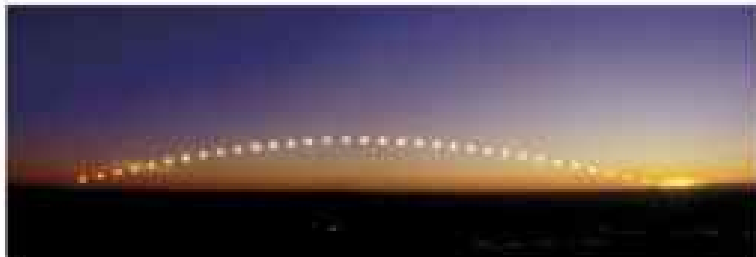
Earth spins on its axis, which is an imaginary line that goes through the North and South poles.

The sun is where Earth's light and heat come from.

Light from the sun

Half of the Earth is in light.

The Earth used to **spin** much **faster** in the distant past, and the **days** were much **shorter**.

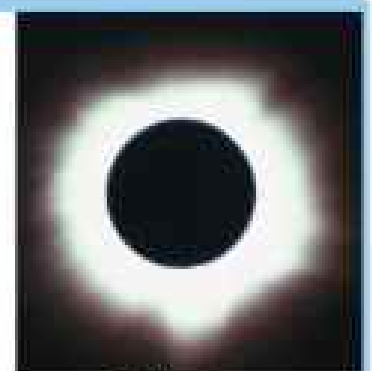


Moving sun

The sun appears to move across the sky during the day as the Earth spins around it. It rises in the east and sets in the west. In the summer, the sun is higher in the sky than in the winter.

Eclipses

The moon circles the Earth. Occasionally, it blocks our view of the sun during the day, and the sky darkens for a few minutes. This event is called a solar eclipse. If the moon blocks all of the sun, it is called a total eclipse, and stars can be seen in the sky.



Total eclipse

Deserts

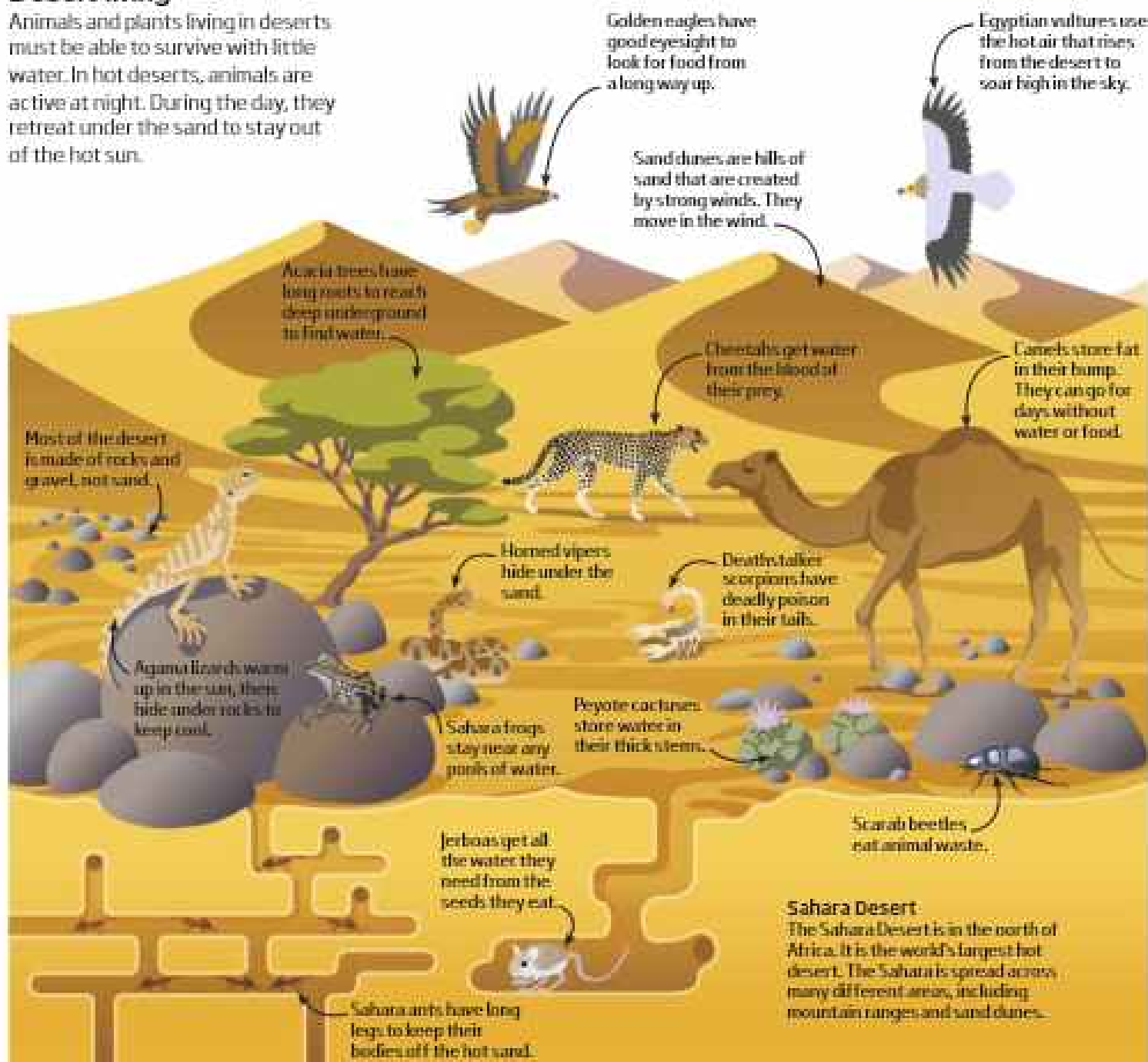
The world's driest areas are deserts. They have less than 10 in (25 cm) of rain a year. Deserts can be sandy, rocky, or even icy. Most deserts have hot days and cold nights. Some animals survive here by getting water from plants or by only moving around after sunset.

SEE ALSO

- ▶ Africa p.12
- ▶ Antarctica p.24
- ▶ Habitats p.126
- ▶ Mountains p.172
- ▶ Plants p.194
- ▶ Reptiles p.210
- ▶ Weather p.271

Desert living

Animals and plants living in deserts must be able to survive with little water. In hot deserts, animals are active at night. During the day, they retreat under the sand to stay out of the hot sun.



Digestion

Digestion is when we eat food and it gets broken down and used by our bodies to give us the energy we need to move and stay healthy. Your digestive system starts with your mouth and ends at your bottom.

SEE ALSO

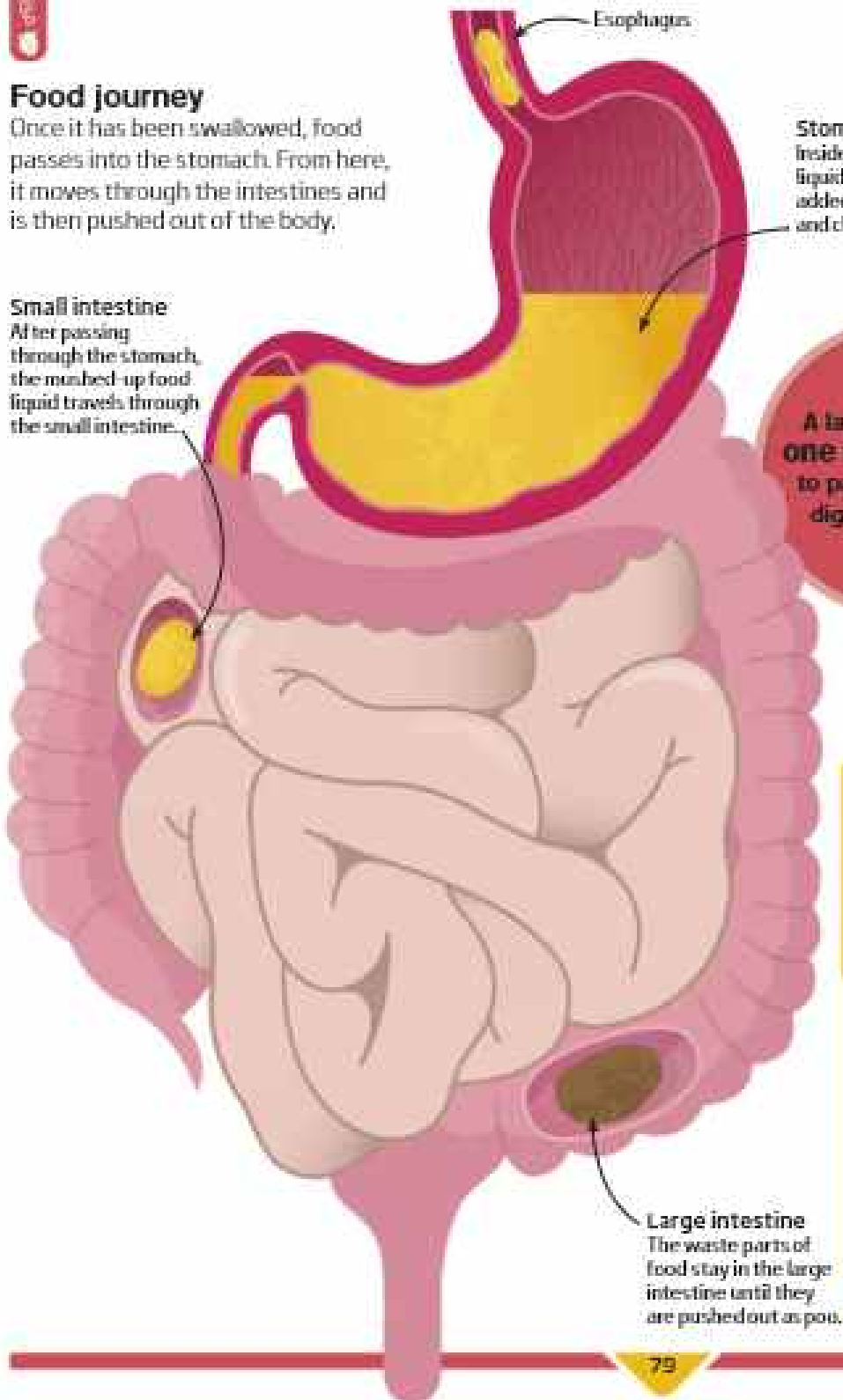
- Food p.106
- Eating pp.104–105
- Human body p.130
- Lungs p.149
- Taste p.249

Food journey

Once it has been swallowed, food passes into the stomach. From here, it moves through the intestines and is then pushed out of the body.

Small intestine

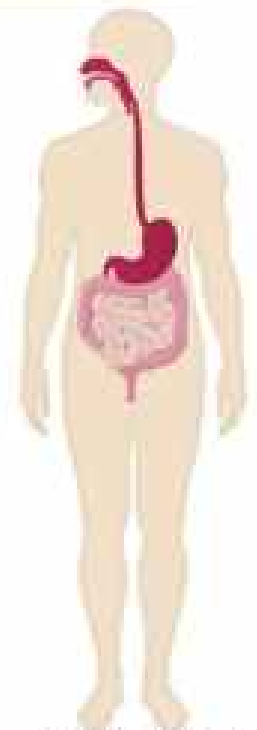
After passing through the stomach, the munched-up food liquid travels through the small intestine.



Stomach

Inside the stomach, liquid chemicals are added to the food and churned around.

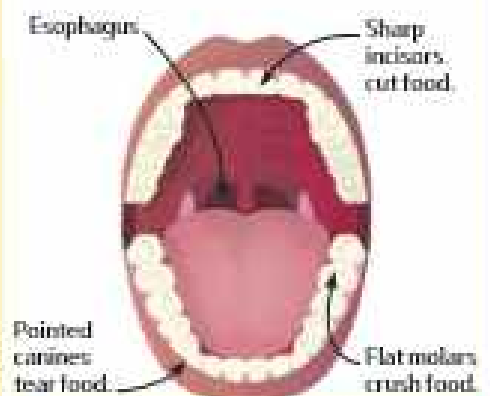
A large meal takes one to three days to pass through the digestive system.



Digestive system

In the mouth

When we chew, food is broken and mixed with saliva. Different teeth do different jobs. When the chewed food is swallowed, it goes down a tube called the esophagus.



Dinosaurs

Dinosaurs are reptiles that lived on Earth for 160 million years, about 225 million years ago. Some were fierce meat eaters, others gentle plant eaters. Scientists have learned about dinosaurs from studying the fossils left behind when the dinosaurs died.

SEE ALSO

- ▶ Asteroids p.30
- ▶ Birds p.39
- ▶ Fossils p.111
- ▶ Prehistoric life p.202
- ▶ Reptiles p.210
- ▶ Rocks and minerals p.214

Ceratopsians

Ceratopsians were plant-eating dinosaurs. They had a protective frill around their head.

Triceratops's frill was used to protect the neck during fights.

The horns were used for defending itself against other dinosaurs.

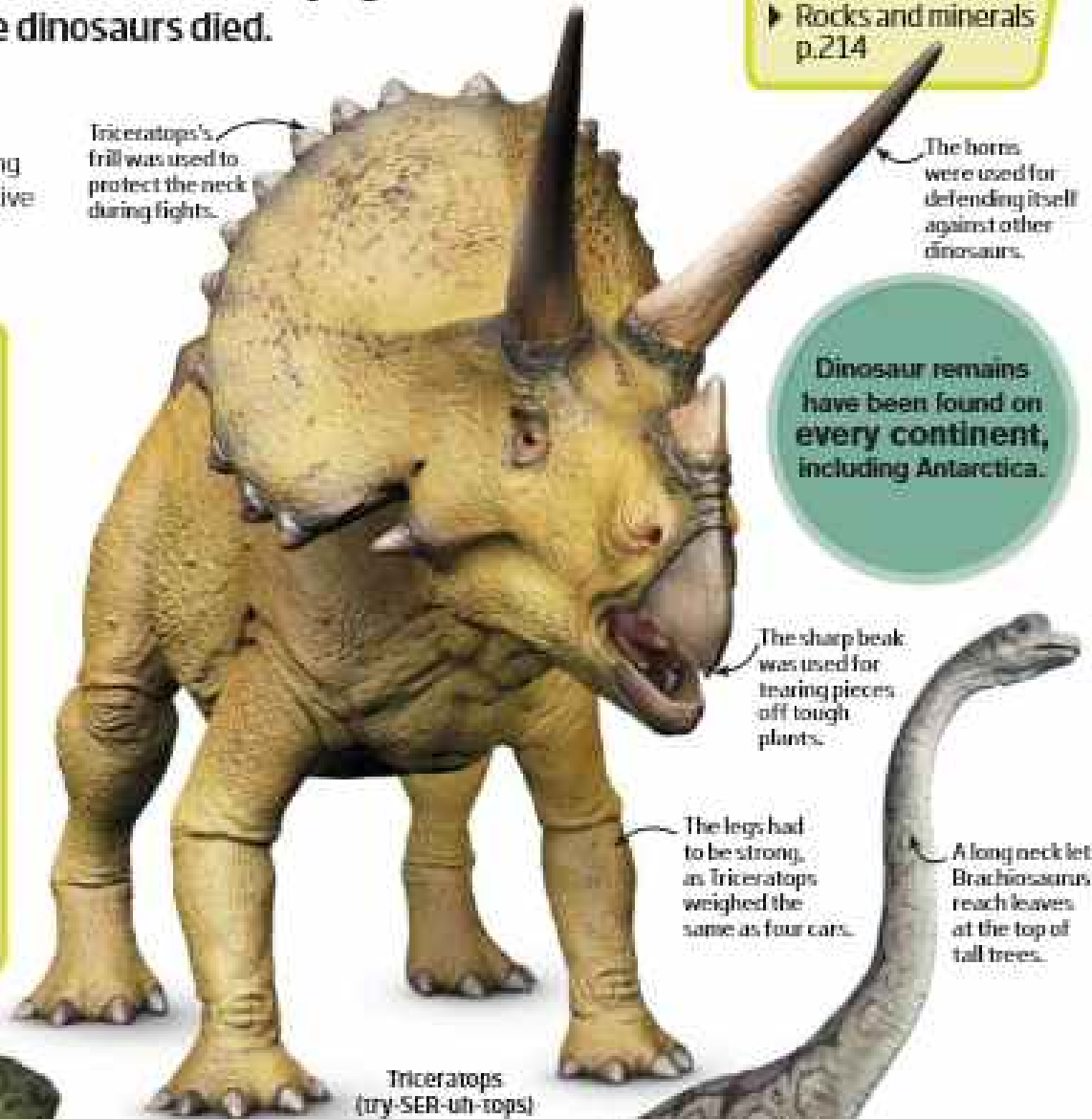
Dinosaur remains have been found on every continent, including Antarctica.

Dinosaur fossils

The remains of dinosaurs have been preserved in rock. Some even show the dinosaur's last meal still inside them.



Archaeopteryx fossil



The sharp beak was used for tearing pieces off tough plants.

The legs had to be strong, as Triceratops weighed the same as four cars.

A long neck let Brachiosaurus reach leaves at the top of tall trees.

Triceratops
(try-SER-uh-topz)

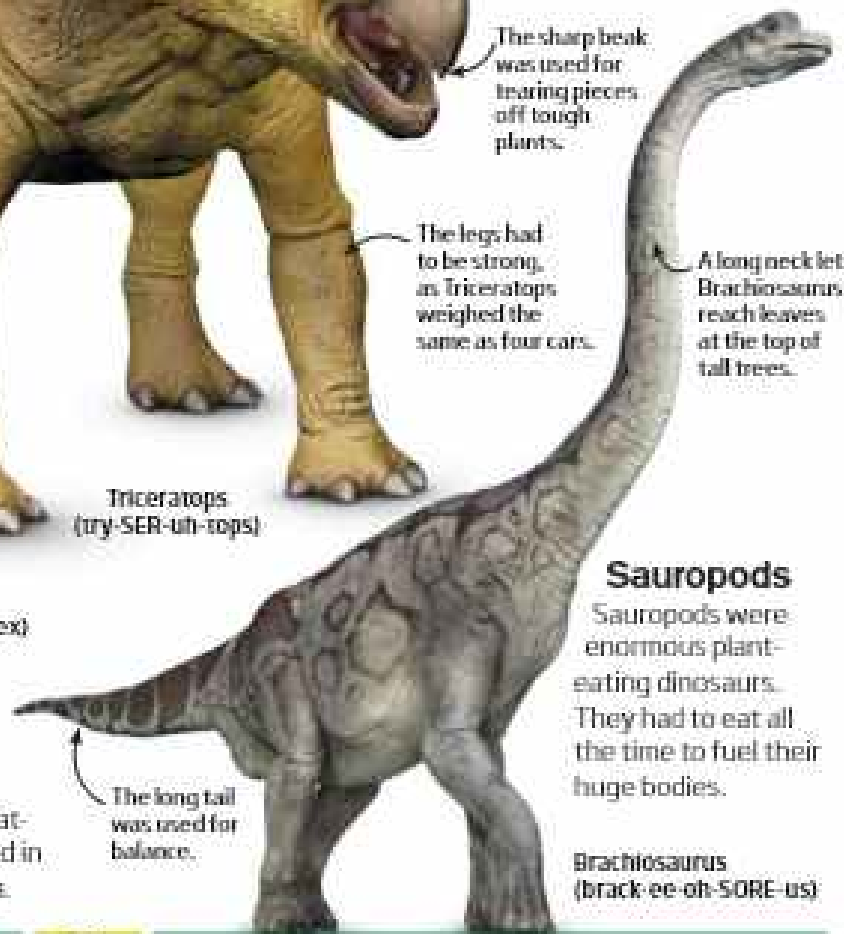


Tyrannosaurus rex
(TIE-ran-oh-SORE-us rex)

Sharp teeth let T. rex tear meat off bones.

Theropods

Theropods were fierce meat-eating dinosaurs. They lived in what is now North America.



The long tail was used for balance.

Sauropods

Sauropods were enormous plant-eating dinosaurs. They had to eat all the time to fuel their huge bodies.

Brachiosaurus
(brack-ee-oh-SORE-us)

Dogs

Dogs are meat-eaters with sharp teeth and excellent senses. They include wild jackals, foxes, and wolves, as well as the tame dogs we keep as pets in our homes. Wild dogs hunt for food or eat animals that have already died.

SEE ALSO

- ▶ Animal families p.21
- ▶ Cats p.54
- ▶ Deserts p.78
- ▶ Hearing p.127
- ▶ Pets pp.152-153
- ▶ Work p.274

Pet dogs

There are lots of different types of dogs. Some are friendly and make good pets. Some are strong and loyal to their owners and are good at guarding things.

There are
more than
300
different
types of pet
dog.

Irish Wolfhound

Beagle

Lhasa Apso



Grey wolf



Fennec fox

Wolves

The grey wolf is the most closely related animal to pet dogs. Wolves live and hunt in groups called packs.

Foxes

These pointy-eared animals can be found in deserts, icy locations, mountains, and even cities. The fennec fox is the smallest fox.

Working dogs

For thousands of years, dogs have been known as man's best friend. This is because they work closely with people. Dogs work in the fields, hunt, and even sniff out people who have been buried in rubble or snow.



Early humans

The first humans were similar to apes, such as gorillas and chimpanzees. Over millions of years, they learned to walk on two legs and got smarter as their brains grew larger. They also lost most of their body hair, slowly becoming more like the humans we are today.

SEE ALSO

- ▶ Africa p.12
- ▶ Evolution p.95
- ▶ Fossils p.111
- ▶ Monkeys and apes p.170
- ▶ Exploration pp.180-181
- ▶ Stone Age p.243

Ancient humans

There were many different types, or species, of human relatives. Some of them lived at the same time and may have met each other regularly.

7 million years ago

Hominins
Early humanlike species, called hominins, developed from apes. They spent a lot of their time in trees, and began walking on two legs.

4 million years ago



2 million years ago

First toolmakers
Homo habilis was one of the first species to use stone tools to help with work. The tools made it easier for them to get food.



3 to 2.5 million years ago

Prehumans
Australopithecus is a species of hominin that learned to walk fully upright, like humans do today.

First fire-makers
Human relatives became steadily more clever and ate more meat. *Homo erectus* may have been using fire to cook food over a million years ago.

200,000 years ago

Modern humans
Modern humans appeared in Africa. The tools they made helped them adapt to other environments. They spread across the world, while other humanlike species died out.



Human evolution

The first humanlike animals were short, had small brains, and lived mostly in trees. Over many years, they began spending more of their lives on the ground.



How do we know?

Ancient humans left behind bones and tools. Scientists study the bones to learn everything from how ancient humans walked to what they ate and what diseases they had. Their tools can reveal information about daily life.



Ancient skull

Earth

Earth is the planet we live on. It is the third planet away from the sun and the largest rocky planet in the solar system. Earth is about 4.5 billion years old. At the moment it is the only planet known to support life.

SEE ALSO

- ▶ Atmosphere p.33
- ▶ Earth's surface p.84
- ▶ Gases p.117
- ▶ Water pp.120-121
- ▶ Solar system p.233
- ▶ Sun p.247

Our home

Earth has all the right conditions for life. It is just the right distance from the sun, it has oceans of liquid water, and a blanket of gases called an atmosphere that protects it from outer space.

Earth has seven large land masses called continents.

After seeing the first pictures from space, scientists nicknamed Earth "the blue marble."

Earth's atmosphere is mostly made of two gases, nitrogen and oxygen.

About 70 percent of the Earth's surface is covered in liquid water.

Habitable

Too cold

Too hot

Earth is here

The white swirls are clouds. Thick patches of white are storms.

Earthrise

On Earth, we see a sunrise and a moonrise as the sun and moon become visible in the sky. When astronauts orbited the moon in 1968, they saw our own planet rising in the sky.

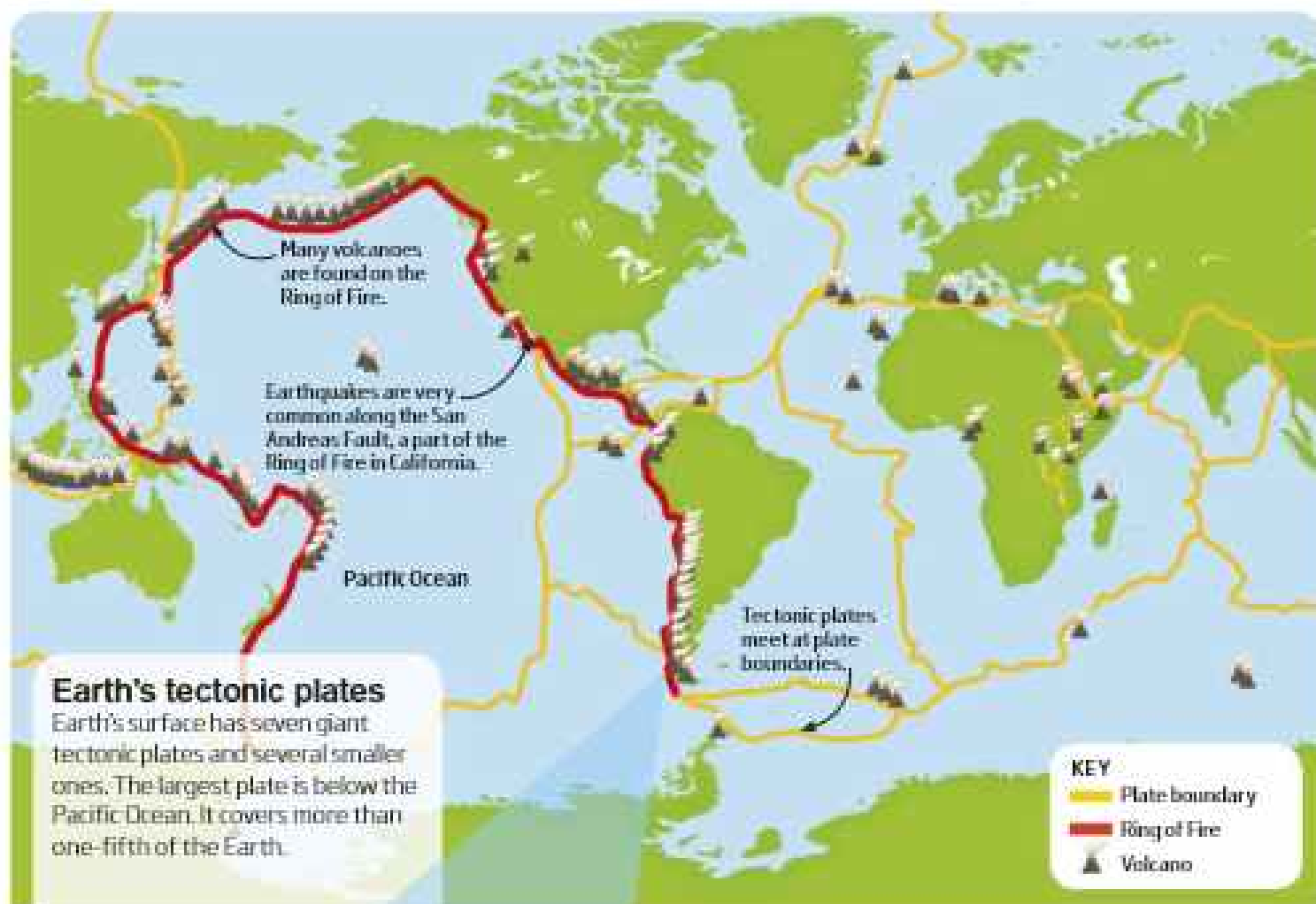


Earth's surface

The outer layer of the Earth is called the crust. It is made up of many pieces, called tectonic plates, which fit together like a giant, ball-shaped jigsaw. Tectonic plates move very slowly, just a few centimeters each year.

SEE ALSO

- ▶ Earthquakes p.85
- ▶ Inside Earth p.135
- ▶ Mountains p.172
- ▶ Oceans and seas p.187
- ▶ Volcanoes p.268
- ▶ World p.275



Earth's tectonic plates

Earth's surface has seven giant tectonic plates and several smaller ones. The largest plate is below the Pacific Ocean. It covers more than one-fifth of the Earth.

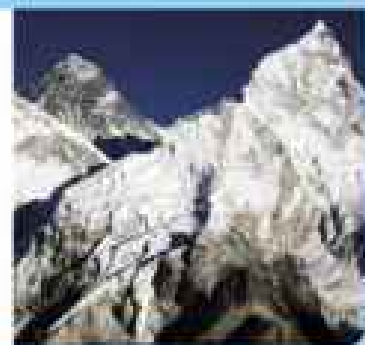
Ring of Fire

Volcanic eruptions and earthquakes are common at the plate boundaries around the Pacific Ocean. This is known as the Ring of Fire.



Mountains

The Himalayas are a mountain range in Asia at the boundary between two tectonic plates. The range formed over millions of years as the plates pushed into each other, forcing the ground up. The mountains are still rising by about $\frac{1}{8}$ in (5 mm) every year.



Earthquakes

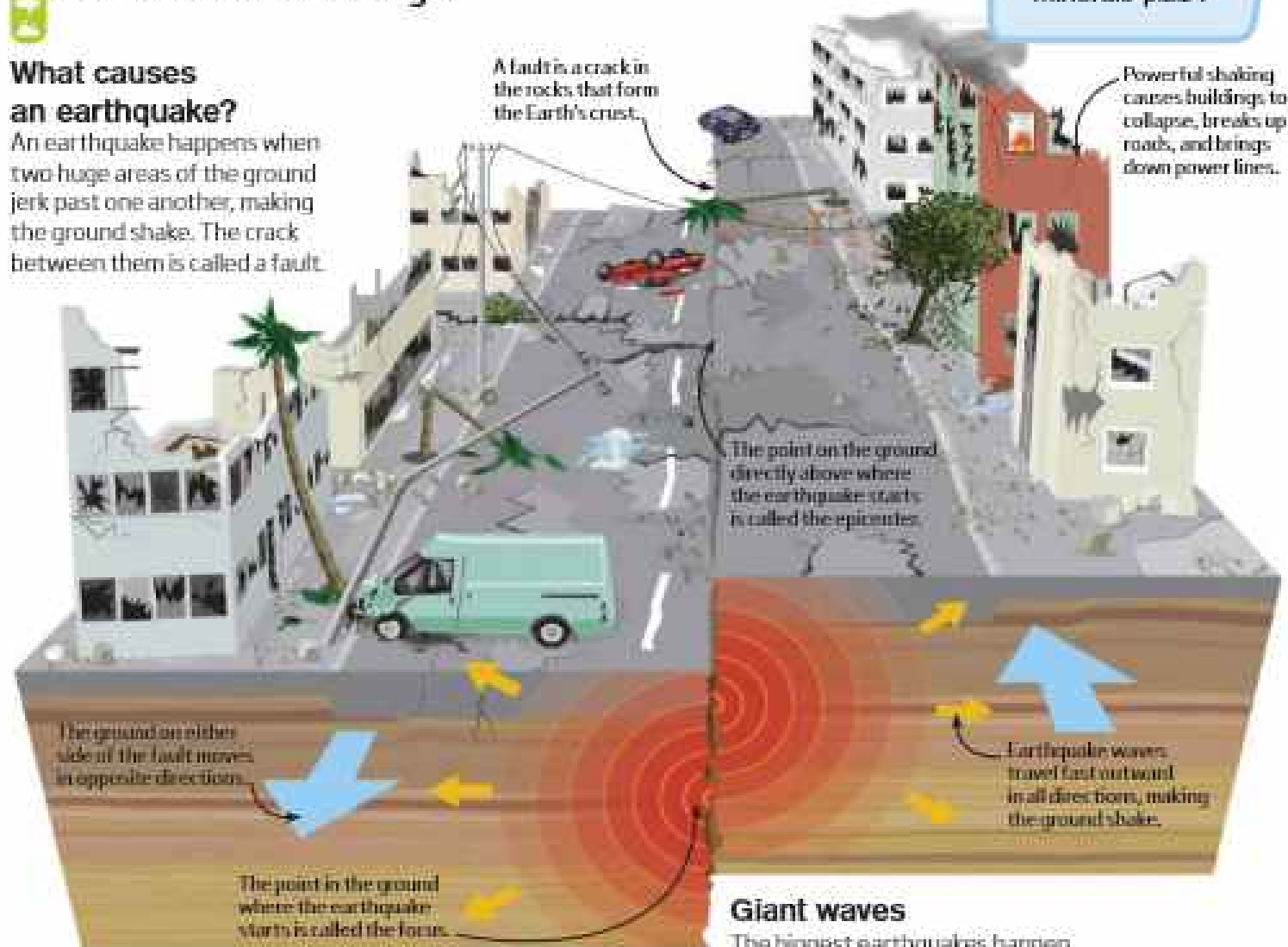
An earthquake is when the ground shakes. Earthquakes happen along cracks in the Earth's crust, known as faults. Small earthquakes can barely be felt by humans, but the biggest can cause enormous damage.

SEE ALSO

- Buildings p.48
- Changing world pp.50-51
- Earth's surface p.84
- Inside Earth p.135
- Rocks and minerals p.214

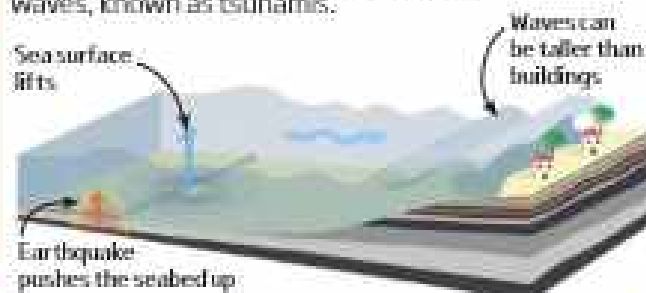
What causes an earthquake?

An earthquake happens when two huge areas of the ground jerk past one another, making the ground shake. The crack between them is called a fault.



Giant waves

The biggest earthquakes happen under the sea. They make the sea bulge upward, forming dangerous giant waves, known as tsunamis.



San Andreas Fault

The San Andreas Fault is a large crack in the ground that slices through California. It marks the join between two huge pieces of the Earth's crust, called plates. Big earthquakes happen on the fault about every 10 years.



Eggs

Some young animals grow and develop inside round objects called eggs. There are different types of eggs. The size of the egg and the length of time it takes to hatch depends on the size of the animal that laid it.

SEE ALSO

- ▶ Amphibians p.15
- ▶ Birds p.39
- ▶ Fish p.101
- ▶ Life cycle p.146
- ▶ Mammals p.154
- ▶ Metamorphosis p.163



Ostrich eggs are the largest in the world.

Bird eggs

Bird eggs are hard and waterproof. They are kept warm, or incubated, by one of the parents. Most bird eggs are kept safe in a nest.

It takes 42 days for an ostrich chick to be ready to hatch.



Ostrich chick

Egg-laying mammals

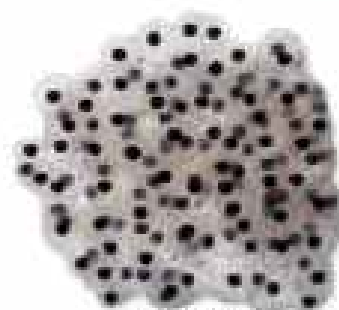
Most mammals give birth to live babies. Only one group, called monotremes, lay eggs. The monotreme shown here is called an echidna.



Tortoise hatchlings are male or female depending on how warm the egg was kept.



Leopard tortoise hatchling



Frogspawn

Fish eggs

Most fish produce lots of eggs and don't look after them. But they do lay the eggs in places like sea grasses to keep them safe.



Dogfish eggs look like leathery bags, which are sometimes called "mermaid's purses."

Reptile eggs

Reptile eggs are soft and leathery. The mother buries them in the ground and leaves them to hatch on their own.

Amphibian eggs

Amphibians such as frogs and toads lay wet eggs in water. When they are ready, the eggs hatch and tadpoles come out.

Electricity

Electricity is the flow of tiny charged particles called electrons. It is used to power lights and electrical appliances, such as kettles and televisions, in our homes, at school, and all around us.

SEE ALSO

- ▶ Atoms p.34
- ▶ Circuits p.59
- ▶ Energy pp.88–89
- ▶ Materials p.157
- ▶ Metals p.162
- ▶ Television p.251

Lightning

Lightning is a type of natural electricity called static electricity. Tiny bits of ice in the clouds rub together, charging up electricity until a big spark of lightning occurs.



Making electricity

We make electricity from different types of energy. One example is how solar panels change the sun's light energy into electricity. Also, wind turbines change the wind's movement energy into electrical energy.



Wind turbines

Using electricity

Appliances like coffee pots and televisions in our homes work when we press a switch to connect them to electricity. Electricity flows through them to make them function.

A laptop charges up using electricity.



The **story** of...

Energy

Energy is power to make things happen. It is everywhere around us. Heat, light, and movement are types of energy. We need energy to make our bodies work and it's what we use to make electricity and power our homes. Energy can be stored and it can change from one form to another.

Fossil fuels

Fossil fuels are made from dead plants and animals squashed underground millions of years ago. Coal, oil, and gas are fossil fuels. We burn these fuels to release heat and this makes electricity in power stations.



Our bodies

Your body needs energy to move, grow, keep warm, and stay alive. The food you eat gets digested and changes inside you to give you the energy you need.

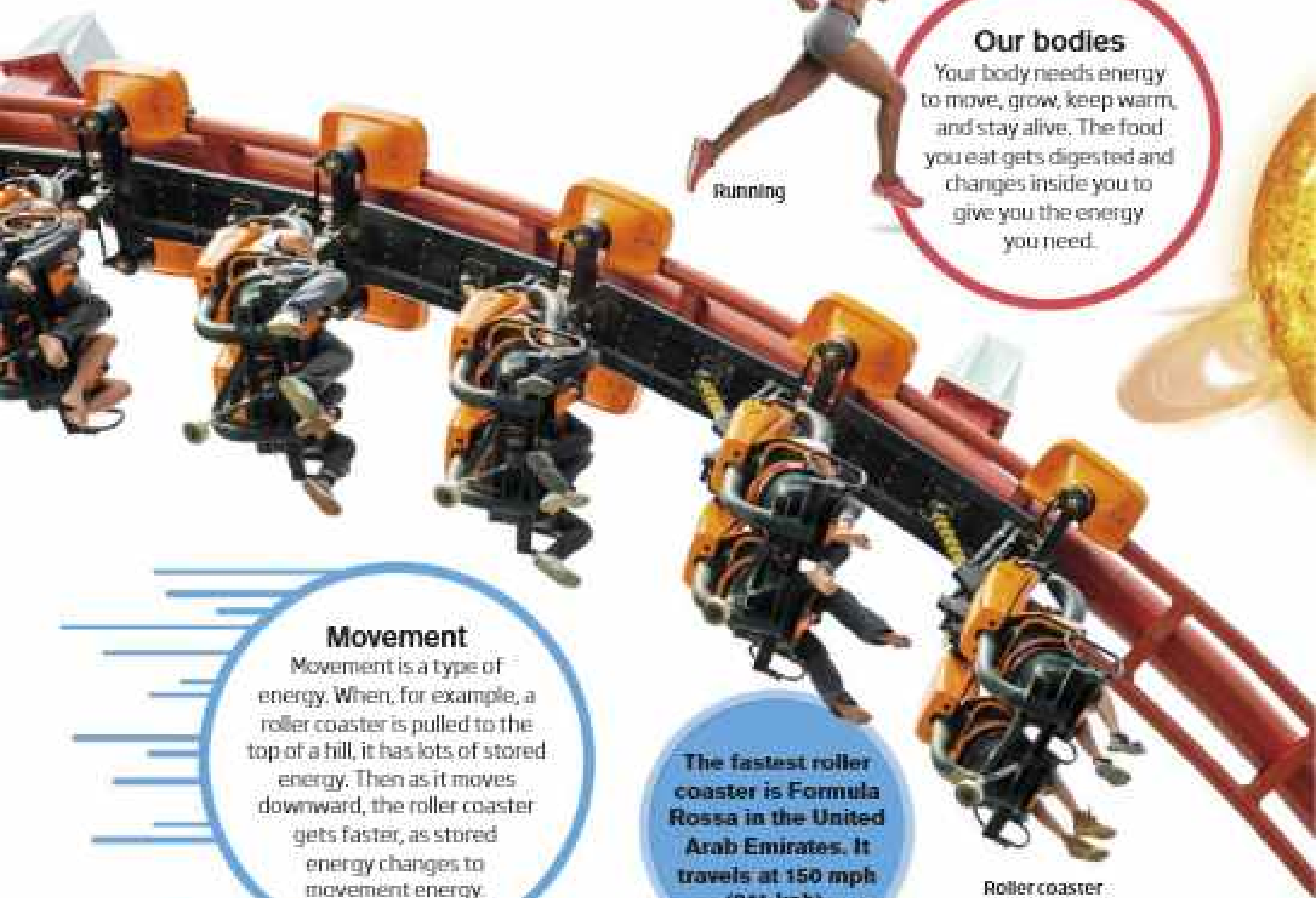


Running

Movement

Movement is a type of energy. When, for example, a roller coaster is pulled to the top of a hill, it has lots of stored energy. Then as it moves downward, the roller coaster gets faster, as stored energy changes to movement energy.

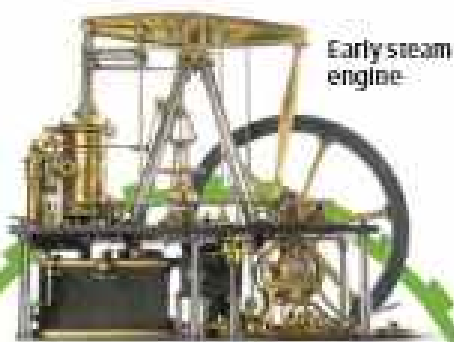
The fastest roller coaster is **Formula Rossa** in the United Arab Emirates. It travels at 150 mph (241 kph).



Roller coaster



Burning coals



Early steam engine

Industrial Revolution

From the late 1700s, people began to use energy in new ways, creating huge industrial growth. Movement energy from water turned wheels to drive machines to weave textiles in mills. Heat energy from steam engines drove trains and machines in factories.

The first steam engine was developed in the 1760s by Scotsman James Watt.



The sun

The sun

Most of the energy we use is from the sun. The sun's light energy changes to heat energy, warming planet Earth. Light helps plants grow and plants provide animals with energy.

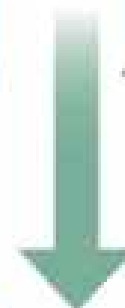
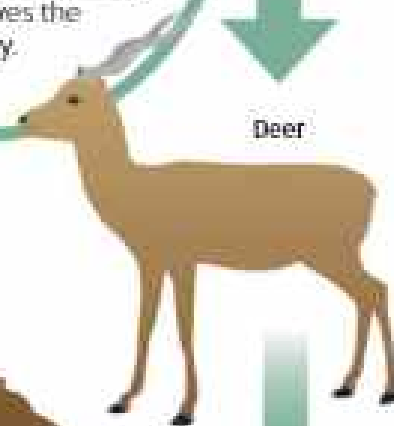
Food chain

Plants take energy from the sun's light and convert it into sugar, a type of stored energy in plants. In this example, the deer eats the stored energy in the plants, which gives the deer energy. The lion eats the deer, and this food gives the lion energy.

Plants



Deer



Lion



Renewable

Renewable energy is energy made from sources that won't run out, such as sunlight, wind, and water. For example, we can use a wind turbine or a water wheel to turn movement energy into electricity.



Wind turbines

The wind turns the blades of the turbine.

Elements

An element is a material that can't be broken down into other materials. All objects are made from tiny particles called atoms, which usually join together in groups. Elements are pure materials, which means they are made from only one type of atom.

SEE ALSO

- ▶ Atoms p.34
- ▶ Electricity p.87
- ▶ Gases p.117
- ▶ Liquids p.148
- ▶ Metals p.162
- ▶ Gold pp.200-201
- ▶ Solids p.234

Elements everywhere

Three-quarters of the elements are part of a group called metals. They are usually solids that conduct electricity. Non-metal elements include gases, such as hydrogen and oxygen, and solids, such as carbon and sulfur.

Aluminum

The soft, light metal aluminum can be made into foil, cans, and airplane parts. It doesn't rust like some metals.

Calcium

The metal calcium is found in rocks, living things, and milk. It helps make bones, teeth, and animal horns.

Helium

The gas helium is used in party balloons because it is lighter than air, so it floats. Helium is made inside stars.



Gold

The valuable metal gold is found in its pure form in nature. It can be hammered into shapes without snapping.

The periodic table

The periodic table lists all the known elements in the universe. There are more than 100 chemical elements and we keep finding more. All the elements have a symbol and are placed in groups according to how they behave and how many particles they have inside them.



The first periodic table, written by Mendeleev

Nitrogen has the symbol N.

Uranium has the symbol U.

Mercury is the only metal that is liquid at room temperature.

Engineering

Engineers use math and science to solve problems. They invent and create machines, buildings, tools, and other useful inventions that make our lives easier. There are different types of engineers that specialize in different areas.

SEE ALSO

- Bridges p.46
- Buildings p.48
- Factories p.97
- Inventions pp.136-137
- Machines p.150
- Materials p.157



Civil
Civil engineers design and build structures such as buildings, bridges, and roads.



Chemical
Chemical engineers turn materials into useful products, including medicines.



Electrical
Electrical engineers create parts for devices such as computers, tablets, and mobile phones.



Mechanical
Mechanical engineers study movement, heat, and energy to help them design new machines and tools.

Engineering process

Engineers develop older technology to make new, better designs. An invention like the wheel has changed over time from the earliest version to a hi-tech modern one.



3D printer

Engineers can now use computer-aided design (CAD) programs to make three-dimensional (3-D) models of their designs. The 3-D printer creates the model using layers of plastic.



3-D printer at work

Engines

Engines power machines by turning fuel into movement. Coal, oil, or electricity is heated to create energy. The energy is then used to turn wheels or parts that move the machine forward. There are three main types of engines.

SEE ALSO

- ▶ Aircraft p.13
- ▶ Cars p.52
- ▶ Factories p.97
- ▶ Forces p.108
- ▶ Machines p.150
- ▶ Ships p.224
- ▶ Trains p.260

Steam engine

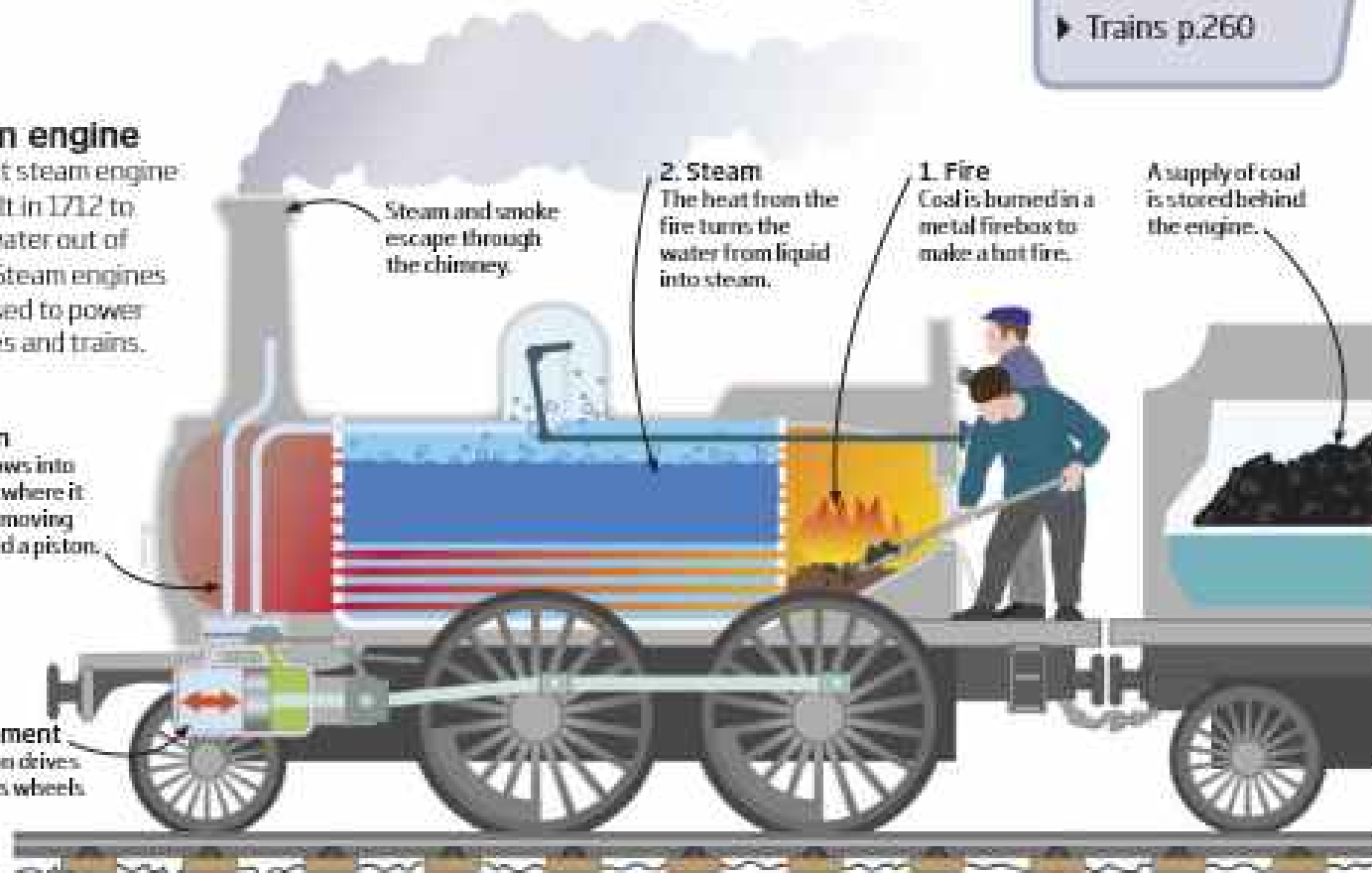
The first steam engine was built in 1712 to pump water out of mines. Steam engines were used to power factories and trains.

3. Piston

Steam flows into this tube where it pushes a moving part called a piston.

4. Movement

The piston drives the train's wheels forward.



Car engine

Car engines burn gasoline or diesel. There are four moving parts called pistons, which move up and down, making the car's wheels turn.



Jet engine

Jet engines are used for aircraft. They work by squashing, heating, and speeding up air. This hot air is blasted out backward, pushing the aircraft forward.

Erosion

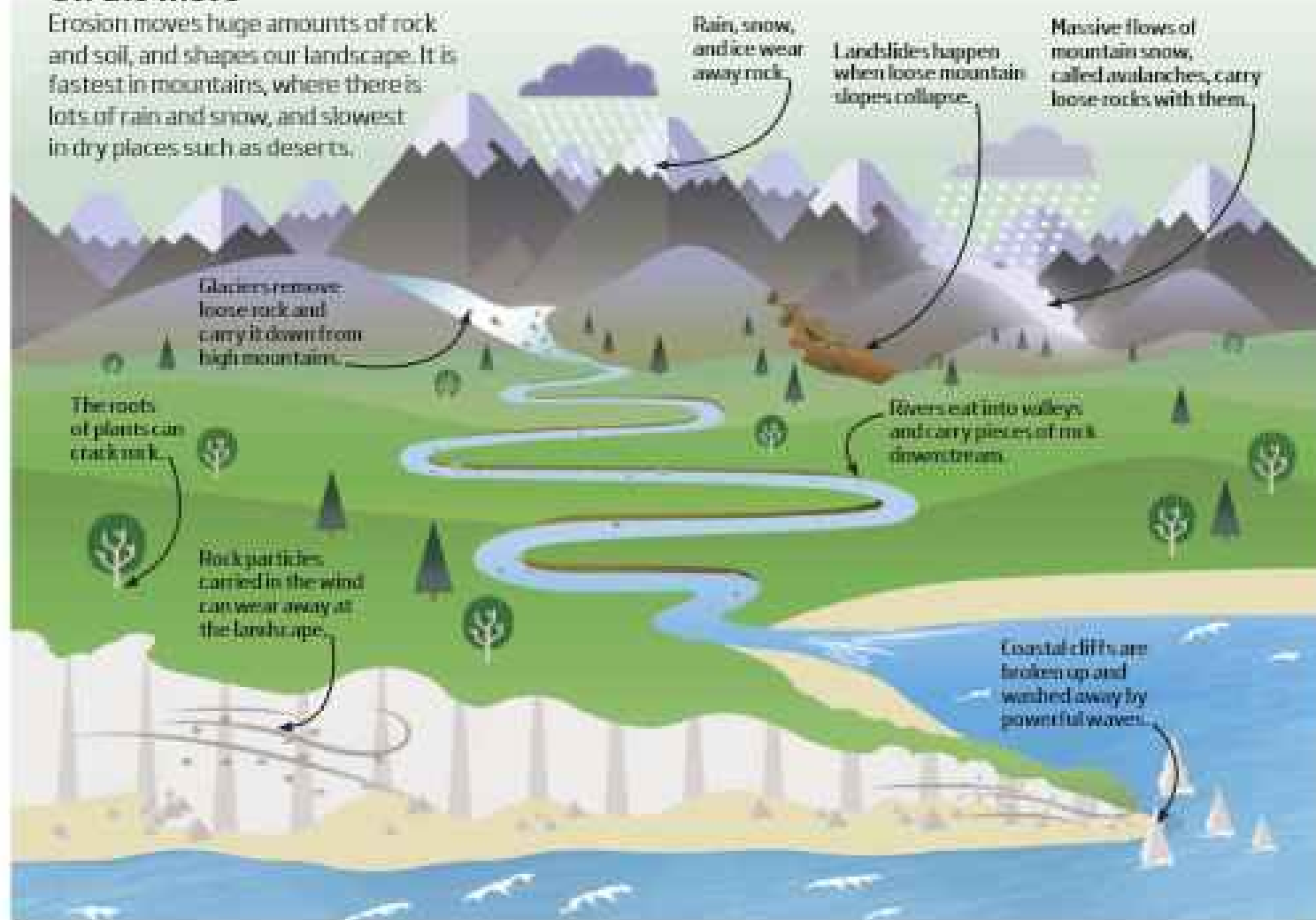
Erosion is the natural movement of rocks, tiny pieces of loose rock, and soil, over long distances. Many different things can cause erosion, including wind, rivers, ice, oceans, and landslides.

SEE ALSO

- ▶ Caves p.55
- ▶ Deserts p.78
- ▶ Glaciers p.122
- ▶ Mountains p.172
- ▶ Rivers p.211
- ▶ Rock cycle p.213
- ▶ Weather p.271

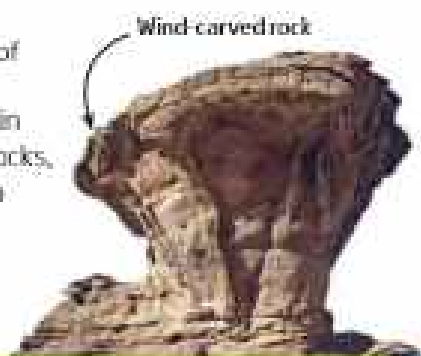
On the move

Erosion moves huge amounts of rock and soil, and shapes our landscape. It is fastest in mountains, where there is lots of rain and snow, and slowest in dry places such as deserts.



Wind sculptures

Over a very long period of time, tiny pieces of rock carried by strong winds in deserts can sandblast rocks, wearing them away into amazing shapes.



Glacier power

Glaciers are rivers of ice that move down from high mountains very slowly. As they move, they pick up rocks that scrape away at the landscape, carving steep-sided valleys and hollows in the ground.



Europe

The continent of Europe is surrounded by ocean, except in the east, where it is joined to Asia. Much of Europe is flat, but there are several high mountain ranges, including the Alps, Pyrenees, and the Carpathians.

SEE ALSO

- ▶ Ancient Greece p.18
- ▶ Ancient Rome p.20
- ▶ Asia p.29
- ▶ World War I p.276
- ▶ World War II p.277

About Europe



Population:
743.1 million



Highest point:
Mount Elbrus



Lowest point:
Caspian Sea



Biggest desert:
Oheila Sahara



Longest river:
Volga

Although it is the
second smallest
continent, Europe
contains nearly **50**
countries.

This powerful
meat-eater is
the biggest
member of the
weasel family.

Beneath the
domes of this
cathedral in
Moscow are
ten separate
churches.

The building
of this cathedral
began in 1882.
It is due to
finish in 2026.

Eiffel Tower

This iron tower in Paris
is 1,063 ft (324 m) high. It
was completed in 1889 and is
the most visited monument
in the world.

The tower is made
up of more than 18,000
pieces of cast iron.

Giant's Causeway

The Giant's Causeway is an area of
hexagonal columns made of ancient
volcanic rock. It is on the coast of
County Antrim in Northern Ireland.



Evolution

In order to survive, an animal needs to change when the weather and food around it changes. This is called adaptation. The theory of evolution is that lots of these tiny changes over millions of years create new types of living things.

SEE ALSO

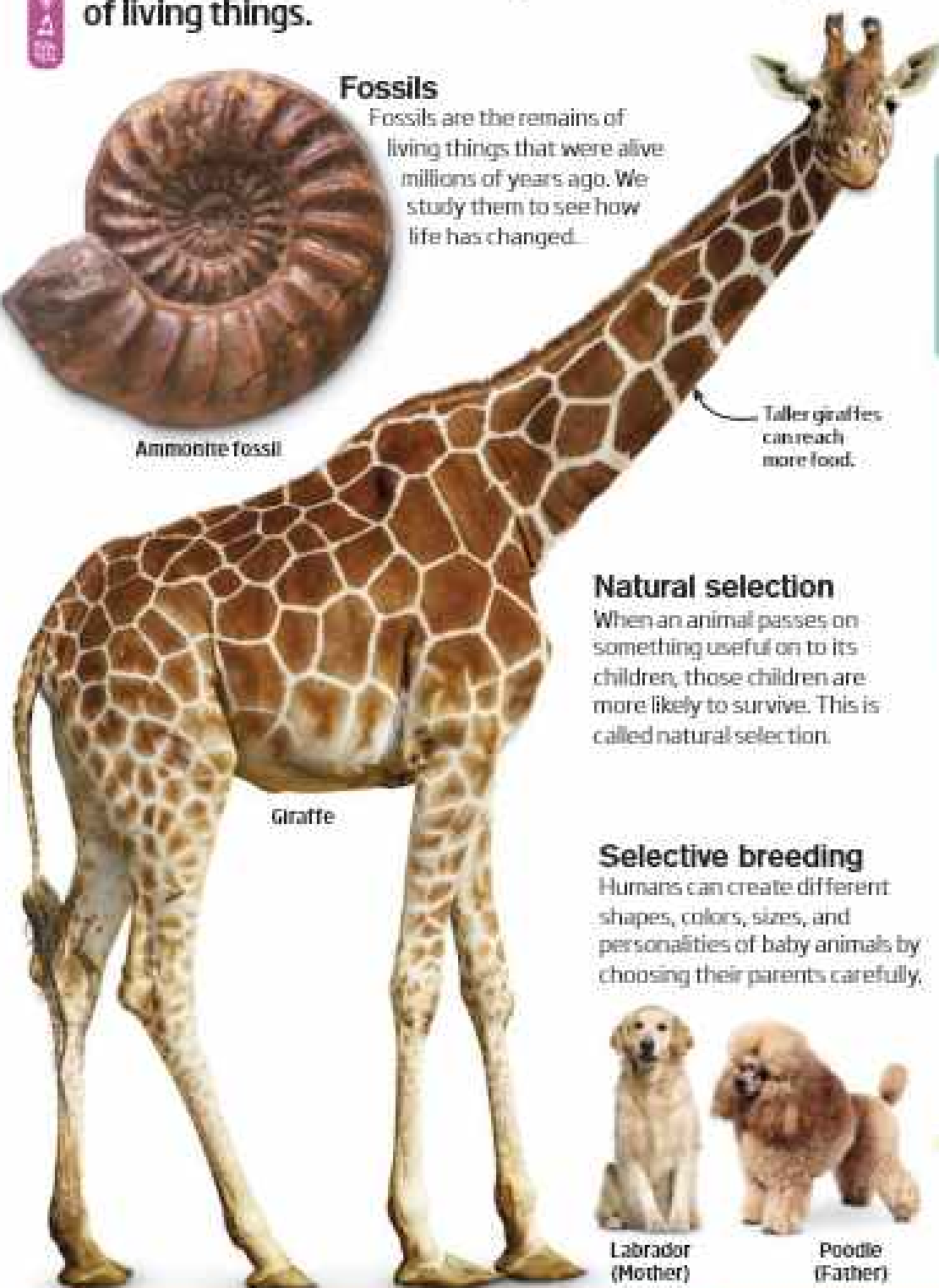
- Dinosaurs p.80
- Dogs p.81
- Fossils p.111
- Genes p.119
- Life cycle p.146
- Prehistoric life p.202

Fossils

Fossils are the remains of living things that were alive millions of years ago. We study them to see how life has changed.



Ammonite fossil



Giraffe

Taller giraffes can reach more food.

Natural selection

When an animal passes on something useful on to its children, those children are more likely to survive. This is called natural selection.

Selective breeding

Humans can create different shapes, colors, sizes, and personalities of baby animals by choosing their parents carefully.



Labrador (Mother)



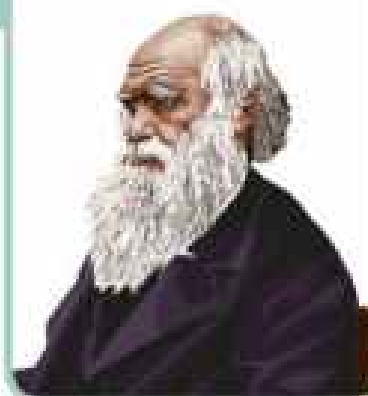
Poodle (Father)



Labradoodle (Child)

Charles Darwin

Scientist Charles Darwin came up with the theory of evolution while traveling around the world investigating living things.



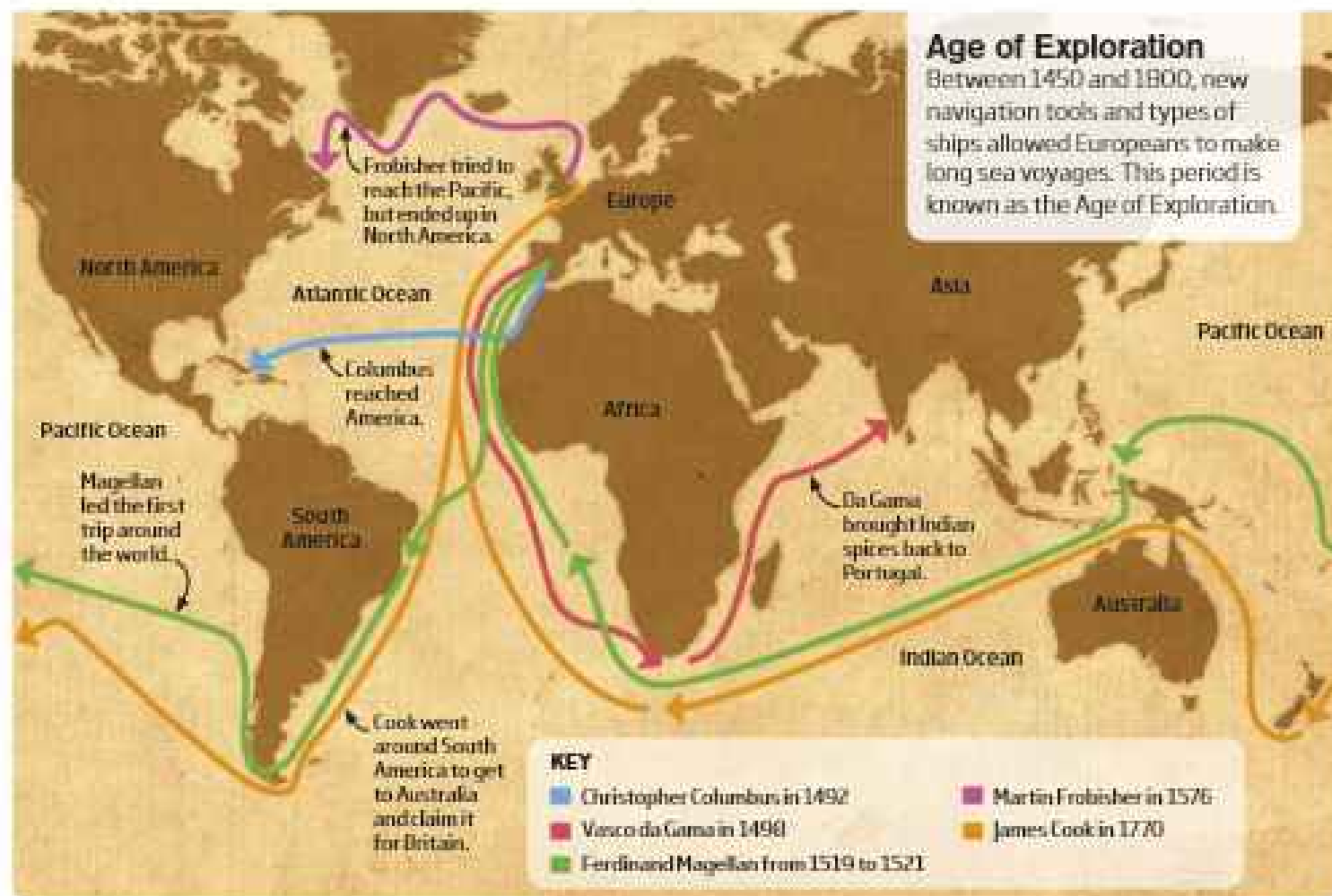
Mammals first appeared and began evolving
220 million
years ago.

Explorers

Explorers travelled to new places to meet different people, find goods for trade, or just to see what was there! They came from all over the world and faced big challenges on their journeys. Some were successful, such as Chinese explorer Zheng He, who reached Madagascar in the 1420s. Others did not complete their missions.

SEE ALSO

- ▶ Europe p.94
- ▶ Maps p.155
- ▶ Exploration pp.180-181
- ▶ Navigation p.182
- ▶ Ships p.224
- ▶ Trade p.257



Trade

Explorers discovered items they had never seen before. Merchants then traded these goods, such as food, spices, and precious metals. Pepper, for example, spread from India around the world.



Magellan

Spanish sailor Ferdinand Magellan set out in 1519 to find a new route to Asia. He left with five ships and 270 men, but only one ship and 18 men made it back.



Factories

Factories are places where people and machines work to make things. When factories make many identical things at the same time, it is called mass production. Almost everything we have, use, and wear comes from a factory.

SEE ALSO

- ▶ Cars p.52
- ▶ Engineering p.91
- ▶ Machines p.150
- ▶ Robots p.212
- ▶ Transportation pp.258-259
- ▶ Work p.274

Assembly line

Things with many parts and different materials are put together at various workstations in a factory. This is called an assembly line.

1. Bodywork

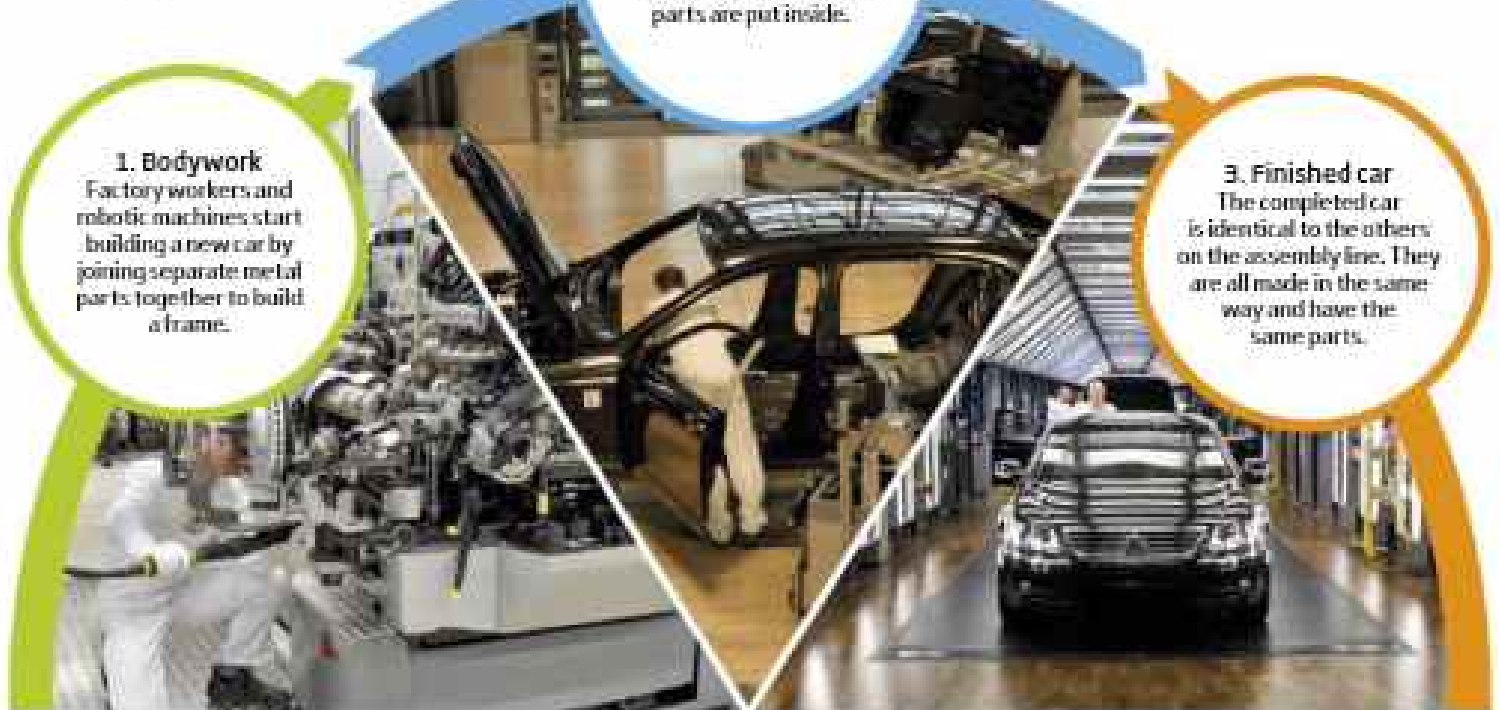
Factory workers and robotic machines start building a new car by joining separate metal parts together to build a frame.

2. Inside and outside

The frame of the car goes to the painting station. The outside is painted, and the seats and other parts are put inside.

3. Finished car

The completed car is identical to the others on the assembly line. They are all made in the same way and have the same parts.



Bottling factories

Drinks are made and bottled in factories. The same ingredients and processes are used to fill thousands of bottles every day.



Oranges enter the factory.



The oranges are squeezed.



The juice is put into bottles.

Farming

Farming is growing plants and raising animals, usually for food. Common crops include cereals, fruit, and vegetables. Farm animals include cows, sheep, pigs, chickens, and even fish. As well as their meat, cows are farmed for their milk and chickens for their eggs.

SEE ALSO

- ▶ Fish p.101
- ▶ Eating pp.104-105
- ▶ Food p.106
- ▶ Fruit and seeds p.115
- ▶ Incas p.132
- ▶ Plants p.194

Cereal farming

Cereal crops such as wheat, corn, and barley are grown in large fields. Rice is also a cereal. It is grown in hot countries in water-covered areas called paddy fields.



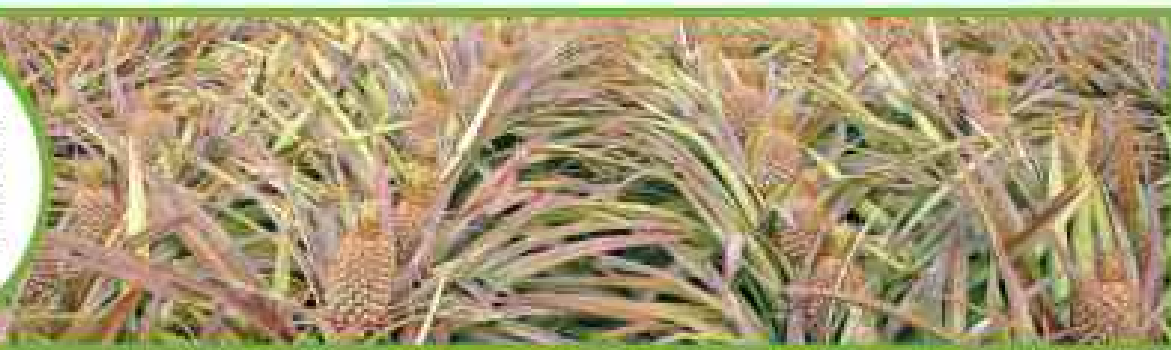
Animal farming

Farmers raise pigs, cows, and chickens in large sheds or outdoors in fields. Sheep, goats, and llamas are often kept on rough or higher ground.



Fruit and vegetable farming

Crops such as pineapples and potatoes are grown outside. Others, such as strawberries and peppers, are grown all year round in glass houses or plastic tunnels.



Fish farming

Much of the fish we eat, such as salmon and cod, is now farmed rather than caught in the wild. The fish are kept in net pens or cages in lakes, rivers, or the sea.



Feelings

Feelings are how we respond to things that happen in and to us. Feelings affect our brains, bodies, and how we behave. Being able to tell other people our feelings is important and helps us to feel connected to each other.

SEE ALSO

- ▶ Art p.28
- ▶ Brain p.45
- ▶ Heart p.128
- ▶ Language p.144
- ▶ Philosophy p.189
- ▶ Taste p.249



Film

A film, or movie, is a series of still images that are quickly played one after the other, so that the pictures seem to move. Films are used to tell stories or show real events. They were first invented in the late 1800s. Live action films are recorded on cameras. Animations are usually drawn by hand or on a computer.

SEE ALSO

- ▶ Art p.28
- ▶ Storytelling pp.42-43
- ▶ Machines p.150
- ▶ Photography p.190
- ▶ Television p.251
- ▶ Theater p.253

Film types

"Genre" is a French word that means type. Films are grouped into genres when they have similar stories or styles. Common genres include action, science fiction, and documentary.

E.T. the Extra-Terrestrial (1982)



Science fiction

Sci-fi films explore themes of science and technology, such as space. "Fiction" means imaginary, and the science is of ten made up.

The Eagle Huntress (2016)



Documentary

Documentaries are a record of real life or actual events. They show the wonders of nature and how people live. This was one of the first film genres.

Spy Kids (2001)



Action

Action films are exciting stories about imaginary heroes or heroines. They use their strength and intelligence to stop people from doing bad things.

Around

2,000

new films are made in India every year.

Animation

Animated films bring drawings or models to life on screen. Drawings are made by hand or on a computer. Stop-motion is a type of animation that uses models. The models are photographed, moved, and photographed again.



The Wizard of Oz (1939)



Musicals

These stories are told through music, song, and dance. They became popular in the 1930s, when films were made with sound and color for the first time.

Silent films

Technology has come a long way since films were invented. The first films were in black and white, and had no sound. Background music was played live at the cinema, and actors used their faces and body gestures to tell a story.



Charlie Chaplin in *A Dog's Life* (1918)

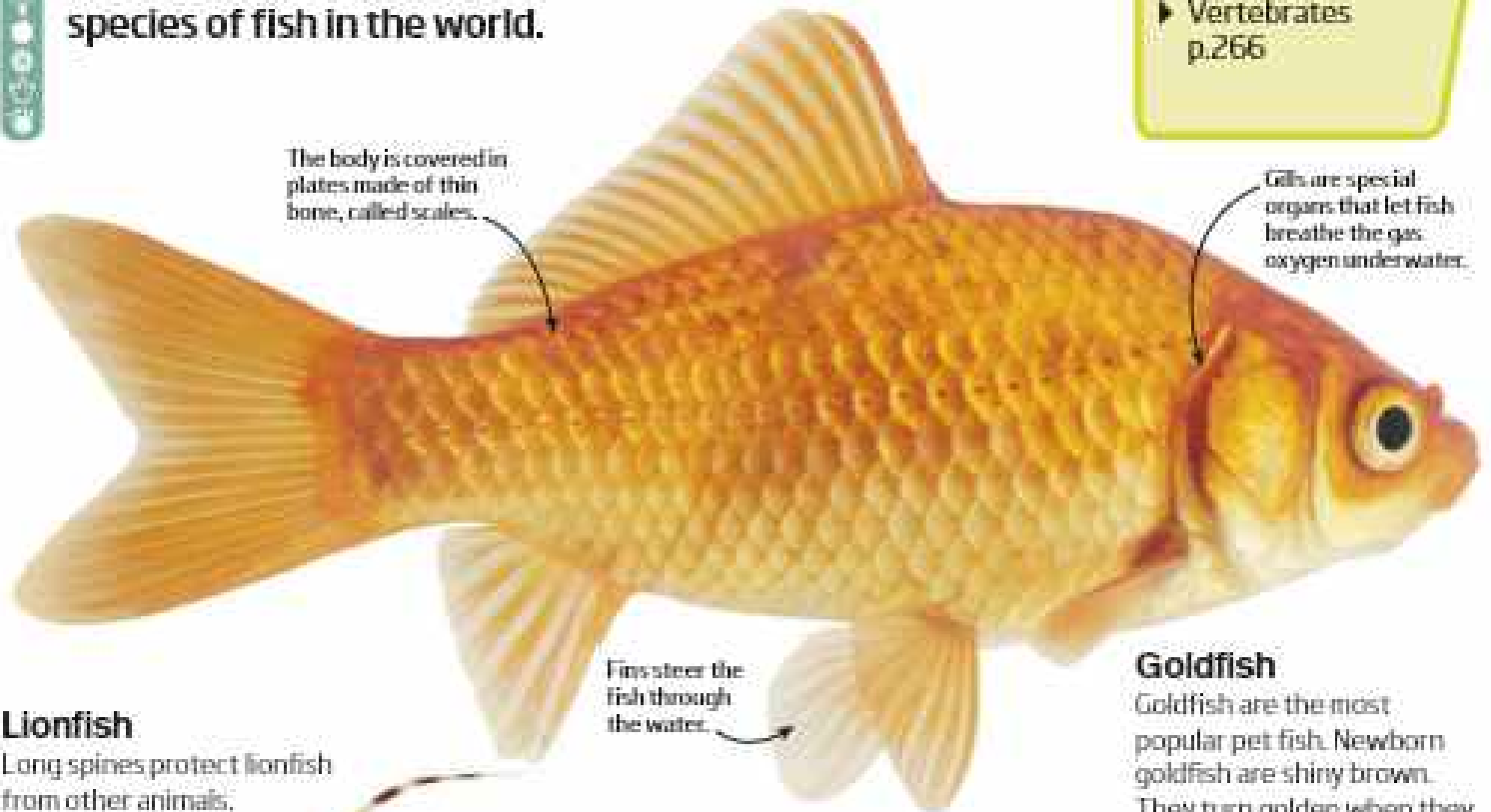
My Neighbor Totoro (1988)

Fish

Fish are animals that live in water. They are able to breathe underwater, and have fins to help them swim around. There are more than 3,000 different species of fish in the world.

SEE ALSO

- ▶ Life cycle p.146
- ▶ Pets pp.152-153
- ▶ Oceans and seas p.187
- ▶ Seashore p.220
- ▶ Vertebrates p.266



The body is covered in plates made of thin bone, called scales.

Gills are special organs that let fish breathe the gas oxygen underwater.

Fins steer the fish through the water.

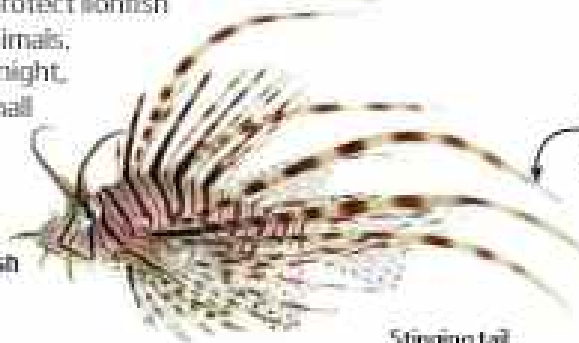
Goldfish

Goldfish are the most popular pet fish. Newborn goldfish are shiny brown. They turn golden when they are a year old.

Lionfish

Long spines protect lionfish from other animals. They hunt at night, feeding on small fish, crabs, and shrimp.

Red lionfish

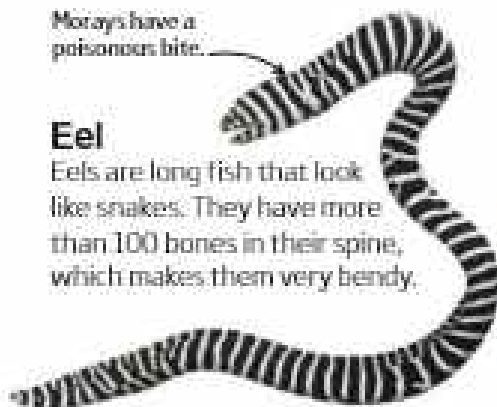


Deadly fins are used to knock out other sea animals.

Morays have a poisonous bite.

Eel

Eels are long fish that look like snakes. They have more than 100 bones in their spine, which makes them very bendy.



Zebra moray eel

Stinging tail has one or two poisonous spines.

Blue spots let other fish know that the stingray is deadly.

Stingray

These fish are found in warm, shallow waters. Most of their time is spent buried in the sand, waiting to pounce on other sea animals.



Blue spotted stingray

Seahorse dad

Most fish don't look after their eggs. Seahorses are different—the male carries the eggs around in a pouch on its belly, until they hatch.



Flags

A flag is a piece of material showing a unique set of colors and symbols. Flags represent a country, city, religion, organization, or sport. The symbols and colors can also represent a message, such as a request for help. Flags are often flown from flagpoles outside buildings to show who the building belongs to.

SEE ALSO

- ▶ Africa p.12
- ▶ Color pp.26-27
- ▶ Asia p.29
- ▶ Governments p.123
- ▶ North America p.184
- ▶ World p.275

National flags

Each country has its own special flag, called a national flag. Most of these have colors or stripes with stars or other symbols placed on top. Each part says something about the country.



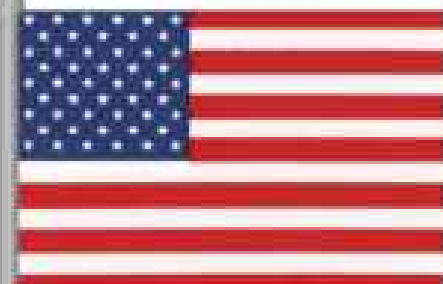
United Kingdom

The upright red cross and the diagonal red and white crosses show the English, Northern Irish, and Scottish flags of the United Kingdom.



China

Red stands for communism, which is the type of government in China. The stars show communism and Chinese unity.



USA

The stars stand for the 50 US states. The stripes are for the original 13 states. The flag is nicknamed "the stars and stripes."

Germany

The black, red, and gold are from the uniforms of German soldiers in the 1800s.

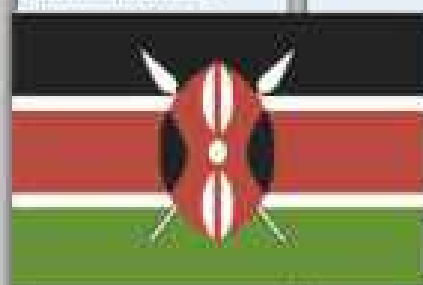


India

The colors stand for ideas such as peace and truth. The central wheel symbol is from the Buddhist religion.

Kenya

The shield is a sign of the Maasai people of Kenya. The color white represents peace.



The first national flag was flown in Denmark in

1478

Signal flags

Flags can be used to send messages. Ships use flags to ask for help if they are in trouble, or to tell other ships to keep out of the way.



Flowers

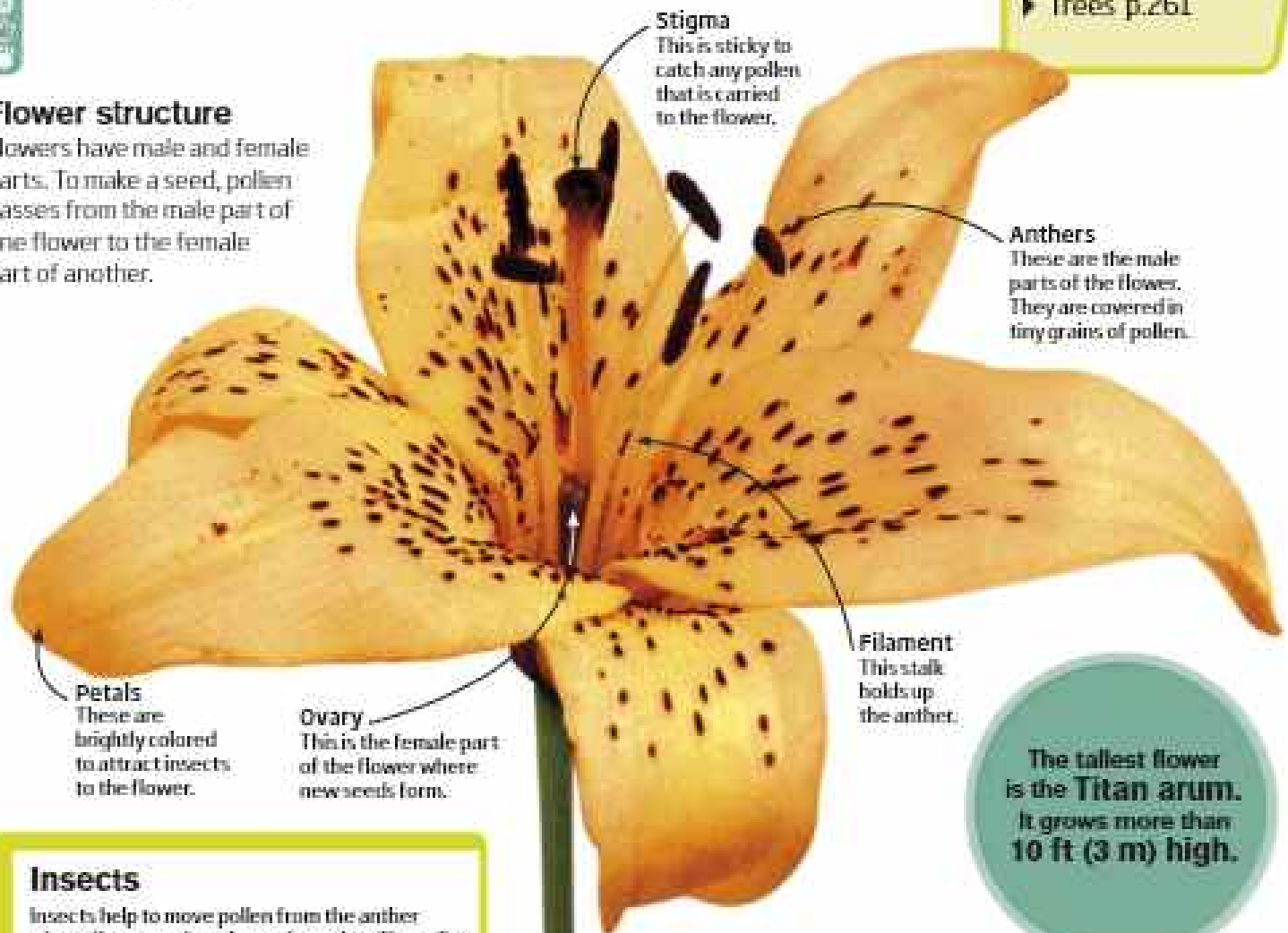
Flowers are a part of a plant. To make new seeds, they swap tiny grains called pollen. Pollen can be spread by wind or insects. Flowers have brightly colored petals to attract insects.

SEE ALSO

- Fruit and seeds p.115
- Habitats p.126
- Insects p.134
- Plants p.194
- Shapes p.222
- Trees p.261

Flower structure

Flowers have male and female parts. To make a seed, pollen passes from the male part of one flower to the female part of another.



Petals

These are brightly colored to attract insects to the flower.

Ovary

This is the female part of the flower where new seeds form.

Stigma

This is sticky to catch any pollen that is carried to the flower.

Anthers

These are the male parts of the flower. They are covered in tiny grains of pollen.

Filament

This stalk holds up the anther.

The tallest flower is the Titan arum. It grows more than 10 ft (3 m) high.

Insects

Insects help to move pollen from the anther of one flower to the stigma of another. The pollen travels on their bodies. Once transferred, the pollen fertilizes the ovary to form seeds.



Flower shapes

Different shapes of flowers attract different insects. Some insects fit down long, narrow flowers, others need big petals to land on.



Dome



Cone-shaped



Regular



Rosette



Bell-shaped

The **story** of...

Eating

When we eat, we take in all the things we need to keep us alive and healthy. Eating food gives us the energy that allows us to think, walk, play, and work. It is also something we love to do.

Food around the world

People used to only eat things that were grown near them. Now, we can eat food from all over the world. Different countries have their own special recipes to make their favorite foods.

Ribs with black-eyed peas and collard greens, from the US



Paella from Spain



Pizza from Italy



Kebabs from Turkey



Dosa from India



Early eating

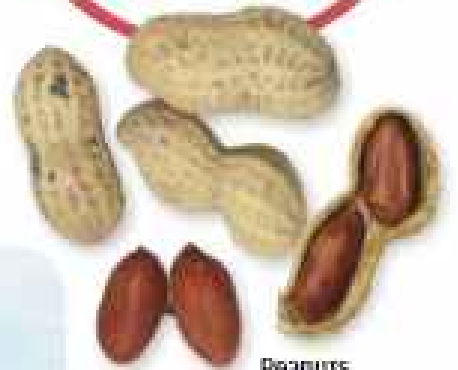
Our ancient ancestors hunted for meat and fish, and foraged for fruit, nuts, and roots. They started using fire to cook food around 400,000 years ago.



Fire

Food allergies

Some people are allergic to certain foods, which means they become ill when they eat them. These foods can include shellfish, peanuts, and milk-based foods.



Peanuts

More than

1.3
billion people
in the world work
as farmers.



Chow mein
from China



Bento box
from Japan

Farming

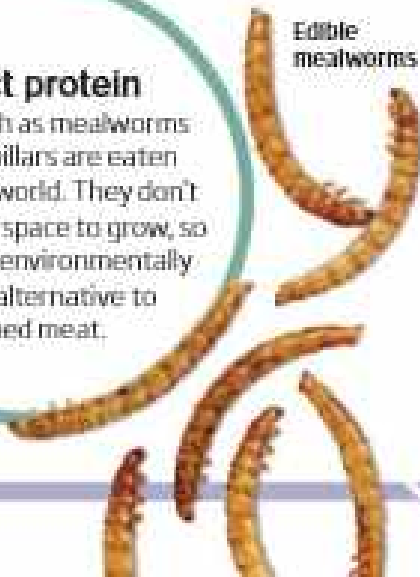
People began farming at least 15,000 years ago. Today, almost half of the world's land is farmed for food. Farmers breed animals for their meat, milk, and eggs. They also grow plants called crops, such as wheat and oats.



Combine harvester cutting wheat

Insect protein

Insects such as mealworms and caterpillars are eaten all over the world. They don't need a lot of space to grow, so they are an environmentally friendly alternative to farmed meat.



Edible
mealworms

Eating in space

Astronaut space food must be easy to eat, light in weight, quick to prepare, and not too messy. Food is often freeze-dried and put in pouches. Water is then added before eating.



Astronaut food

Food

Humans need to eat the right mixture of different food groups to keep all the parts of the body working properly. Food gives us energy to move, grow, and repair our bodies. Types of food give our bodies the different goodness we need.

SEE ALSO

- Digestion p.79
- Energy pp.88-89
- Eating pp.104-105
- Food chains p.107
- Gases p.117
- Plants p.194

Food groups

There are five main food groups that give us the nutrients and vitamins we need to be healthy.

Fruit and vegetables

Fruit and vegetables have fiber, which helps break down our food. They also contain vitamins and minerals that help our bodies work properly.

Drinking

Water transports the goodness we get from food into and around our body. It then flushes waste materials out of our bodies.



Fat and sugar

Sugar and fat give our bodies energy. We can get fat from food like cheese and nuts, and sugars from fruits. Too much fat and sugar is bad for us.

Energy

When we eat, our body changes energy in food into the energy we need to move and grow. Stored energy is turned into movement energy in our bodies.

Carbohydrates

Bread, rice, cereal, and pasta contain carbohydrates, which give our bodies energy.

Dairy foods

Milk, yogurt, cheese, and butter contain calcium to help teeth, nails, and bones grow.

Protein

Meat, fish, eggs, and beans are high in protein, which is needed to grow and repair our bodies.



Food chains

A food chain is the passing along of energy from food. Only plants can make their own food. All animals are part of a food chain, either eating plants or other animals. All animals need energy to grow, survive, and reproduce.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Conservation p.72
- ▶ Eating pp.104–105
- ▶ Food p.106
- ▶ Habitats p.126
- ▶ Photosynthesis p.191

Energy on the move

Energy moves along a food chain. Each animal in the chain gets energy from what it eats. The arrows show how the food energy is passed along.

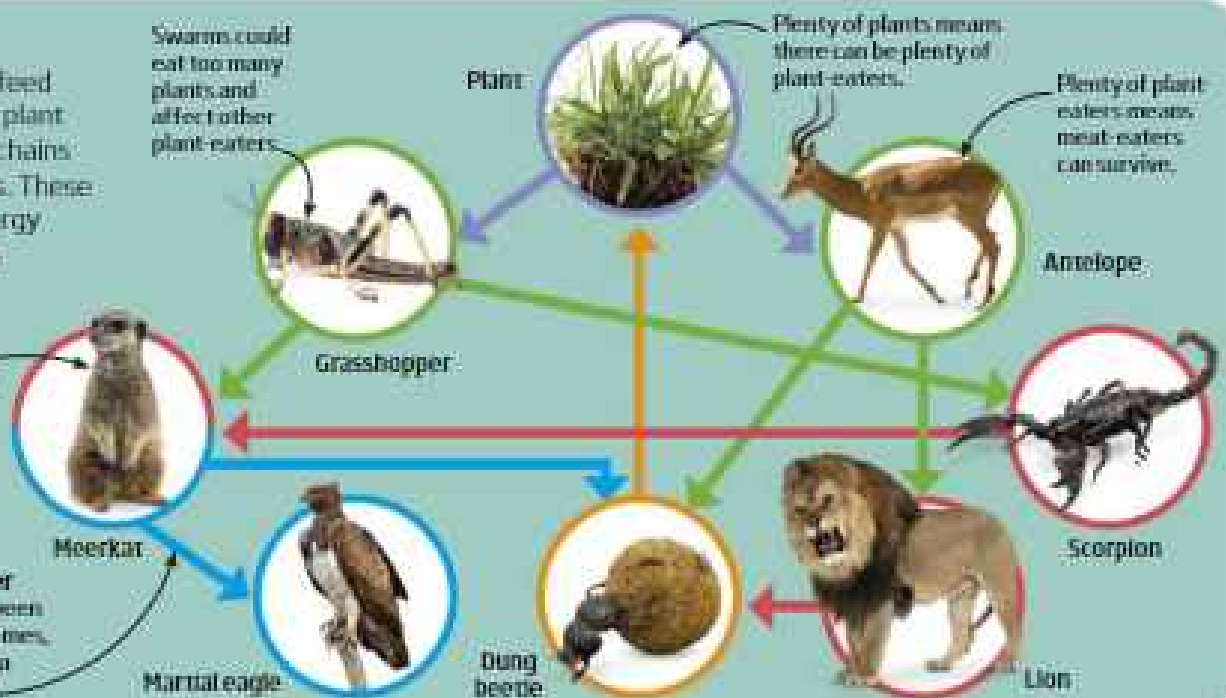


Food web

Animals don't just feed on a single type of plant or animal, so food chains become food webs. These can show how energy is passed around a whole habitat.

Carnivores eat a great variety of food to survive. Meerkats eat grasshoppers, scorpions, and other small animals.

Tertiary consumer
If food energy has been transferred three times, the animal is called a tertiary consumer.





Forces

A force is a push or a pull. Forces can start things moving, speed them up, or slow them down. Some forces work through touching. Others, such as gravity, work invisibly and can affect objects a long way away.

SEE ALSO

- Friction p.114
- Gravity p.125
- Magnets p.151
- Measuring p.159
- Solar system p.233
- Sun p.247

Pushing force

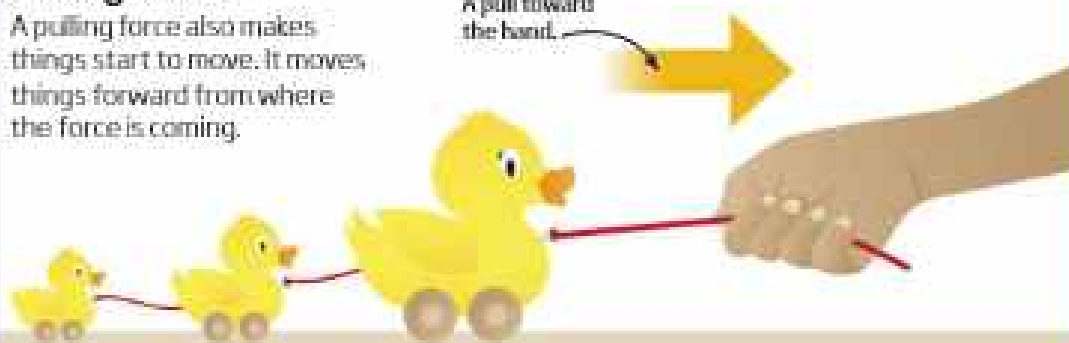
A pushing force makes things start to move and can also speed them up. Your hand applies a pushing force when it moves a toy car.



Pulling force

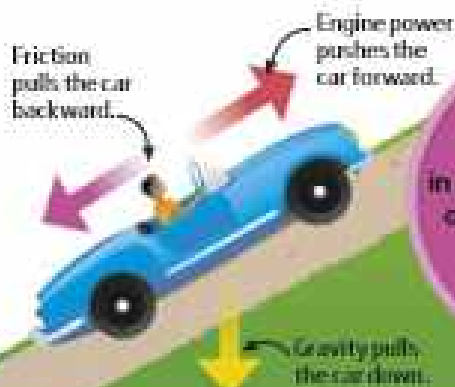
A pulling force also makes things start to move. It moves things forward from where the force is coming.

A pull toward the hand.



Balanced forces

Pushes and pulls can act in different directions at the same time. If they are balanced, the object moves at a steady speed, or is still. If one force is bigger, the object gets faster or slower.



When you sit still in a chair, the forces on you are perfectly balanced.

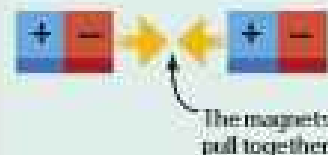
Gravity

Gravity pulls objects down toward the Earth. It's the force that stops us from floating off into space.



Magnetism

Magnetism is a force that pulls objects toward a magnet, or pushes them away. Opposite magnetic forces attract each other.



Friction

Friction is a force that slows down movement. It is created when two surfaces touch.



Forests

Forests are places where many trees grow. Forests grow in a wide range of places all over the world. There are different types, depending on the temperature and how much rain they get. Different trees, plants, and animals live in each type of forest.

SEE ALSO

- ▶ Animal homes p.23
- ▶ Fruit and seeds p.115
- ▶ North America p.184
- ▶ Rain forests p.204
- ▶ Seasons p.221
- ▶ Trees p.261

Deciduous forests

These forests have four seasons, with warm summers and cold winters. Many trees drop their leaves in autumn and grow them back in spring. These are called deciduous trees.



Coniferous forests

These forests are found in cold, northern parts of the world. The trees have needlelike leaves. Their branches slope so that any snow slides off. These are called coniferous trees.



Fossil fuels

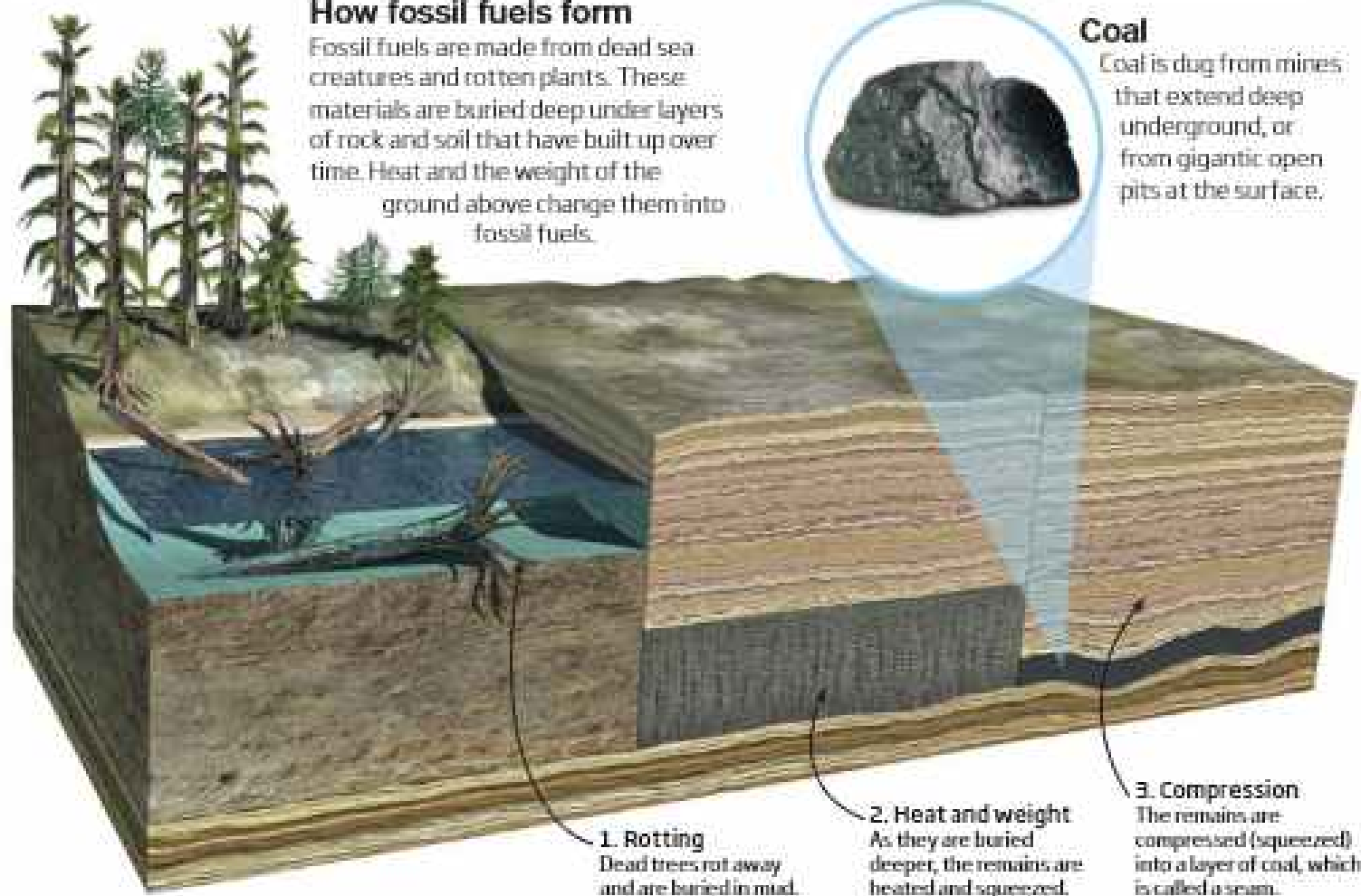
Fossil fuels are natural materials that formed underground millions of years ago. We dig them up or pump them out of the ground so that we can burn them to make energy to power vehicles or to make electricity. There are three types of fossil fuel: coal, crude oil, and natural gas.

SEE ALSO

- Carbon cycle p.49
- Climate change p.60
- Dinosaurs p.80
- Fossils p.111
- Industrial Revolution p.133
- Pollution p.198

How fossil fuels form

Fossil fuels are made from dead sea creatures and rotten plants. These materials are buried deep under layers of rock and soil that have built up over time. Heat and the weight of the ground above change them into fossil fuels.



Coal

Coal is dug from mines that extend deep underground, or from gigantic open pits at the surface.

Generating electricity

Fossil fuels are burned in power stations to make electricity. We have used fossil fuels to make electricity for many years, but burning them harms the environment.



Cooling towers at a power station.

Gas and oil

Crude oil is extracted from the earth by drilling. It is used to power vehicles and make plastics. Gas is also released through drilling. It is used for heating buildings. There are limited amounts of fossil fuels – if we keep using them, they will eventually run out.

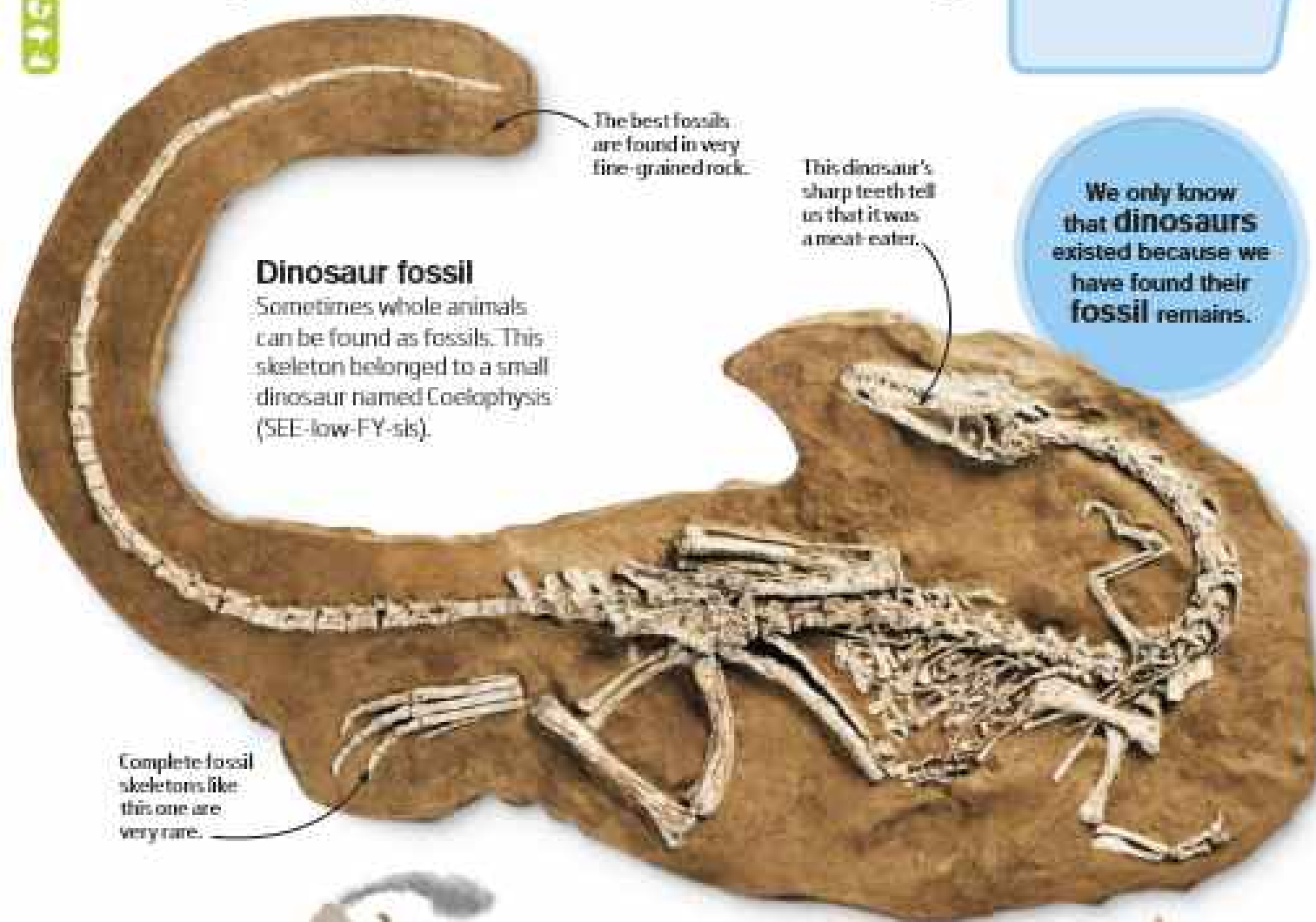


Fossils

Fossils are the remains of plants and animals from long ago. They are usually bones or shells that have turned into stone. Some fossils are so small that we cannot see them without special equipment. Others are as tall as a building.

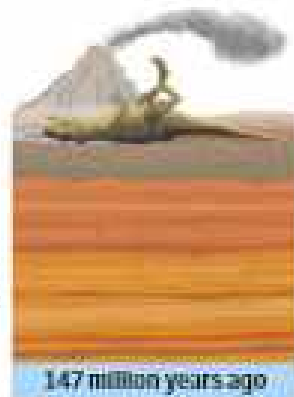
SEE ALSO

- ▶ Dinosaurs p.80
- ▶ Fossil fuels p.110
- ▶ Prehistoric life p.202
- ▶ Rock cycle p.213
- ▶ Skeleton p.228



How fossils are made

To become a fossil, an animal or plant needs to be covered up soon after it dies. Turning into a fossil takes millions of years.



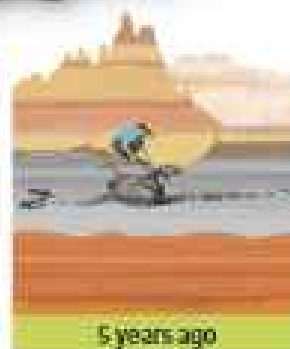
Death
A dinosaur dies and its body sinks into the soft mud by a river.



Burial
Layers of mud, sand, and ash cover the dinosaur, and its flesh rots away.



Turning to stone
The skeleton of the dinosaur slowly turns from bone to stone.



Discovery
Millions of years later, a scientist uncovers the fossil skeleton.



Fractions

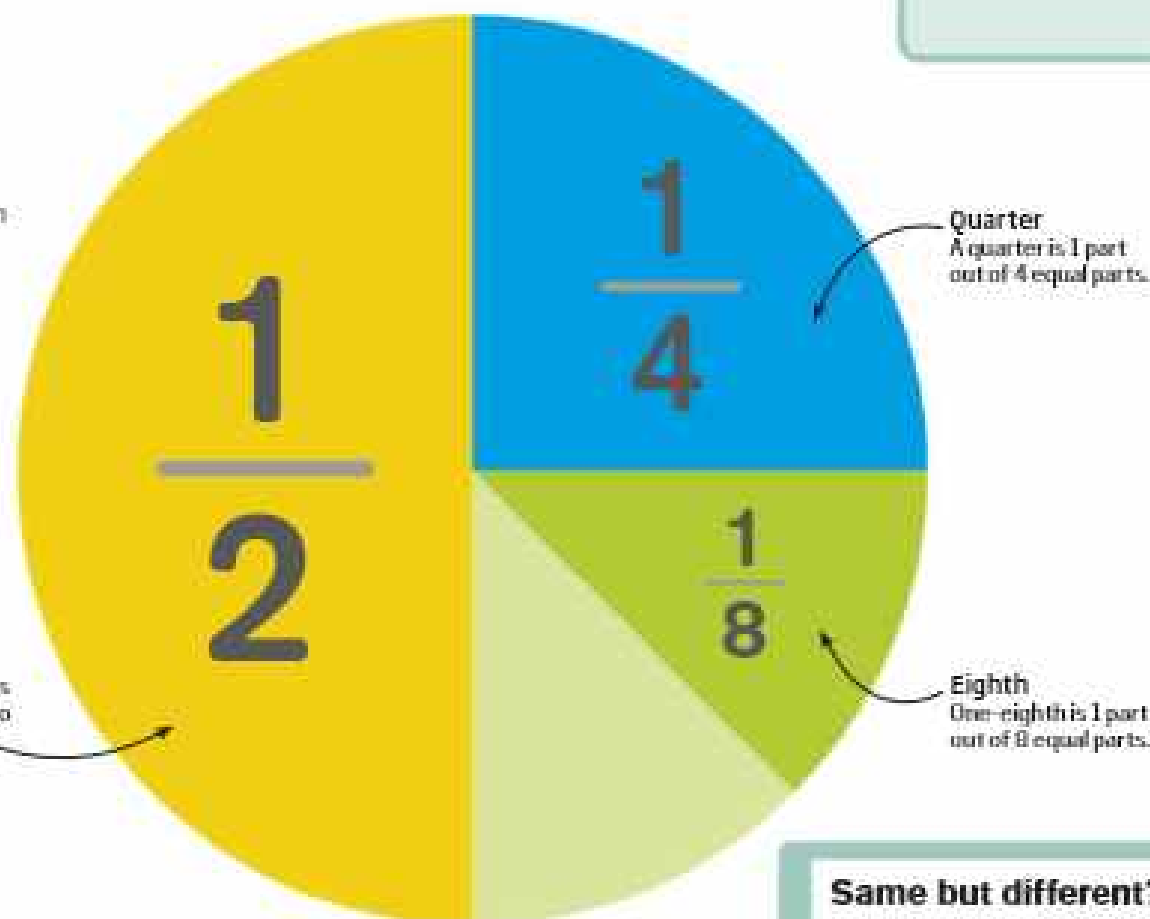
Fractions are parts of whole numbers. They are written as one number over another number. The number on the top is usually smaller than the one on the bottom. Fractions can be used on their own or with whole numbers.

SEE ALSO

- ▶ Astronomy p.32
- ▶ Clocks p.61
- ▶ Measuring p.159
- ▶ Numbers p.185
- ▶ Shapes p.222
- ▶ Symmetry p.248

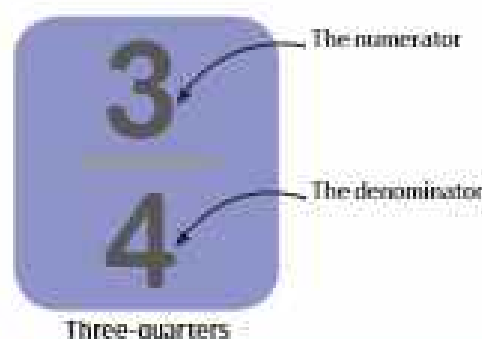
Common fractions

A whole number can be split into any number of parts to make a fraction. Here are some of the fractions we use most often.



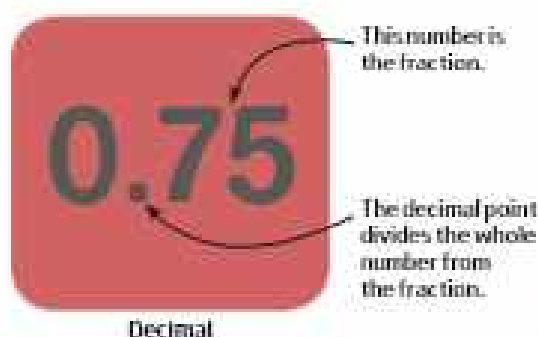
Parts of a fraction

The top number in a fraction is called the numerator. The bottom number is called the denominator. They are divided by a line.



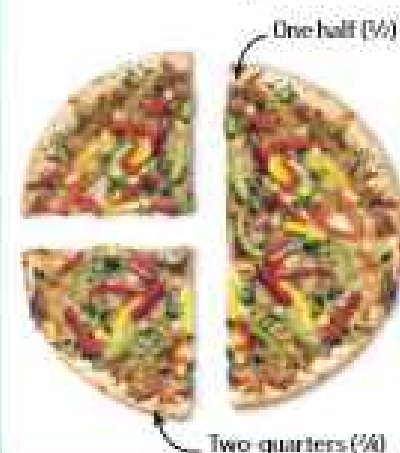
Decimals

Fractions can also be written as decimals. The number to the left of the decimal point is a whole number. The number to the right is the fraction.



Same but different?

Fractions can be written differently but mean the same. A half is the same amount as two quarters.



French Revolution

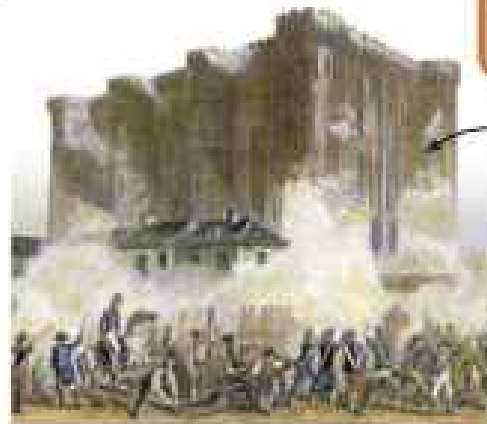
For years, France was ruled by all-powerful kings, who were very rich and could do whatever they wanted. The ordinary people of France were poor. Between 1789 and 1799, the people overthrew the king and changed laws.

SEE ALSO

- Buildings p.48
- Money p.69
- Europe p.94
- Governments p.123
- Law p.145
- War pp.278-279

Marie Antoinette

The luxurious lifestyle of Marie Antoinette, the French queen, made many ordinary people angry. They thought she did not care about their problems.



The Bastille was used to keep prisoners of the king.

The Bastille

The Bastille was a royal prison in Paris. On July 14, 1789, a crowd attacked it and stole weapons that had been stored inside.

Louis was beheaded using a machine called the guillotine.



King Louis XVI was killed in 1793.

General Antoine-Joseph Santerre was a leader in the Revolution.

End of the monarchy

The king and queen tried to run away from France dressed as servants. They were caught and later executed in front of crowds of people.



Friction

Friction is a force that slows down moving things by pulling against the direction of their movement. It is created between two surfaces as they move past each other. Different surfaces produce different amounts of friction.

SEE ALSO

- Forces p.108
- Water pp.120-121
- Gravity p.125
- Materials p.157
- Temperature p.252

Grip or slip?

Rough surfaces grip better because they create more friction than smooth surfaces. Smooth surfaces slip past each other because they create less friction than rough surfaces.



Rubber sole

Snow boots have rough rubber soles that grip, so that the climber's feet don't slip.

Producing heat

If we rub our hands together there is friction between them. They start to get warm because friction produces heat.



Rubber grip on snow boots



Icy surface

Lots of friction
The icy surface and the sole of the boot have lots of friction between them.

Smooth bottom of the ski



Icy surface

Not much friction
The snow and the ski have little friction between them and the ski slides easily.

Skis

Smooth skis slide over an icy surface very easily. They are made of many different materials, including wood and plastic.

Fruit and seeds

Seeds store everything a new plant needs to start growing, including food. A fruit protects the seeds inside it. It also gives a plant a way to move its seeds away to somewhere they can grow.

Fruit

Fruit forms from the flowers of some plants. Fruit usually tastes sweet, so people and animals like to eat it.

Apple seeds sit in a core in the middle of the fruit.

Apple seeds can take up to **80 days** to begin to grow.

Apple

Peapod

Peas are the seeds inside a peapod.

Leaves unfold and shoot straightens.

Leaves start to make food for the plant.

How seeds grow

Many plants grow from seeds. With water, the right temperature, and soil, a seed can begin to grow into a plant.

Bean seed starts to swell.

Roots grow to anchor the plant in the soil.

Shoot bursts through the soil.

Shoot begins to grow upward, toward the light.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Flowers p.103
- ▶ Eating pp.104-105
- ▶ Plants p.194
- ▶ Trees p.261
- ▶ Weather p.271

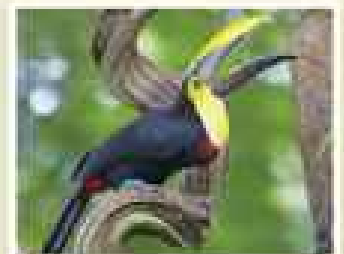
Spreading seeds

Plants spread, or disperse, their seeds in different ways.



By wind

Some plants have "winged" seeds that help their seeds fly away in the wind.



By animal

Animals eat fruit, and then poop out the seeds away from the plant.



Exploding pods

Some plants have seed pods that explode, flinging the seeds into the air.

Galaxies

Galaxies are enormous groups of stars, planets, dust, and gas that are held together by the force of gravity. They come in different sizes and shapes, including spiral, elliptical, and irregular.

SEE ALSO

- ▶ Gases p.117
- ▶ Milky Way p.167
- ▶ Physics p.192
- ▶ Shapes p.222
- ▶ Stars p.242
- ▶ Universe p.263

New stars form in the spiral arms.

Spiral galaxy

Spiral galaxies are disk-shaped with spiral arms. They are the most common type of galaxy. On average, they contain more than 100 billion stars. Our Milky Way is a spiral galaxy.

Scientists think that galaxies are mostly made of a material called "dark matter," which we can't see.

All stars rotate around the center of the galaxy.

The middle of the galaxy bulges outward.

Elliptical galaxy

Elliptical galaxies are rounded in shape, and are usually made up of older stars. They are generally larger than spiral galaxies. Elliptical galaxies contain lots of stars but little gas or dust.



Irregular galaxy

Small galaxies that do not have a clear structure are known as irregular galaxies. These may be created by two galaxies colliding. They contain lots of young stars, dust, and gas.



Gases

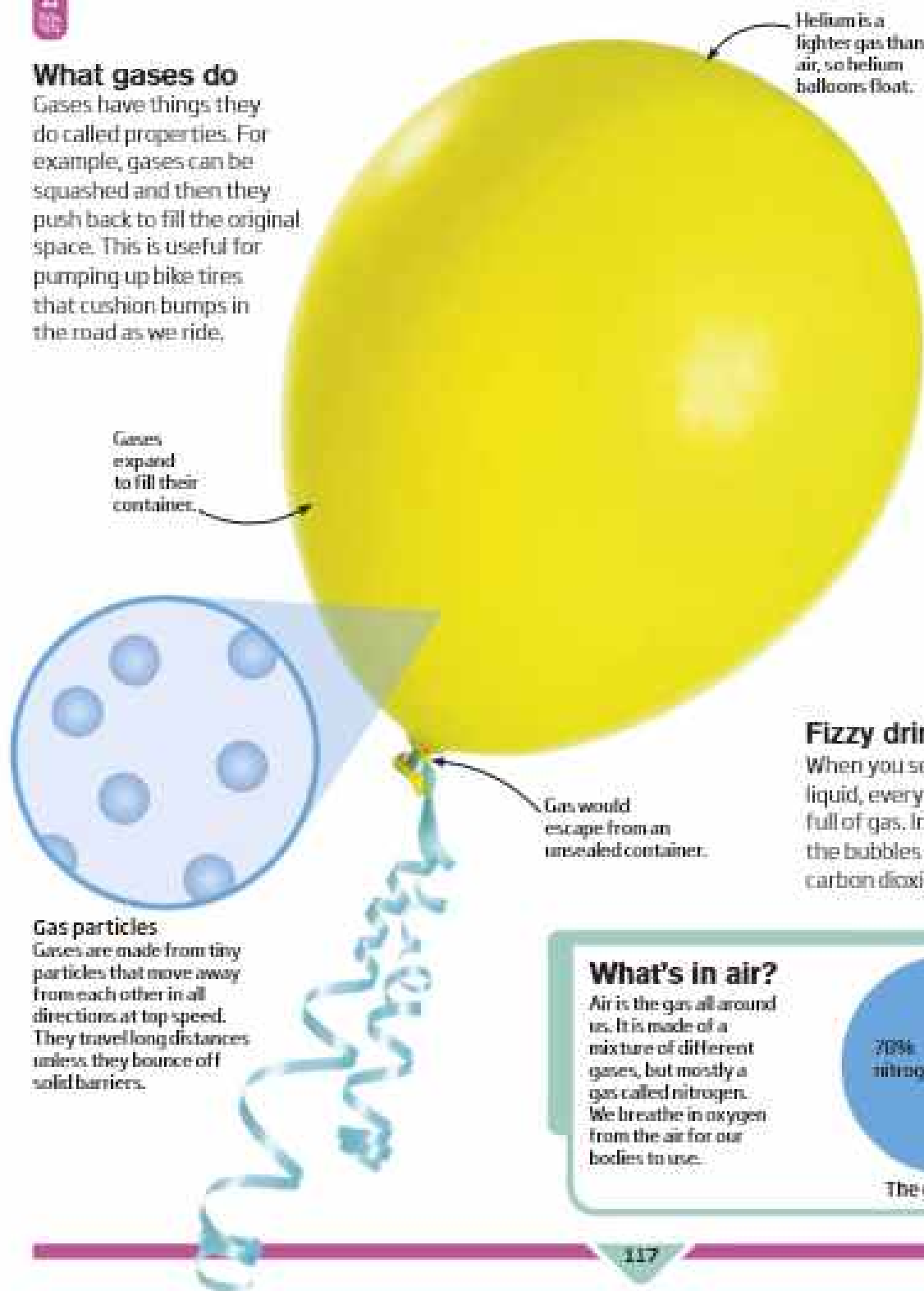
Gases are all around us—we are surrounded by a mixture of gas called air. We can keep them in sealed containers but if we open the container, the gas escapes and spreads out. Most gases are invisible.

SEE ALSO

- Changing states p.57
- Elements p.90
- Liquids p.148
- Lungs p.149
- Mixtures p.168
- Solids p.234

What gases do

Gases have things they do called properties. For example, gases can be squashed and then they push back to fill the original space. This is useful for pumping up bike tires that cushion bumps in the road as we ride.



Helium is a lighter gas than air, so helium balloons float.

Gases expand to fill their container.

Gas would escape from an unsealed container.

Gas particles

Gases are made from tiny particles that move away from each other in all directions at top speed. They travel long distances unless they bounce off solid barriers.



Blowing bubbles

Soap bubbles contain a little bit of air that pushes out toward the bubble. The soap bubble mix is stretched but pushes back, squashing the air into a sphere.

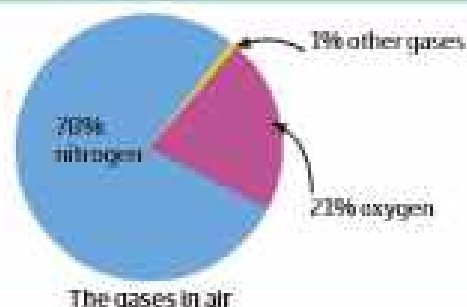
Fizzy drinks

When you see bubbles in a liquid, every one of them is full of gas. In a fizzy drink, the bubbles are the gas carbon dioxide.



What's in air?

Air is the gas all around us. It is made of a mixture of different gases, but mostly a gas called nitrogen. We breathe in oxygen from the air for our bodies to use.



Gemstones

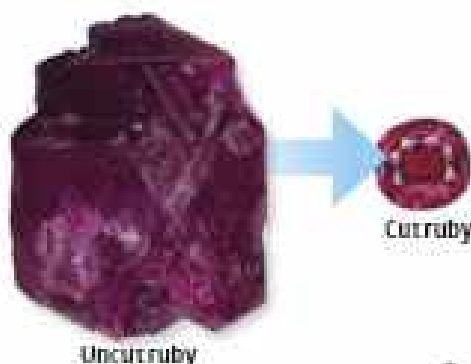
Gemstones are minerals that can be cut and polished to make jewels. They are often set in pieces of jewelry, such as rings, brooches, and even crowns. Many gemstones, such as rubies, are brightly colored. Others, such as diamonds, are usually colorless.

SEE ALSO

- ▶ Elements p.90
- ▶ Metals p.162
- ▶ Money p.169
- ▶ Precious metals p.199
- ▶ Gold pp.200-201
- ▶ Rocks and minerals p.214

Cutting gemstones

Gemstones are found as minerals. To become sparkly jewels, they must be cut into shape. Very sharp tools are used to carefully cut the stone into exactly the right shape.



Jewelry

Gems are often set in precious metals, such as gold and silver. They are used to make brooches, earrings, and other pieces of jewelry.



Different colors

The colors of gemstones are often caused by substances in minerals known as impurities. Sapphires and rubies are both types of a mineral called corundum, with impurities that make them blue or red.

Diamonds are made entirely from carbon squeezed under enormous pressure deep within the Earth.

Genes

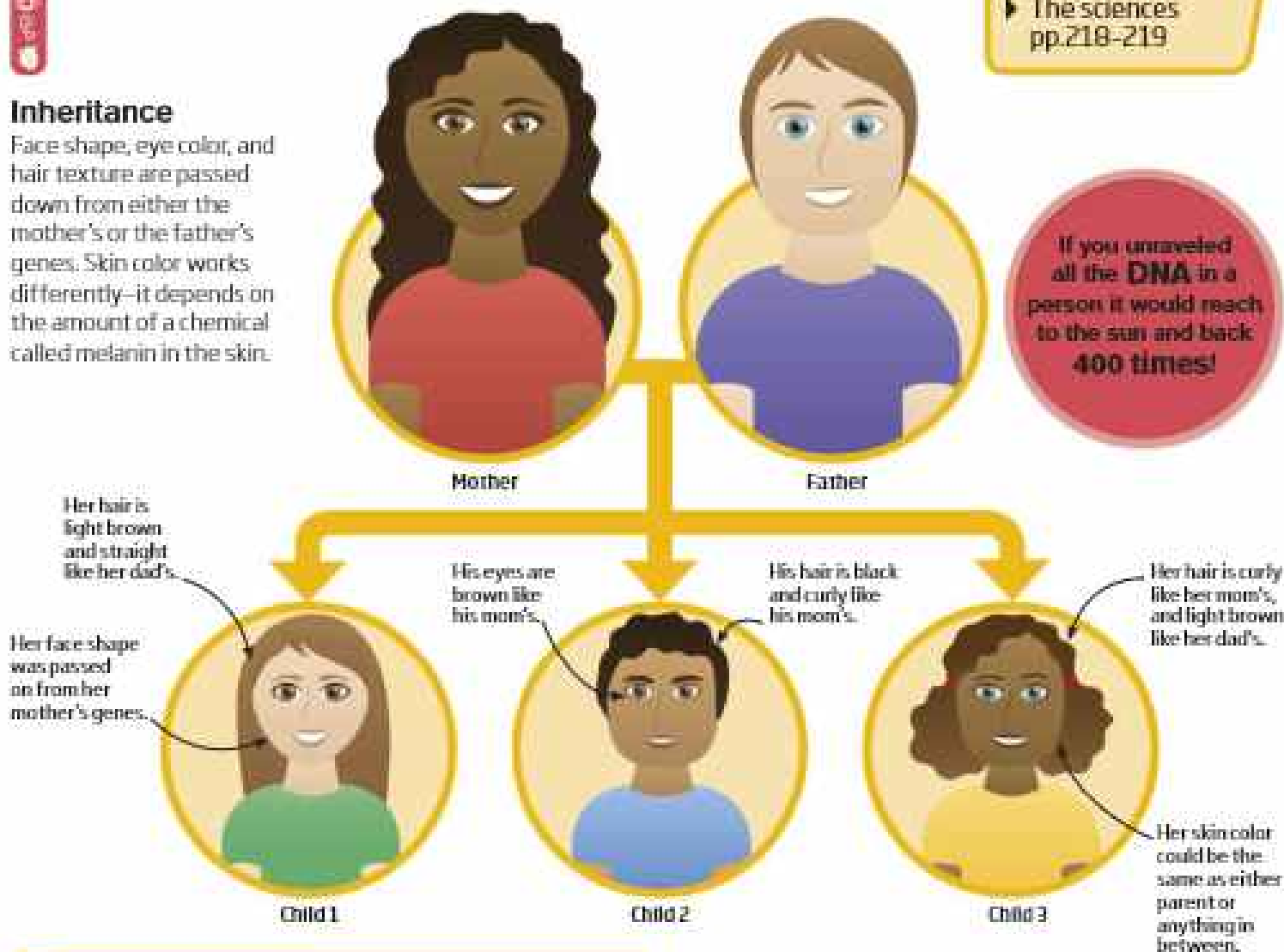
Genes are the instructions that make people the way they are. They include things like skin color, hair color, and height. Half our genes come from our mother and half from our father.

SEE ALSO

- ▶ Body cells p.41
- ▶ Evolution p.95
- ▶ Human body p.130
- ▶ Life cycle p.146
- ▶ Mammals p.154
- ▶ The sciences pp.218-219

Inheritance

Face shape, eye color, and hair texture are passed down from either the mother's or the father's genes. Skin color works differently—it depends on the amount of a chemical called melanin in the skin.



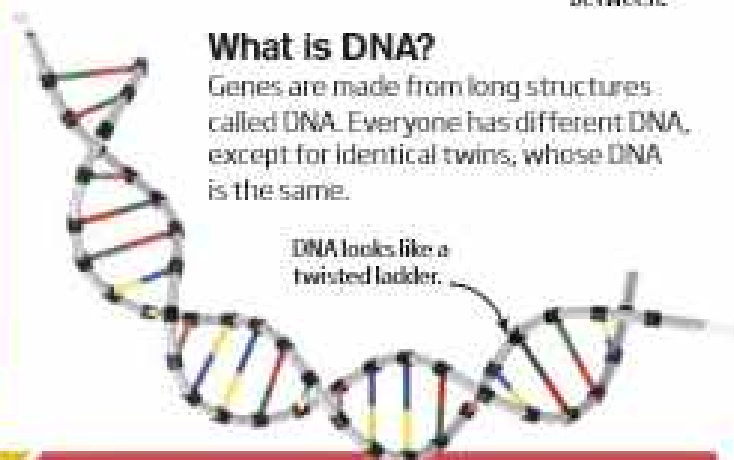
Clones

Most animals get a mix of different genes from their parents. Clones have exactly the same genes as another animal. Dolly the sheep was the first cloned mammal. Her genes were taken from a single cell of a female sheep.



What is DNA?

Genes are made from long structures called DNA. Everyone has different DNA, except for identical twins, whose DNA is the same.

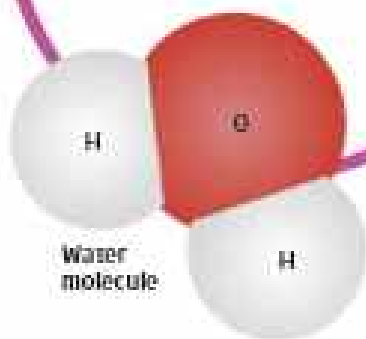


Water

Water is a clear, colorless substance that is found all around us. It forms Earth's oceans, lakes and rivers, snow and ice, and the clouds drifting overhead. All living things—including us—are mostly made of water, so without it our planet would be lifeless.

What's in water?

Water is made up of tiny particles called molecules. Each one has an oxygen (O) atom joined to two smaller hydrogen (H) atoms, so water is also known as H_2O .



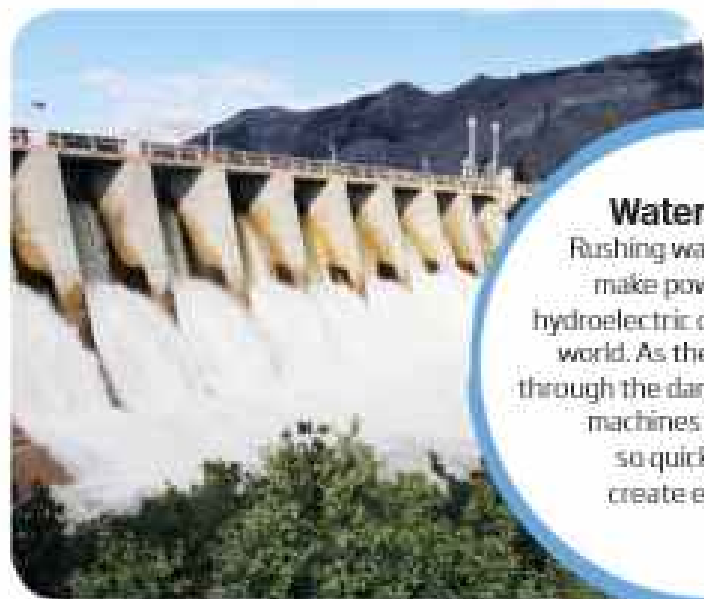
Fish breathe using special organs called gills to extract oxygen from the water.

More than 96 percent of all the water on Earth's surface is too salty to drink.

Blue planet

Nearly three-quarters of the Earth is covered by oceans and seas. Rivers and lakes stretch across the land's surface. The polar regions, near the North and South Poles, lie beneath frozen water, ice, and snow.

Planet Earth



Hydroelectric dam

Water power

Rushing water is used to make power at giant hydroelectric dams around the world. As the water moves through the dam, it turns turbine machines that spin so quickly they create electricity.

About 90 percent of all frozen water on Earth is in Antarctica, around the South Pole.

Only 2.5 percent of the Earth's water is fresh. It is found mostly in rivers, lakes, and glaciers.

Stone for grinding wheat into flour

Grain grown in Mesopotamia

Land of the rivers

The first cities were built in Mesopotamia (now mainly Iraq) around the Tigris and Euphrates rivers. The rivers allowed goods and people to move around and provided essential water to drink, cook with, and use to water crops.

Stone sickle for cutting crops

Kayaking

Water sports

Without water there would be fewer ways for us to have fun. There would be nothing to swim in or surf on, and no kayaking or sailing. There would be no snow for skiing or sledding and we could never build a snowman.

We lose **water** from our bodies through **sweating** when we are active and play sports.

Bad for the environment

More than 200 billion plastic water bottles are used around the world each year. Making the bottles releases lots of harmful gases into the air, and only one in five bottles is recycled—the rest are thrown out as garbage.

Plastic water bottles

Glaciers

Glaciers are slow-moving rivers of ice. They form high up in mountains or regions near Earth's poles and flow downhill. The front of a glacier may melt to make a river or lake. If a glacier reaches the coast, huge blocks of it break off and float away into the ocean as icebergs.

SEE ALSO

- ▶ Antarctica p.24
- ▶ Arctic p.25
- ▶ Changing world pp.50–51
- ▶ Climate change p.60
- ▶ Erosion p.93
- ▶ Mountains p.172

How are glaciers made?

Glaciers are made from snow that builds up and turns into ice. Eventually, there is so much ice that it starts to flow downhill.

The dark streaks on the surface are rocks carried along by the glacier.

A lake of melted ice often forms at the front of a glacier.

Glaciers can wear down the sides of mountains, making them pyramid-shaped.



Signs of glaciers

Some parts of the world were once colder and covered in glaciers. As the climate warmed, the glaciers melted away. However, they left signs in the landscape that they were once there.



U-shaped valley

Glaciers carve steep-sided valleys into flatter U-shaped ones as ice and rock grind the hillside away.



Sharp ridge

An arête is a sharp ridge of rock that separates two valleys that once had glaciers in them.



Giant rocks

Glaciers can pick up giant rocks and dump them far from where they came. They are called erratics.

Governments

A government is an official group of people that runs a country. Governments keep their people safe through rules called laws. They often try to keep the peace with other countries, and can help provide services such as schools and hospitals. Most governments try to help people lead better lives.

SEE ALSO

- ▶ Law p.145
- ▶ Medicine p.160
- ▶ Trade p.257
- ▶ School pp.272-273
- ▶ Work p.274
- ▶ World p.275

How a government works

Each country has its own system of government that looks after its people. Large countries have different levels of government, while smaller countries have simpler systems.

Local governments

Local governments run smaller areas within countries. They look after local issues, such as roads and libraries.

Head of state

One person is in charge of each country, such as a president or queen. They represent it when meeting with other countries.

National government

The national government looks after the whole country. It makes laws and has people in charge of different areas, such as education.

Electorate

People in many countries choose their governments by voting in elections. They are the electorate.

Types of government

Most countries in the world are democracies. They elect (choose) who is in government. Other countries have heads of state who aren't chosen.



Democracy

In a democracy, people vote for a government and a head of state to rule them and take decisions for them.



Monarchy

A monarchy is a family system in which the job of the head of state is passed from the king or queen to their child or relative.



Dictatorship

Dictators often rule by force. They use an army to make people do what they say.

Constitutions

A constitution is a written document that sets out the aims and values of a country and how it should be ruled. The US constitution was written in 1787.



Grasslands

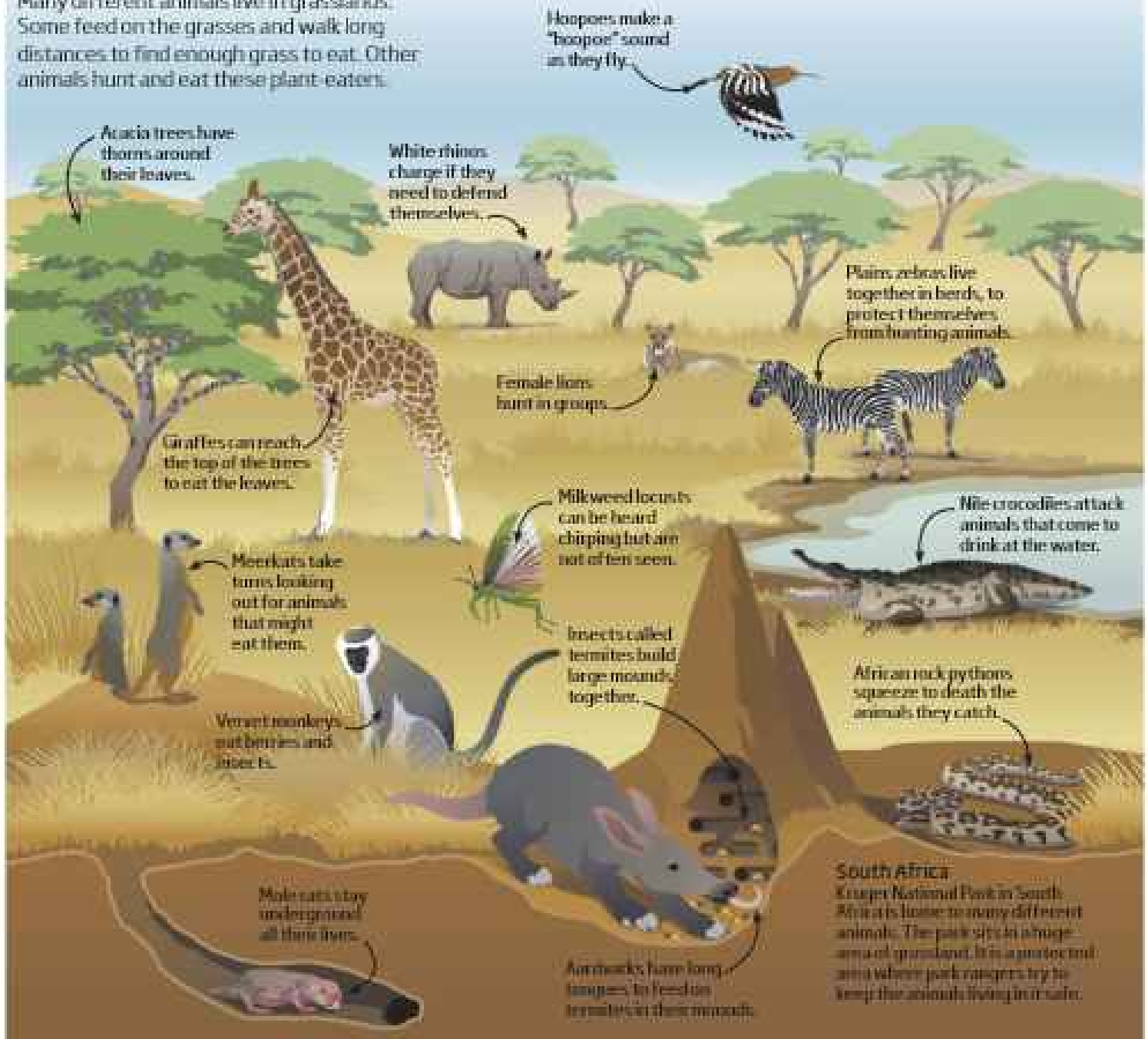
Grasslands are large areas covered in grass with just a few trees. They are dry, but get more rain than deserts do. Grasslands are known as savannas in Africa, steppes in Russia, prairies in North America, and pampas in South America.

SEE ALSO

- ▶ Africa p.12
- ▶ Animal homes p.23
- ▶ Birds p.39
- ▶ Deserts p.78
- ▶ Migration p.166
- ▶ Monkeys and apes p.170

Grassland life

Many different animals live in grasslands. Some feed on the grasses and walk long distances to find enough grass to eat. Other animals hunt and eat these plant-eaters.



Gravity

Gravity is an invisible force that pulls us back down toward Earth when we jump in the air. If we throw a ball up it comes down because of gravity. Without it, we would float off into space.

SEE ALSO

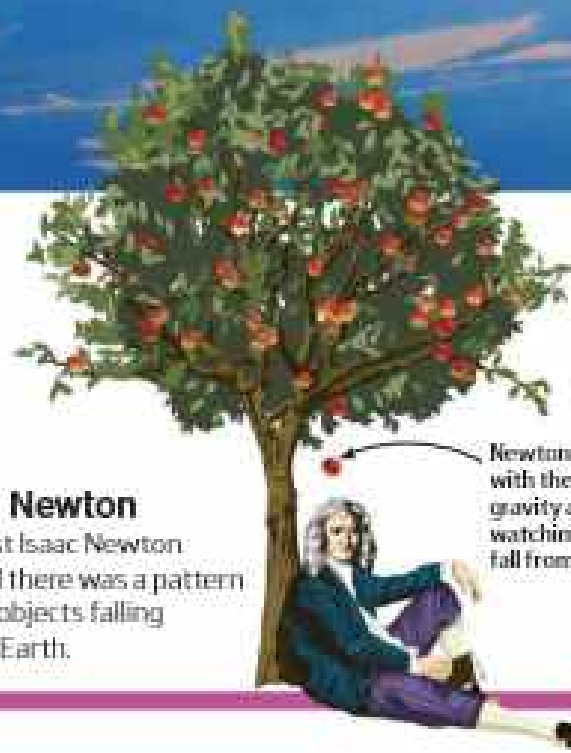
- Forces p.108
- Gases p.117
- Measuring p.159
- Moon p.171
- Solar system p.233
- Sun p.247

Falling to Earth

Earth's gravity pulls things toward it. When a skydiver jumps out of a plane, gravity starts to pull him down. Eventually he will use a parachute to slow his fall.

Air pushes up on the skydiver as he falls.

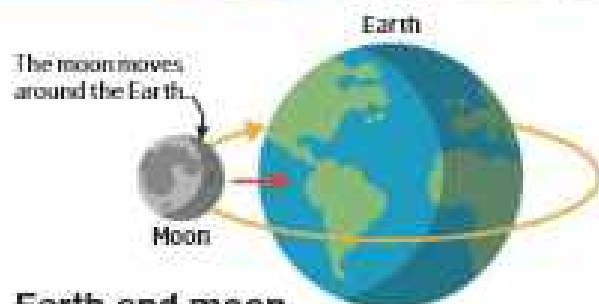
Gravity pulls the skydiver down toward Earth.



Isaac Newton

Scientist Isaac Newton realized there was a pattern behind objects falling toward Earth.

Newton came up with the theory of gravity after watching apples fall from a tree.



Earth and moon

The strong pull of Earth's gravity keeps the moon moving around it. Without gravity, the moon would disappear into space.

Habitats

A habitat is the place a plant or animal lives. Habitats around the world have different temperatures and landscapes. Animals and plants have certain features that allow them to survive in their habitats.

SEE ALSO

- ▶ Coral reefs p.74
- ▶ Deserts p.78
- ▶ Forests p.109
- ▶ Grasslands p.124
- ▶ Polar habitats p.197
- ▶ Seashores p.220

Tundra

This cold habitat has very few trees, plants are small, and animals often move away after the short summer.

Polar regions

Very few plants and animals can survive in the freezing temperatures of this icy habitat.

Coniferous forest

This habitat is full of coniferous trees, which keep their needlelike leaves all year round.

Tropical rain forests cover **7 percent** of Earth, but are home to **more than half** the world's plants and animals.

Desert

Rocks and sand cover this very dry habitat. Animals and plants have to be able to survive with very little water.

Rain forest

Trees grow quickly in this warm, rainy habitat. They provide food and homes for thousands of different animals.

Grassland

Grasslands have more rain than deserts, but not enough for many trees to grow. Most animals here eat grass.

Deciduous forest

This habitat has four seasons. The trees drop their leaves in autumn and grow them again in spring.

Ocean

This saltwater habitat covers 70 percent of Earth's surface. Some animals live deep down.

Hearing

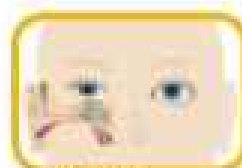
Hearing happens when our ears receive a sound. Sound is a vibration that travels through the air and into our ears. The sound travels into the hidden parts of the ear and our brain identifies what we hear.

SEE ALSO

- ▶ Brain p.45
- ▶ Communication p.69
- ▶ Human body p.130
- ▶ Music pp.176-177
- ▶ Radio p.203
- ▶ Sound p.235

The ear

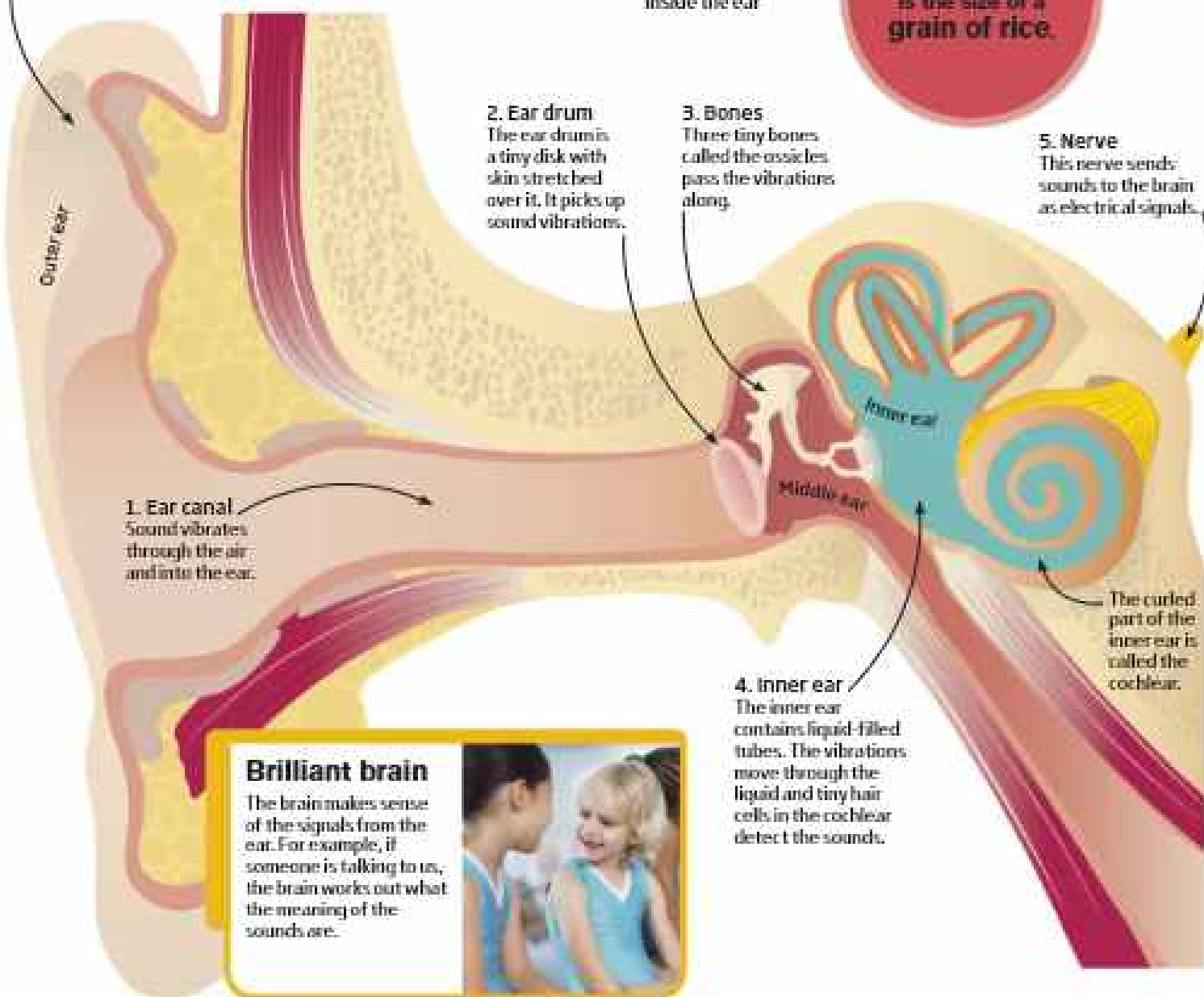
The ear is much bigger than it looks from the outside. The inner ear and outer ear are hidden inside our heads.



Inside the ear

The smallest bone in your ear is the size of a grain of rice.

The ears are shaped like cups to catch sound.



Brilliant brain

The brain makes sense of the signals from the ear. For example, if someone is talking to us, the brain works out what the meaning of the sounds are.



Heart

The heart is a pump about the size of a fist. It is mainly made of muscle and it pushes blood around the body by squashing itself in and out around 80 times a minute. Blood is a liquid that carries oxygen and food around the body. If the heart stops beating, the body stops working.

SEE ALSO

- ▶ Body cells p.41
- ▶ Brain p.45
- ▶ Feelings p.99
- ▶ Human body p.130
- ▶ Lungs p.149
- ▶ Medicine p.160

Inside the heart

The heart pumps blood every second of the day. The right side of the heart sends blood to the lungs. The left side sends blood to the rest of the body.

Atrium

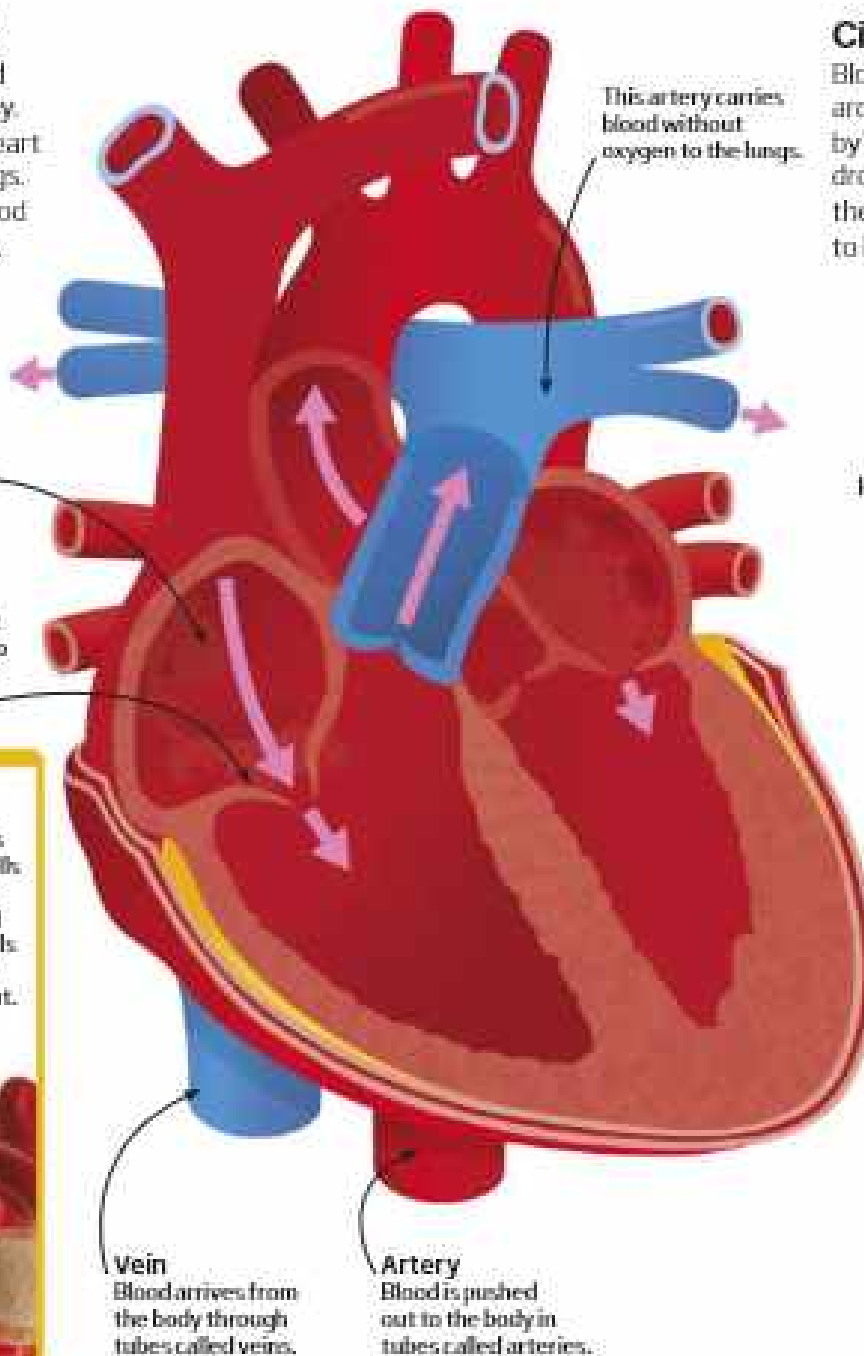
The heart has two "chambers" on each side, called atriums.

Valve

Valves are gates that only open one way, so blood can only travel in one direction.

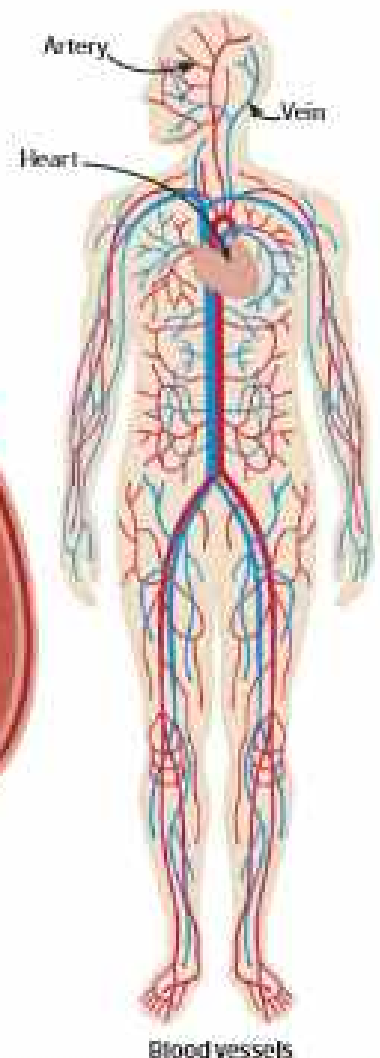
Blood

Blood contains tiny parts called cells. Red blood cells carry oxygen and waste gas. White blood cells kill germs. Broken bits of cells stick together to make a scab when the body is cut.



Circulation

Blood with oxygen in it travels around the body, pumped along by the heart. The oxygen is dropped off in different places, then comes back to the heart to be pushed to the lungs.



Hibernation

For many animals, finding enough food in winter is hard. Some animals migrate, or move, to warmer areas. Others survive by going into a deep sleep, called hibernation. They wake up again in the springtime when there is more food.

SEE ALSO

- ▶ Amphibians p.15
- ▶ Animal groups p.22
- ▶ Day and night p.77
- ▶ Mammals p.154
- ▶ Migration p.166
- ▶ Seasons p.221

Dormouse

These small mammals make cosy nests under leaves on the forest floor, or at the bottom of hedges.



The furry tail wraps around the mouse's face to keep it warm.

Bats hibernate upside down in caves or trees.



Bat

Bats enter a deep hibernation. Their heart rate drops from 400 to 25 beats per minute.

Dormice can curl up and sleep for up to seven months in a year.



Wood frog

Wood frog

This frog's body freezes in the winter and its heart stops beating. When the weather gets warmer, its heart starts beating again and the frog thaws out.

Do bears hibernate?

Bears sleep in the winter, but not very deeply, so they can be woken easily. This is called a torpor. It is like hibernation, but without such a deep sleep.



Human body

The body is made from lots of different parts called organs. Each organ has a different job to do. Organs work with muscles and other parts of the body to make important things happen, such as breathing, digestion, and movement.

Body systems

Organs that are linked together are called systems. Each system has its own job, but they work together, too.

SEE ALSO

- Biology p.38
- Body cells p.41
- Carbon cycle p.49
- Heart p.128
- Lungs p.149
- Monkeys and apes p.170

Body ingredients

The human body is made up of tiny parts called cells. The cells of the body contain lots of different ingredients. They all have different uses in the body.



Calcium helps muscles to work and the heart to keep beating.



One-fourth of our bodies is carbon. It is also found in diamonds!



A tiny amount of iron is found in the body. It makes your blood red.



Tears contain sodium chloride, which is the same as table salt.



Phosphorus helps make bones strong. Matches burn using phosphorus.



Over half of the body is made of water. Water is found in our blood and cells.

Respiratory system
The lungs bring air into the body and send waste air out. They supply oxygen to the blood.

Nervous system
The brain thinks and controls body movement. Signals are sent to the brain through nerves.

Circulatory system
The heart pumps blood around the body in a loop to transport oxygen and food.

Digestive system
The stomach and intestines break down the food so it can be used to power the body.

Urinary system
The kidneys clean the blood and make urine from the waste. The bladder stores the urine.

Muscular system
The muscles move all the parts of the body. They make the heart pump and the lungs breathe.

Skin and hair system
The skin is a waterproof layer that protects the body from germs and sunshine. Hair keeps us warm.

Skeletal system
The skeleton is a frame of bones that protects the inner organs of the body. It also allows movement.

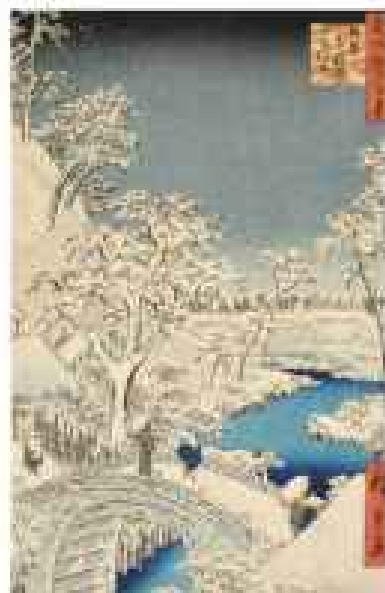
Imperial Japan

During the Edo period, from 1603 to 1868, Japan was ruled by a series of powerful men called shoguns. Each ruled from the city of Edo, which is now Tokyo.

Noble samurai

Samurai were warriors who fought for a powerful lord and followed strict rules. Their way of life was called *Bushido* (bu-shi-do), "the way of the warrior."

The main weapon was a sword called a *katana*.



Art

Poets, painters, writers, and craftspeople made beautiful works of art during this time. This print from 1857 shows a traditional Japanese scene of trees and a river in the snow.

A horned helmet was part of a samurai's armor.



SEE ALSO

- ▶ Art p.28
- ▶ Dance p.76
- ▶ Knights p.142
- ▶ Musical instruments p.175
- ▶ Theater p.253
- ▶ War pp.278-279

Making music

Music has always played an important part in Japanese culture. This instrument, called a *shamisen* (sha-mee-sen), is like a guitar. It provided background music for dance performances and puppet theater.



Shamisen have three strings and a square body.

Strict command

Edo society's chain of command was strict. The emperor was the leader, but the shogun was in charge of the lords, who were called *daimyo*. The *daimyo* were in charge of the samurai army.



Incas

The Inca people lived in the mountains along the west coast of South America. Between 1438 and 1532 CE, their wealthy empire was the largest in the world. Society was well organized and everyone knew their place.

Inca society

At the head of Inca society was an emperor called Sapa Inca, meaning "Great Inca." Most people were peasant farmers who worked for the emperor and in return were given food and housing.

The ancient Inca city of Machu Picchu sits in the Andes Mountains, in Peru.

The Sapa Inca was treated like a god. He was carried in a chair by servants.

Houses were made of stone and had thatched roofs.

Everybody had to bow down to the Sapa Inca.

Farmers planted potatoes.

Ullmas were used for carrying goods and for their wool.

Maize, or corn, was an important food.

SEE ALSO

- ▶ Aztecs p.35
- ▶ Crafts p.75
- ▶ Farming p.98
- ▶ Maya p.158
- ▶ Gold pp.200-201
- ▶ South America p.236



Sun god

This gold disk shows Inti, the Inca god of the sun. Every year the Incas held a nine-day festival of eating, drinking, and sacrifice to honor the sun god.

Inca crafts

The Incas made sacred objects from gold and silver to be used in temples. Craft workers also made fine pieces from clay, leather, and feathers.



Industrial Revolution

The Industrial Revolution was a time of huge growth in industry brought about by the introduction of new machinery. Factories were built to make products using large numbers of machines. The revolution began in Britain during the 1760s and later spread around the world.

SEE ALSO

- ▶ Buildings p.48
- ▶ Engines p.92
- ▶ Factories p.97
- ▶ Inventions pp.136-137
- ▶ Machines p.150
- ▶ Pollution p.198

Factory work

Factories contained rows of machines that made things in large amounts, such as textiles, iron and brass goods, pottery, and glassware. These machines were powered first by water and then by steam.

Steam trains transported goods and fuel, such as coal, to and from the factory.

Chimneys belched out smoke all day.

Steam engines were used to power ships.

Barges traveled on canals carrying goods between towns.

Workers lived in small, overcrowded houses.

Small towns grew up quickly around the factories.

Horsedrawn carts carried supplies to the factory.

Child labor

Children as young as four worked in the factories and mines. The work was often dangerous and dirty. Children also swept chimneys and ran errands. They worked up to 80 hours a week, but were paid very little money.



Boys working in a cotton mill in Georgia, 1900

Insects

Insects are the largest group of animals, and are found all over the world. They have three pairs of legs and a tough outer covering, called an exoskeleton. Many insects have wings and can fly.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Eggs p.86
- ▶ Invertebrates p.139
- ▶ Metamorphosis p.163
- ▶ Migration p.166

Butterflies

A butterfly changes a lot in its life cycle. It starts as an egg, then hatches into a caterpillar. It wraps into a chrysalis before hatching again as a butterfly.

Swallowtail butterfly

Feelers called antennae help the butterfly to smell nectar and stay balanced.

Many insects have wings, which let them fly around.

There are more than
900,000
different types
of insects
in the world.

Body parts

Insects' bodies are divided into three parts: the head, the thorax, and the abdomen. They have three pairs of legs attached to the thorax and one pair of antennae on their heads.



Patches of color on the butterfly's wing tell other animals that it is not good to eat.

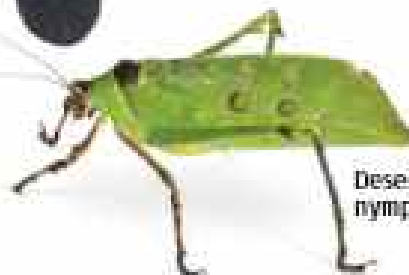
Red wood ant



Ants

Ants live together in groups called colonies, with thousands of workers and a queen. They are small but strong—ants can lift 20 times their body weight.

Desert locust nymph



Grasshoppers

These insects can leap 20 times their body length. They can also fly fast, reaching speeds of up to 8 mph (13 kph).

Stag beetle



Beetles

Beetles are found on land and in water all over the world. They have hard, shiny outer wings that close over a soft inner set of wings, to protect them.

Inside Earth

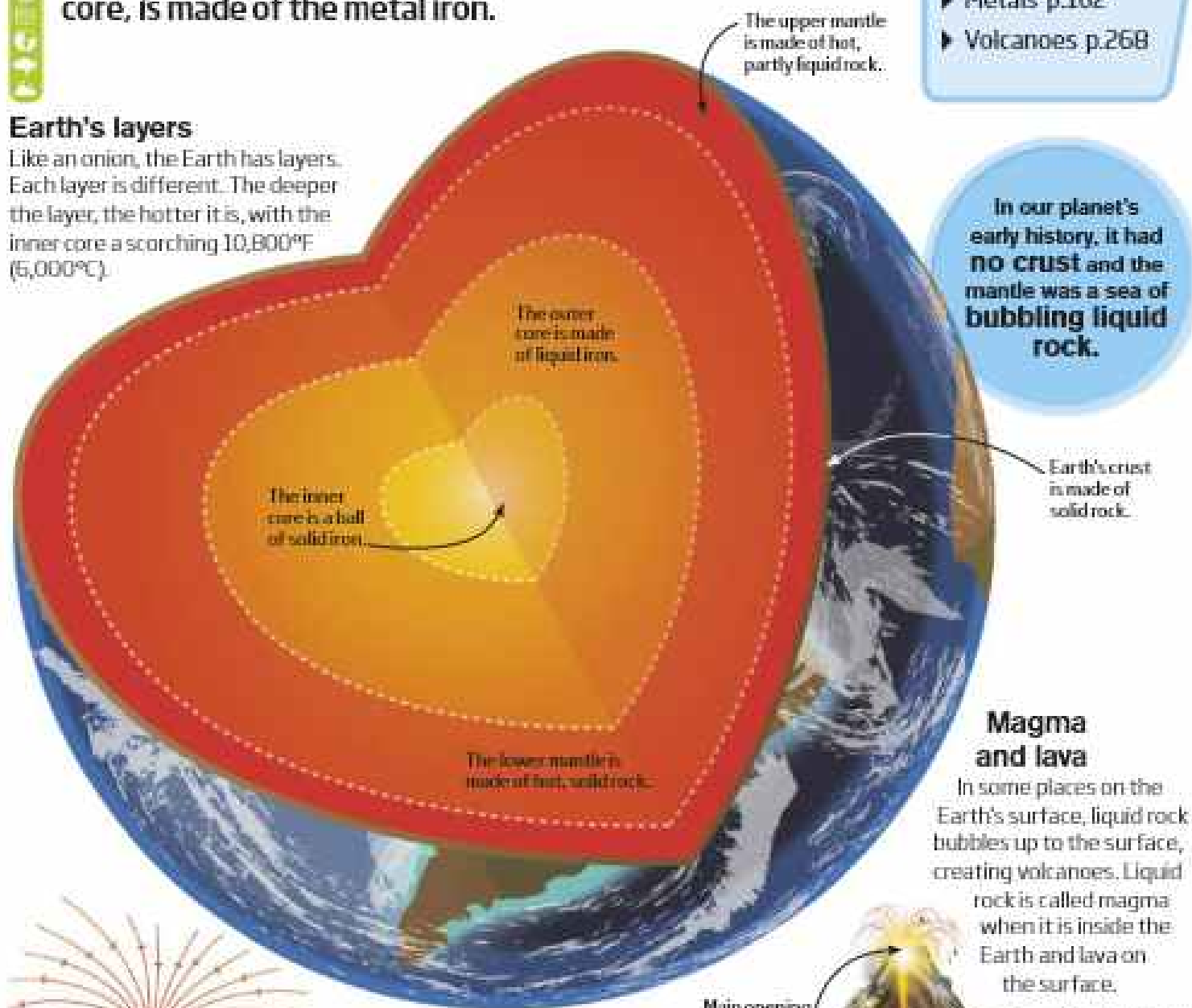
The Earth is made up of four layers. We live on its outer layer, or crust. The crust floats on a layer of extremely hot rock called the mantle. Below this, our planet's center, or core, is made of the metal iron.

SEE ALSO

- Compass p.70
- Earth p.83
- Earth's surface p.84
- Magnets p.151
- Metals p.162
- Volcanoes p.268

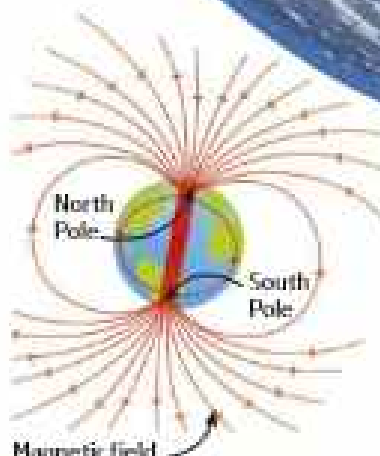
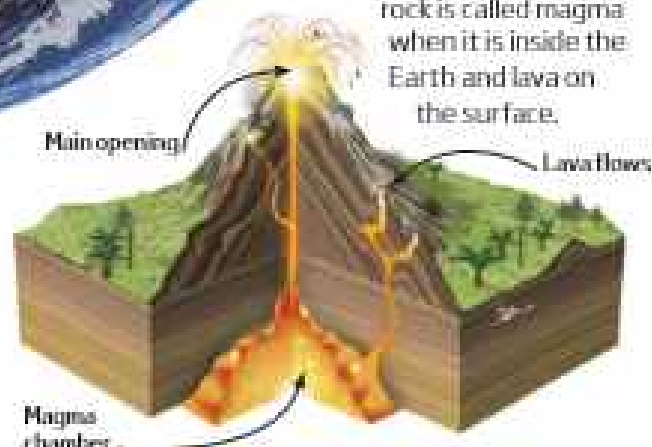
Earth's layers

Like an onion, the Earth has layers. Each layer is different. The deeper the layer, the hotter it is, with the inner core a scorching 10,800°F (6,000°C).



Magma and lava

In some places on the Earth's surface, liquid rock bubbles up to the surface, creating volcanoes. Liquid rock is called magma when it is inside the Earth and lava on the surface.



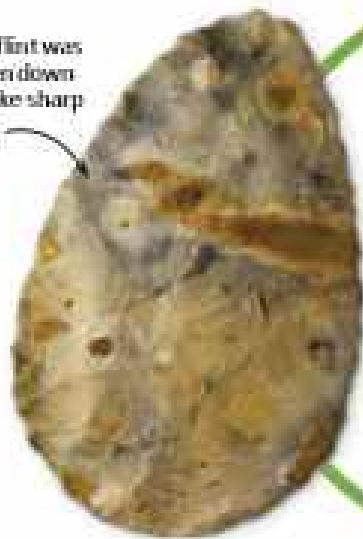
Magnetic Earth

Because the Earth's outer core is liquid, it moves as the Earth spins. This creates a magnetic field around the Earth. This field keeps out harmful energy waves from space. It also lets us find directions when we use a compass.

Inventions

Inventions are new ideas that solve problems or prove useful to us. Thousands of years of brilliant brainpower have resulted in things and ideas that have transformed people's lives. Even in the modern world, people keep coming up with new ideas and inventions.

Hard flint was broken down to make sharp tools.



Hand axe

The hand axe was the first stone tool in prehistoric times. People dug for flint, which they shaped into handcrafted axes to cut meat, chop wood, scrape skins, and protect themselves. Hand axes were used for more than a million years.

A rod, or axle, links the wheels together—turning the axle uses much less energy than turning the wheel.



Wheel

The wheel was invented more than 5,000 years ago in Mesopotamia (part of modern-day Iraq). Wooden discs were attached to animal carts to carry heavy loads. Later, wheels became lighter, which made them faster and smoother.

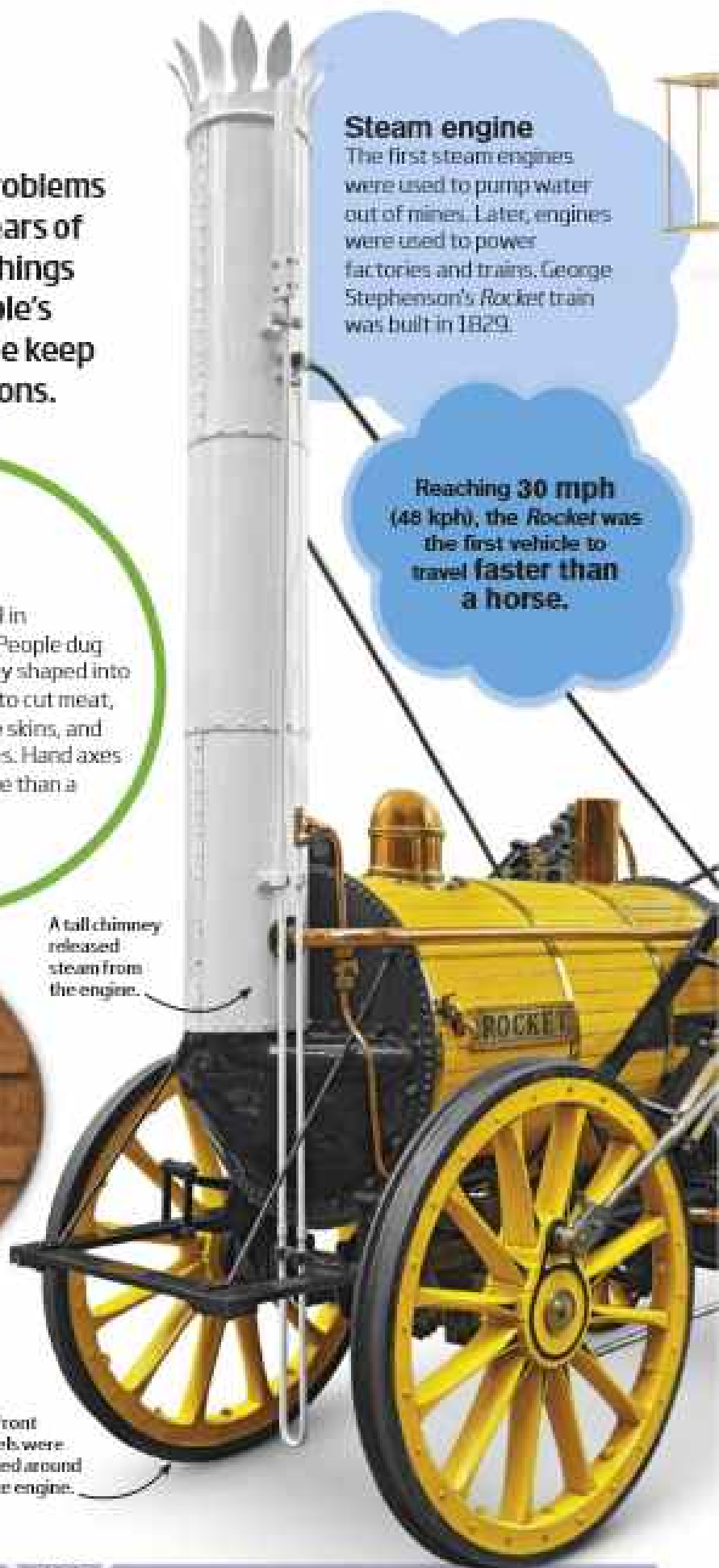
Steam engine

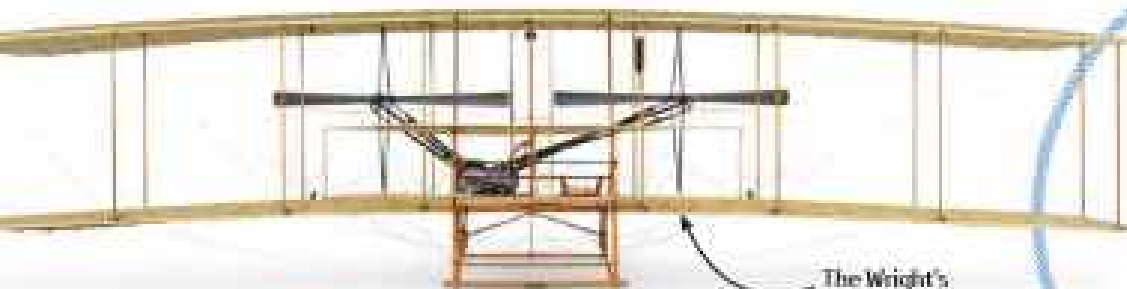
The first steam engines were used to pump water out of mines. Later, engines were used to power factories and trains. George Stephenson's *Rocket* train was built in 1829.

Reaching 30 mph (48 kph), the *Rocket* was the first vehicle to travel faster than a horse.

A tall chimney released steam from the engine.

The front wheels were pushed around by the engine.





The Wright's Flyer, built in 1903.

Flight

In 1903, American brothers Wilbur and Orville Wright built a glider with a light engine. Although Flyer took off for only 12 seconds over 120 ft (37 m), it was the first-ever powered flight.

Plastics

Plastics are cheap to make, easy to shape, and last a long time. They can be hard, soft, or bendy. The first human-made plastic was made by Belgium-born chemist Leo Baekeland in 1905. Now plastics are all around us.



Antibiotics

Infection-fighting antibiotics battle bacteria inside the body. They were discovered by accident in 1928, when Scottish scientist Alexander Fleming noticed bacteria dying around mold in his lab. The bacteria-killing chemical penicillin has since saved many lives.



Antibiotic pills



Thomas Edison (1847–1931) was an American master of invention, patenting

1,093

ideas in his lifetime, including **batteries** and **light bulbs**.

Computers

Computers are electric machines that follow instructions. Modern computers can do billions of sums every second. We use them to find, store, and share information. The idea of a computer was invented by English engineer Charles Babbage in the 1830s.

The first electronic computer, **ENIAC**, was huge, filling a room measuring 50 ft (15 m) long.





Internet

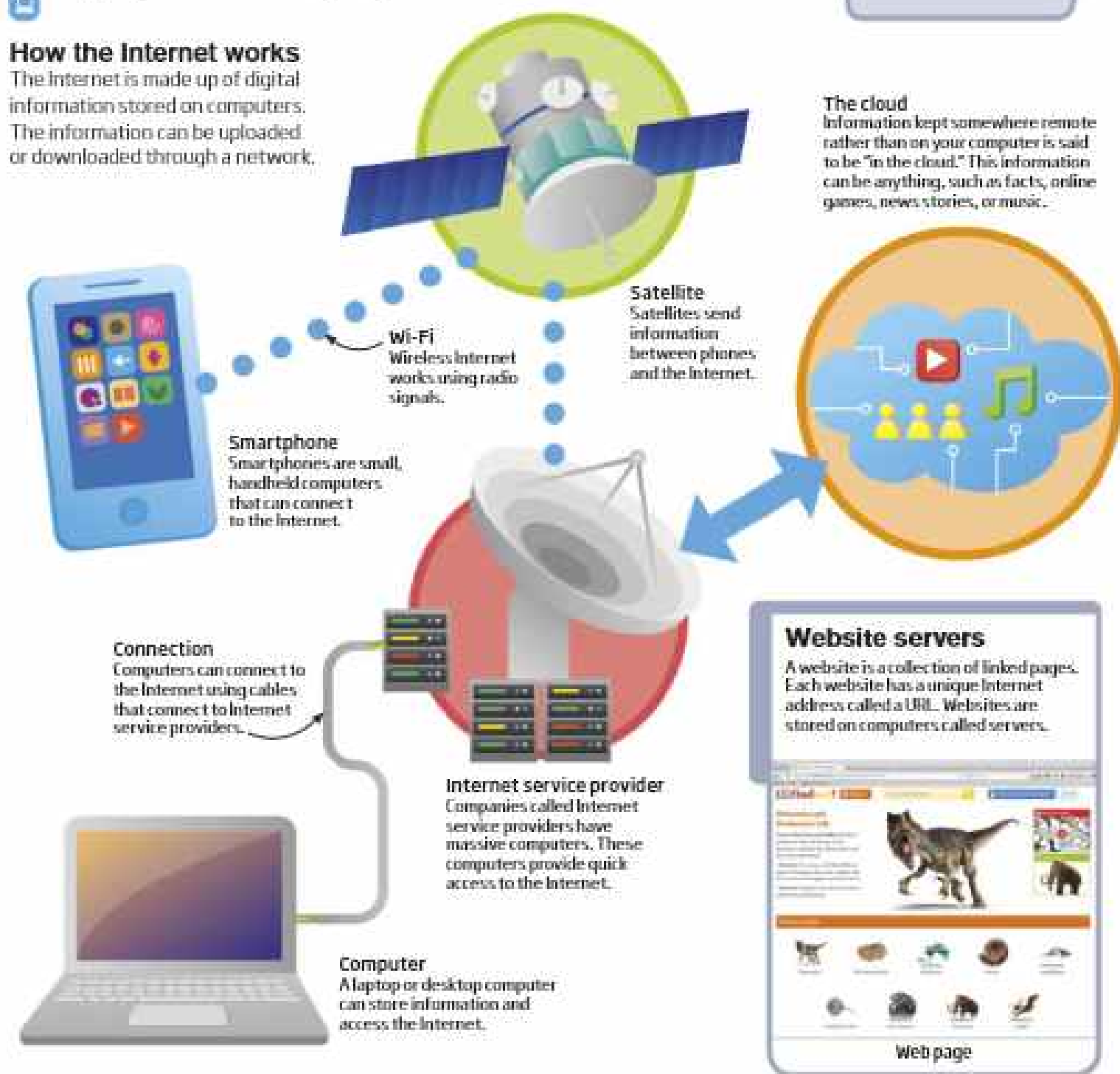
The Internet is a network that links together computers all over the world. We use it to learn things, for entertainment, and to communicate with each other. The Internet was first thought of in 1962. Now, billions of people use it every day.

SEE ALSO

- ▶ Codes pp.66–67
- ▶ Communication p.69
- ▶ Computers p.71
- ▶ Satellites p.215
- ▶ Telephones p.250

How the Internet works

The Internet is made up of digital information stored on computers. The information can be uploaded or downloaded through a network.



Invertebrates

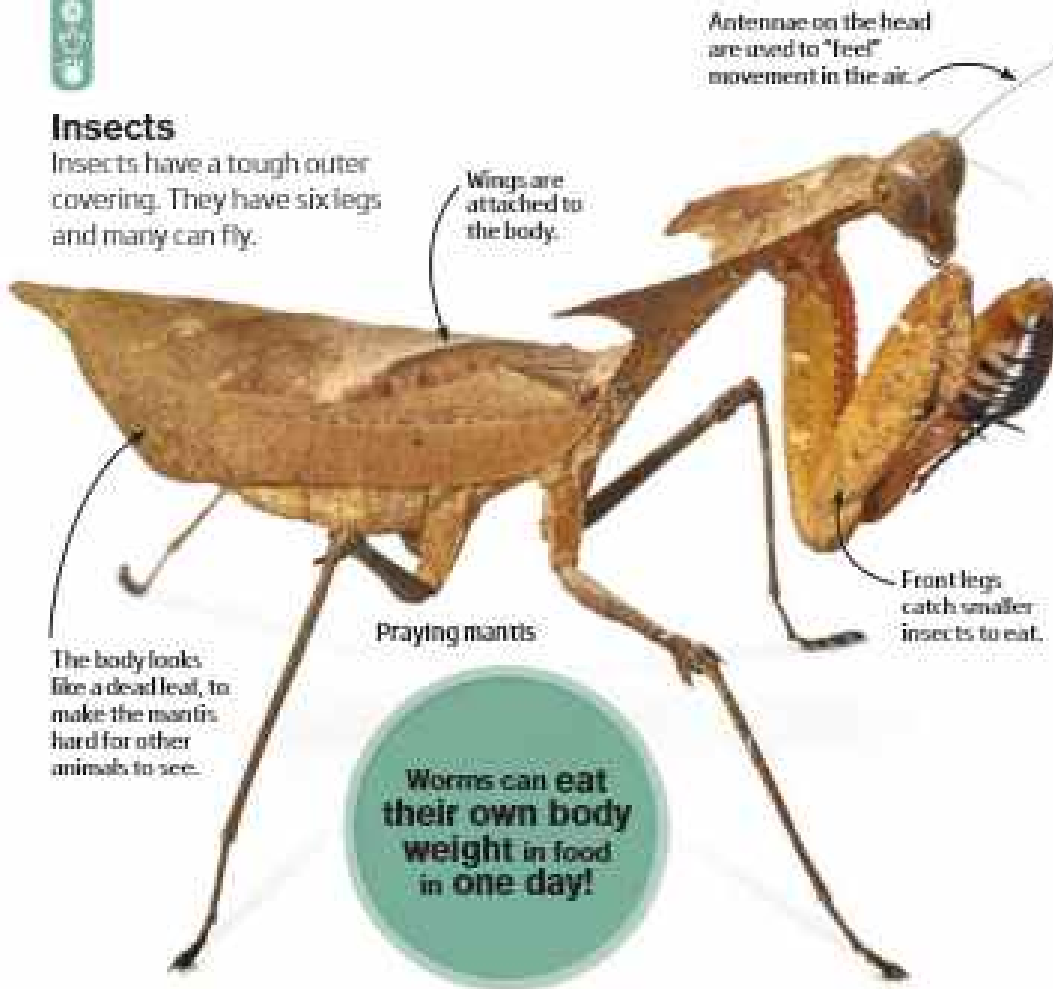
Invertebrates are animals that don't have backbones. They are divided into many smaller groups, such as insects and mollusks. Ninety-eight percent of all animals are invertebrates.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Animal homes p.23
- ▶ Habitats p.126
- ▶ Insects p.134
- ▶ Vertebrates p.266

Insects

Insects have a tough outer covering. They have six legs and many can fly.



The hard shell protects the snail's soft body.



Mollusks

Mollusks have soft bodies with no skeleton and no legs. They live in water or wet soil.



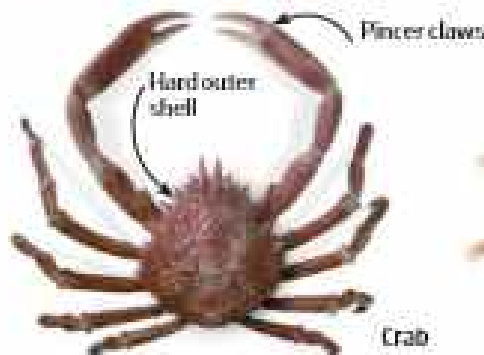
Worms

Worms have long soft bodies with no legs. Their bodies are split into many parts, or segments.



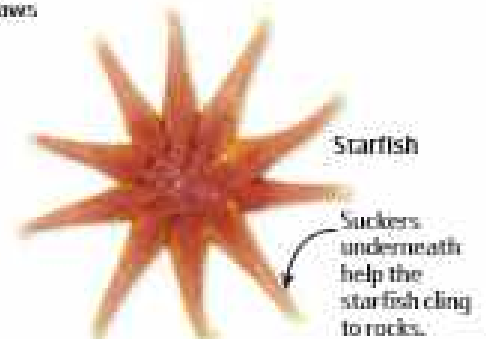
Arachnids

These animals have eight legs, not six like insects. Arachnids include spiders, scorpions, and mites.



Crustaceans

Crustaceans have hard outer shells, and more than eight legs. Most of them live in water.



Echinoderms

These animals have bodies split into equal parts around a disk in their middle. They all live in the sea.

Iron Age

During the Iron Age, people began using iron tools and weapons instead of bronze ones. It started about 3,200 years ago and lasted for around 1,000 years, depending on the area. Iron tools were stronger and more durable than anything before.

SEE ALSO

- ▶ Bronze Age p.47
- ▶ Farming p.98
- ▶ Metals p.162
- ▶ Rocks and minerals p.214
- ▶ Trade p.257
- ▶ Vikings p.267

Tools

Iron tools allowed farmers and builders to work better. They were able to grow more crops, clear land for farms, and build bigger buildings.



This iron sickle would have been attached to a wooden handle and used to cut wheat.

Iron cutting edge

Making a sword was very difficult. The best swordmakers were well paid.

Danish Iron sword from the Viking period of 800–1100 AD

Weapons

Iron weapons were lighter and cheaper to make than weapons made from bronze. Well-made iron weapons were also stronger and sharper than bronze ones. Having good weapons meant societies could build powerful armies.

Hill fort

Hill forts are Iron Age villages built on the top of hills. They had earth or stone walls around them. Hill forts allowed people to see the enemy approaching and prepare themselves for attack.

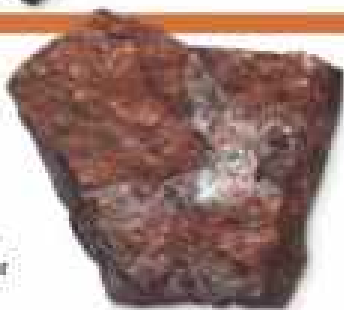


The walls at the base of the fort acted as a protective barrier.

Iron Age hill fort in Dorchester, UK

Making iron

Making iron items is a process that needs great skill and care. The iron must be made extremely hot before it can be shaped into a weapon or a tool. How we shape iron today is similar to how Iron Age craftsmen performed the task.



1. Dig it out
Iron is dug from the ground in lumps, called iron ore.



2. Heat it up
The iron ore is heated to high temperatures to make it melt.



3. Pour it in
The melted iron is poured into a shaped mold and left to cool.

Jupiter

Jupiter is the largest planet in the solar system, and the fifth planet from the sun. It is a "gas giant" made of hydrogen and helium. Jupiter does not have a solid surface like Earth.

SEE ALSO

- ▶ Astronomy p.32
- ▶ Atmosphere p.33
- ▶ Elements p.90
- ▶ Gases p.117
- ▶ Solar system p.233
- ▶ Solids p.234
- ▶ Storms p.246

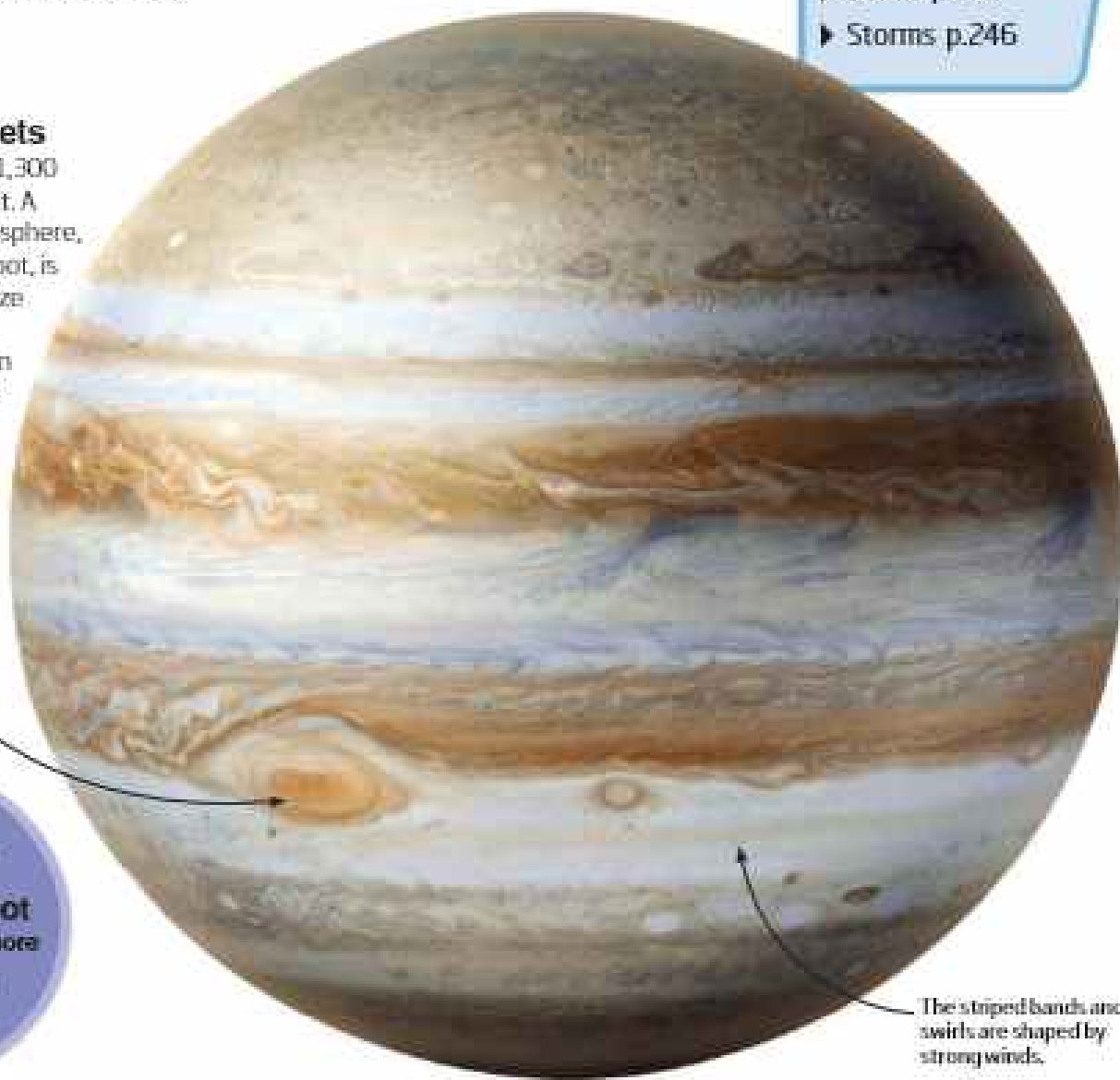
King of the planets

Jupiter is so large that 1,300 Earths could fit inside it. A huge storm in its atmosphere, called the Great Red Spot, is more than twice the size of Earth. Jupiter is the third brightest object in our night sky after the moon and Venus.

Scientists have been watching the Great Red Spot storm since 1830.

Winds swirling around the Great Red Spot reach speeds of more than 250 mph (400 kph).

The striped bands and swirls are shaped by strong winds.



The Galilean moons

Jupiter has more than 60 moons of different sizes. Its four largest moons are called Io, Europa, Ganymede, and Callisto. They are known as "the Galilean moons" because they were discovered by the Italian scientist Galileo Galilei in the 17th century.



Io



Europa



Ganymede



Callisto

Knights

Knights were men who owned big pieces of land in Europe between the 600s and the 1600s, a time period known as the Middle Ages. They trained from age seven to be fighters and lead armies.

Armor

Knights wore suits of armor to protect them from enemy weapons, such as swords. The first armor was made of mail, which is rings of metal linked together. Later armor also included sheets of shaped steel.

Curved armor protected the knight's elbow.

Squires

Each knight had a squire to assist him. Squires looked after their knight's armor, sharpened his weapons, and cared for his horse. Some squires later became knights.

Knights wore colorful outfits for jousting.

Mail was heavy to wear, but offered good protection.

SEE ALSO

- Castles p.53
- Clothing pp.62-63
- Europe p.94
- Flags p.102
- Metals p.162
- Sports p.239
- War pp.278-279

Knights carried their shield on a strap when not using it.

Steel gloves were made of more than 40 metal pieces.

A belt held the knight's sword and dagger.

Designs

Each knight had a special design passed down to him by his father. Knights put their design on their shield so others would recognize them in battle. Men called heralds recorded which knight had which design.



Jousting

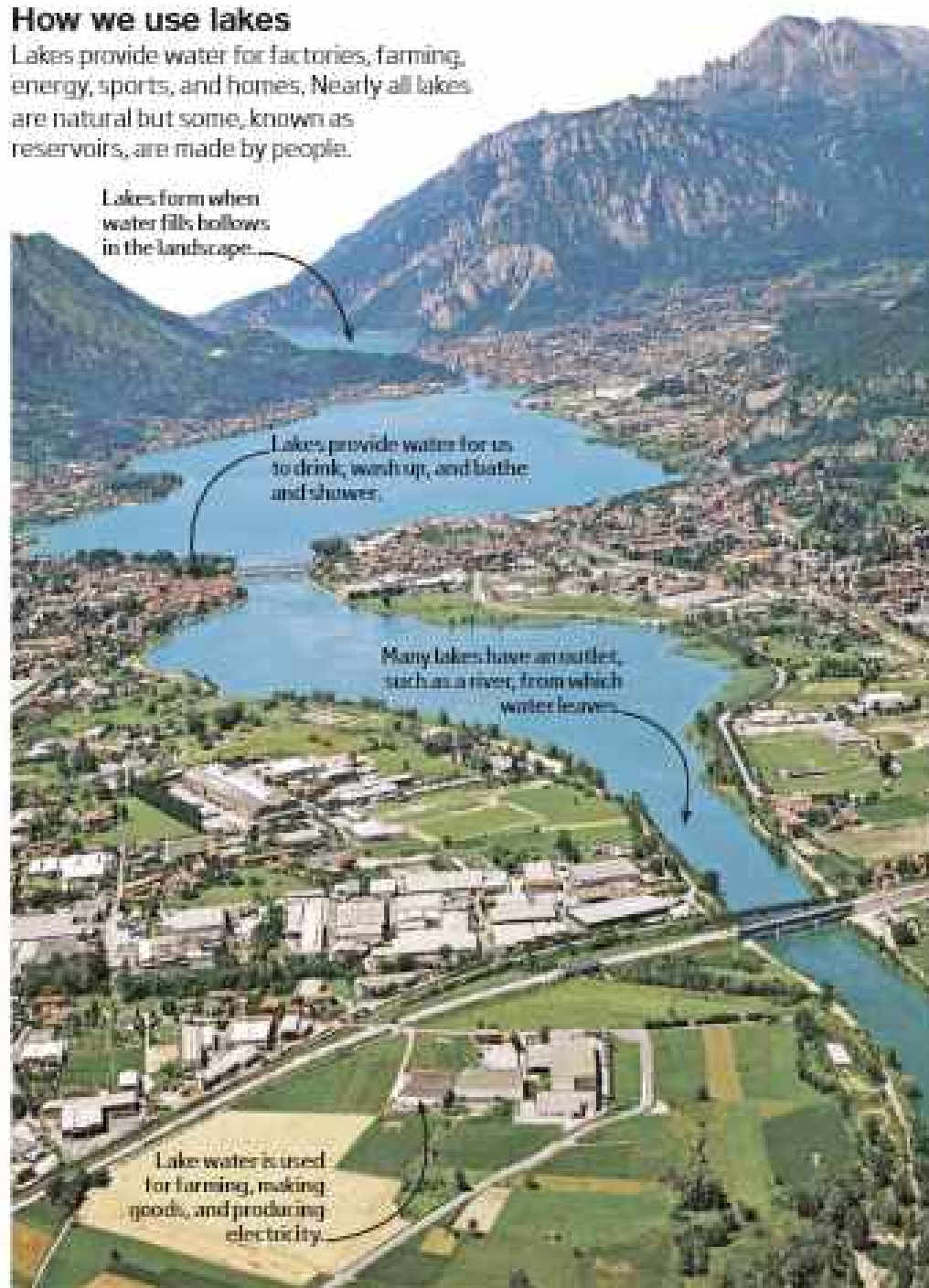
At jousts, crowds watched as knights on horseback raced at each other holding blunt, wooden sticks called lances. They scored points by hitting the other knight or knocking him off his horse.

Lakes

A lake is a large body of water surrounded by land. Most lakes contain fresh water, but some are salty. Lakes are usually found in high areas or near large rivers. The water in lakes comes from streams or rivers around them.

How we use lakes

Lakes provide water for factories, farming, energy, sports, and homes. Nearly all lakes are natural but some, known as reservoirs, are made by people.



SEE ALSO

- ▶ Climate change p.60
- ▶ Factories p.97
- ▶ Farming p.98
- ▶ Water pp.120-121
- ▶ Rivers p.211
- ▶ Water cycle p.270

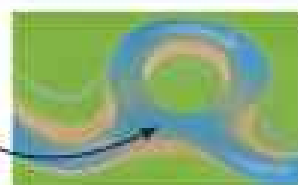
Oxbow lakes

Rivers sometimes change their course. When they do, bends in them may become cut off to form U-shaped bodies of water called oxbow lakes.

A deep bend, or meander, forms in the course of the river.



The river makes a shortcut and changes its course.

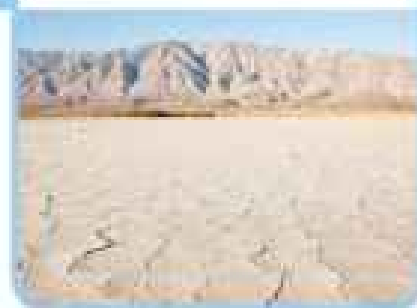


An oxbow lake forms as the river's new course separates from it.



Wet or dry?

Lakes can dry out if there is a long period without rain, called a drought, or if the climate changes. They may reappear when the rain returns or stay dry forever.



Language

Language is how people communicate. Across the world, people speak many different languages. There are more than 7,100 in total, some spoken by millions of people and others by just 100. These languages often sound very different from one another, and can look different when written down.

SEE ALSO

- ▶ Ancient Rome p.20
- ▶ Storytelling pp.42-43
- ▶ Books p.44
- ▶ World p.275
- ▶ Writing p.280

The most-spoken languages

There are almost 7.5 billion people in the world today. About one-third of them speak one of five languages, which are used by countries all over the world.



Sign languages

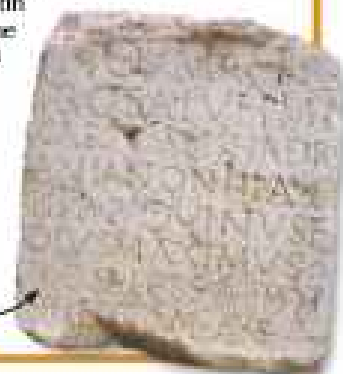
People sometimes use hand shapes, body movements, and facial expressions to talk, rather than spoken language. This is called a sign language. It is useful if someone cannot hear or speak.



Dead languages

Some languages die out when their speakers start using another language. Latin was once spoken in the Roman Empire, which included much of Europe. It's now a dead language, but can still be read and written.

Latin words from the Roman Empire



Law

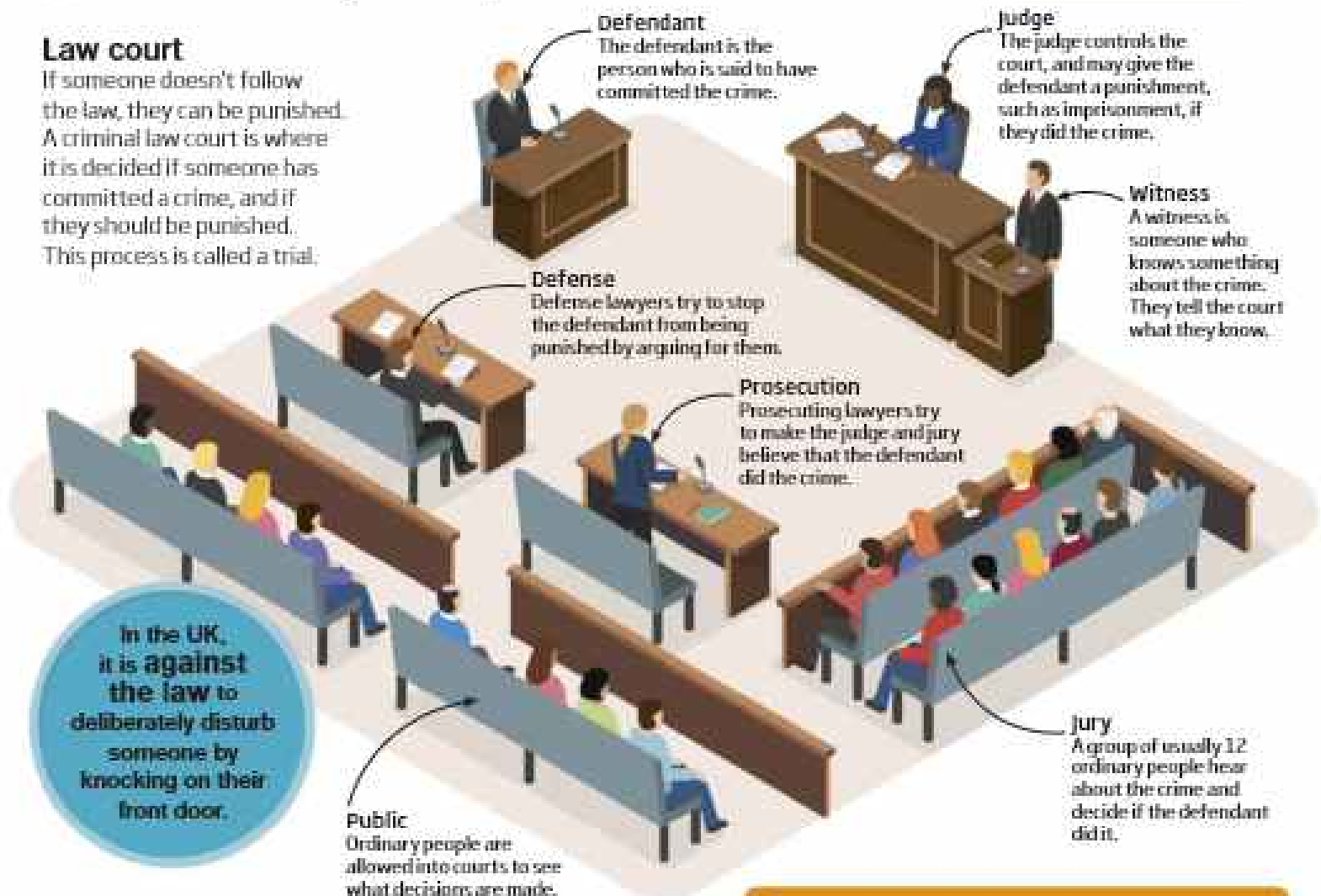
Laws are official rules for people to follow. The government writes laws for the whole country. Some laws stop people from doing bad things (crimes), such as stealing. These are called criminal laws. Other laws try to improve life, such as making sure people are paid a fair amount of money for doing work.

SEE ALSO

- ▶ Changing world pp.50-51
- ▶ Codes pp.66-67
- ▶ Governments p.123
- ▶ Trade p.257
- ▶ Work p.274
- ▶ World p.275

Law court

If someone doesn't follow the law, they can be punished. A criminal law court is where it is decided if someone has committed a crime, and if they should be punished. This process is called a trial.



Police

The police make sure people follow laws. They capture people who might have broken laws. This is called putting someone under arrest.



First laws

One of the earliest sets of laws was drawn up by King Hammurabi of Babylon, who ruled from 1792 to 1749 BC. There are 282 laws about the family, trade, and wages.

Hammurabi's laws

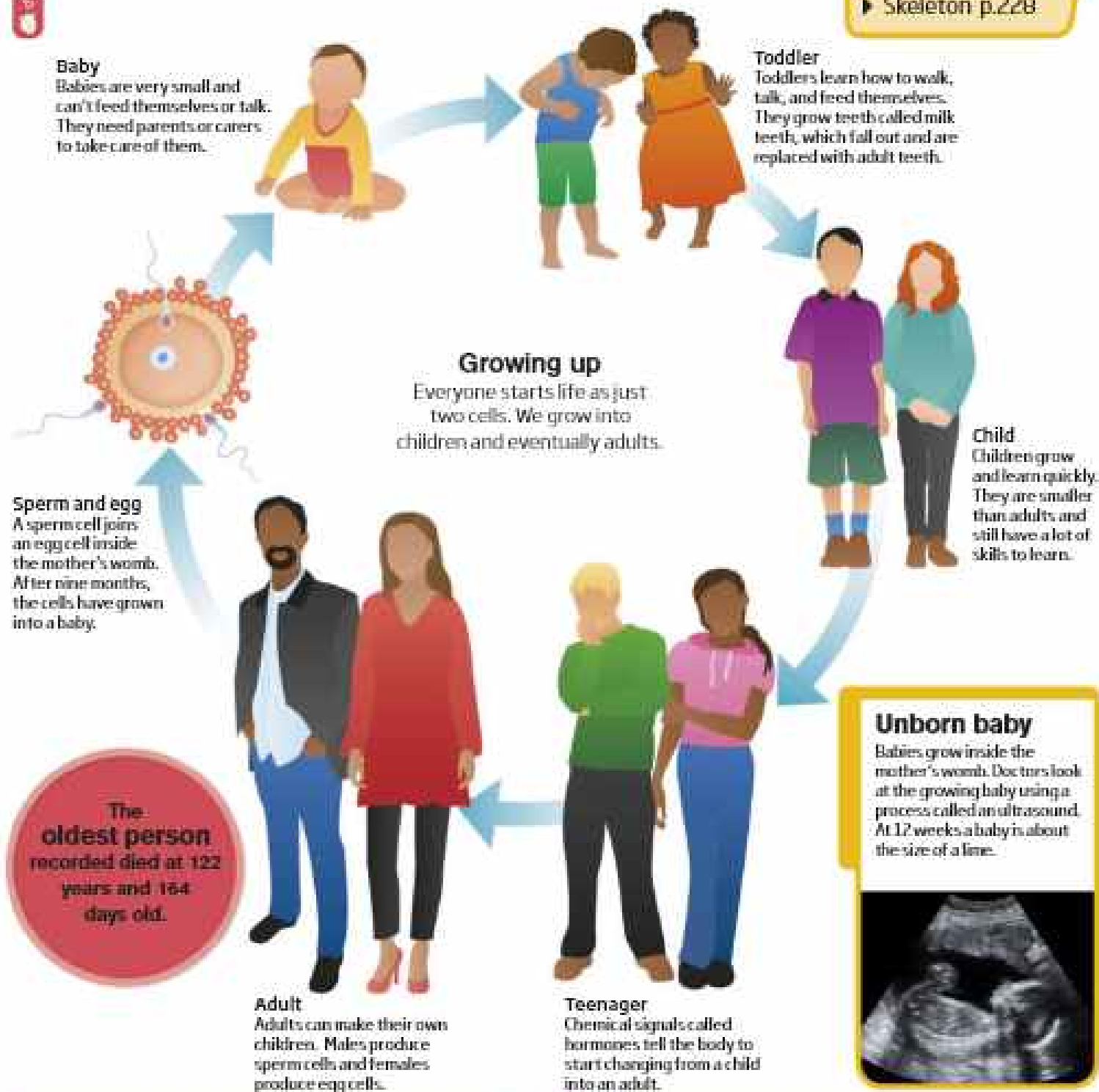


Life cycle

Like all animals, humans are born, grow up, and can have children of their own. This is called a life cycle. There are many different stages that a human goes through on their journey from birth to adulthood.

SEE ALSO

- ▶ Body cells p.41
- ▶ Cells p.56
- ▶ Genes p.119
- ▶ Human body p.130
- ▶ Metamorphosis p.166
- ▶ Skeleton p.228



Light

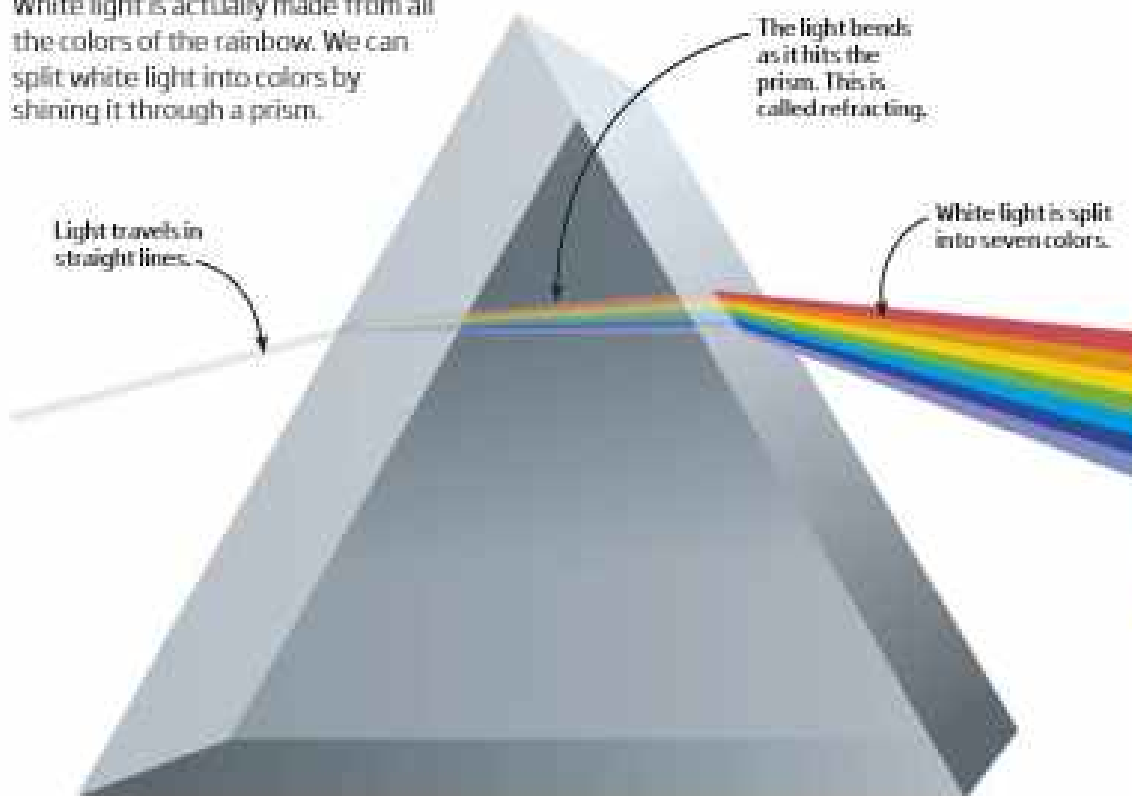
Light is a type of energy that bounces off objects and into our eyes—we need it to see things. Light can change into other types of energy such as heat or electricity. Dark is the absence of light.

SEE ALSO

- ▶ Color pp.26–27
- ▶ Energy pp.88–89
- ▶ Invertebrates p.139
- ▶ Materials p.157
- ▶ Sight p.226
- ▶ Sun p.247

White light

White light is actually made from all the colors of the rainbow. We can split white light into colors by shining it through a prism.



Light sources

The main source of light on Earth is the sun. Artificial light sources include light bulbs, candles, and oil lamps. Some animals, such as jellyfish and fireflies, produce their own light.



The sun is a natural light source.



A candle is an artificial light source.



Some jellyfish can glow in the dark.



Shadows

A shadow is a dark area where light is blocked by an object. It takes the shape of the object blocking the light.



Reflection

More light bounces off shiny surfaces like mirrors or still water. This is called a reflection.



Liquids

Liquids are runny materials. They can be poured into things, and move to fit the shape of their container. Liquids make a pool, not a pile. We drink liquids including water. When a liquid cools, it can become a solid. Heating a liquid can create a gas.

SEE ALSO

- Atoms p.34
- Changing states p.57
- Gases p.117
- Rivers p.211
- Solids p.234
- Water cycle p.270

What liquids do

The things a liquid can do are called its properties. For example, liquids can be heated to cook things. Liquids are good at mixing because of the way they flow and move. Some liquids flow more easily than others, but they all change shape to fit their containers.



Liquids can be poured.

Liquids move to fit the shape of their container.

Liquid particles

Liquids are made from tiny particles that move around quickly and stick together in groups. When they cool, they slow down to eventually become solid.

Around
75 percent
of your brain
is made of
water.

Dissolving

Some materials dissolve (disappear into) liquids. If we add salt to water, it becomes salty water. We can't see the grains of salt because they have dissolved into the water to make salt water.



Water

Salt

Salt water

Water

Two-thirds of the Earth's surface is covered in water. Water is essential because living things need water to survive. Animals drink water and plants take water in from the ground or air. Most living things are made up of at least half water.



Lungs

We breathe air in and out of our lungs. The lungs take oxygen from the air and transfer it to the blood. Every part of the body needs oxygen, so the lungs collect it and the blood carries it around the body.

SEE ALSO

- ▶ Brain p.45
- ▶ Gases p.117
- ▶ Heart p.128
- ▶ Human body p.130
- ▶ Skeleton p.228
- ▶ Sound p.235

Lungs

The lungs are two spongy bags filled with tubes and air sacs. The air sacs are where gases change places. Oxygen is breathed in and carbon dioxide is breathed out.

Bronchi

These two air tubes connect the trachea to the lungs.

Bronchioles

The air goes into these tiny tubes. Each one ends in air sacs called alveoli.

Diaphragm

This muscle changes the shape of the lungs so we can breathe in and out.

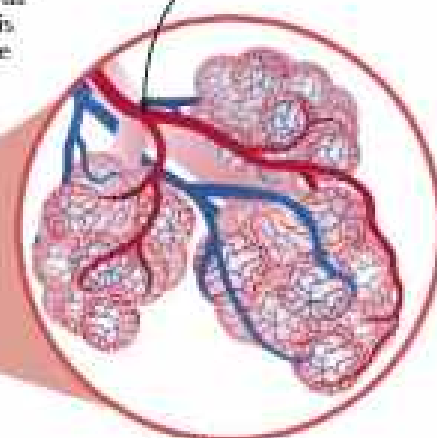
Nose

Air enters and leaves our bodies through our nose and mouth.

Trachea

This tube carries air into our lungs. It is also known as the windpipe.

Oxygen is taken into the blood and carbon dioxide is sent out of the blood in the air sacs.



Alveoli, or air sacs

Breathing

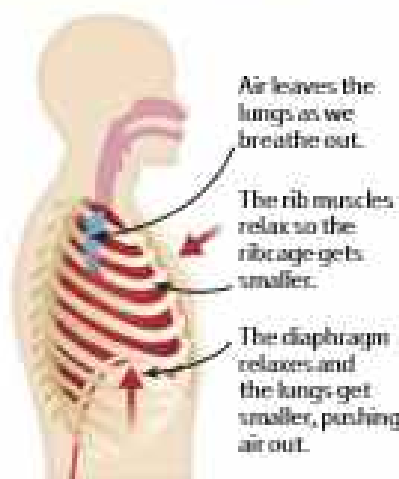
Muscles work together to let us breathe in and out. We have a diaphragm muscle below the lungs and more muscles around the ribcage. They change the size and shape of the lungs.



Air enters the lungs as we breathe in.

The rib muscles pull up to make the ribcage bigger.

The diaphragm pulls down to pull air into the lungs. The lungs get bigger.



Air leaves the lungs as we breathe out.

The rib muscles relax so the ribcage gets smaller.

The diaphragm relaxes and the lungs get smaller, pushing air out.

Voice box

The voice box is in the throat. It stops food from getting into the lungs and makes us cough if any food gets in. The voice box also lets us speak and sing.



Machines

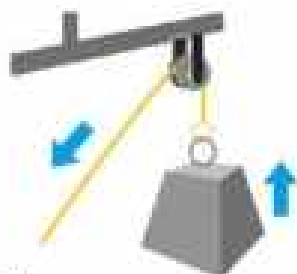
Machines help us do things. They usually perform tasks that are too big, small, boring, long, or dangerous for people to do. Most modern machines are powered by electricity or gas.

SEE ALSO

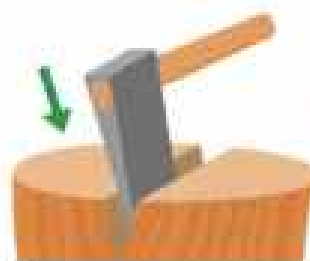
- ▶ Aircraft p.13
- ▶ Bicycles p.36
- ▶ Electricity p.87
- ▶ Energy pp.88-89
- ▶ Engines p.92
- ▶ Robots p.213
- ▶ Trains p.260

Simple machines

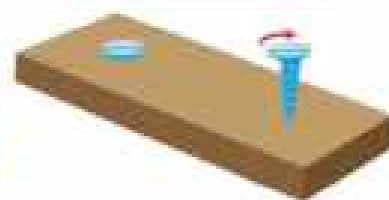
Simple machines reduce the effort needed to do things. A person must operate the machine, but only a small amount of effort is required to move heavy objects.



Pulley
A pulley uses a rope or chain looped over a wheel to lift heavy loads.



Wedge
Made of wood or metal, this triangular tool can be used to push things apart.



Screws
These sharp metal pins are used to attach things. As the screw is turned, it moves down and around.



Big machines

Backhoes are made up of several simple machines. They are powered by engines.

Why we use machines

Machines are usually more efficient and reliable than human workers. This is because they can do things without ever becoming bored, tired, slow, or distracted.



Neat work
A sewing machine can sew more neatly and more quickly than most people can by hand.



Repetitive work
Automatic cash machines don't sleep, so they can give out money 24 hours a day.



Dangerous work
Robots are used to study active volcanoes, so people do not have to put themselves at risk.

Magnets

Magnets are objects that other magnets and some metals stick to. They have two sides, or ends, called poles. The area all around the magnet where the magnet acts is called the magnetic field.

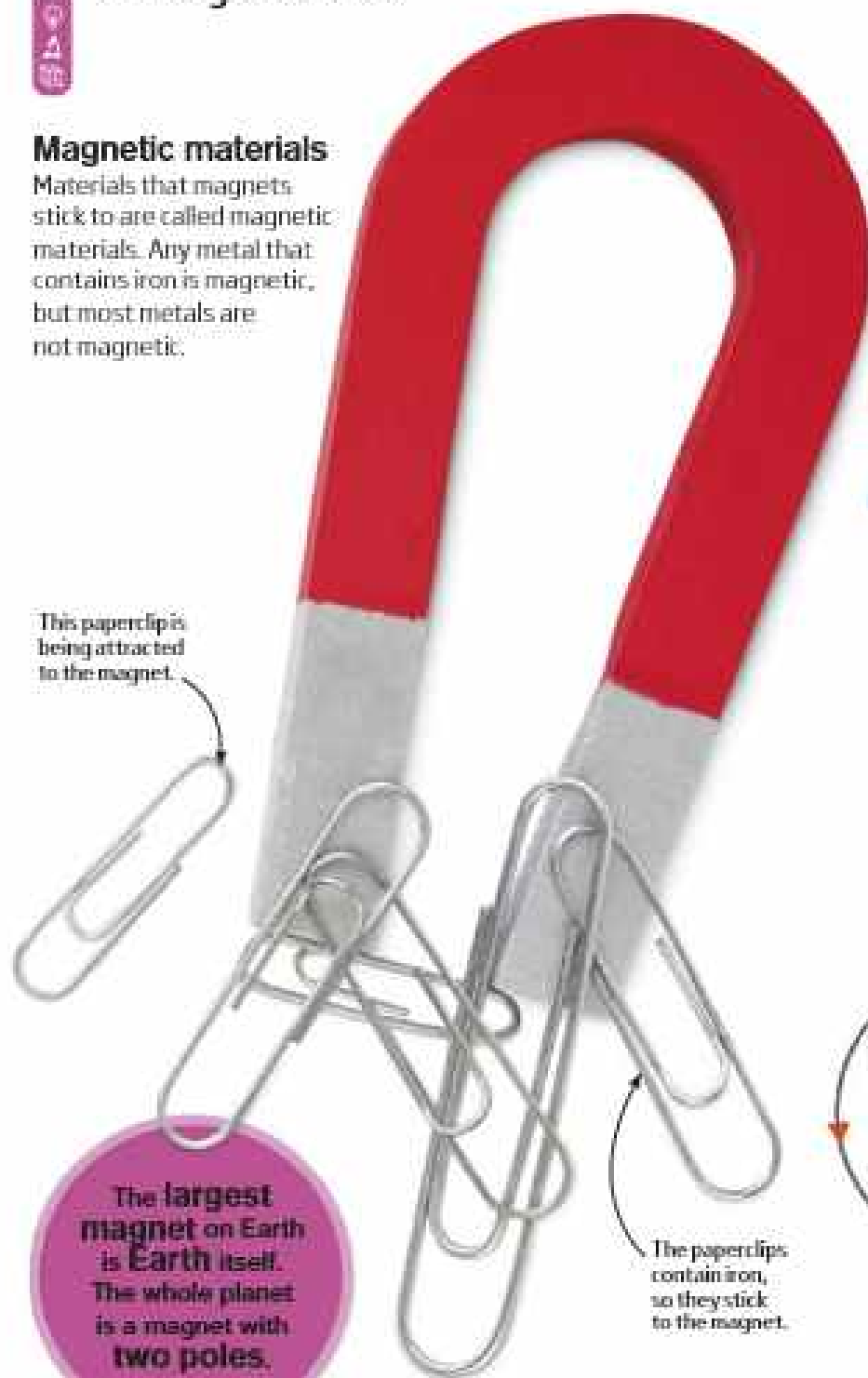
SEE ALSO

- ▶ Compass p.70
- ▶ Earth p.83
- ▶ Electricity p.87
- ▶ Inside Earth p.135
- ▶ Forces p.108
- ▶ Materials p.157

Magnetic materials

Materials that magnets stick to are called magnetic materials. Any metal that contains iron is magnetic, but most metals are not magnetic.

This paperclip is being attracted to the magnet.

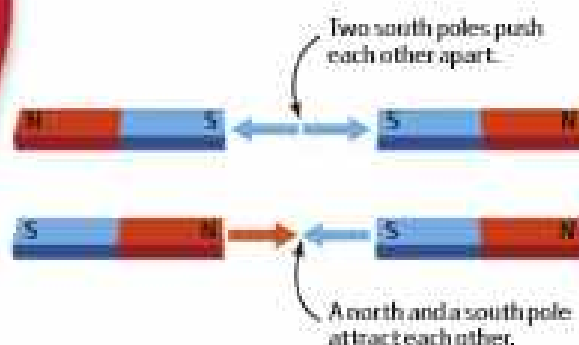


The largest magnet on Earth is Earth itself. The whole planet is a magnet with two poles.

The paperclips contain iron, so they stick to the magnet.

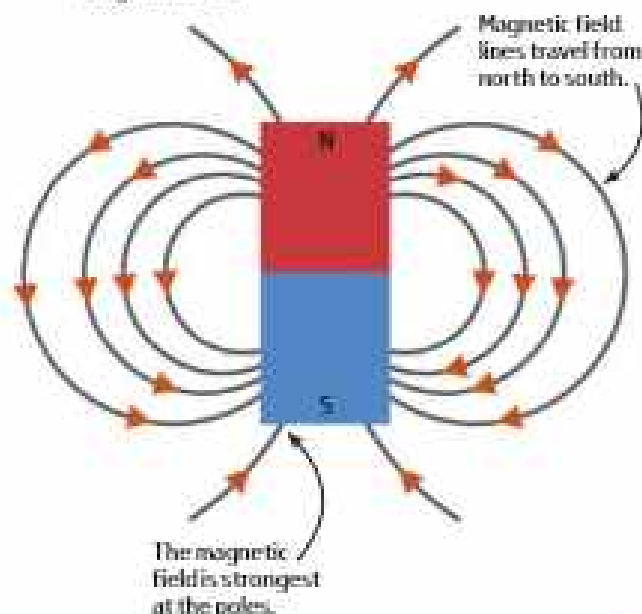
Magnetic force

Two identical poles repel each other, pushing each other away. Two opposite poles attract each other and stick.



Magnetic field

The magnet acts on things that are nearby. The region where the magnet attracts or repels magnetic materials is called the magnetic field.



Pets

Pets have become an important part of humans' lives. Many animals are kept as companions, for work, or to help people go about their daily lives. It is estimated that 44 percent of homes in the world have a pet.



Canaan dog

First pets

Dogs were the first animals to be kept as pets. They were used for hunting—helping early humans to catch food. Ancient art from 12,000 years ago shows humans and dogs together.

Pets big and small

Pets aren't just dogs and cats—we keep all sorts of animals as pets. They vary from big dogs and horses to small snakes and hamsters. Each species needs a special diet and room to exercise.



Bronze statue of a cat from ancient Egypt

Holy cats

The ancient Egyptians loved cats. They caught mice, rats, and snakes, which kept people's homes clean. Cats were also believed to have special powers for guarding children. The punishment for killing a cat was death.



Bearded dragon

Dog



Goldfish



Snake



Hamster



Gerbil

Helpful pets

Dogs are good at keeping people company. They are also easy to train, and can be used to help people with disabilities.

A dog can be a person's eyes or ears, and help them get around.



Guide dogs are specially trained to help people who need help seeing.



Pets in space

For years, animals have helped scientists answer questions about how humans would survive in space. Dogs Belka and Strelka (above) were sent into space on the *Sputnik 5* in 1960. They returned safely to Earth using a parachute.

Budgie



Cat



Rabbit



Guinea pig



Tarantula



Not pets

It is illegal to keep some animals. Wild animals, such as monkeys, can even be dangerous. Before getting a pet, make sure you know that the animal has come from a good home and has not been taken from the wild.

Mammals

Mammals are animals that have body hair and feed their babies on milk made by the females. They are warm-blooded, which means their bodies stay the same temperature. There are many different groups of mammals.

SEE ALSO

- ▶ Animal families p.21
- ▶ Animal groups p.22
- ▶ Food chains p.107
- ▶ Habitats p.126
- ▶ Vertebrates p.266

Mammal babies

Mammals give birth to live babies. Parents feed and care for their young until they can look after themselves.

A mother elephant is pregnant for nearly **two years** before its baby is ready to be born.

Cheetahs have sharp hearing to help them catch animals to eat.



Cheetah

Meat-eaters

Meat-eating animals are called carnivores. They hunt other animals for food.

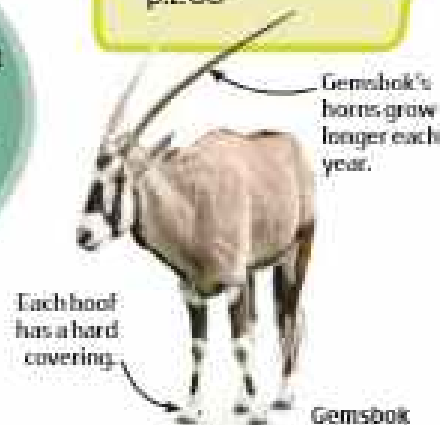


The pouch is a warm place for the baby.

Red kangaroo

Pouched mammals

Some animals, called marsupials, look after their babies in a special pouch. The baby stays in the pouch, drinking milk until it is big enough to leave.



Gemsbok's horns grow longer each year.

Each hoof has a hard covering.

Gemsbok

Plant-eaters

Animals that eat plants are called herbivores. They have special teeth for cutting and chewing leaves.



Asian elephants

Fur or hair on the body keeps mammals warm.

Dolphins

Not all mammals live on land. Dolphins are mammals that live in water. They come up to breathe through a blowhole at the top of their head.



Maps

A map is a detailed picture of what the ground looks like from above, as if you were flying over it. Maps tell us how big an area is and what can be found there. Maps can be of anything, from the whole world to the insides of buildings.

SEE ALSO

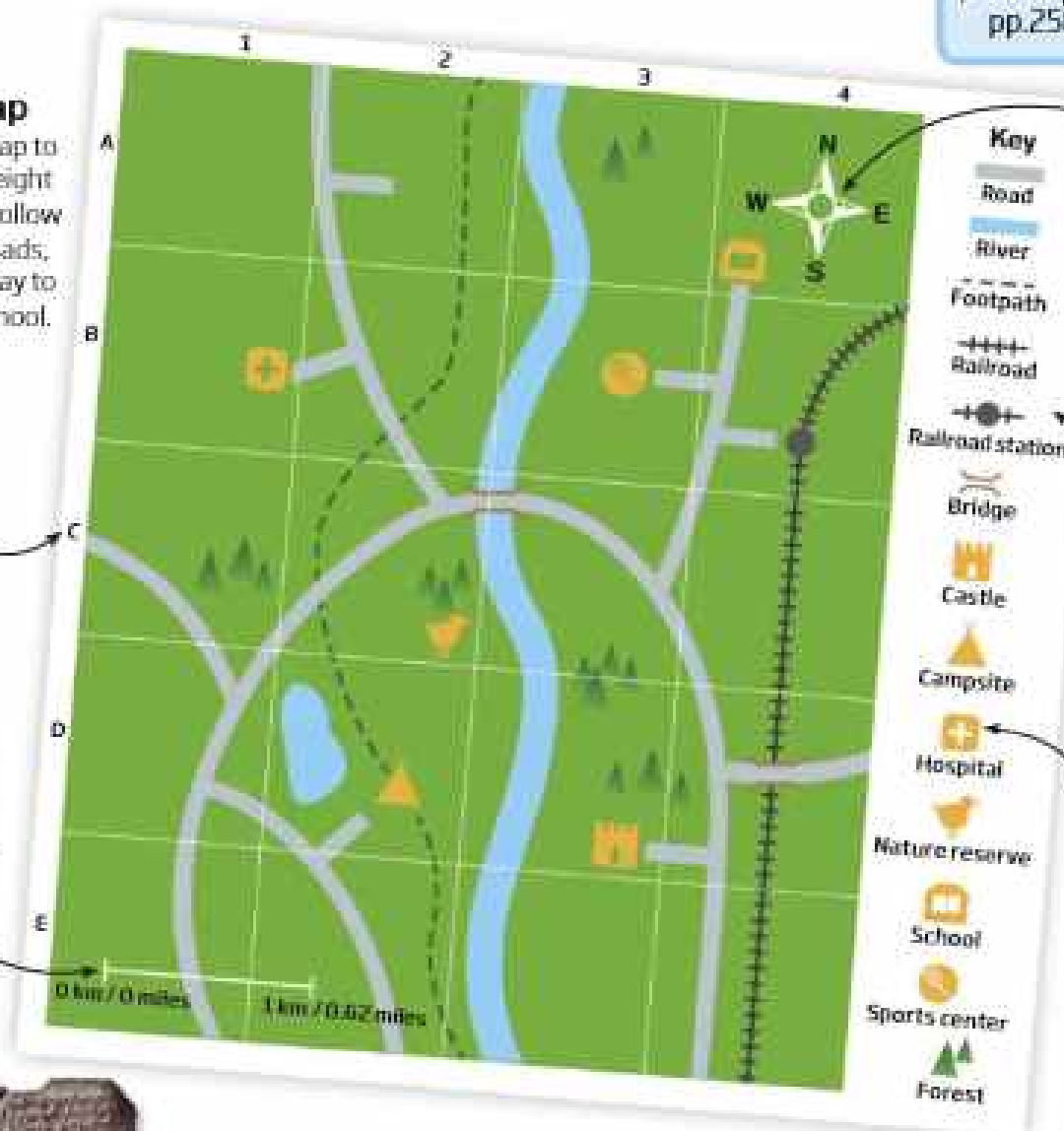
- ▶ Compass p.70
- ▶ Explorers p.96
- ▶ Measuring p.159
- ▶ Exploration pp.180-181
- ▶ Navigation p.182
- ▶ Transportation pp.258-259

Using a map

We can use a map to work out the height of the land, to follow roads and railroads, or to find our way to a hospital or school.

The map is divided into a grid with squares for different areas.

The scale bar shows the real-life distance between points on the map.



This shows which direction the top of the map is pointing toward. This is usually north, or "N."

A key shows which real-life features the map's lines and symbols represent.

Symbols are used for different features, such as buildings and campsites.



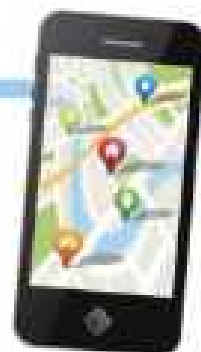
Ancient maps

Maps were less accurate in the past. This 2,500-year-old stone map shows how people in Babylon (modern-day Iraq) saw the world.

Babylon is shown at the center of the world.

The end of paper maps?

Paper maps are still around today, but fewer people use them. Most cars now have GPS to guide the driver, and digital maps can be viewed on mobile phones or laptops.



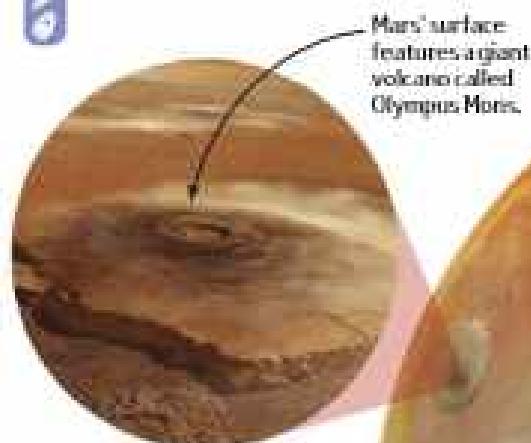
Phone map

Mars

Named after the Roman god of war, this rocky planet has huge volcanoes, ice caps, and deep canyons. It was once a wet, warm world where water flowed. Now, Mars is a cold, dry world covered in craters.

SEE ALSO

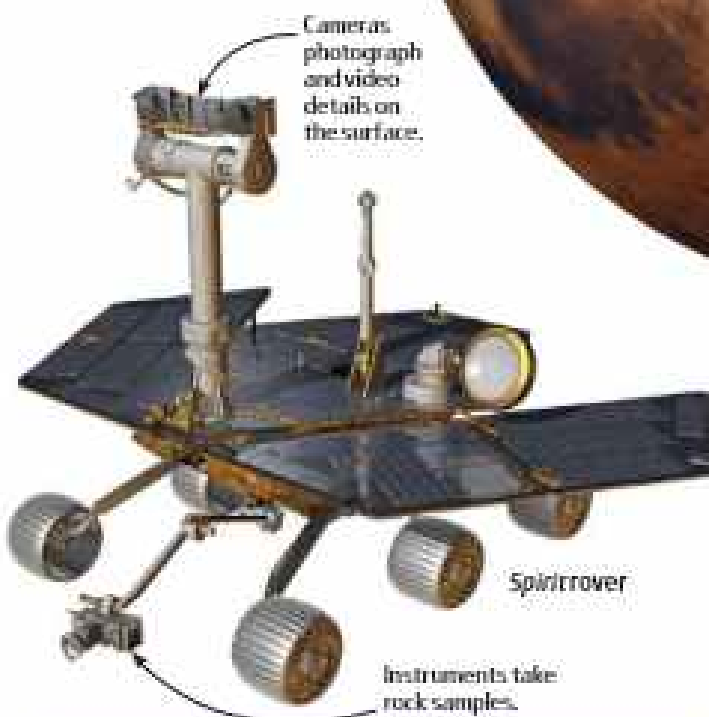
- ▶ Ancient Rome p.20
- ▶ Asteroids p.30
- ▶ Elements p.90
- ▶ Rocks and minerals p.214
- ▶ Space travel p.237
- ▶ Volcanoes p.268



Mars' surface features a giant volcano called Olympus Mons.

Red planet

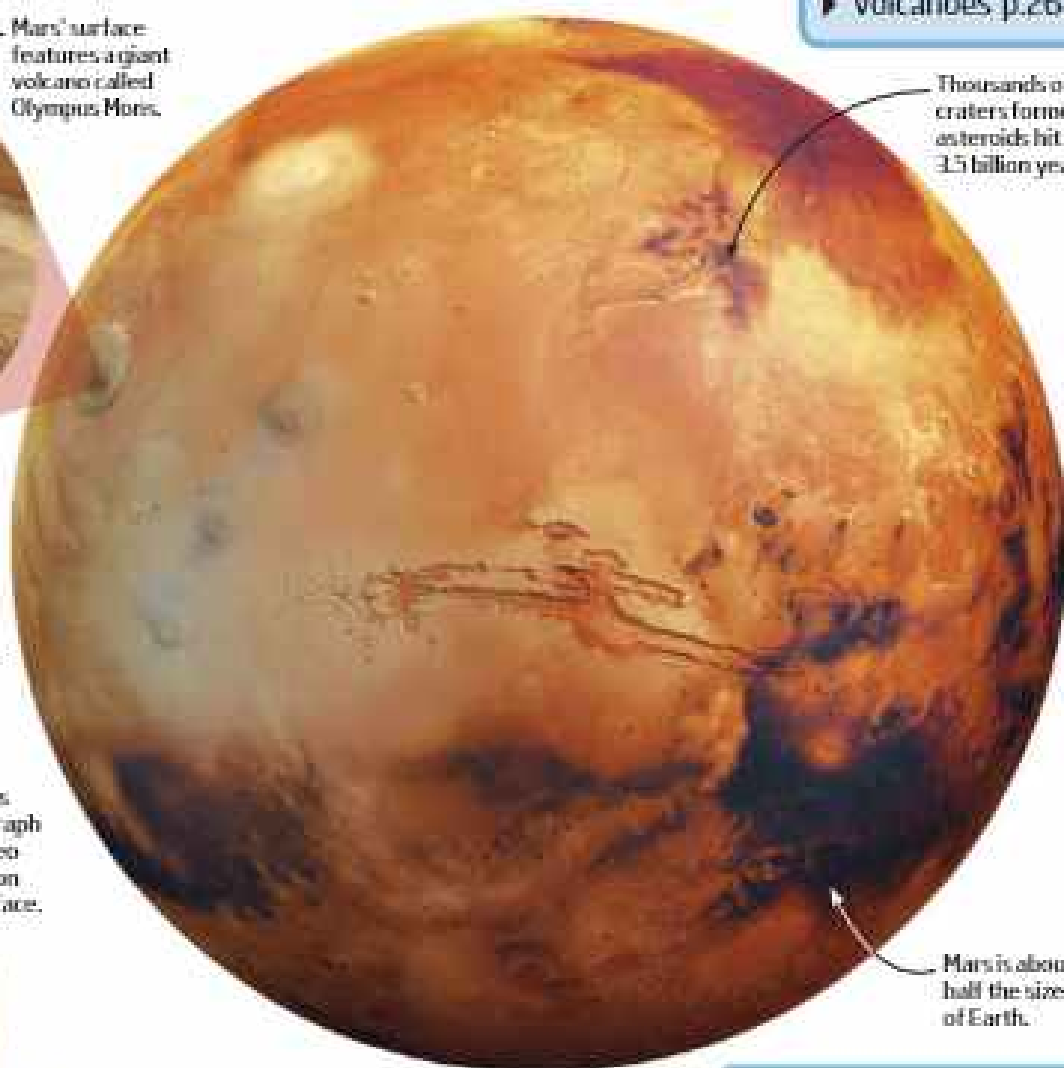
Mars is called the red planet because its surface is covered in a layer of reddish dust. When the wind blows, this dust enters the atmosphere and turns the sky red.



Camera photograph and video details on the surface.

Spirit rover

Instruments take rock samples.



Thousands of craters formed when asteroids hit Mars 3.5 billion years ago.

Mars is about half the size of Earth.

Mission to Mars

Since 1976, spacecraft have visited Mars to study its surface. Two twin rovers called *Spirit* and *Opportunity* arrived on Mars in 2004, and *Opportunity* is still exploring.

Moons of Mars

Mars has two tiny moons called Deimos and Phobos. These rocks may have started as asteroids. Phobos is the largest, measuring 16 miles (27 km) in length.



Deimos



Phobos

Materials

Materials are what all things in the world are made of. We describe them using their properties, which are how they behave and what they do. They can be hard, bendy, waterproof, or magnetic. Materials will either float or sink, and electricity can pass through some of them.

SEE ALSO

- ▶ Atoms p.34
- ▶ Buildings p.48
- ▶ Electricity p.87
- ▶ Gases p.117
- ▶ Liquids p.148
- ▶ Plastic p.195
- ▶ Solids p.234

Solubility

Solubility is how easily something dissolves (mixes into) liquid. If you put salt in water, the salt dissolves into it, so salt is soluble. Soluble materials can be solids, liquids, or gases.

Sand is insoluble—it does not dissolve in water.



This purple powder dissolves into the water.



Conducting heat

Metals are good at conducting heat, which means when something hot touches them, the heat transfers into them. Wood, plastic, and rubber are poor conductors of heat. Both properties are useful in saucepans.

The rubber handle stays cool, so we can pick up the pan.

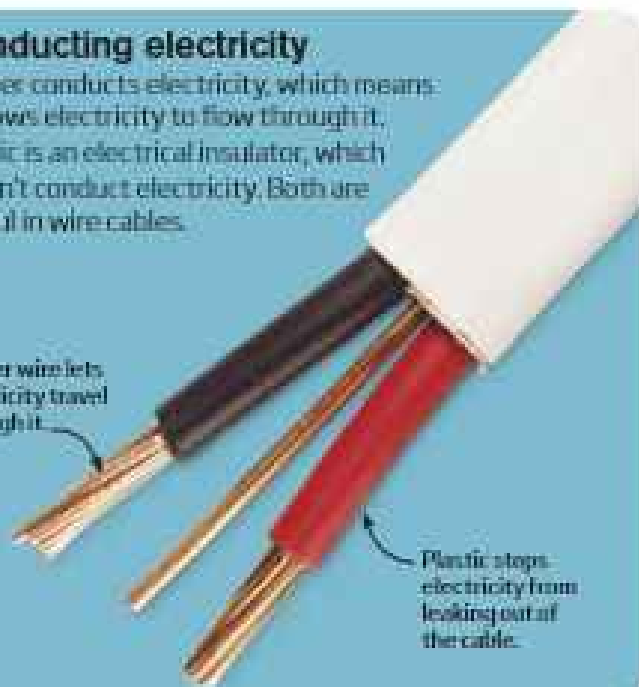


The flame causes the metal pan to heat up.

Conducting electricity

Copper conducts electricity, which means it allows electricity to flow through it. Plastic is an electrical insulator, which doesn't conduct electricity. Both are useful in wire cables.

Copper wire lets electricity travel through it.



Plastic stops electricity from leaking out of the cable.

Flammability

Flammability is how easily something catches fire and burns. Dry wood is highly flammable, which means it catches fire and burns easily, giving off heat.

Dry wood catches fire and burns easily.

Non-flammable stones stop the fire from spreading.



Maya

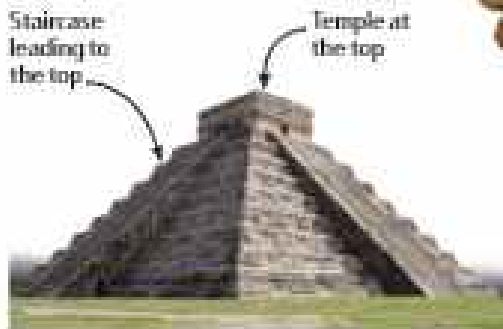
The Maya people lived in Central America from 1000 BCE to 1600 CE. They built great cities from stone and farmed maize, beans, and squash. The Maya had many gods, who they built temples for. They were skilled mathematicians and developed a calendar.

SEE ALSO

- ▶ Art p.28
- ▶ Aztecs p.35
- ▶ Farming p.98
- ▶ Incas p.132
- ▶ Games pp.240-241
- ▶ Religion p.208

Mayan gods

The Maya worshipped many different gods. They believed the gods controlled the world around them, including animals and the weather.



Mayan pyramid at Chichén Itzá, Mexico

Temples

Many Mayan temples were built on top of stone pyramids. Priests at the temples sacrificed animals, sang, and danced to honor the gods.

Mayan sport

The Mayans played a sacred ball game. Players hit a large rubber ball to different parts of a court to win points. They could only use their forearms and hips.



Mayan gods were often shown wearing large hats decorated with holy symbols.

Chaac was shown with the nose and fangs of a snake.

Balls of incense were burned to honor the gods.

Chaac was the brother of the SUN. When he cried, his tears fell as rain.

Chaac, the Mayan rain god

Measuring

We measure things to find a number that shows the size or amount of something. Being able to measure things lets us record and compare them. We use many different types of measurements, as well as various tools for measuring.

SEE ALSO

- ▶ Ancient Egypt p.17
- ▶ Astronomy p.32
- ▶ Clocks p.61
- ▶ Earth p.83
- ▶ Numbers p.185
- ▶ Volume p.269

Measuring tools

Different tools let us measure time, size, distance, speed, weight, and volume. Volume is the amount of three-dimensional space something takes up.



Weighing through history

People have always wanted to be able to compare amounts of things. The ancient Egyptians invented their own, very accurate system of weights and simple scales to measure the goods they bought and sold.



Big and small

All objects can be measured one way or another. A box of eggs fits easily on weighing scales, but the Earth is so big it can only be weighed using complicated scientific calculations.

A box of six eggs weighs about 10 oz (300 g).



Earth weighs 13.2 billion trillion lbs (5.9 trillion trillion kg).

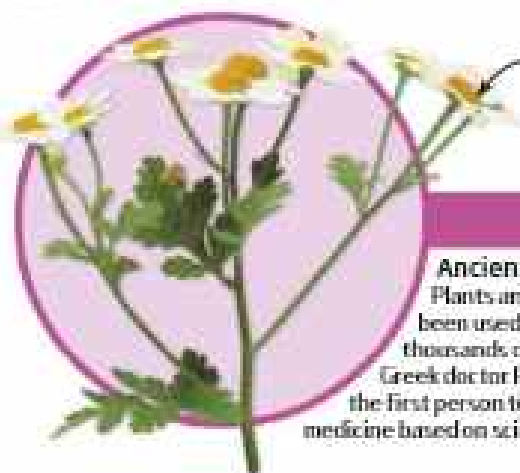


Medicine

Medicine is used to treat and prevent sickness. Medicine can be made from plants or from chemicals in a science lab. Doctors are trained to discover different problems and find the right treatment or medicine to make people better.

SEE ALSO

- ▶ Ancient Greece p.18
- ▶ Biology p.38
- ▶ Chemistry p.58
- ▶ Human body p.130
- ▶ Science p.217
- ▶ Sickness p.225



Chamomile was used to treat fevers in ancient Greece.

Ancient medicine
Plants and herbs have been used as medicine for thousands of years. Ancient Greek doctor Hippocrates was the first person to teach about medicine based on science.

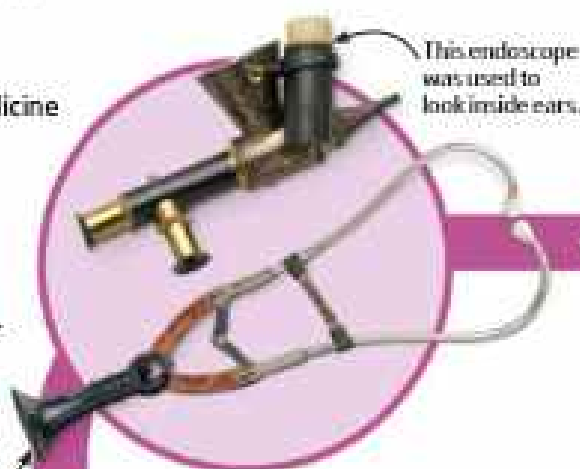
Medieval medicine
Too much blood was thought to make people sick, so leeches were used to suck out small amounts.

Medicinal leech



This sickly royal patient has been covered in leeches to try to cure him.

Early modern medicine
Doctors used new inventions to find out what was wrong with patients. Bad smells were thought to cause sickness, so patients breathed in nice smells like herbs.

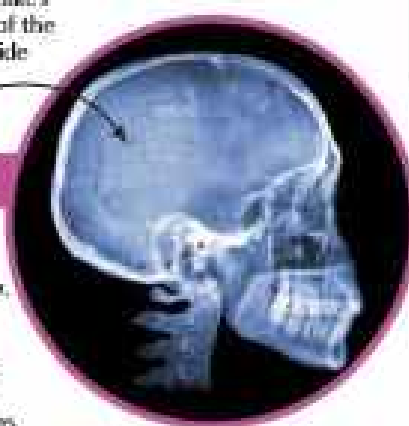


This endoscope was used to look inside ears.

A patient's heart could be heard with the newly invented stethoscope.

Smallpox was the first infectious disease to be wiped out by humans, in **1980**.

Modern medicine
Doctors use machines like X-rays to look inside the body. We know tiny germs and viruses cause sickness. We can kill germs with antibiotic medicines, and we can stop viruses with vaccine injections.



An X-ray takes a picture of the bones inside the body.

Doctors

Doctors find out what is wrong with patients by looking at their bodies, asking questions, and testing things from the body, such as the blood. They can then treat them.



Mercury

Mercury is the smallest planet in the solar system. Despite this, it can often be seen from Earth at sunrise and sunset. The average temperature is a scorching 332°F (167°C) because this planet is closest to the sun.

SEE ALSO

- ▶ Ancient Rome p.20
- ▶ Asteroids p.30
- ▶ Water pp.120-121
- ▶ Moon p.171
- ▶ Solar system p.233
- ▶ Sun p.247

Fast mover

This planet takes its name from the speedy Roman messenger god. Mercury moves faster across Earth's sky and faster around the sun than the other planets.

Craters were made when asteroids hit Mercury billions of years ago.

Mercury is a dry, rocky planet with no liquid water.

Large solar panels kept MESSENGER working by turning the sun's rays into electricity.

Temperatures on Mercury can soar to 800°F (430°C) in the day and drop to -290°F (-180°C) at night.

Exploring Mercury

Between 2011 and 2015, the robotic space probe MESSENGER explored the surface of Mercury. The information collected allowed scientists to make complete maps of Mercury for the first time.



Tiny planet

Mercury is a very small planet. It is only slightly bigger than Earth's moon. The planets Jupiter and Saturn both have moons that are bigger than Mercury.

Metals

We find metals in rocks. Metals can be strong or bendy, and they let electricity pass through them. These features are useful for making many things, from wires to buildings. Metals can be used on their own or mixed together.

SEE ALSO

- ▶ Bicycles p.36
- ▶ Elements p.90
- ▶ Iron Age p.140
- ▶ Liquids p.148
- ▶ Magnets p.151
- ▶ Meteorites p.164

Bicycle basics

A bicycle is made from a combination of strong and bendy metals. The type of metal used for each part depends on what it needs to do.



Metal from mines

People dig tunnels underground to find metal in places called mines. Usually, the metal found in mines is not pure, which means it has rock and gas mixed in it. These materials need to be separated from the metal before we can use it.



1. Ore
An ore is a rock that contains metal. The ore is discovered and dug out of the mine.



2. Melting
The ore is heated to melt and remove the metal. Chemicals are added to help remove gases.



3. Cooling
The metal cools into a solid. It can be heated up and hammered into useful shapes.

Metamorphosis

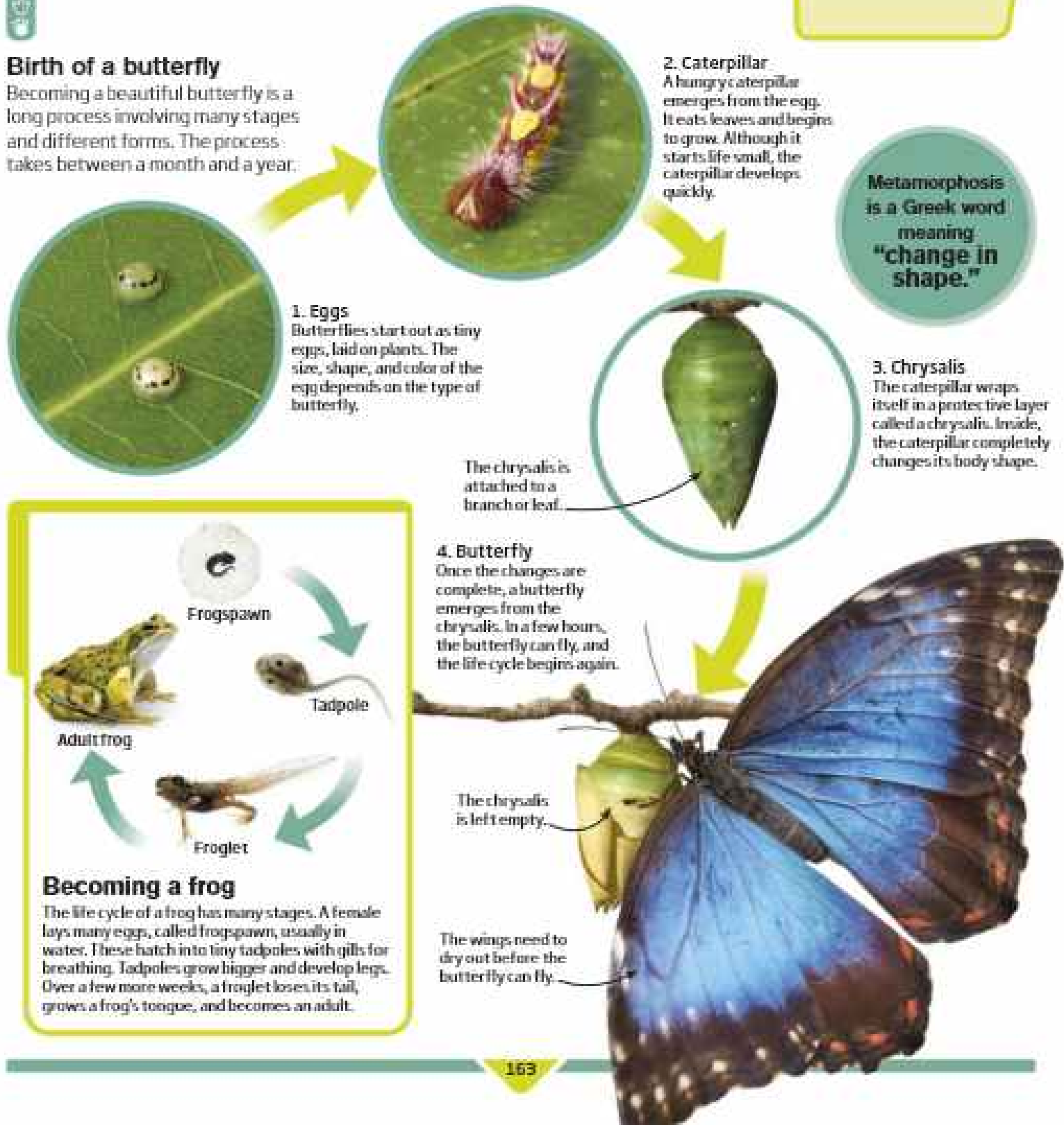
Some animals go through amazing changes between birth and adulthood. Their appearance changes so much that their fully grown shape is completely different to the newborn one. This process is called metamorphosis.

SEE ALSO

- ▶ Amphibians p.15
- ▶ Animal groups p.22
- ▶ Eggs p.86
- ▶ Insects p.134
- ▶ Life cycle p.146

Birth of a butterfly

Becoming a beautiful butterfly is a long process involving many stages and different forms. The process takes between a month and a year.



Meteorites

Meteorites are pieces of space rock (asteroids and comets) that reach the surface of the Earth. They come in many different sizes, from tiny pebbles to rocks the size of a house. Only big meteorites can create craters when they hit the ground.

SEE ALSO

- ▶ Asteroids p.30
- ▶ Atmosphere p.33
- ▶ Comets p.68
- ▶ Metals p.162
- ▶ Rocks and minerals p.214
- ▶ Solar system p.233

Space rocks

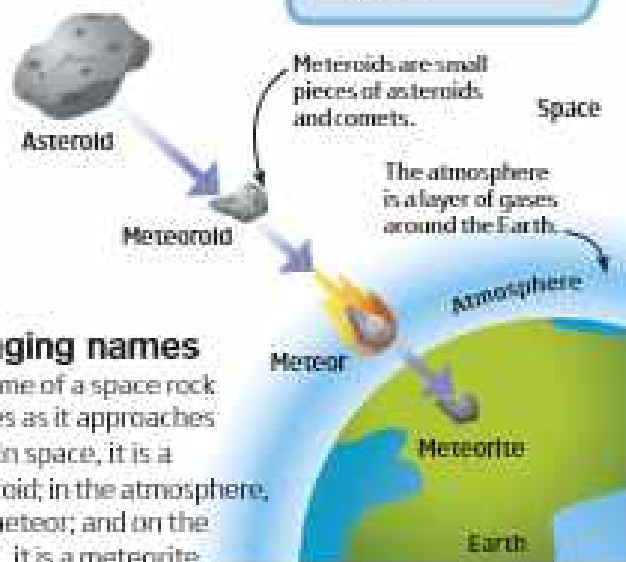
Meteorites are made of materials that are also found on Earth. There are three main types:

Stony
Most meteorites that are found are stony. They come from the crusts of asteroids.



Stony-iron
Stony-iron meteorites are a mixture of metal and rock. They are very rare.

iron
Iron meteorites are made of iron and nickel metals. They come from the cores of asteroids.



Changing names

The name of a space rock changes as it approaches Earth. In space, it is a meteoroid; in the atmosphere, it is a meteor; and on the ground, it is a meteorite.



Crash site

If a meteorite makes it through Earth's atmosphere, it crashes into the surface. The hole it leaves is called an impact crater. This huge crater is found in Winslow, Arizona.

Microscopic life

Microorganisms are very tiny living things. They are all around us—in the air, in our bodies, and in water. Most microorganisms are so small that they can only be seen through a magnifying machine called a microscope.

SEE ALSO

- ▶ Body cells p.41
- ▶ Food chains p.107
- ▶ Inventions pp.136–137
- ▶ Invertebrates p.139
- ▶ Sickness p.225

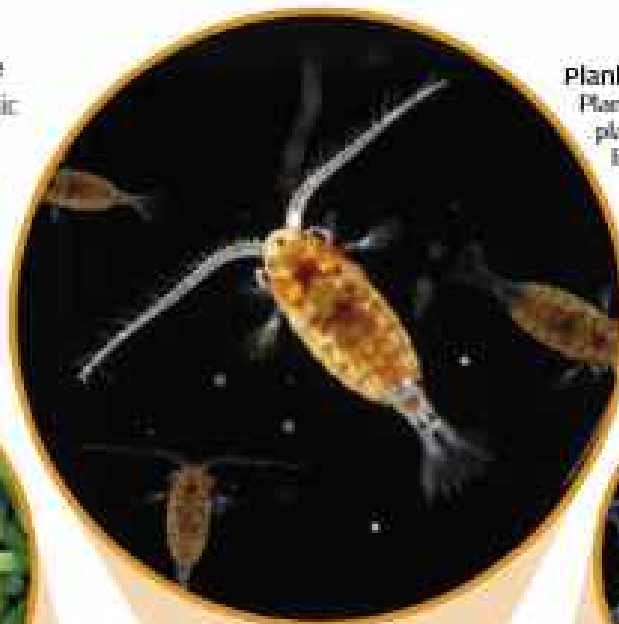
Types of microscopic life

There are many kinds of microscopic life. Some are harmful and spread illness. Others are helpful, such as the bacteria in our stomach, which break down our food.



Bacteria

Some bacteria help our bodies use food. Others cause illness, such as cholera and tetanus.



Plankton

Plankton are microscopic plants and animals that live in water.



Virus

Viruses attack the cells of plants and animals, causing sickness.



Dust mites look like tiny bugs.



Scientists put samples on glass slides to examine them in closer detail under a microscope.

Dust mites

These microscopic bugs live all around us. They live in people's homes and eat flakes of dead skin that drop off us.

Humans have trillions of bacteria in their bodies to help them survive.

Microscope

Microscopes use lenses to enlarge, or magnify, things. This lets us look at things much smaller than what we can see with just our eyes.



Migration

Some animals make long journeys each year. These journeys are called migrations. Animals migrate to find water, to spend winter in warmer places, or to find the best place to mate and have their babies.

SEE ALSO

- ▶ Birds p.39
- ▶ Insects p.134
- ▶ Mammals p.154
- ▶ Metamorphosis p.163
- ▶ North America p.184
- ▶ Seasons p.221

Monarch butterfly

Monarch butterflies fly thousands of miles to get from North America to Mexico. Butterflies that arrive in Mexico hatch from eggs laid by butterflies in North America. These live until the spring, when they can lay eggs on their own.

- Key**
- Autumn
 - Spring
 - Summer

Autumn
As the temperature drops and there is less food for the butterflies, the young ones begin their long migration south to warmer areas.

Summer
Once the caterpillars have turned into butterflies, they are ready to fly farther north in large groups, to mate and lay eggs.

Spring
The butterflies travel north to lay their eggs in the warm spring air, and then die. There will be plenty of leaves for the caterpillars to eat when they hatch.

Winter
Huge numbers of butterflies come together in the forests in the winter to rest.

Thousands of butterflies migrate together.



Caribou

These hoofed animals from the Arctic travel in enormous herds. They walk up to 30 miles (50 km) a day for three months to spend their summer in open areas and winter in forests.



Arctic tern

These small birds have the longest migration of all animals. They fly back and forth between the North and South Pole areas. They fly for eight months of every year.



Milky Way

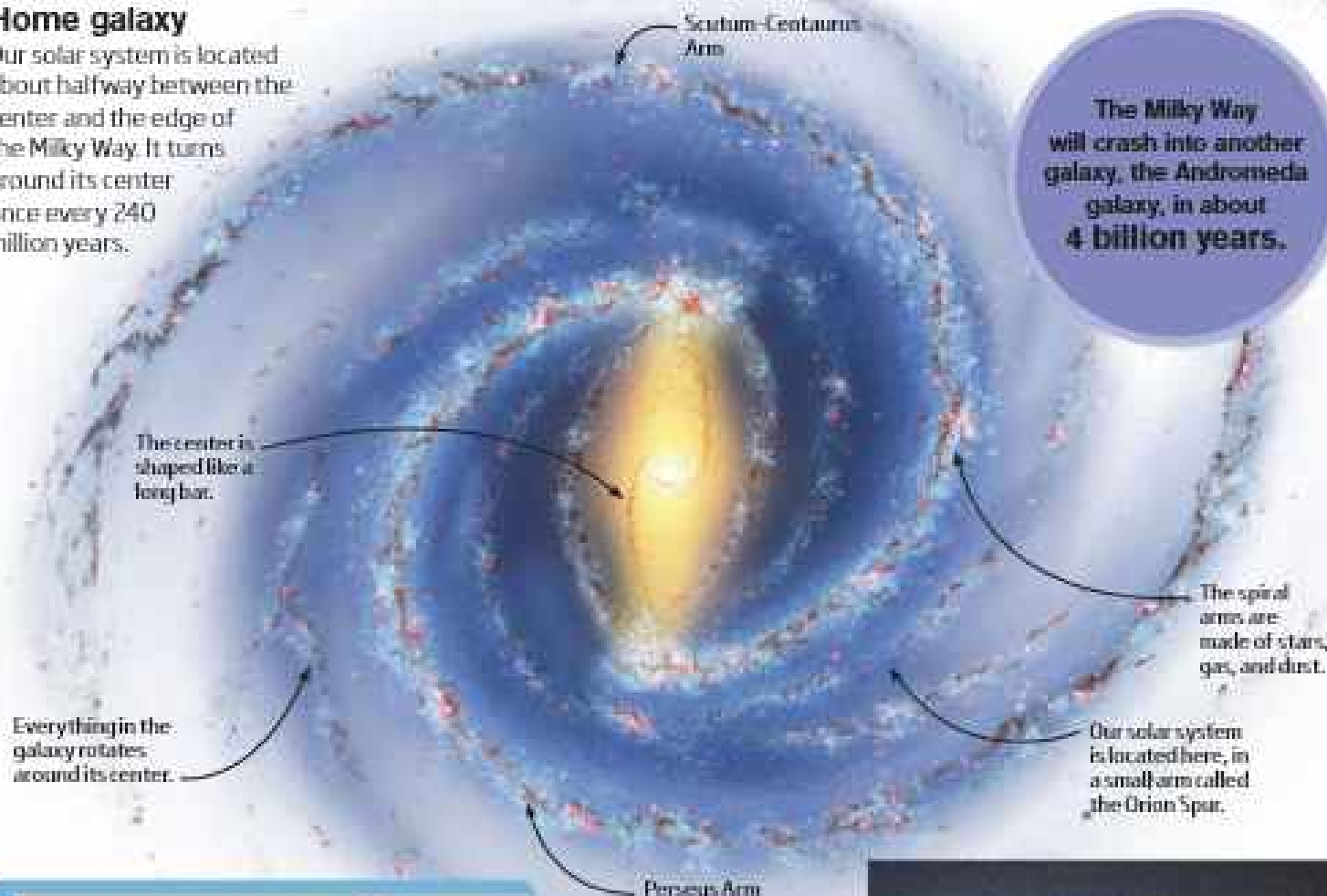
The Milky Way is a galaxy, or group of stars. It contains more than 200 million stars, including our sun. Astronomers think that the Milky Way is shaped like a spiral, with two main arms.

SEE ALSO

- ▶ Astronomy p.32
- ▶ Constellations p.73
- ▶ Earth p.83
- ▶ Galaxies p.116
- ▶ Solar system p.233
- ▶ Stars p.242
- ▶ Universe p.263

Home galaxy

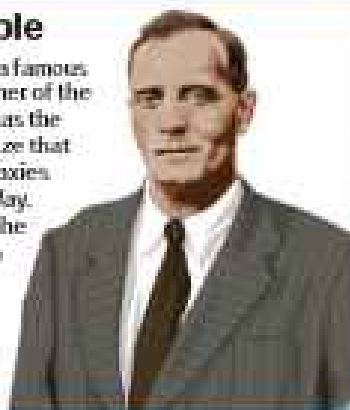
Our solar system is located about halfway between the center and the edge of the Milky Way. It turns around its center once every 240 million years.



The Milky Way will crash into another galaxy, the Andromeda galaxy, in about 4 billion years.

Edwin Hubble

Edwin Hubble was a famous American astronomer of the 20th century. He was the first person to realize that there are other galaxies beyond the Milky Way. He also measured the distances between galaxies.



View from Earth

From Earth, we can see the Milky Way as a faint white band across the night sky. The light we see is created by billions of shining stars.



The Milky Way from Earth



Mixtures

A mixture is made when we mix different materials together and they can be easily separated back out into their original parts. Mixtures can be made from solids, liquids, and gases. There are three main ways of separating mixtures.

SEE ALSO

- ▶ Changing states p.57
- ▶ Gases p.117
- ▶ Liquids p.148
- ▶ Rocks and minerals p.214
- ▶ Solids p.234

Sifting

We can use a sifter to separate large solids from small ones, or solids from liquids. A sifter is made from crisscrossed wire with small holes that let some solids through.



Filtering

Filters are made from materials with tiny holes in them. The holes catch solids that are too big to fit through, but let liquids pass.



The liquid changes to gas and the solid will be left behind.

Evaporation

Some solids such as salt dissolve (disappear) when they are mixed into a liquid. If we heat the liquid, we can take out the solid in a process called evaporation.

Heat causes the liquid to boil and change to gas.



Compounds

Some materials that are joined together can't be easily separated by sifting, filtering, or evaporation. These materials are not mixtures, they are compounds. Iron and sulfur make the compound iron sulfide.



Money

We exchange money for things we want to buy, such as food, clothes, and electricity. Money has a number value, and it is made up of coins and bills. Some objects are expensive and worth more money than others.

SEE ALSO

- ▶ Measuring p.159
- ▶ Metals p.162
- ▶ Numbers p.185
- ▶ Plastic p.195
- ▶ Precious metals p.199
- ▶ Work p.274

Currency

The different units of money used around the world are known as currency. In the US, for example, the currency is the dollar, and in Japan the currency is the yen.

Early currency

Before coins were invented, people exchanged other things as money, including cattle, salt, grain, and even shells.

A type of shell that was once widely traded.



Cowry shell

Ancient coins

The first coins were used nearly 3,000 years ago. They were made from gold and silver. Different coins were made across the ancient world.



Chinese Han Dynasty coin



Ancient Greek coin



Roman Emperor Antonius Pius



Ancient Egyptian coin



British penny



American cent



Danish krone



European euro



Indian rupee



Japanese yen



Mexican peso



South African rand

A springbok—the national animal of South Africa.

Digital money

Banks store money for people in bank accounts. People can pay into or take money out of their bank account. They can use a card or phone to spend the money from the bank in a shop.



Value

An object that takes a lot of time to make or uses expensive materials is said to have a high value. It will cost more money than something that is quick and easy to make and uses cheaper materials.



High-value sports car



Low-value toy car



Earning money

People exchange their time for money, too. This vet gets paid for the time she spends making animals better. She goes to work and earns money.



Vet at work

Monkeys and apes

Apes, monkeys, and lemurs belong to a group of animals called primates, which also includes humans. Most primates are smart and like to play. Primates are the only animals with hands that can grab things.

SEE ALSO

- ▶ Africa p.12
- ▶ Early humans p.82
- ▶ Habitats p.126
- ▶ Rain forest p.204
- ▶ South America p.236
- ▶ Vertebrates p.266

Apes

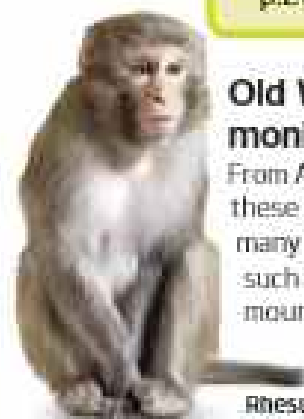
Apes do not have tails and can stand more upright than monkeys. They use their huge, strong arms to climb trees and hang from branches.



Chimpanzee

Chimpanzees live in groups of up to

120
animals.

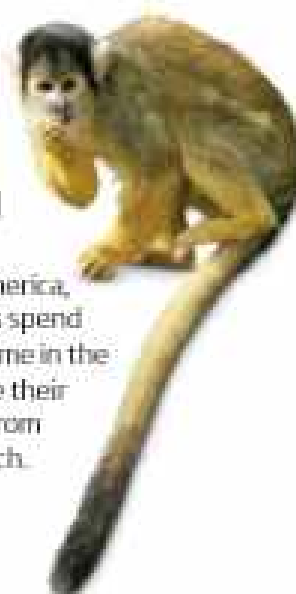


Rhesus macaque

Old World monkeys

From Africa and Asia, these monkeys live in many different places, such as swamps and mountain forests.

Squirrel monkey



New World monkeys

From South America, these monkeys spend most of their time in the trees. They use their tails to swing from branch to branch.

Lemurs

Lemurs are only found on the African island of Madagascar. Most of them live in trees, and are good climbers.



Ring-tailed lemur

Tool use

Chimpanzees are some of the smartest animals in the world. They use tools to open hard nuts or find insects to eat. Young chimpanzees learn how to use tools from the older members of their group.



Moon

The moon is a round, rocky, airless “body” that circles the Earth. It is the most familiar object in the sky after the sun. People have visited the moon, but not since 1972.

SEE ALSO

- ▶ Asteroids p.30
- ▶ Atmosphere p.33
- ▶ Comets p.68
- ▶ Earth p.83
- ▶ Solar system p.233
- ▶ Tides p.254



The dark areas are where there used to be seas of liquid rock.

Rocky body

The moon is large and rocky with a dusty, airless surface. It is about one-quarter the width of the Earth.

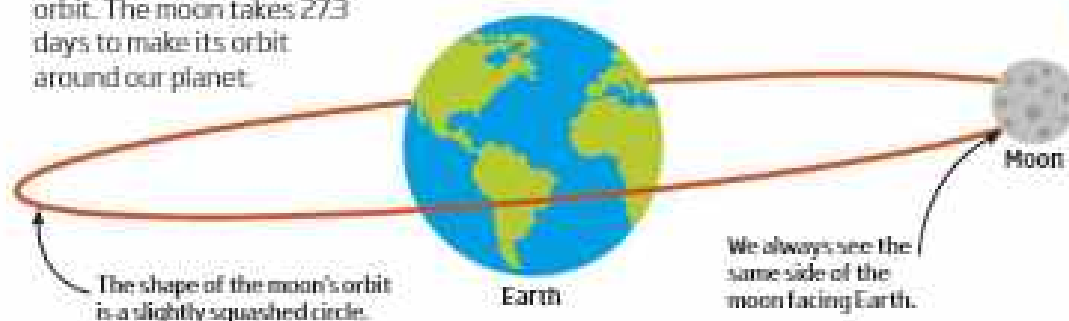


Man on the moon

The moon is the only object in the solar system that has been visited by humans. American Apollo space missions landed 12 people on the moon between 1969 and 1972.

Moon's orbit

The moon travels around the Earth. This is called an orbit. The moon takes 27.3 days to make its orbit around our planet.

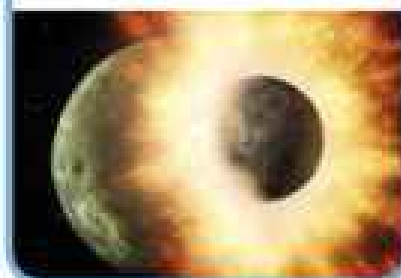


The moon's surface is covered with pits left by space rocks that crashed into it.

We always see the same side of the moon facing Earth.

Moon creation

Scientists think the moon was created when a small planet, Theia, crashed into the Earth 4.5 billion years ago. As a result, rocks on the moon are similar to rocks on Earth.



Mountains

Mountains are tall, rocky features on the Earth's surface. They usually have very steep sides and tower over the surrounding landscape. The tops of mountains are called summits, and they are often covered in snow, even in summer.

SEE ALSO

- ▶ Earth's surface p.84
- ▶ Evolution p.95
- ▶ Glaciers p.122
- ▶ Rock cycle p.213
- ▶ Rocks and minerals p.214
- ▶ Volcanoes p.268

Mountains around the world

Every continent has mountains. Most mountains are in rows, or ranges, that can be thousands of miles long.

The Andes mountain range runs the length of South America.

The summit

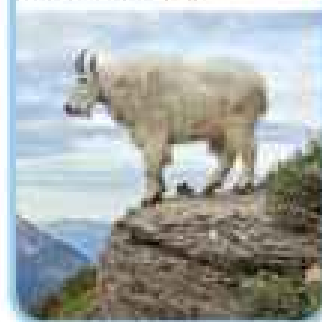
How are mountains made?

Most mountains are formed over millions of years, as huge pieces of the Earth's crust push into each other. Where they meet, the ground is forced up, making mountain ranges.

The pointed shape of the **Matterhorn** was made by slow-moving rivers of ice thousands of years ago.

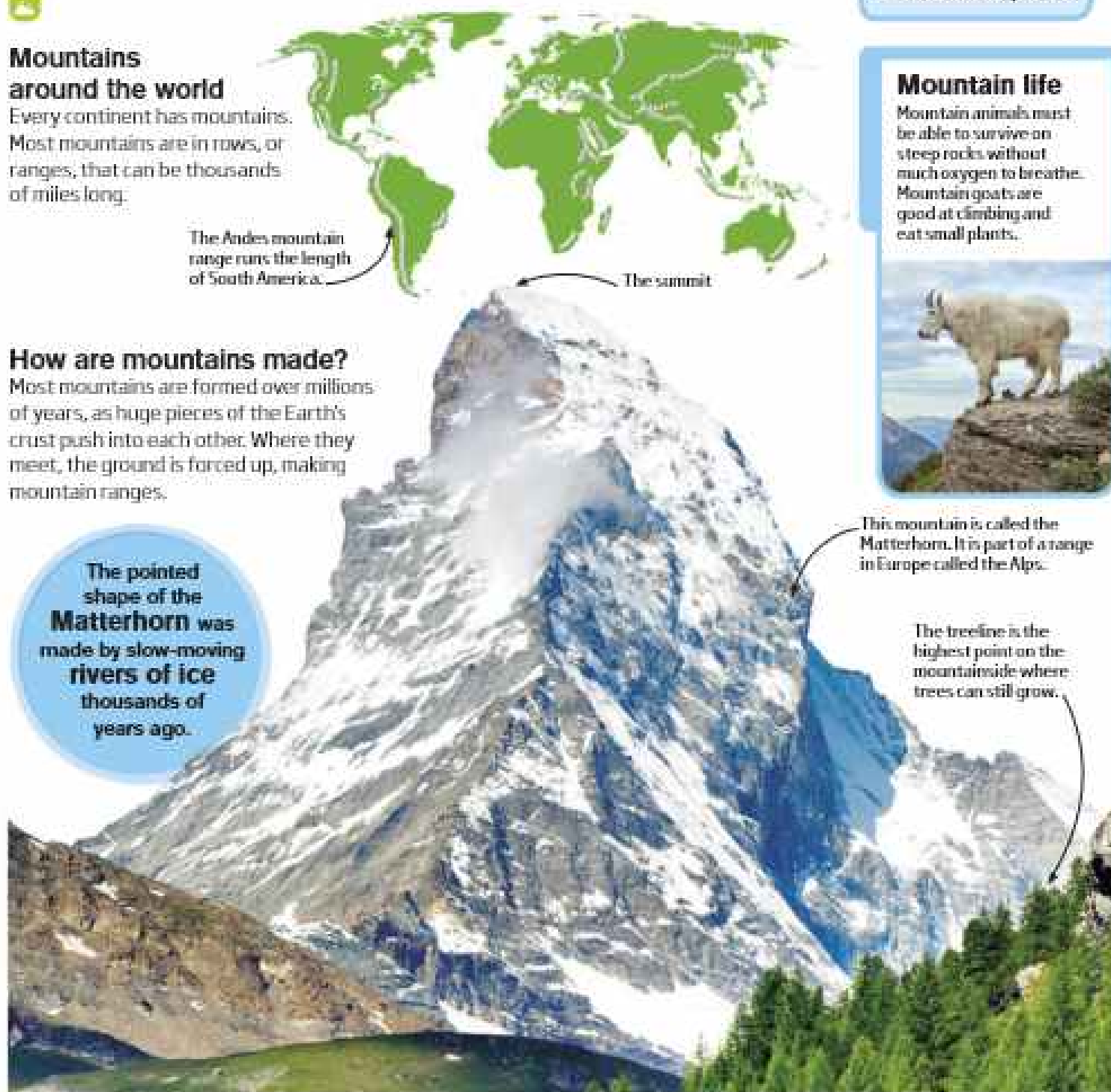
Mountain life

Mountain animals must be able to survive on steep rocks without much oxygen to breathe. Mountain goats are good at climbing and eat small plants.



This mountain is called the Matterhorn. It is part of a range in Europe called the Alps.

The treeline is the highest point on the mountainside where trees can still grow.



Muscles

Muscles are stretchy cords that pull parts of the body to make them move. They work in teams. Some muscles work without us thinking, others move when we choose to move them. Every time we blink, smile, or move, it is with the help of muscles.

SEE ALSO

- ▶ Body cells p.41
- ▶ Feelings p.99
- ▶ Human body p.130
- ▶ Skeleton p.228
- ▶ Sports p.239
- ▶ Games pp.240-241

Muscular system

Most of our muscles are wrapped around the bones of our skeletons, forming the muscular system. They move our bodies by pulling on the bones.

Muscles are attached to bones by tendons.

The biggest muscle in our body is in our bottom. It is called the gluteus maximus.

Bicep muscle

Tricep muscle

Working in pairs

Muscles only pull—they can't push. To lift your arm up, biceps pull and triceps relax. When triceps pull and biceps relax, the arm moves back down again.

Stomach muscles are called abs.

Upper thigh muscles are called quads.



Face muscles

The muscles in our face move the eyes and mouth and help us to express our feelings to others. For example, we show we are happy by smiling.

You use

300

different muscles just to stand up.

Exercise

The more we move our muscles, the stronger they get. After exercise, the body repairs any damage to muscle cells by making new muscle fibers. This is why muscles become bigger and stronger the more you use them.

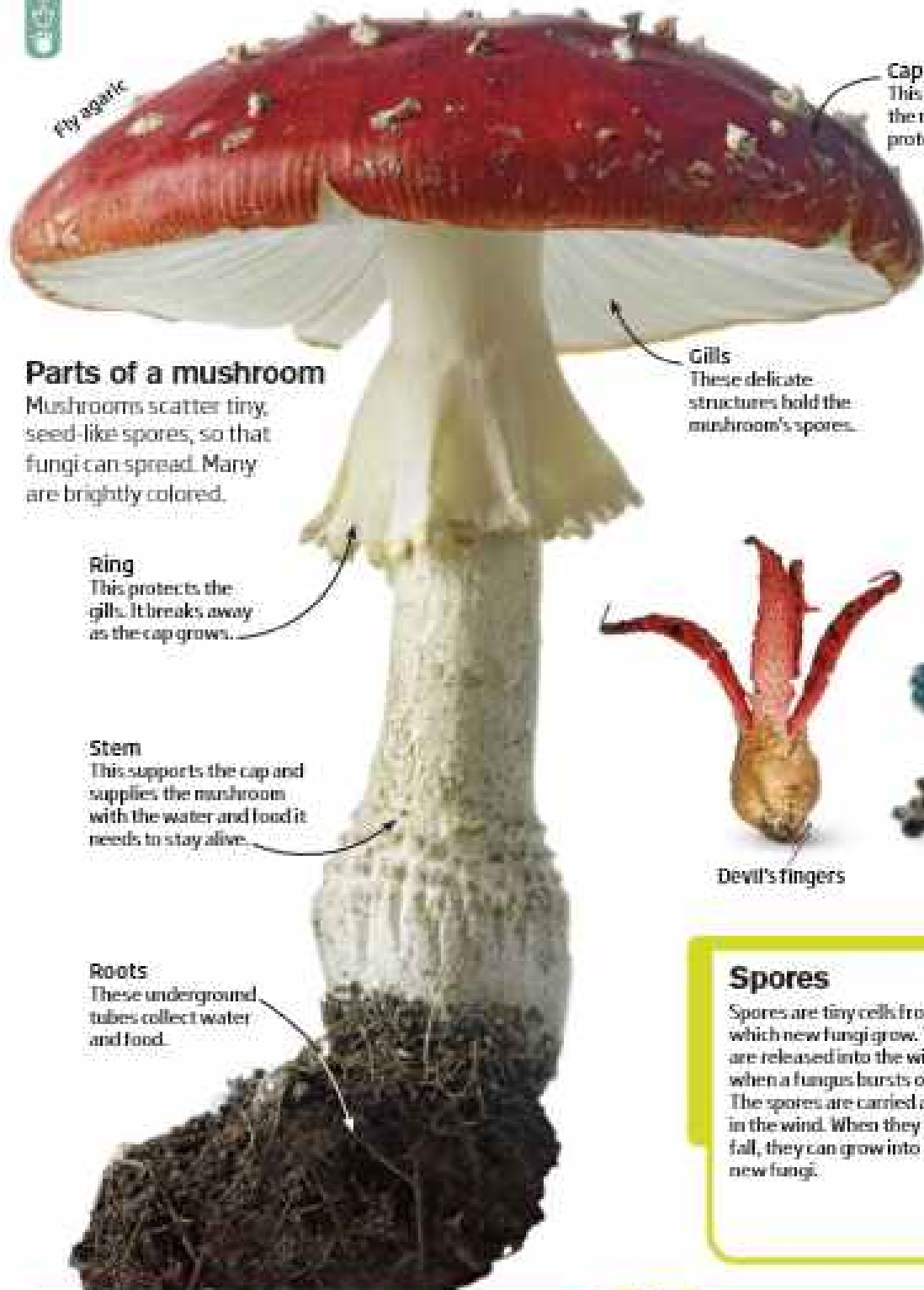


Mushrooms

A mushroom is the fruit of a fungus. Fungi are neither animals nor plants. They feed on living and dead animals and plants. Many fungi are very poisonous—you should not touch or pick them.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Color pp.26–27
- ▶ Fruit and seeds p.115
- ▶ Life cycle p.146
- ▶ Plants p.194



Fly agaric

Cap
This is the head of the mushroom. It protects the gills.

Gills
These delicate structures hold the mushroom's spores.

Parts of a mushroom

Mushrooms scatter tiny, seed-like spores, so that fungi can spread. Many are brightly colored.

Ring
This protects the gills. It breaks away as the cap grows.

Stem
This supports the cap and supplies the mushroom with the water and food it needs to stay alive.

Roots
These underground tubes collect water and food.

All parts of the fly agaric mushroom are **poisonous**. It was used to kill flies in medieval times.

Fungi

There are many different types of fungus. Most of them grow in damp places, such as grassy fields and shady woodlands.



Devil's fingers



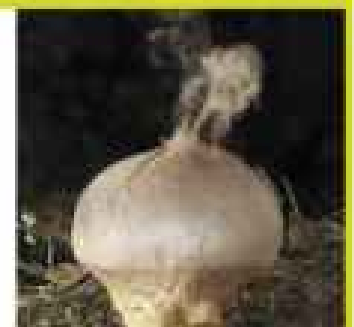
Green elf cup



Yellow jelly antler

Spores

Spores are tiny cells from which new fungi grow. They are released into the wind when a fungus bursts open. The spores are carried away in the wind. When they fall, they can grow into new fungi.



Puffball

Musical instruments

An object used to make musical sounds is called a musical instrument. Musical instruments make sounds in different ways—some have strings that vibrate, others a hole to blow into or a surface to beat. We put musical instruments into four groups based on how they make sound.

SEE ALSO

- ▶ Dance p.76
- ▶ Hearing p.127
- ▶ Music pp.176-177
- ▶ Orchestra p.188
- ▶ Radio p.203
- ▶ Sound p.235

Strings

The sound of stringed instruments comes from their vibrating strings. Players pluck the strings with their fingers or move a bow across them.



Wind

Wind instruments such as the trumpet or flute are made of tubes of wood or metal, which might be straight or looped. Musicians play wind instruments by blowing into them.



Keyboard

Musicians play instruments such as pianos and synthesizers by pressing keys on a keyboard. Piano keys cause a tiny hammer to hit a string, which produces a particular sound called a note.

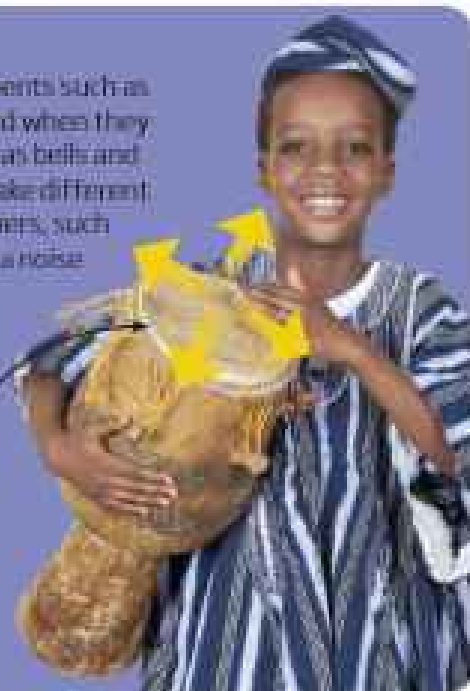
A modern grand piano has 88 keys.



Percussion

Percussion instruments such as drums make a sound when they are hit. Some, such as bells and xylophones, can make different notes (sounds). Others, such as rattles, produce a noise when shaken.

Traditional drum heads are made from animal skin.



The **story** of...

Music

Since the earliest times, people have felt the need to make music. We can express our feelings by singing or playing instruments. Musicians organize sound into tunes and regular patterns known as rhythms. A steady rhythm can inspire people to dance.



Musicians
rehearsing
in Brazil

Playing together

Many people enjoy getting together to make music. Players and singers perform in concerts or just for fun.



The American
Symphony
Orchestra
performs in
New York

Classical music

Most music performed in concert halls is known as classical music. It is played by orchestras or groups of musicians called ensembles, and sung by choirs. Classical music began hundreds of years ago, but is still written, played, and enjoyed today.

Pop and rock

Most of the music we hear on the radio is rock or pop music. Before pop music, most music was classical or traditional. Pop music introduced electronic instruments, a strong beat, and words that are easy to sing along to. It quickly became popular all over the world.

This part changes the vibrations of the strings into an electric signal.

Singing

Singing is an important part of music-making all over the world. Singing helps us to express our feelings. A singer can sing alone, or with others as part of a choir.



Kalengo drum
from Nigeria



Around the world

There are different types of music and musical instruments all over the world. Singing styles are varied too. African music is often very rhythmic and exciting, while Asian music emphasizes the tune.



Pan flute
from South America

The longer the pipe, the lower the sound it makes.

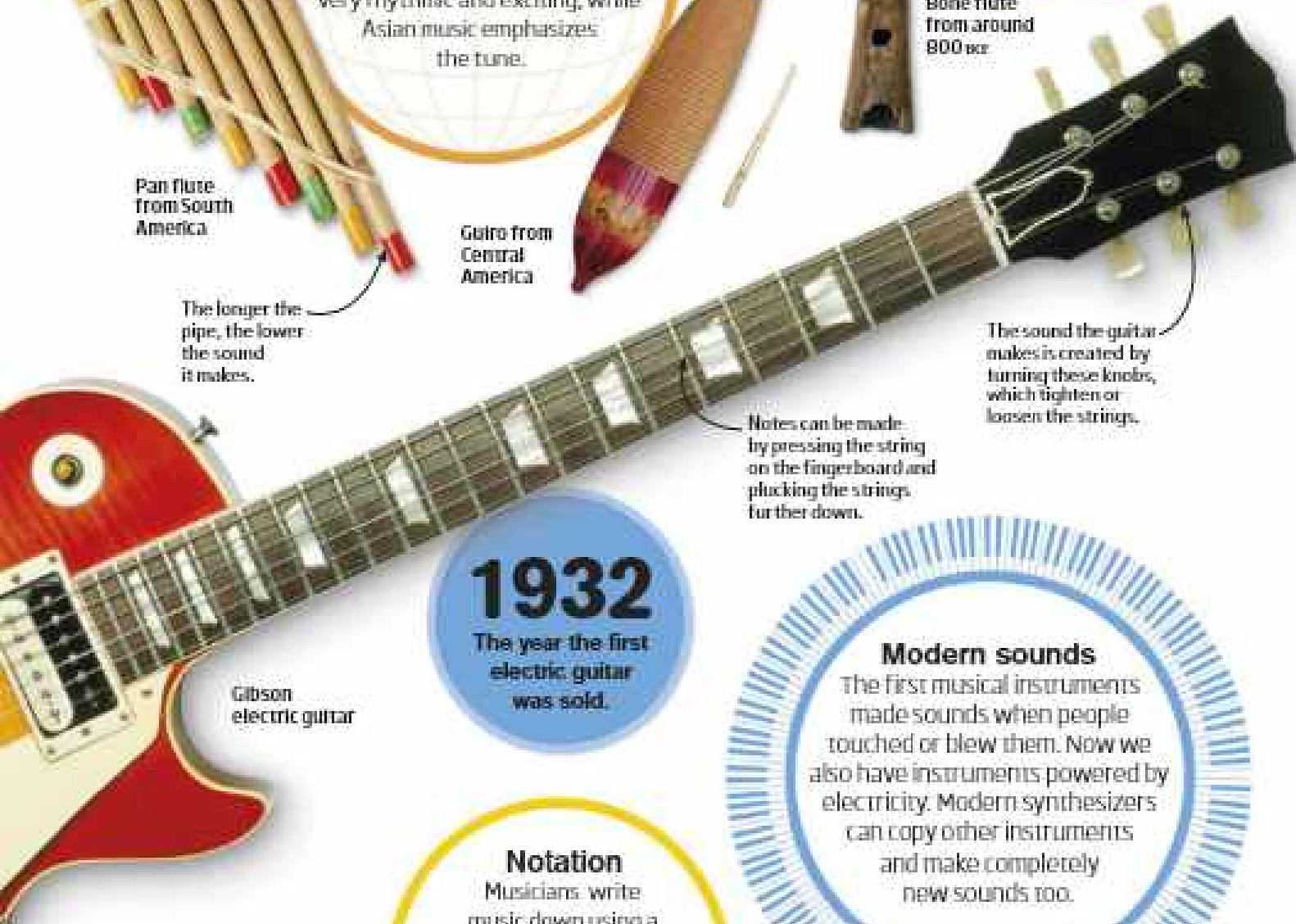
Güiro from
Central America



First instruments

The first instruments were probably rattles and drums made of wood or bone. Instruments that make a sound when you blow them appeared more than 40,000 years ago.

Bone flute
from around 800 BC



Gibson
electric guitar

Notes can be made by pressing the string on the fingerboard and plucking the strings further down.

The sound the guitar makes is created by turning these knobs, which tighten or loosen the strings.

1932

The year the first electric guitar was sold.

Notation

Musicians write music down using a system of symbols called musical notation. The dots on and between the lines tell the performer which notes to play or sing.



This is the music for "Twinkle Twinkle Little Star."

Modern sounds

The first musical instruments made sounds when people touched or blew them. Now we also have instruments powered by electricity. Modern synthesizers can copy other instruments and make completely new sounds too.



Synthesizer

Myths and legends

Myths and legends are stories. In the past, people invented myths to answer big questions, such as where our world came from. Unlike myths, legends are often based on real events, but the details have changed a lot over time, so there is not much truth left!

SEE ALSO

- ▶ Ancient Egypt p.17
- ▶ Ancient Greece p.18
- ▶ Storytelling pp.42–43
- ▶ Books p.44
- ▶ Writing p.280

Mythical creatures

Myths often include strange creatures, which sometimes have a mix of features from different animals. Mythical creatures can be terrifying monsters, or friendly beasts like the Chinese dragon.

Minotaur

The minotaur is a scary monster with a human body and the head of a bull. He appears in an ancient Greek myth.



Chinese dragon

The Chinese dragon has four legs and a long, snakelike body. In China, dragons are a symbol of good luck.



Griffin

The griffin is part lion and part eagle. In Greek myths, griffins stand guard over treasures.



Legendary heroes

Many myths and legends tell the stories of brave people called heroes. Hua Mulan is the hero of a Chinese legend. She pretends to be a man and takes her elderly father's place as a soldier.

This is a modern statue of Hua Mulan. Her story has been told in many books and films.

"Myth" comes from the Greek word "mythos," which simply means "story."



Creation myth

Many myths are about how the world was created. An Egyptian myth says that the first people in the world were shaped out of clay on a potter's wheel, by a ram-headed god called Khnum.

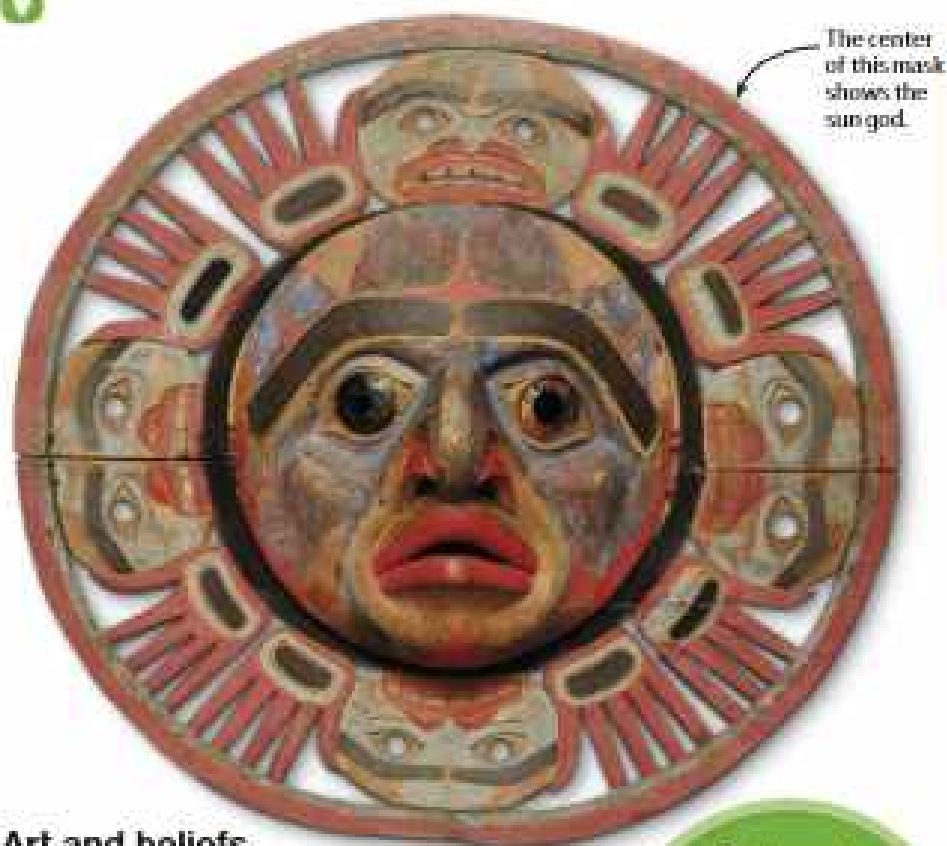


Native Americans

People first moved from Asia into the Americas more than 25,000 years ago. When Europeans first arrived in the Americas in the late 1400s, there are thought to have been 50 million people already living there in tribes. These people are known as Native Americans.

SEE ALSO

- ▶ Arctic p.25
- ▶ Art p.28
- ▶ Dance p.76
- ▶ North America p.184
- ▶ Religion p.208
- ▶ Homes pp.244-245



The center of this mask shows the sun god.

Cultural areas

There were once hundreds of Native American tribes, each with its own traditions. This map shows the ten Native American cultural areas. The tribes of each area of ten shared similar customs or ways of life.



KEY

- | | |
|-----------|-------------|
| Plateau | Southeast |
| Northwest | Southwest |
| Arctic | Plains |
| Subarctic | Great Basin |
| Northeast | California |

Art and beliefs

Native Americans had many gods and beliefs. Some rituals involved dance, and people often expressed their beliefs through art. For example, the Bella Coola tribe used this carved mask for dance ceremonies.

By the early 16th century, the Native American population had fallen to only 400,000 due to disease brought by the Europeans.



Finding food

Some tribes grew crops such as potatoes, corn, or tomatoes. Other tribes relied on hunting wild animals, such as buffalo, or gathering plants.

Homes

Native Americans lived in different kinds of homes. Northeast farmers had longhouses, which were homes built for several families. Plains hunters used tents called tipis.



The **story** of...

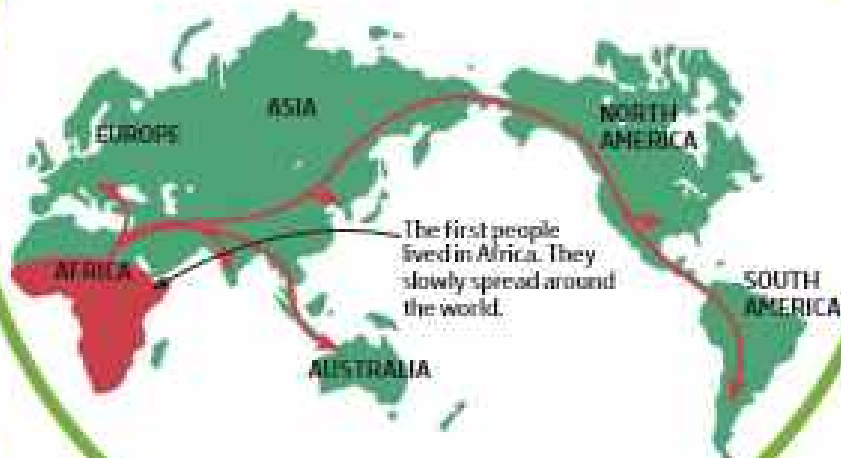
Exploration

Humans have explored the land, sea, and sky, and we are starting to explore space. Since our first travels on land, we have made new technology to let us sail and fly. Distant countries can work together because of world exploration. We might even live in space one day!

Leaving Africa

Humans first lived in Africa.

They started to leave this continent in large groups between 80,000 and 70,000 years ago. They went by foot to the nearby continent of Asia, and later traveled by boat to Australia.



Trade

In the past, people traveled many miles by land and sea to buy things from other countries. Merchants found new routes between distant lands, and bought or traded items such as spices. These goods were then brought home to sell.



Ginger



Cinnamon



Cloves

Vasco da Gama found the **first sea route** from Europe to India in the 15th century.

Reaching the poles

The freezing North and South poles were unexplored until the early 1900s. The first explorers traveled in sleds pulled by dogs and wore fur for warmth.



American explorer Robert Peary, in 1909

This ship carried Christopher Columbus to an island off the coast of America in 1492.

The Santa Maria

Some deep-sea fish live for more than

200

years.

Age of exploration

In the 1400s, Europeans traveled by ship to distant places they had never been to before, including America. These long trips were called expeditions.

Wind blew into the sails to power the ship.

Europeans first sailed all the way to Australia in the 17th century.



A deep-sea hatchet fish

Under the sea

Oceans are many miles deep in places. These black depths are largely unexplored. The few crafts that have reached the bottom of ocean trenches have found mysterious new creatures.

Around the world

As technology advances, new opportunities for exploration open up. The first flight around the world was made in 1924 by a plane that ran on fuel. In 2016, *Solar Impulse 2* made the first around-the-world flight using sunlight converted into power.

Solar cells on the wings of *Solar Impulse 2* use sunlight to power the plane.

In 1961, Yuri Gagarin became the first man in space.

Space exploration

A Russian spacecraft called *Sputnik 1* first flew around the Earth in 1957. Humans soon made it into space, too, and landed on the moon in 1969. Since then, we've used robots to visit every planet in the solar system, as well as comets.



Navigation

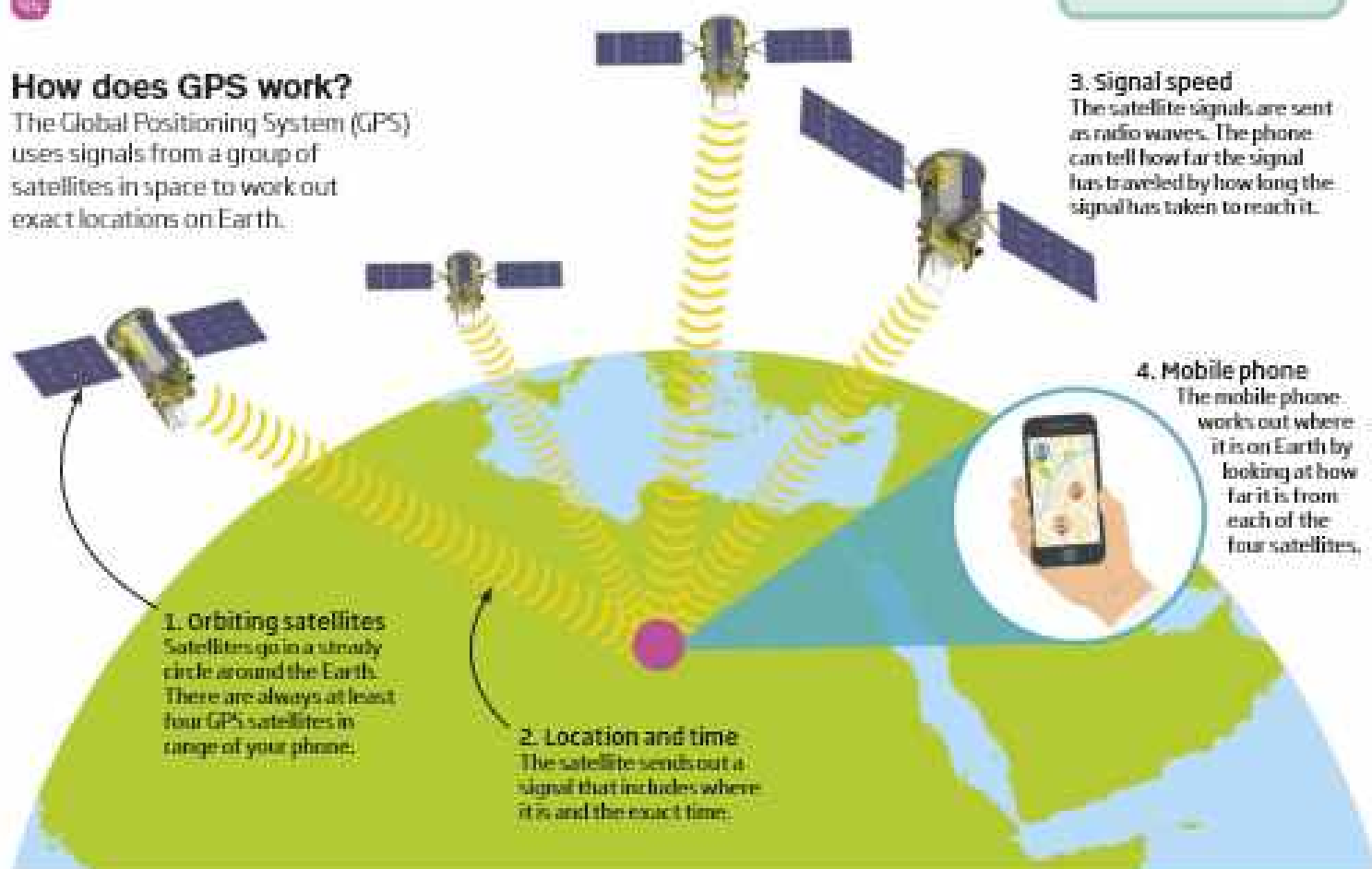
Navigation is finding where you are and where you are going. We can use the sun, stars, a compass, and paper maps to navigate. Today, most modern transportation receives signals from satellites in space to show where it is on Earth. This is called GPS.

SEE ALSO

- ▶ Compass p.70
- ▶ Constellations p.73
- ▶ Light p.147
- ▶ Maps p.155
- ▶ Radio p.203
- ▶ Satellites p.215

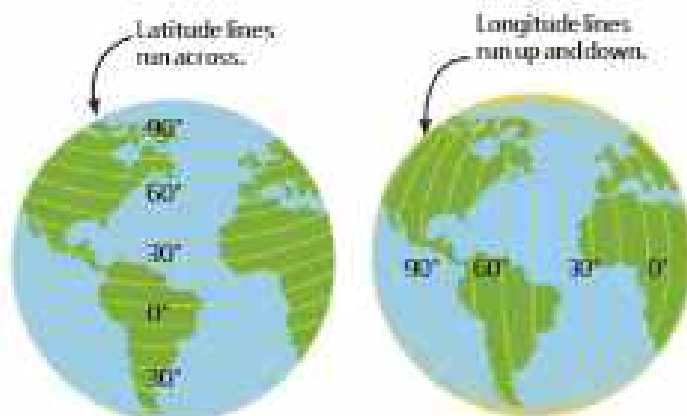
How does GPS work?

The Global Positioning System (GPS) uses signals from a group of satellites in space to work out exact locations on Earth.



Latitude and longitude

Maps of Earth have grid lines of latitude and longitude to show where places are. Every place has a unique latitude and longitude number in degrees.



Map and compass

Without GPS, we can still navigate using a paper map and a compass. The compass shows the direction North, which helps us work out which direction we need to go.



Neptune

Neptune lies in the freezing cold, dark outer parts of the solar system. It is the eighth planet and the farthest planet from the sun. Neptune is often called the "twin planet" of Uranus because it is made up of similar ices and gases. It is four times larger than Earth.

SEE ALSO

- ▶ Atmosphere p.33
- ▶ Gases p.117
- ▶ Liquids p.148
- ▶ Pluto p.196
- ▶ Solar system p.233
- ▶ Uranus p.264

Wispy clouds made of frozen methane gas are found high in Neptune's atmosphere.

The windy planet

Neptune is known as "the windy planet" because it has powerful winds in the outer layers of its atmosphere. Winds can reach speeds of more than 1,500 mph (2,400 kph), about twice the speed of sound.

Neptune gets its blue color from methane gas in the upper atmosphere.

Scientists think that Neptune's largest moon **Triton** is an icy minor planet that got caught by Neptune's gravity.

Hydrogen and helium gases make up most of Neptune's atmosphere.

The sun

God of the sea

Neptune was named after the Roman god of the sea, who is usually pictured carrying a three-pronged spear called a trident. The planets Mercury, Venus, Mars, Jupiter, and Saturn are also named after Roman gods.



Beyond Neptune

There are thought to be several thousand icy "minor planets" that orbit the sun beyond Neptune. The first, and largest, minor planet that was discovered is Pluto.



North America

North America stretches from the icy Arctic in the north to tropical Central America in the south. Huge areas of grassland, called prairies, cover much of North America. There are also mountains, forests, deserts, and some of the world's biggest lakes.

SEE ALSO

- ▶ American West p.14
- ▶ Arctic p.25
- ▶ Aztecs p.35
- ▶ Native Americans p.179
- ▶ South America p.236

About North America



Population:
579 million



Highest point:
Denali



Lowest point:
Badwater Basin



Biggest desert:
Great Basin Desert



Longest river:
Missouri

Statue of Liberty

This statue towers 305 ft (93 m) over New York Harbor. Completed in 1886, it was a gift from the people of France to the people of the US.



The United States is home to the remains of more types of dinosaur than any other country.

Mississippi River

The Mississippi is a huge river in North America. Ships use it for transporting goods, and tourists ride on its riverboats called paddle steamers.





Numbers

Numbers are symbols that can be used to show amounts, sizes, distances, and times. The ordinary numbers we count with are called “whole numbers” or “natural numbers.” All math requires us to use numbers.

SEE ALSO

- ▶ Clocks p.61
- ▶ Coding p.65
- ▶ Codes pp.66–67
- ▶ Fractions p.112
- ▶ Measuring p.159
- ▶ Temperature p.252

Whole numbers

The numbers 0 to 9 are used to build all the bigger numbers. They are called **whole numbers** because they are not split into smaller amounts.

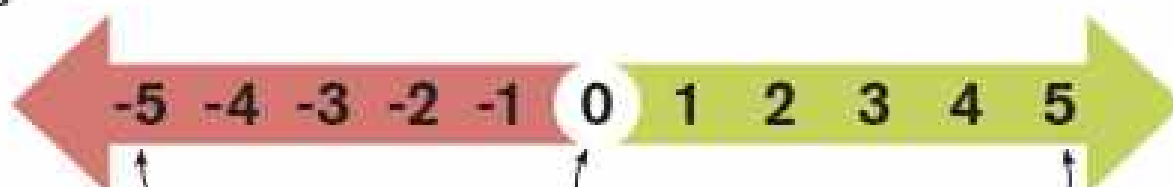


Zero represents nothing.

Whole numbers are also called “positive” numbers.

Negative numbers

Numbers less than zero are called “negative” numbers. They are used to show things less than zero, such as cold temperatures.



Negative numbers are shown with a minus sign.

Zero is neither positive nor negative.

Positive numbers have no symbol in front.

Algebra

Algebra is a type of math that uses letters to stand for numbers or amounts. We can use algebra to work out the value of unknown amounts.

$$2 + a = 5$$

The “a” represents a mystery number. We can work out what it is by taking 2 away from 5.

An equals sign means both sides of the equation have the same value.

Place value

The place where a number is written in a longer number shows how much it is worth. The lowest worth, the unit, is on the right.



Thousands
This shows how many thousands the number has.

Hundreds
This shows how many hundreds the number has.

Tens
This shows how many tens the number has.

Units
Units are numbers from 0–9.

Oceania

Oceania is a continent made up of Australia, New Zealand, Papua New Guinea, Fiji, and other islands in the tropical Pacific Ocean. It is home to some of the world's most unusual wildlife, including kangaroos, koalas, duck-billed platypuses, and kiwis.

SEE ALSO

- ▶ Asia p.29
- ▶ Birds p.39
- ▶ Coral reefs p.74
- ▶ Deserts p.78
- ▶ Mammals p.154
- ▶ Sports p.239
- ▶ World p.275

This island, called New Guinea, is split between Oceania and Asia.

Male birds of paradise have brightly colored feathers, which they show off to female birds.



These caterpillars live underground, where they eat roots. They can grow to 4 1/2 in (12 cm) long.

This fierce dog-sized meat-eater lives only in Tasmania.

Uluru

This giant sandstone rock feature towers over the landscape of central Australia. Uluru is a special place for the aboriginal people who have lived in Australia for thousands of years.

The kiwi uses its strong legs for running and fighting.



Kiwis

The kiwi is a flightless bird that lives in New Zealand. It is about the same size as a chicken, but its egg is six times the size of a chicken's egg.

About Oceania

- Population: 40.3 million
- Highest point: Mount Wilhelm
- Lowest point: Lake Eyre
- Biggest desert: Great Victoria Desert
- Longest river: Murray

Oceans and seas

More than two-thirds of our planet is covered by oceans and smaller seas. They contain most of the world's water and are full of life of all shapes and sizes. Some of the deepest parts of the ocean have still not been explored.

SEE ALSO

- ▶ Coral reefs p.74
- ▶ Water pp.120-121
- ▶ Exploration pp.180-181
- ▶ Seashore p.220
- ▶ Tides p.254

Ocean depths

Oceans are divided into different zones according to depth. The deepest part is more than 6 miles (10 km) beneath the surface.

Sunlit zone

This zone receives lots of sunlight and is the layer of the ocean that contains most plants and animals.

Twilight zone

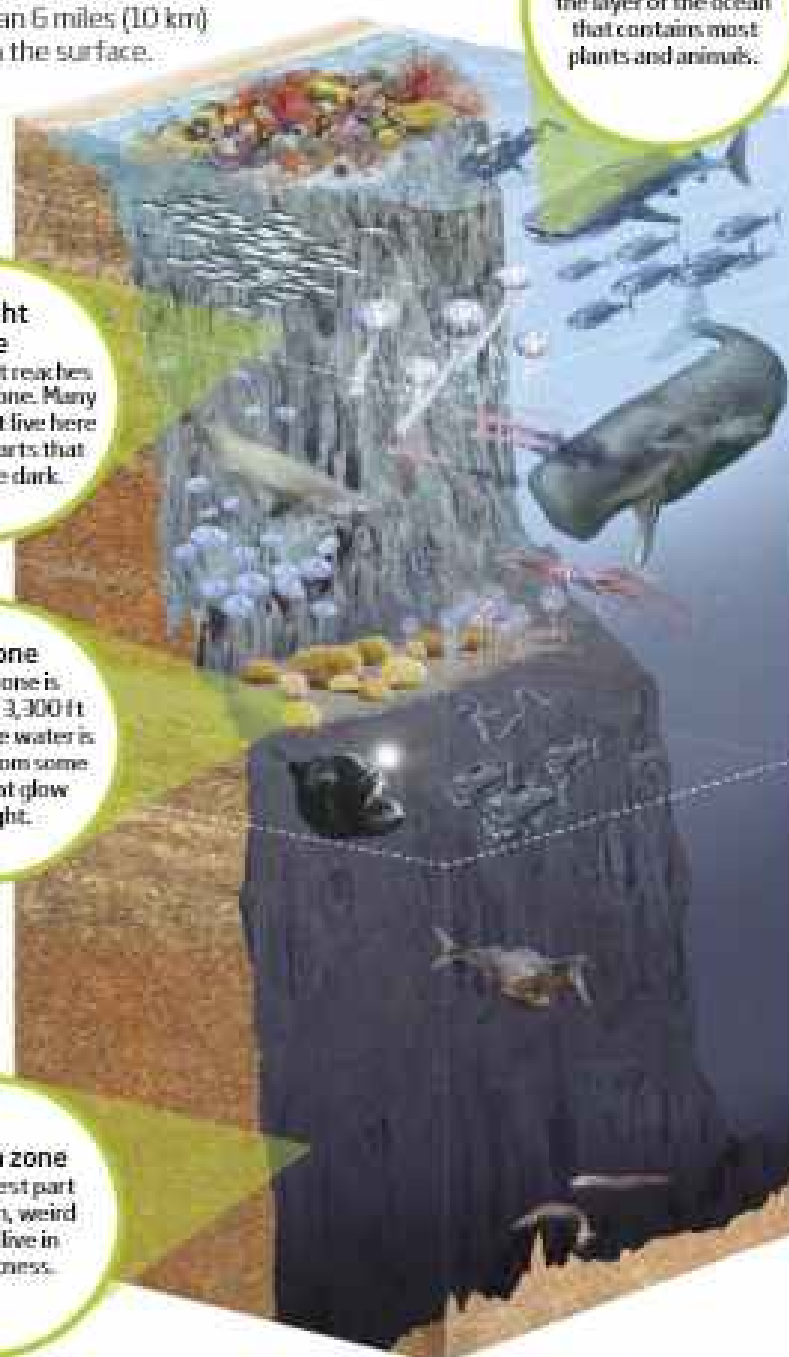
Little sunlight reaches the twilight zone. Many creatures that live here have body parts that glow in the dark.

Dark zone

The dark zone is deeper than 3,300 ft (1,000 m). The water is dark apart from some animals that glow with light.

Deep-sea zone

In the deepest part of the ocean, weird creatures live in total darkness.



The world's oceans

The Earth has five oceans. The largest is the Pacific, which holds half of the world's salt water. The smallest is the Arctic Ocean, which is partly frozen.

Ocean smokers

In places, hot water bursts from the seabed, creating chimneylike structures known as smokers. The water that comes out from the smokers can be white or black, depending on what minerals the water around them contains.



Orchestra

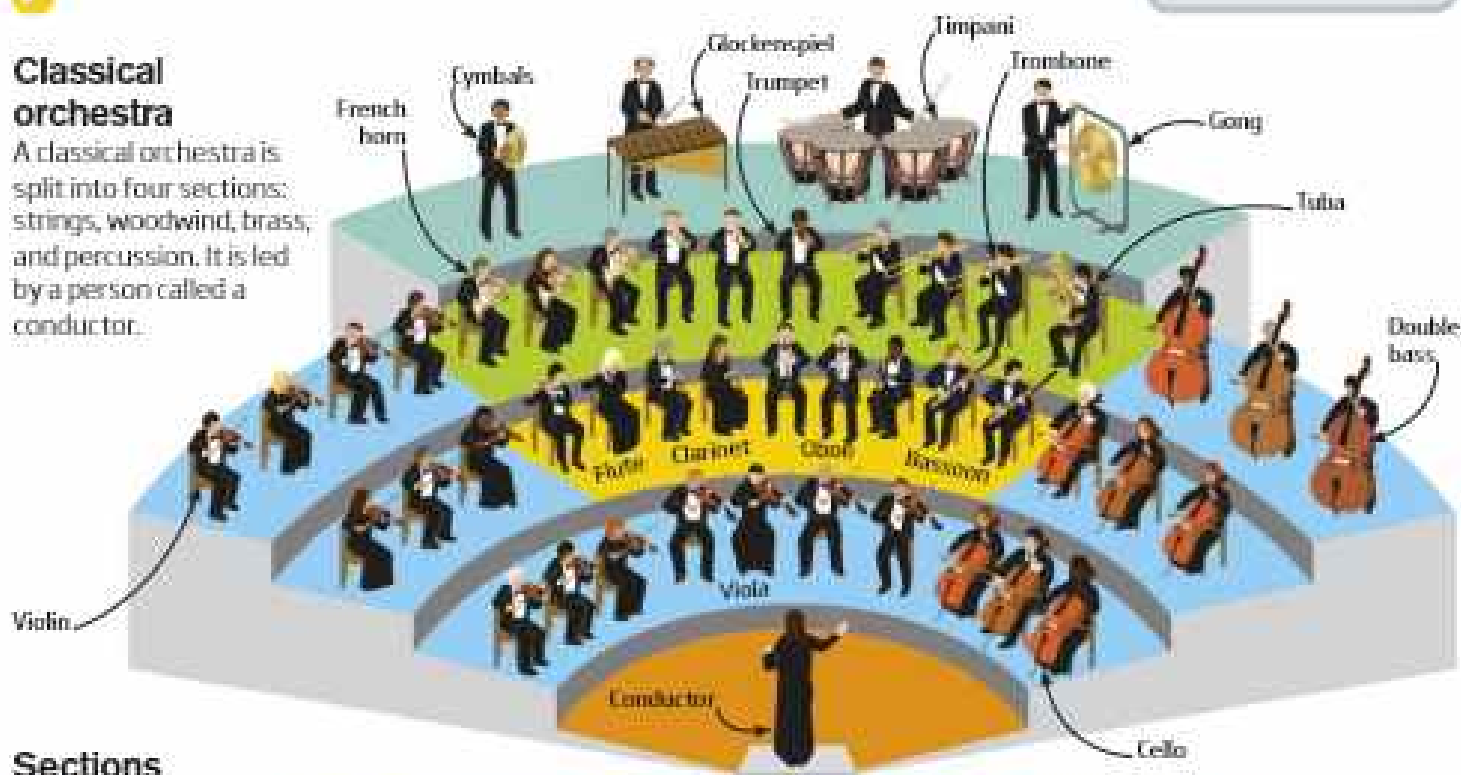
An orchestra is a large group of people playing different musical instruments together. Orchestras were created to play classical music, such as complicated pieces called symphonies. Orchestras often create music for films, and sometimes play non-classical pieces, such as pop music.

SEE ALSO

- Dance p.76
- Film p.100
- Musical instruments p.175
- Music pp.176-177
- Radio p.203
- Sound p.235

Classical orchestra

A classical orchestra is split into four sections: strings, woodwind, brass, and percussion. It is led by a person called a conductor.



Sections



Strings

Every classical orchestra has stringed instruments. There are usually two groups of violins, as well as violas, cellos, and double basses.



Woodwind

Many classical orchestras include instruments of the woodwind family, such as flutes and bassoons. They usually sit behind the strings.



Conductor

Using hand gestures and a stick called a baton, the conductor makes sure the musicians play together at the right time and speed.



Brass

Full orchestras include a brass section, which is usually behind the woodwind. Trumpets, trombones, French horns, and tubas are brass instruments.



Percussion

There are many percussion instruments. Percussion usually seen in a classical orchestra includes the timpani, bass and side drums, and cymbals.

Playing together

There are different types of orchestras around the world. They often use very different instruments from the classical orchestra, and might have fewer people playing.

The gamelan
In Java and Bali, people play percussion instruments, such as xylophones, in an orchestra called a gamelan.



Chinese orchestra
A Chinese orchestra uses traditional Chinese instruments. There are woodwind, percussion, and stringed sections.

Chinese gong



Philosophy

Philosophy is a way of trying to understand things by asking questions and thinking of answers. It was first studied thousands of years ago, when people wanted to find out about the world and their own lives. People who try to find answers to these questions are known as philosophers.

SEE ALSO

- ▶ Ancient China p.16
- ▶ Ancient Greece p.18
- ▶ Governments p.123
- ▶ Religion p.208
- ▶ Science p.217

Asking questions

To find out about the world, philosophers ask all sorts of questions. They ask about things such as what is real around us, and what is the best way to live our lives.

Thinking of answers

Philosophers try to come up with answers to these questions. By thinking hard about their answers, they can decide whether they are true or false.

What makes me who I am?

Why do things exist?

How do I know what is true?

Right and wrong

An important part of philosophy is deciding what makes something right or wrong. For example, we all know that stealing is bad. Philosophers ask why it is bad.

Equality

Men and women are often treated differently. For example, men are generally paid more. Philosophers try to explain how everybody should be treated equally (the same).

First philosophers

Western philosophy began in ancient Greece. The Greek city of Athens was home to many of the most important early philosophers, such as Socrates, Plato, and Aristotle.



Statue of Plato



Photography

A photograph is a still image taken using a machine called a camera. Photographs give us a visual record of who we are and what we have done. These images can capture important people and events in history, or private moments in your own life.

Cameras

The first photographic cameras were invented in France in the 1800s, but were large and hard to use. Cameras are now small enough to fit inside mobile phones.



Daguerreotype cameras were the first cameras to go on sale to the public in 1839.

Early cameras

The first cameras took pictures on metal sheets that had been made sensitive to light. It took many minutes to take a photo.



The film is rolled up inside the camera.

Film camera

Later cameras use strips of light-sensitive plastic film. When the film is exposed to light, a picture forms.

Digital camera

Modern digital cameras produce images made up of millions of tiny points of color that are displayed on screens.



These cameras can connect to computers to transfer images.



Camera phone

Many people now take photographs using tiny digital cameras built into their mobile phones. These photos can also be transferred to and viewed on computers.

The first photograph was taken in 1816 by French inventor Nicéphore Niépce.

SEE ALSO

- ▶ Art p.28
- ▶ Computers p.71
- ▶ Film p.100
- ▶ Inventions pp.136-137
- ▶ Telephones p.250
- ▶ Television p.251

Taking pictures

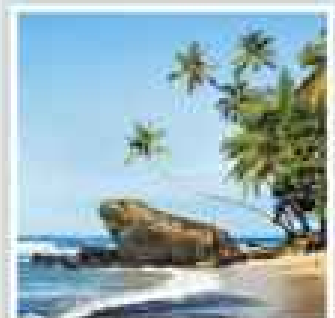
Photographs fall under different categories called genres. Pictures you take of yourself are called selfies, and other genres include animals and travel.



Selfie



Pet portrait



Vacation snap



Photosynthesis

Plants make their own food. They do this by taking in the sun's light energy, as well as water from the ground and gas from the air. Producing energy in this way is called photosynthesis.

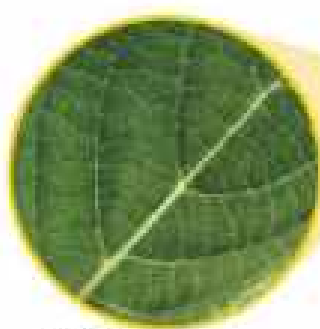
SEE ALSO

- Carbon cycle p.49
- Cells p.56
- Gases p.117
- Light p.147
- Plants p.194
- Temperature p.252

How do plants make food?

The plant combines carbon dioxide with water to make sugar. The energy it needs to do this is supplied by the sun's light.

The **oxygen** that plants give out is the same gas that **humans and animals** need to breathe in.



Leaves

A chemical in the leaves called **chlorophyll** absorbs light energy from the sun. Chlorophyll makes plants look green.

Roots

The plant roots take in water and minerals from the soil.

The stem is strong to support the plant and move it toward light.

Water

The plant needs water to survive. Water travels up the stem into the plant.



Sunlight

Plants need energy from sunlight to complete photosynthesis. They grow toward the light.

Carbon dioxide

Holes in the plant's leaves take in a gas called carbon dioxide.

Oxygen

Oxygen is released as waste when carbon dioxide, water, and light react together in the plant.

Conditions

Temperature, light, and water need to be just right for a plant to survive. If these conditions change too much and don't suit the plant it will start to die.



Without light

Without water



Physics

Everything in the universe that weighs something is called "matter." Physics is a science that looks at how matter moves and interacts. This includes energy, forces, magnets, light, heat, waves, and sound.

SEE ALSO

- ▶ Circuits p.59
- ▶ Electricity p.87
- ▶ Energy pp.88–89
- ▶ Forces p.108
- ▶ Magnets p.151
- ▶ The sciences pp.218–219

Understanding weather

Physics lets us predict weather, by studying how patterns of heat and cold create wind.

Weather balloons collect weather information

Nuclear physics

Atoms are the tiny particles that everything is made from. Splitting them releases energy we can use.

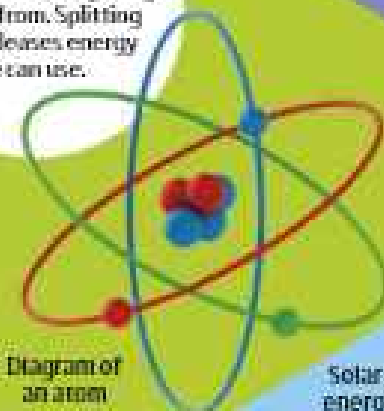
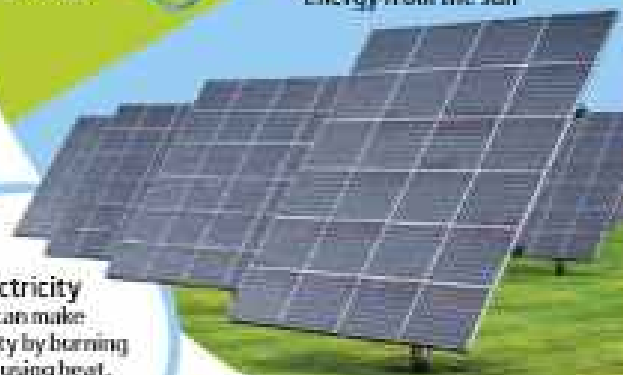


Diagram of an atom

Solar panels collect energy from the sun



Electricity

We can make electricity by burning fuel or using heat, sunlight, wind, or water.

Physics around us

Discoveries from physics are used in everyday life. This is called applied physics.

Medical equipment

Scanners, heart monitors, and X-ray machines were all invented by people using physics.



Heart rate monitor

Computers

Tiny wires connect parts inside machines and computers so that they can carry out tasks.

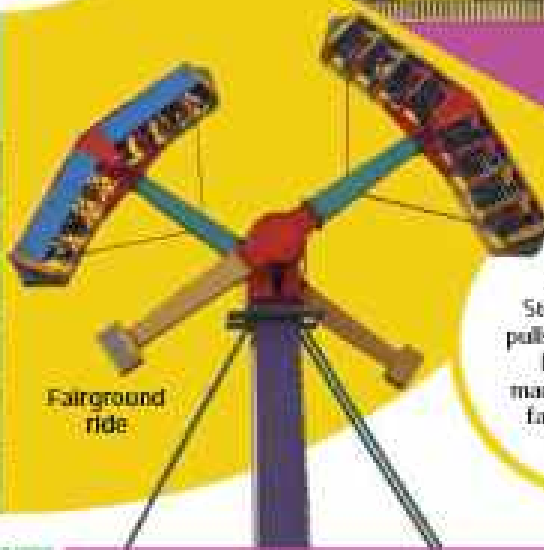
Circuit board from inside a computer



Mechanics

Studying pushes, pulls, and movement lets us design machines, including fairground rides.

Fairground ride



Pirates

Pirates were criminals who attacked ships to steal goods, using force and violence to get their way. They are often now remembered as jolly villains who buried treasure and relaxed on tropical islands, but the reality was not so pleasant.

SEE ALSO

- ▶ Clothing pp.62-63
- ▶ Explorers p.96
- ▶ Flags p.102
- ▶ Maps p.155
- ▶ Oceans and seas p.187
- ▶ Ships p.224



Pirate flag

Pirates put symbols of death on their flags to frighten people. These symbols included skulls, bones, and skeletons. This flag was flown by famous pirate Jeremiah Cocklyn.

Large hats kept pirates sheltered from the sun and rain.

Clothes were made of wool, linen, and canvas.

Pirate life

Pirates could be at sea for weeks at a time. To keep from getting irritable, which could lead to fights among the crew, they passed the time with music, games, food, and drink.

The years between 1690 and 1725 are known as the golden age of piracy.

Swords called cutlasses had short, curved blades.

Pirate ship

Pirate ships had to be fast so they could catch other ships or escape from trouble. They were armed with cannons for fighting. This type of ship is called a sloop.

Large sails helped the ship move quickly.

Leather shoes closed with small brass buckles.

Blackbeard

The most famous pirate of all was nicknamed Blackbeard. He attacked ships on the American coast until he was finally killed in a battle with the British Navy in 1718.



Plants

Plants are living things that make their own food using energy from the sun. Most plants stay in one place, with roots that fix them in the ground.

SEE ALSO

- ▶ Flowers p.103
- ▶ Food p.106
- ▶ Fruit and seeds p.115
- ▶ Insects p.134
- ▶ Photosynthesis p.191
- ▶ Trees p.261

Types of plants

There are four main groups of plants. Some have flowers, like hibiscus. Others don't, such as conifers and mosses.

Conifer

Plants that have seeds in cones are called conifers. They are mainly trees.



Moss

These leafy plants grow where it is damp and dark.



Fern

Ferns don't have flowers. Their leaves start off very small and uncurl as they grow.



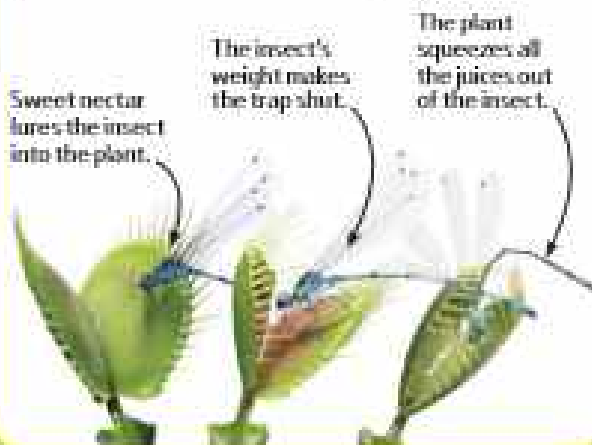
Flowering plant

Most plants have flowers, which make seeds.



Meat-eating plants

Some plants get extra energy by catching and eating animals, such as insects. Some even catch frogs! The plant shown here is called a Venus flytrap.



Plastic

Plastic is a useful material that we make, but it can also be found naturally. It can be colored and molded into shapes without breaking. It is waterproof, so it can be used for packaging and to carry liquid. It's also strong enough to make rope.

Plastic everywhere

We use plastic to make many everyday objects, including toys, glue, cars, bags, computers, tents, and clothes like fleeces and raincoats. For most items, the plastic is heated to become liquid, then poured into molds to set.



SEE ALSO

- ▶ Atoms p.34
- ▶ Carbon cycle p.49
- ▶ Electricity p.87
- ▶ Gases p.117
- ▶ Liquids p.148
- ▶ Materials p.157
- ▶ Recycling p.205

Useful plastic

Plastic is a useful material because of how it behaves. For example, it doesn't let electricity through, and is strong and long-lasting.



Insulation

Plastic is an insulator, which means electricity and heat can't flow through it. It keeps electricity inside cables.



Hard to break

Plastic containers are harder to break than glass or pottery ones. They are also easier to make and safer to use.



Long-lasting

Plastic lasts a long time. It can build up in the environment, so we turn it into new plastic by recycling it.

25

plastic bottles can be recycled to make one polyester fleece.

Making plastic

Natural plastic is found in plants, trees, insects, animal horns, and milk. Artificial plastic is made from oil, coal, and natural gas found underground. All plastic contains an element called carbon.



Pluto

Pluto is a dwarf planet that travels around the sun at the edge of our solar system, beyond Neptune. It has one giant moon called Charon, and four small moons.

Pluto's surface is covered with ice.

SEE ALSO

- ▶ Earth's surface p.84
- ▶ Glaciers p.122
- ▶ Moon p.171
- ▶ Neptune p.183
- ▶ Solar system p.233
- ▶ Volcanoes p.268

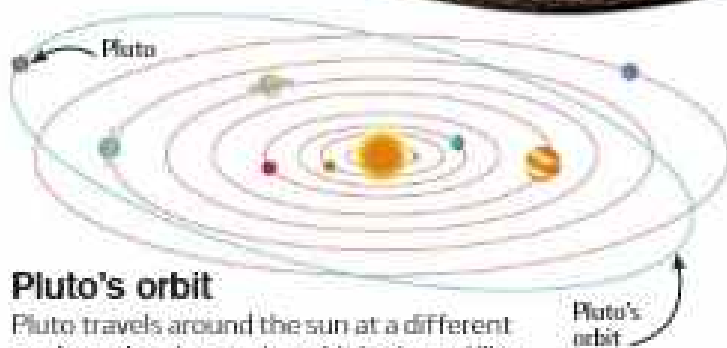
Ex-planet

Pluto used to be considered the ninth planet in our solar system. When other similar small "planets" were discovered, astronomers downgraded Pluto to a dwarf planet.

Ice volcanoes

Pluto may have cryovolcanoes—volcanoes that erupt an icy slush of water and gases.

A feature known as Wright Mons is thought to be a cryovolcano.



Pluto's orbit

Pluto travels around the sun at a different angle to the planets. Its orbit is shaped like an elongated circle. It takes Pluto 248 Earth years to circle the sun.

Dwarf planets

Dwarf planets are similar to planets but smaller, which means they share their path around the sun with other objects, such as asteroids and comets.



Polar habitats

Polar habitats are snow- or ice-covered areas found in the coldest places on Earth—the Arctic in the north and Antarctic in the south. There are no trees and only very few plants. Animals have to survive in freezing temperatures.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Antarctica p.24
- ▶ Arctic p.25
- ▶ Earth p.83
- ▶ Habitats p.126
- ▶ Oceans and seas p.187

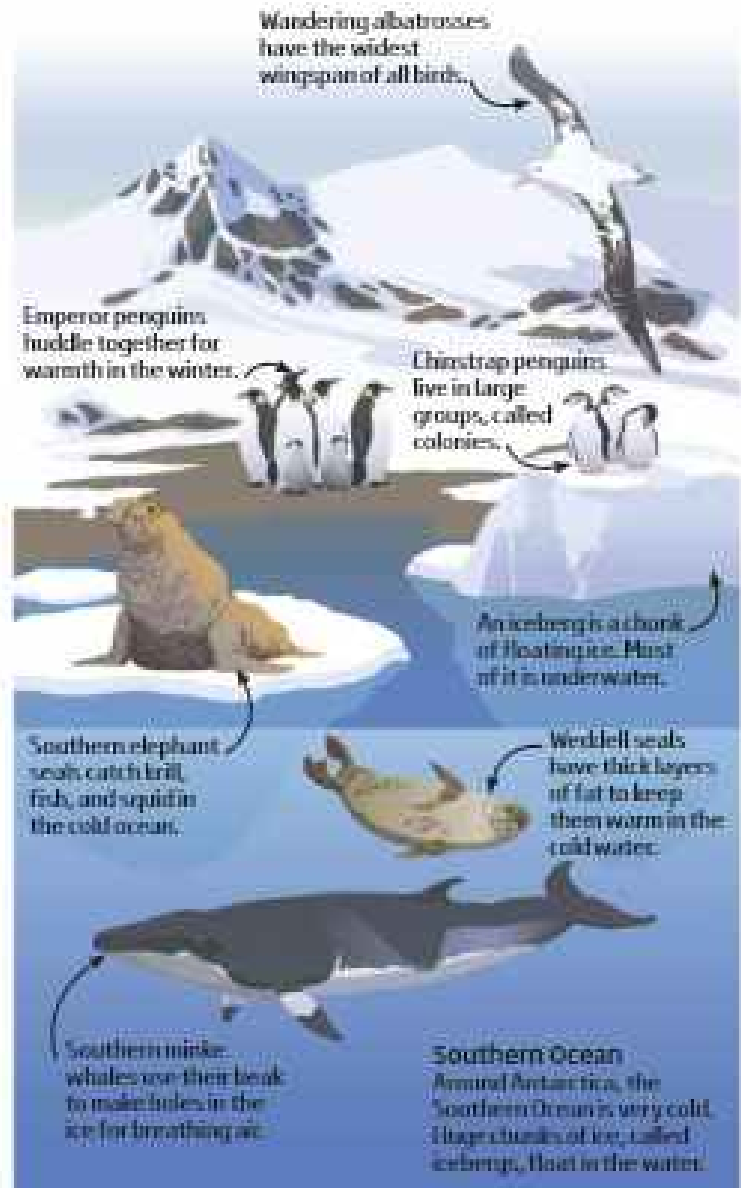
Arctic

Around the North Pole at the top of the world is the frozen Arctic Ocean. The Arctic also includes the northern tips of Canada, Russia, Greenland, and Norway.



Antarctic

Around the South Pole at the bottom of the world is the large landmass, or continent, of Antarctica. The Antarctic is the coldest and windiest place on Earth. It has no large land animals.



Pollution

Pollution happens when dirty or harmful substances are released into the world around us. Pollution kills wildlife, causes health problems for humans, spoils the countryside, and makes the world around us dirty. It is even making our planet warmer.

SEE ALSO

- ▶ Cars p.52
- ▶ Climate change p.60
- ▶ Factories p.97
- ▶ Farming p.98
- ▶ Industrial Revolution p.133
- ▶ Recycling p.205

Air pollution

Cars, trucks, factories, and power stations pump out gases into the air. These can cause illness, poison rivers and oceans, and heat up our world.



Land pollution

Poisons from garbage dumps seep into the ground and then into rivers. Chemicals used in farming kill insects, such as bees, and can make people ill.



Water pollution

Plastic garbage is washed into the sea and swallowed by sea life. Harmful waste from factories and sewage from homes pollute rivers and sea water.



Garbage at sea

When waste plastic is dumped in the sea, it is carried away by currents into gigantic, floating garbage patches. The biggest patch is in the North Pacific Ocean. It is called the Great Pacific garbage patch and is bigger than the US.



Precious metals

Precious metals are rare and worth a lot of money. They are found in the ground as pure metal or combined with other elements in rocks. Silver and gold are the best known and have been treasured for thousands of years. Other precious metals include platinum and beryllium.

SEE ALSO

- ▶ Aircraft p.13
- ▶ Elements p.90
- ▶ Metals p.162
- ▶ Money p.169
- ▶ Gold pp.200-201
- ▶ Rocks and minerals p.214

Gold

Pure gold is a very soft metal. To make it harder—so that we can make useful objects out of it—we combine gold with small amounts of other metals.

A mobile phone contains just a tiny amount of gold—about 0.001 oz (0.025 g).

Gold earrings

This ancient Egyptian burial mask is covered in gold foil.

The highest quality flutes are made of solid silver.

Cutlery

Silver is used in many of today's batteries.

DVD

Mirror

Platinum bar

Pieces of jewelry are often made from platinum.

The most valuable coins have always been made of gold.

Catalytic converters in cars contain platinum. It makes exhaust fumes less poisonous.

Platinum

Just a few hundred tons of platinum are produced each year. Because it is so rare, it is used in very small amounts.

Platinum is used in pacemakers. These devices can keep a person's heart beating.

Watch

Beryllium

Beryllium is a steel-gray precious metal. It is a vital component of computers, cars, aircraft, phones, medical equipment, and many other hi-tech gadgets.

Fighter aircraft

The **story** of...

Gold

Gold is a precious metal that has been used to make jewelry and decorations since ancient times. It is rare and very expensive. But that's just the start of its story. Gold has a glittering history and is still popular all over the world.



Meteor shower

When Earth first formed, gold and other metals sank deep into the core. Other gold, found near the surface of our planet, arrived later from space. Asteroids rained down in a powerful storm, bringing gold with them.

This gold mask is thought to show the Greek hero Agamemnon.

Digging for gold

In the past, a single nugget could change the life of a gold-digger. When gold was found in the US in the 1800s, it started a "gold rush." Thousands of people, known as prospectors, traveled to the US hoping to find gold and become rich.



Gold nugget

Large gold mines have created huge holes deep into the ground.

Gold mine

Pieces of gold found loose on the Earth's surface can be picked up by hand. Bigger quantities deep underground must be dug out in a process called mining.

Modern mines use heavy machines to dig chunks of rock that contain traces of gold.



25

percent of all the world's gold is stored in a New York City vault.

Centuries of sparkle

Gold was the first metal to be discovered and used by humans. It is beautiful and shiny, as well as soft and bendy. This makes it ideal for shaping into jewelry, including delicate rings, bracelets, and necklaces.





The world's oldest coins were made from electrum—a mixture of gold and silver.

Golden money

The first gold coins were made by King Croesus in 564 BCE. All coins used to be made of precious metals, including gold and silver. However, modern coins are usually made of cheaper metals, such as copper, nickel, and zinc.

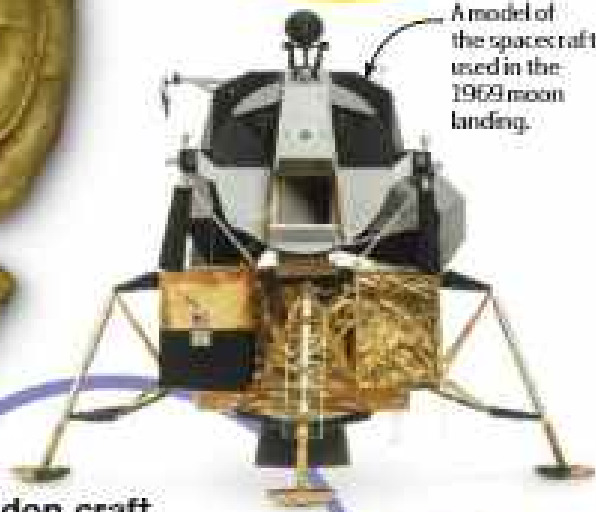


Sacred manuscripts were decorated with gold leaf.



Gold leaf

For centuries, gold has been used to decorate religious buildings, works of art, and objects. As well as using the solid metal, gold can be thinned down into very fine sheets called gold leaf. Gold leaf is used to decorate books and paintings.



A model of the spacecraft used in the 1969 moon landing.

Golden craft

Space scientists use sheets of gold foil to cover parts of some spacecraft and satellites they send into space. The foil protects them by reflecting the sun's harmful rays during their journey.

The visors of space helmets are coated in a thin layer of gold to keep astronauts cool and safe.

The mask was discovered by archaeologists in a burial tomb in 1876.

Prehistoric life

Earth has changed a lot over many millions of years. It has not always been home to plants, animals, and people. Many early living things no longer exist, so we only know about them from their remains. The distant past is known as "prehistory."

SEE ALSO

- Dinosaurs p.80
- Early humans p.82
- Earth p.83
- Fossils p.111
- Habitats p.126
- Oceans and seas p.187



Oceans
The first life was in the oceans. There were underwater plants and early animals.



Ammonites were animals with shells that lived in water.



Dinosaurs were the main land animal in prehistoric forests.

Forests
As Earth warmed up, plants grew on land and forests provided food for different types of animals.



Ice ages
In times when the Earth cooled down, most of it was covered in ice. Animals had to adapt to survive.



Woolly mammoths had thick fur coats to help them keep warm in the ice ages.

Stone Age
After the last ice age, Earth warmed up to how it is today. A great variety of plants and animals live in many different habitats, such as deserts, forests, and polar regions.



Early humans invented ways to hunt and gather food, and survive longer.



Radio

Radios pick up signals and turn them into sounds we can hear. They do this by using invisible waves that carry sound information. Thousands of different radio stations play music, news, and drama to listeners all over the world.

SEE ALSO

- ▶ Atmosphere p.33
- ▶ Books p.44
- ▶ Communication p.69
- ▶ Hearing p.127
- ▶ Navigation p.182
- ▶ Television p.251

How radio works

Radio towers turn sounds into radio waves. Radios pick up these waves and turn them back into sounds that you can hear.

1. Radio tower

Radio towers have antennas on the top of them. These antennas send out radio waves.



Radio waves travel at the speed of light—about 186,000 miles (300,000 km) a second!

2. Radio waves

Invisible radio waves carry sounds from radio towers to radios in the home.



3. Radio antenna

An antenna is a thin metal rod that picks up radio waves.

4. Speaker

The radio's speaker plays the waves as sounds.

Radio towers are tall to make sure that radio waves can travel over buildings.



Digital radio

Instead of radio waves, digital radio uses digital signals that don't get broken up like radio waves do. This means they sound better.



Remote control

Wireless gadgets use radio waves to contact each other. For example, a remote control can tell a toy car how to move around.



Remote control toy car

Rain forests

Rain forests are forests with tall trees and lots of rain. Tropical rain forests are hot places. They are home to nearly half of all animals and plants in the world. The trees are so thick with leaves that very little sunlight reaches the forest floor.

SEE ALSO

- Birds p.39
- Forests p.109
- Habitats p.126
- Materials p.157
- Plants p.194
- Trees p.261
- Weather p.271

Amazon rain forest

The world's largest rain forest is the Amazon in South America. This rain forest surrounds the Amazon River. Its trees and plants provide food and shelter for many animals.

This large blue Morpho butterfly is brown on the other side of its wings, so it can hide when resting.

Canopy layer

This is a thick layer of leaves and branches. Most rain forest animals live here.

Emerald tree boas squeeze animals to death before eating them whole.

Jaguars hunt other animals and climb trees to rest and eat.

Understory

Shrubs and new trees grow in this hot, dark layer of the rain forest.

Forest floor

This dark, damp layer is covered in dead leaves that have fallen from above.

The leaves of bromeliads catch water that small animals come to drink.

Toucan toucans use their long beaks to reach fruit to eat.

Howler monkeys get together and howl each morning.

A praying mantis waits for other insects to come close, then grabs and eats them.

Some trees have giant buttress roots that help them soak up water quickly.

The bright red leaves surrounding the flowers of the heliconia shrub look like lobster claws.

This long-tongued giant anteater can eat 30,000 ants a day.

Capybaras are good swimmers and eat water plants.

This many-legged centipede can kill bigger frogs, spiders, and snakes.



This harpy eagle hunts for animals in the treetops.



This large blue Morpho butterfly is brown on the other side of its wings, so it can hide when resting.

Emergent layer
Only the tallest trees reach this top layer of the rain forest.



Recycling

Recycling means reusing garbage or making it into new things rather than burning it or burying it in the ground. Everything from paper, glass, metal, and plastic to phones and computers can be recycled. The more we recycle our waste, the less we damage our planet.

SEE ALSO

- Changing world pp.50-51
- Climate change p.60
- Computers p.71
- Metals p.162
- Plastic p.195
- Pollution p.198

Paper and cardboard

Old paper and cardboard can be mashed up in water, then rolled flat and dried to make new paper products.



Food waste

Leftover food can be fed to animals, such as pigs or chickens, or used to make compost to help plants grow.



Plastic

Most of the plastic we throw away can be chopped up, melted down, and re-formed into new items.



Metal

Cans can be melted down. They are either remade into more cans or used to make other metal goods.



Glass

Bottles and jars made of glass can be cleaned and used again or melted down to make new glass objects.



Electronics

Devices such as phones and laptops can be repaired or have their valuable metal parts taken out to be reused.



Festivals

Festivals are celebrated across the world. Some festivals, like Eid or Christmas, mark important religious events. Others, like the Chinese New Year, mark the passing of time. Festivals are often a time of joy. People might put up lights, dance, or give each other presents.

Christmas

Every year at Christmas, Christians celebrate the birth of Jesus Christ. Some go to church and sing carols or special songs about the event. Most people give presents to each other and have a large meal with special food. Many people spend Christmas with friends and family.



Christmas trees are often finished with a star or angel on top.

Presents are placed below a decorated Christmas tree.



The "Elegant Skull" is a symbol of the Day of the Dead.

Day of the Dead

In Mexico, people celebrate the Day of the Dead for three days at the start of November. This ancient festival is held to remember friends and family who have died. People build shrines to the dead and bring gifts of food and drink to their graves.



Elegant Skull statue



A Chinese Dragon puppet that is used in festival celebrations.

Chinese New Year

Celebrations for the Chinese New Year last for 15 days. Families clean their houses to sweep away bad luck, decorate their windows and doors with red paper shapes, and enjoy a big feast together. Some people throw firecrackers to make a big noise, and dragons dance in the streets.

At New Year, Chinese parents give their children money in red packets.

Dancers control the colorful dragons using long sticks.



Muslims say the special Eid prayer outside a mosque.

Eid

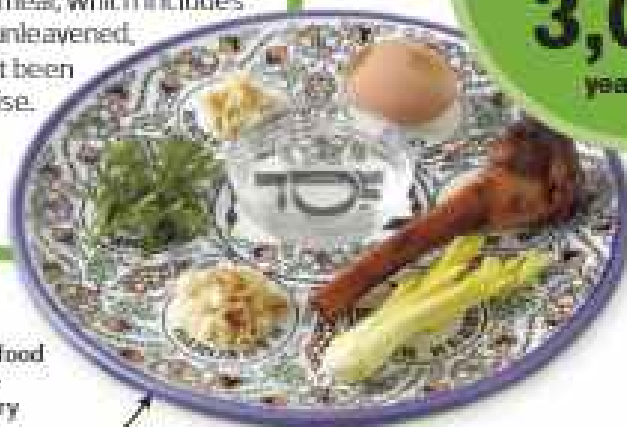
Muslims celebrate the Eid al-Fitr holiday at the end of Ramadan. During Ramadan, they fast, or go without food during the day. At Eid, Muslims give money to charity, pray together, visit friends and family, and enjoy a feast.

Passover

Jewish people celebrate Passover to remember the escape of the Jews from slavery in Egypt. The festival lasts for seven or eight days and is marked with the special seder, or meal, which includes matzo—a flat, or unleavened, bread that hasn't been given time to rise.

Passover is one of the most important Jewish festivals, and is more than

3,000
years old.



The special food on the plate tells the story of the Jews' escape from slavery.

Fireworks light up the sky at Diwali, the festival of lights.

Diwali

The Hindu festival of Diwali takes place every fall in the north of the world and every spring in the south. It celebrates the victory of light over darkness and good over evil. People light up their homes and public places and set off fireworks.



Religion

A religion is a set of beliefs. Religions often try to explain the world, such as how it came to exist. Religion is an important part of many people's lives. They agree with their religion's teachings, and try to behave by its rules, for example, about how to treat other people. Many religions have central figures that are prayed to, called gods.

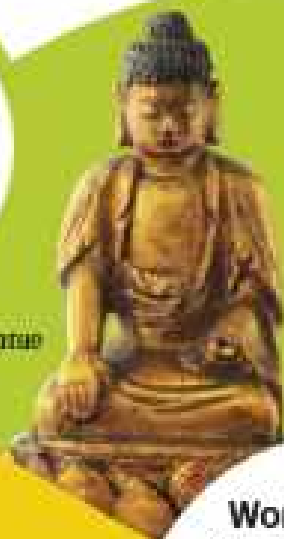
SEE ALSO

- ▶ Ancient India p.19
- ▶ Dance p.76
- ▶ Festivals pp.206–207
- ▶ Turkish Empire p.262
- ▶ World p.275

Buddhism

Buddhists follow the teaching of Buddha, who lived in India in the fifth century BCE. They meditate, which involves thinking deeply. They believe that they will be reborn after death.

Buddha statue



Special candlestick called a Menorah



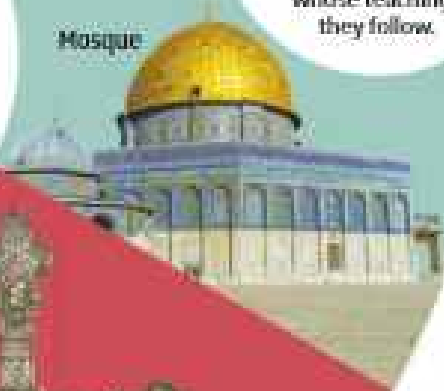
Judaism

Judaism dates back 4,000 years. Jews believe in one God who created the world. They trace their history back to the Hebrew people of what is now Israel.

Islam

The followers of Islam are called Muslims. They believe that the Quran, their holy book, is God's word told through his messenger Muhammad, whose teachings they follow.

Mosque



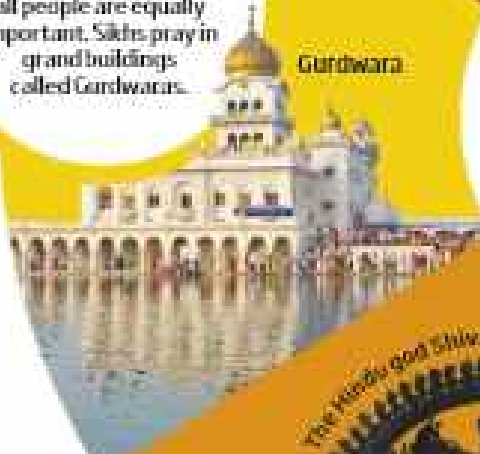
World religions

More than 75 percent of believers follow Buddhism, Islam, Christianity, or Hinduism. Judaism and Sikhism also have worldwide followings. There are many other religions, but fewer people follow them.

Sikhism

Sikhs follow the teachings of Guru Granth Sahib. They believe that all people are equally important. Sikhs pray in grand buildings called Gurdwaras.

Gurdwara



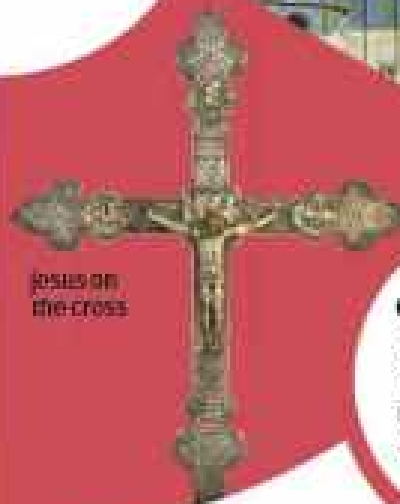
Hinduism

Hinduism began in India more than 2,500 years ago. Hindus pray to many gods and believe that life is a circle of birth, death, and rebirth.

The Hindu god Shiva



Jesus on the cross



Christianity

Christians have one God. They believe that their God's son Jesus lived on Earth 2,000 years ago. He died nailed to a cross so that Christians could have life after death.

Renaissance

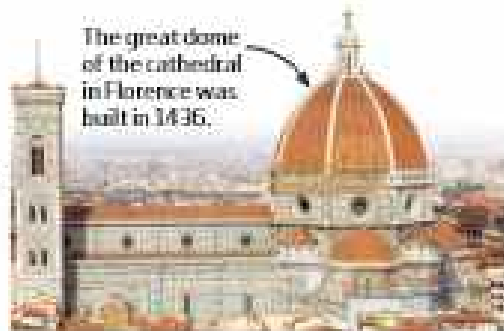
Italy experienced great change in science and art between 1400 and 1600. This movement then spread across Europe. It was called the “Renaissance,” which means “rebirth,” because it looked at ideas that originally came from ancient Rome and Greece.

SEE ALSO

- ▶ Aircraft p.13
- ▶ Ancient Rome p.20
- ▶ Art p.28
- ▶ Inventions pp.136-137
- ▶ Religion p.208
- ▶ Writing p.280

Buildings

Renaissance builders copied ancient styles to produce larger, more elegant buildings. These buildings often had columns and domes.



The great dome of the cathedral in Florence was built in 1436.



Science

Scientists began to carry out experiments for the first time. They made important discoveries about space, science, and medicine.

This flying machine was designed by Leonardo da Vinci. It never actually flew.



Art

Renaissance artists used a more realistic style than previous artists. They tried to show light and shade and came up with new types of paint and materials.

This painting, by Pietro Perugino, shows Jesus giving the keys of Heaven to St. Peter.

The people at the front of the painting appear larger than those farther away. This is called perspective.

Reptiles

Reptiles are scaly-skinned, cold-blooded animals. Most reptiles lay soft, leathery eggs. A baby reptile grows inside the egg and then, when it's ready, breaks its way out. There are four main groups of reptile.

SEE ALSO

- ▶ Amphibians p.15
- ▶ Antarctica p.24
- ▶ Deserts p.78
- ▶ Dinosaurs p.80
- ▶ Eggs p.86
- ▶ Evolution p.95
- ▶ Sun p.247

All reptiles have scaly skin.

Some lizards can move their eyes to look in two directions at once.

Lizards

These reptiles have many skills. This chameleon can change the color of its skin. Other reptiles can run up walls, or break off their tails to escape danger.

Crocodiles and alligators

These giant reptiles have existed since before dinosaurs. They hide under water, then spring up to catch their prey and drag it under.

Crocodiles and alligators have strong jaws.

Tortoises and turtles

Turtles live in water and tortoises live on the land. Their shells protect them, but are heavy, so they move very slowly out of the water.

Tortoises have hard shells.

Reptiles live on every continent except Antarctica.

Snakes

All snakes swallow their prey whole. They smell using their tongues. Some snakes have poisonous bites, but most are not dangerous to people.

Basking

Reptiles are cold-blooded animals. They get the heat their bodies need from the world around them, lying in the sunshine to warm up. They hide in the shade to keep themselves from overheating.



Rivers

Starting as tiny streams, rivers flow from mountaintops down to the ocean. They provide a home to lots of wildlife. People transport things along rivers, grow crops next to them, and even use rivers for spare-time activities, such as sailing and fishing.

SEE ALSO

- Farming p.98
- Water pp.120-121
- Glaciers p.122
- Lakes p.143
- Water cycle p.270
- Weather p.271

River system

Water flows down the mountains, making small streams. These streams join up to form rivers, which wind along and then eventually flow into the ocean.

Small streams flow downhill and come together to make rivers.

Rain and snow fall to make streams.

Some rivers start from lakes.

Dams use water movement to create electricity.

On flatter ground, rivers follow wide, steady bends, called meanders.

Farms use river water to grow crops.

Rivers provide nearby towns and cities with water.

Rivers become wider and deeper downstream.

River habitats are home to many kinds of wildlife.

The "mouth" of a river is where it joins the ocean.

Mighty river

The further rivers get from the mountains, the deeper and more slow-moving they become. Eventually they spill out into a lake or the ocean.

Fishing

Water sports

Rivers are used to transport people and goods to the ocean and beyond.

Robots

Robots are computer-controlled machines that do jobs for us. They can be used in many ways, such as helping doctors, building things, and doing jobs that would be too dangerous for people to do.

SEE ALSO

- Computers p.71
- Factories p.97
- Machines p.150
- Medicine p.160
- Space travel p.237

Types of robot

Robots are carefully designed to do their jobs. Each type of robot has its own unique look.

Eye sensors allow the robot to "see."

Humanoid robot

Some robots are designed to look a bit like people. This robot is called NAO. It can dance and speak.

Robots can see and feel, but not like humans. Their sensors use code to interpret feedback.

Each "finger" has a different function.

Hand sensors allow NAO to "feel" objects.

Medical robot

Robots are capable of performing very delicate movements. They can help doctors operate on their patients.

Robots in space

NASA uses robots to perform dangerous tasks in outer space. This robot mends things on the International Space Station.

Foot sensors are used for walking, climbing stairs, and detecting obstacles.

Factory robot

Robots are strong, and are good at doing the same thing over and over again. This makes them useful in factories.

Motors allow the robot to move up and down.

Rock cycle

Rock may be very hard, but it does not last forever. It is constantly being worn away by wind, water, and ice. At the same time, new rock is being made at the bottom of the sea and by volcanoes. This is called the rock cycle.

Recycled rocks

Tiny pieces of volcanic rock are washed into the sea and settle on the sea bed. They become buried by new layers of rock particles that get squashed together to form new rock. As this new rock is buried deeper, it gets hotter and melts to form magma. Magma rises and cools to form new volcanic rock.

Breaking down

Volcanic rock is broken up by wind, rain, and ice. Tiny particles of the broken rock are washed into the ocean by rain.

Settling

Pieces of volcanic rock settle on the seabed. They are buried and squashed to make a type of rock called sedimentary rock.

Changing

If sedimentary rock is buried deep enough, it is changed by heat and the weight of the rock above it into metamorphic rock.

Melting

Deep in the Earth, where it is extremely hot, rock melts to form magma. In some places, magma rises to the Earth's surface, forming volcanoes.

Cooling

Magma erupts from volcanoes. It cools and hardens to form a type of rock called volcanic, or igneous, rock.

SEE ALSO

- ▶ Earth's surface p.84
- ▶ Erosion p.93
- ▶ Inside Earth p.135
- ▶ Mountains p.172
- ▶ Rocks and minerals p.214
- ▶ Volcanoes p.268



Heated and squashed

Any rock can be changed into metamorphic rock by the heat and the weight of the rock above it. Slate, which is used to make roof tiles, is a common metamorphic rock.



Lots of layers

Most rocks being formed today are sedimentary rocks. They are made from tiny pieces of older rocks that build up on the sea bed in layers. Sandstone is a common sedimentary rock.

Rocks and minerals

SEE ALSO

- ▶ Earth's surface p.84
- ▶ Elements p.90
- ▶ Gemstones p.118
- ▶ Metals p.162
- ▶ Rock cycle p.213
- ▶ Volcanoes p.268

Rocks

Types of rock have different names depending on how they were formed. The three types are called sedimentary, igneous, and metamorphic.

Sedimentary rock

When particles of minerals settle and are squashed together, they slowly become sedimentary rock.



Metamorphic rock

Under great heat and pressure, metamorphic rock is formed deep inside the Earth.



Igneous rock

This type of rock is formed when melted rock cools down and becomes solid.



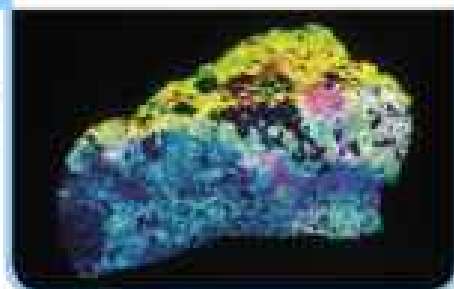
Minerals

A mineral is a naturally occurring solid. It is made from chemical elements—simple substances that cannot be broken down further. Minerals grow together to form rocks.

There are nearly **4,000** different types of minerals on Earth.

Glowing rocks

Some rocks look plain in daylight, but their minerals change color under special "ultraviolet" light. The glowing minerals in this rock are called calcite and willemite.



Serpentine

Amethyst

Garnet

Opal

Satellites

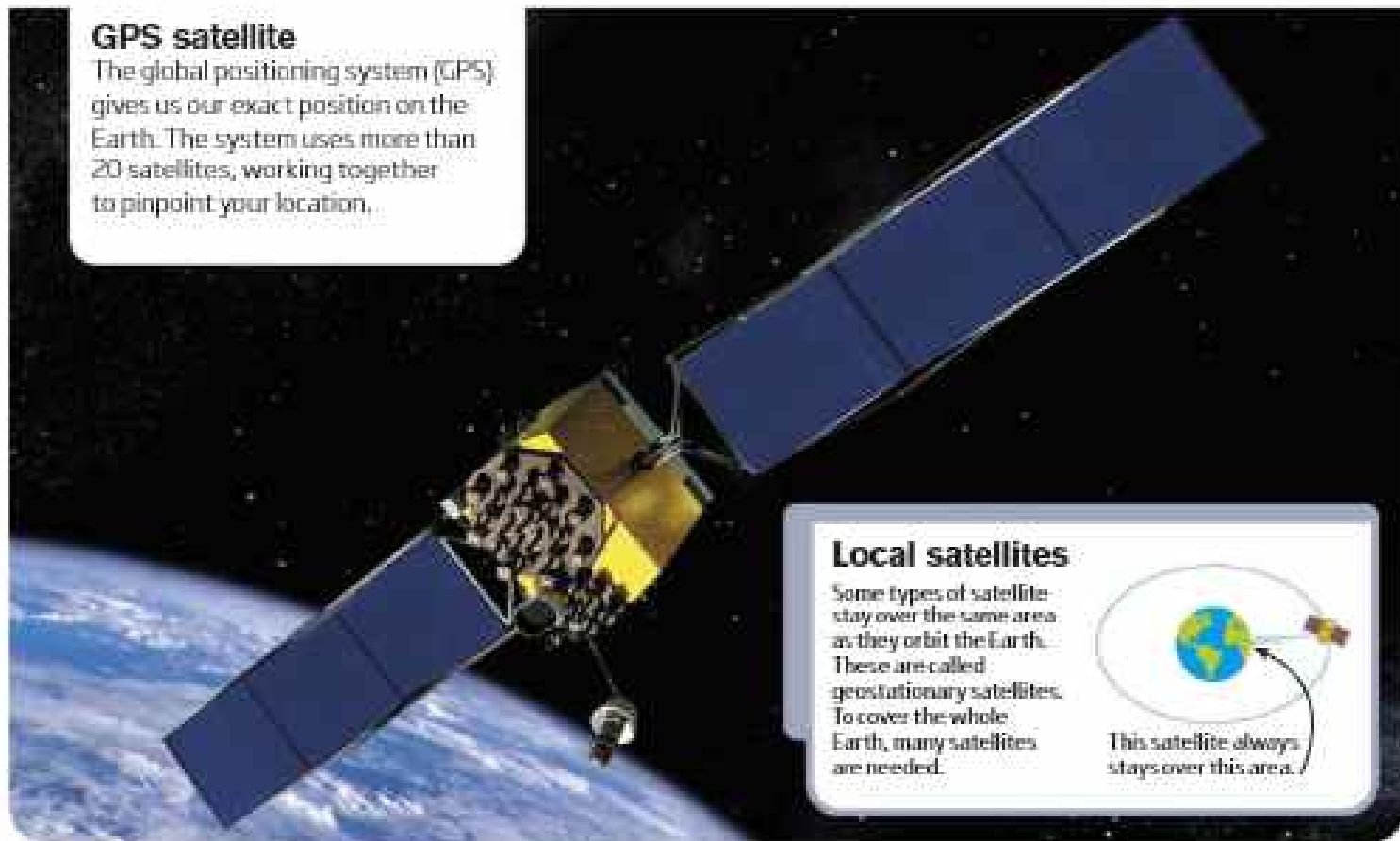
Satellites are objects that go around, or orbit, something bigger. There are more than 2,000 artificial satellites orbiting the Earth. They are used for a variety of different jobs. Some track the weather while others let us communicate with each other.

SEE ALSO

- ▶ Astronomy p.32
- ▶ Clouds p.64
- ▶ Communication p.69
- ▶ Internet p.138
- ▶ Solar system p.235
- ▶ Universe p.263

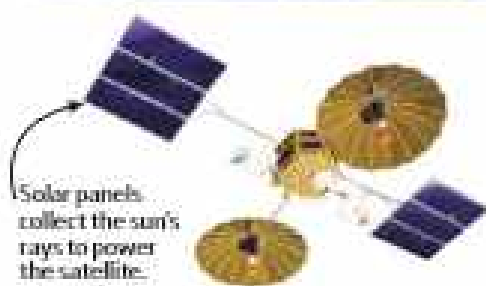
GPS satellite

The global positioning system (GPS) gives us our exact position on the Earth. The system uses more than 20 satellites, working together to pinpoint your location.



Local satellites

Some types of satellite stay over the same area as they orbit the Earth. These are called geostationary satellites. To cover the whole Earth, many satellites are needed.



Solar panels collect the sun's rays to power the satellite.



Photos of the Earth are sent to weather stations.



When this flap is open, the telescope takes photos.

Communication satellite

These satellites pick up signals and send them to other places in the world. They are used for phone calls and live video communications.

Weather satellite

These satellites take pictures of the clouds and measure land and sea temperatures. This information helps us to study and forecast the weather.

Hubble space telescope

This satellite points away from the Earth and takes detailed images of the universe. It lets us see farther into space than we could from the Earth.

Saturn

Saturn is the second largest planet in the solar system after Jupiter and the sixth farthest planet from the sun. It is a "gas giant," mainly made of hydrogen and helium, and is best known for its rings.

SEE ALSO

- ▶ Astronomy p.32
- ▶ Atmosphere p.33
- ▶ Jupiter p.141
- ▶ Moon p.171
- ▶ Exploration pp.180-181
- ▶ Solar system p.233

Ringed planet

Saturn is surrounded by huge rings that stretch over a vast distance. However, they are only a few hundred feet thick.



Icy rings

Saturn's rings are made of chunks of ice, rocks, and dust.

Gaps between rings

The gaps between the rings are areas with less ice and dust.



Moons of Saturn

Like Jupiter, Saturn has more than 60 moons. Titan, the largest moon, has lakes of liquid methane and a thick atmosphere. Enceladus, Saturn's sixth largest moon, shoots out water from its south pole.



Titan



The unmanned spacecraft Cassini

Huygens probe

Cassini-Huygens

The Cassini spacecraft explored Saturn between 2004 and 2017. A probe called Huygens, carried by Cassini, landed on the surface of Saturn's moon Titan in 2005.

Science

Science is the search for truth and knowledge. It's about understanding the world and learning how and why things work the way they do. Science includes doing experiments to test predictions and collect evidence. We divide science into three main areas: chemistry, biology, and physics.

SEE ALSO

- ▶ Astronomy p.32
- ▶ Biology p.38
- ▶ Chemistry p.58
- ▶ Medicine p.160
- ▶ Physics p.192
- ▶ The sciences pp.218-219

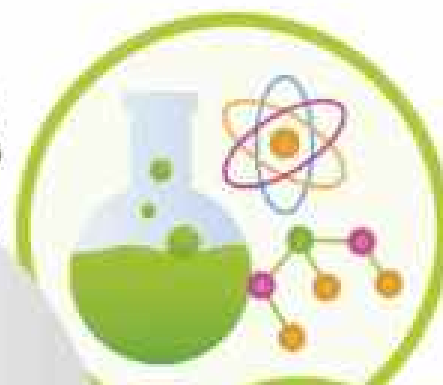
Biology

The study of living things and their surroundings is called biology. It includes the human body, plants, and animals.



Physics

Physics studies light, sound, forces, waves, magnets, electricity, energy, and the planets.



Chemistry

Looking at what things are made of is known as chemistry. This includes the tiny building blocks of all materials, called atoms.



Scientists

Scientists are people who investigate the world to answer questions and find solutions to problems. They do experiments to see if ideas are right or wrong, and they share information.

Inventions

Studying science helps us to create new things. For example, if we understand movement, we can design better cars. If we understand the body, we can invent medicines to help fight off sickness.

Copy of Edison's light bulb



Thomas Edison invented the electric light bulb in 1879, while studying electricity.

Science isn't just facts in a book—it's a whole way of thinking and discovering.

The sciences

For thousands of years, people have been observing the world around them and coming up with ideas to explain why things behave the way they do. Science is about answering questions with ideas, evidence, and experience.



Notches around the edge were lined up with objects in the sky.

Some of the earliest doctors were ancient Egyptian women,

5,000
years ago.

Child making a wave in a bottle



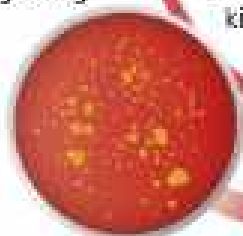
Experiments

In ancient Greece, a man called Aristotle said that people should look at nature and carry out experiments to find answers to their questions. Scientists do experiments to test their ideas and to make new discoveries.

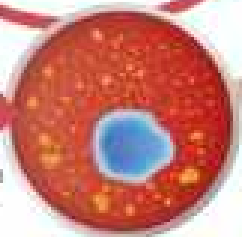
Curing diseases

In 1928, Scottish scientist Alexander Fleming noticed how a mold called penicillin killed bacteria. This discovery led to a type of medicine called antibiotics. They have been killing germs in the human body ever since.

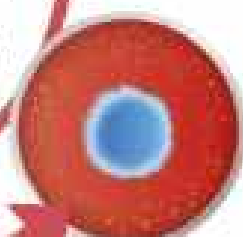
1. Bacteria growing



2. Penicillin introduced



3. Bacteria dies



Compasses helped explorers to travel the world and discover new ideas.

Early compass

Renaissance

Starting in the 15th century, the Renaissance was an explosion of ideas that transformed science and art in Europe. Leading experiments, gathering evidence, and sharing ideas became popular—this led to new inventions and discoveries.



Nebra Sky Disk

Renaissance artist and scientist **Leonardo da Vinci** sketched pictures of humans and animals he had cut up.

Accidental inventions

Many great scientific discoveries have been made by accident. For example, German scientist Wilhelm Conrad Röntgen accidentally discovered X-rays in 1885. He was sending electricity through tubes of gas when he noticed that he could see what was inside a nearby box. He then used this finding to take pictures of the bones inside his wife.



Early X-ray

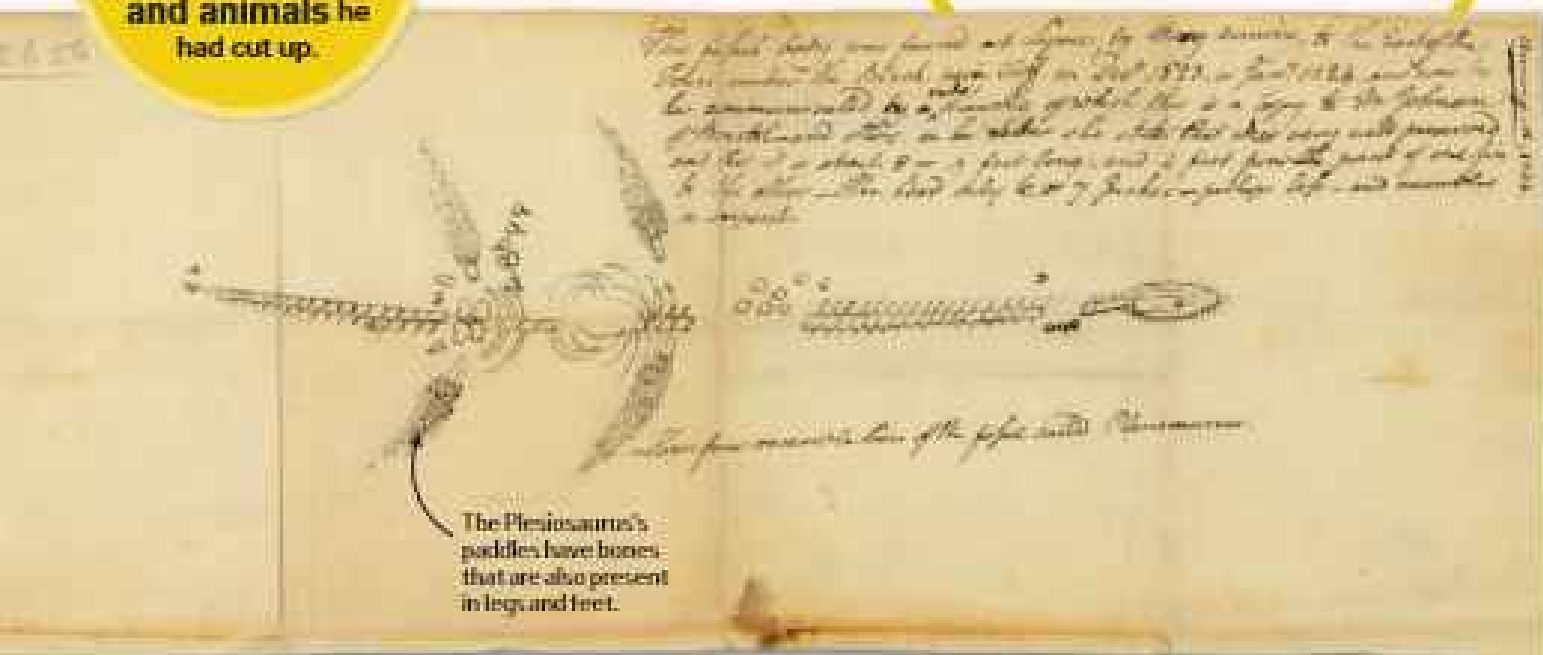
X-rays pass through skin and flesh but bounce back off bones.

The night sky

One of the earliest sciences was studying the movements of the moon, planets, and stars. This is now called astronomy. Moon calendars were first made about 10,000 years ago. Around 4,000 years ago, the Nebra Sky Disk was used to track seasons and the sun's position.

Scientific drawings

Many scientists create beautiful, precise drawings to record their findings. Mary Anning was a famous British fossil hunter who lived in the 1800s. Mary found her first fossils as a child and made drawings of them. Her discoveries helped scientists to understand how life existed in the oceans millions of years ago.



The Plesiosaurus's paddles have bones that are also present in legs and feet.

Plesiosaurus drawn by Mary Anning in 1824

Seashore

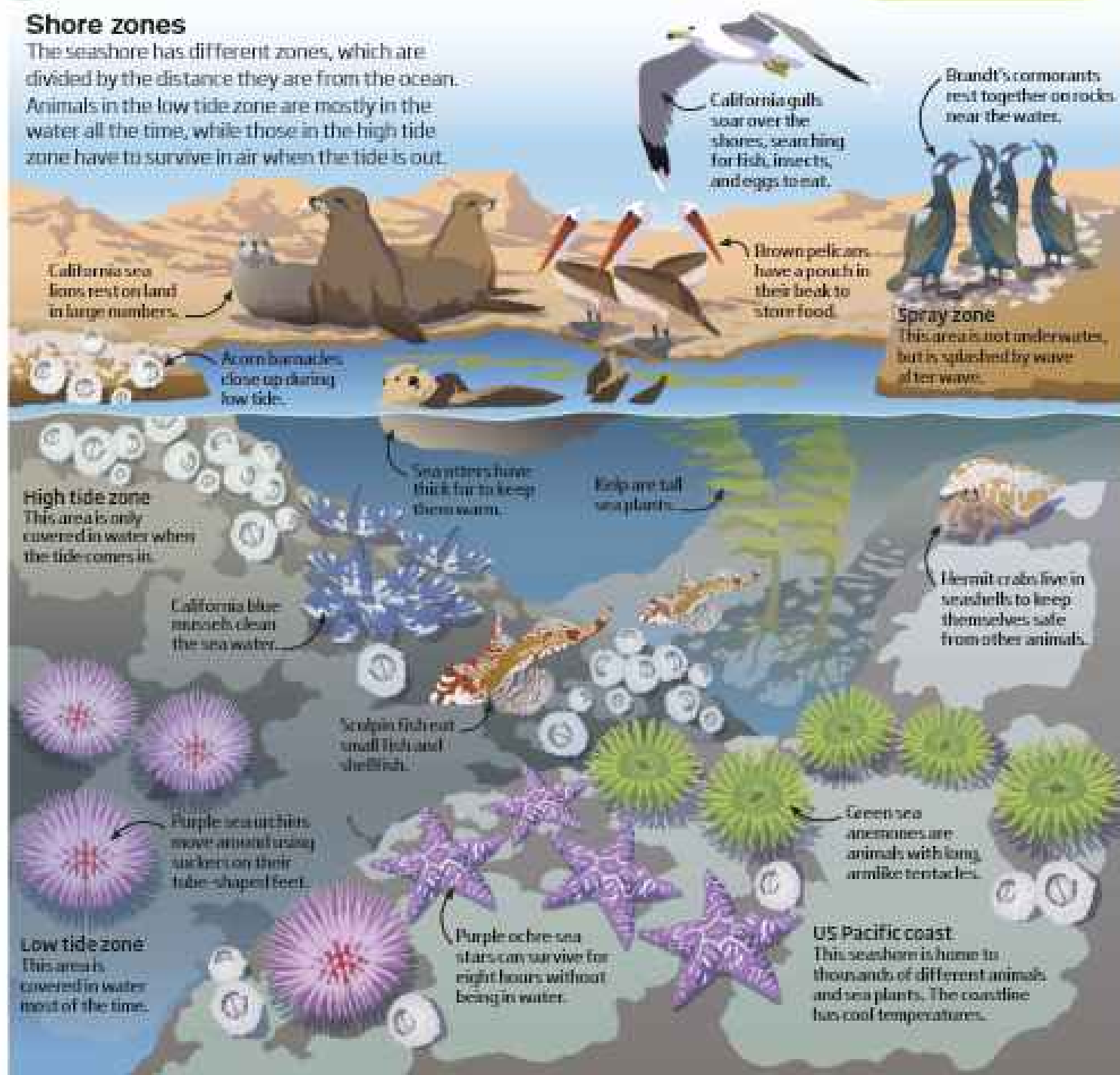
The area where land meets the sea or ocean is called the seashore. It may be sand, mud, or rock. Animals and plants that live here have to be able to survive crashing waves and the water level changing twice a day.

SEE ALSO

- ▶ Birds p.39
- ▶ Habitats p.126
- ▶ Invertebrates p.139
- ▶ North America p.184
- ▶ Oceans and seas p.187
- ▶ Tides p.254

Shore zones

The seashore has different zones, which are divided by the distance they are from the ocean. Animals in the low tide zone are mostly in the water all the time, while those in the high tide zone have to survive in air when the tide is out.



Seasons

In many parts of the world, the year has four seasons. These seasons are winter, spring, summer, and autumn. The lives of plants and animals, the weather, and the hours of daylight all change from one season to the next. In some hot parts of the world, there are just two seasons.

SEE ALSO

- ▶ Climate change p.60
- ▶ Day and night p.77
- ▶ Hibernation p.129
- ▶ Solar system p.233
- ▶ Trees p.261
- ▶ Weather p.271

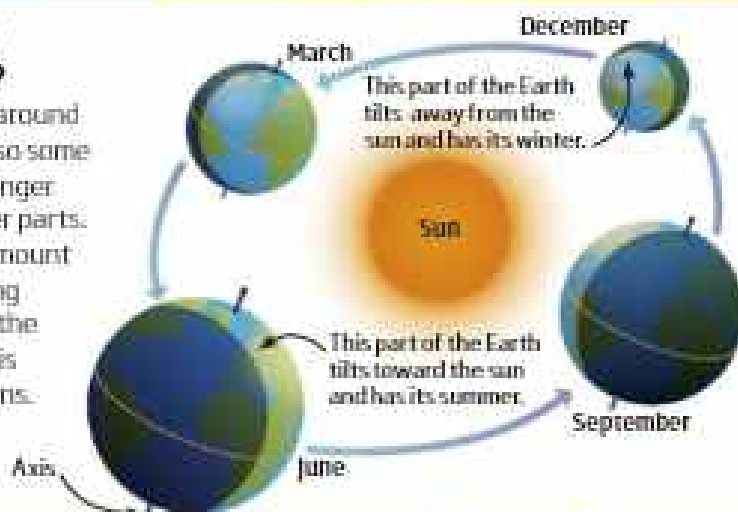
Changing seasons

In the cold winter, plants stop growing. In spring, they begin to grow again and baby animals are born. Summer sees the hottest weather, and then in autumn leaves change color and fall off the trees.



What causes the seasons?

The Earth travels around the sun. It is tilted so some parts of it get stronger sunlight than other parts. As it moves, the amount of sunlight reaching different parts of the Earth changes. This gives us our seasons.



Monsoons

Tropical parts of the Earth are warm all year round. There are often just two seasons—a dry season and a rainy season, called a monsoon.

Shapes

Shapes are areas with different outlines. In math, there are two types of shape. Two-dimensional shapes have length and width, while three-dimensional shapes have length, width, and height. Shapes can be made of straight or curved lines, or a mixture such as in a semicircle.

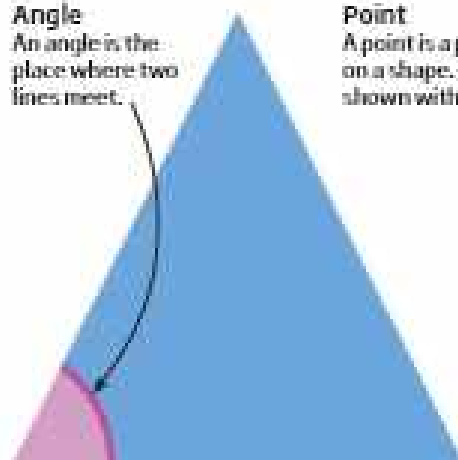
SEE ALSO

- Constellations p.73
- Flags p.102
- Maps p.155
- Numbers p.185
- Symmetry p.248
- Volume p.269

What makes a shape?

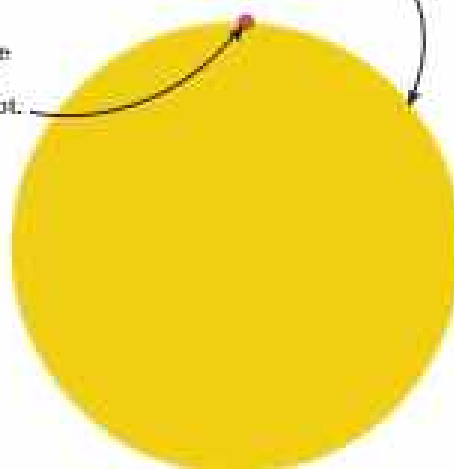
A shape is named depending on the number of sides and angles it has. Regular shapes have sides that are all the same length.

Angle
An angle is the place where two lines meet.



Triangles have three sides and three inside angles adding up to 180° .

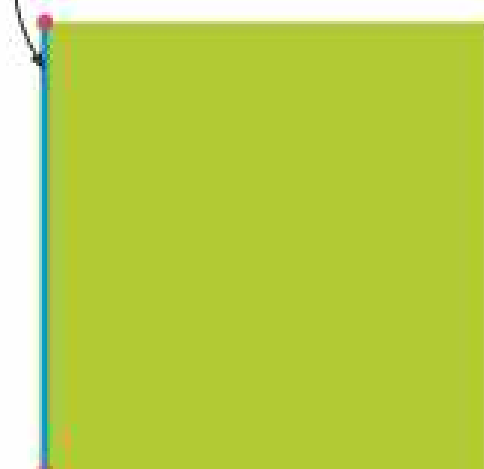
Point
A point is a place on a shape. It is shown with a dot.



Circles have one long, round side.

Edge
The side of a shape is called an edge.

Line
A straight line is the shortest distance between two points.



Squares have four equal sides and four right angles.

Angles

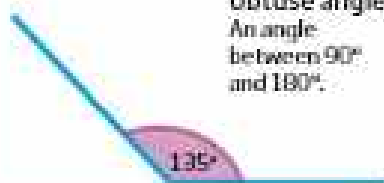
Angles are measured in degrees ($^\circ$) out of 360. Angles have different names depending on how big they are.



Acute angle
An angle less than a right angle (90°).



Right angle
An angle of exactly 90° .



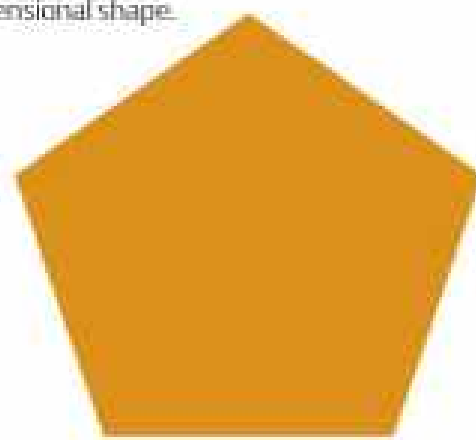
Obtuse angle
An angle between 90° and 180° .



Reflex angle
An angle between 180° and 360° .

Polygons

"Polygon" is the name for any flat, two-dimensional shape.



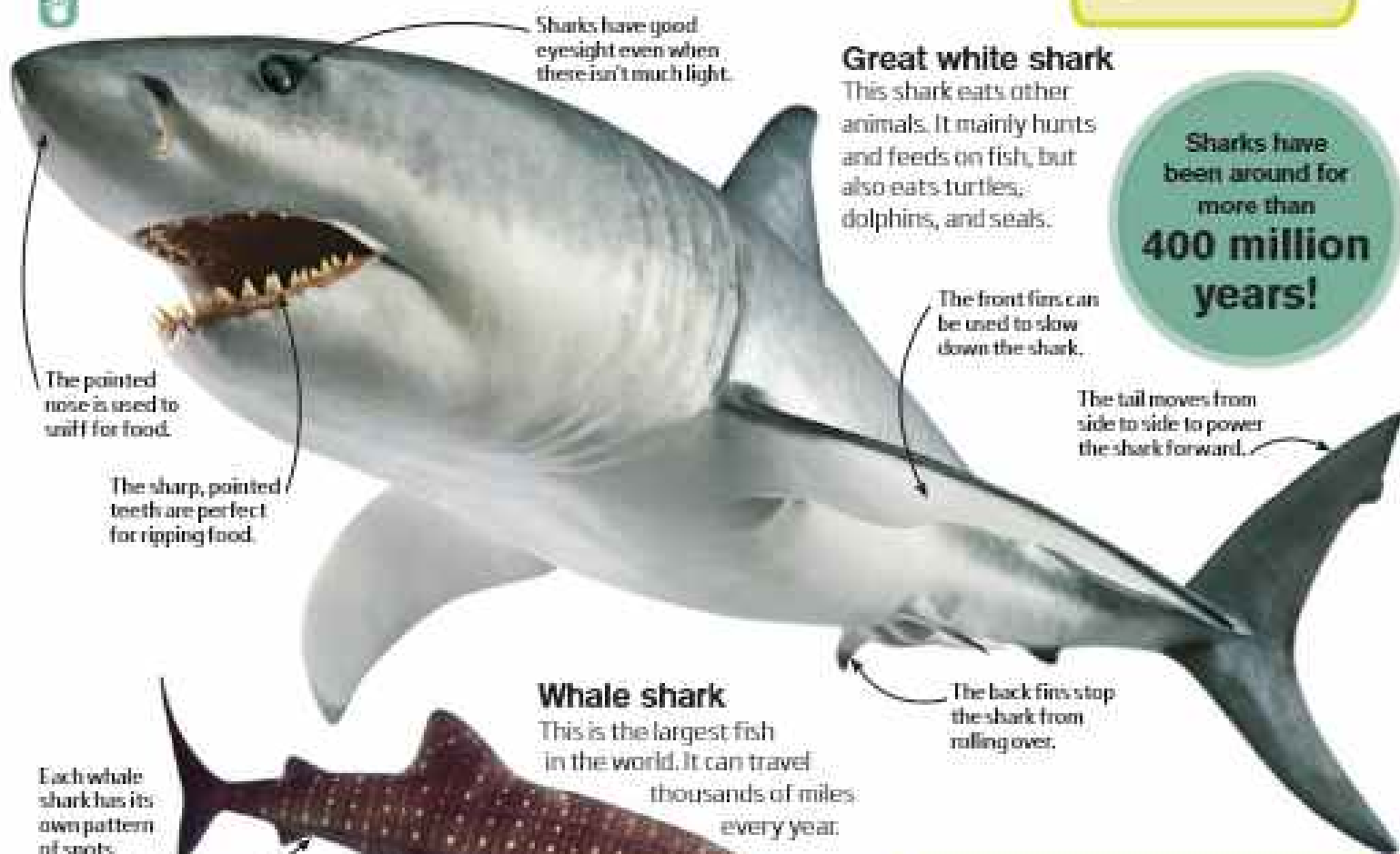
Pentagons have five equal sides and five inside angles.

Sharks

Sharks are a type of fish. Almost all of them eat meat. They are found in every ocean and some rivers. There are more than 400 types of sharks. Most are active during the day, but some hunt at night.

SEE ALSO

- Conservation p.72
- Fish p.101
- Food chains p.107
- Oceans and seas p.187
- Prehistoric life p.202



Sharks have good eyesight even when there isn't much light.

The pointed nose is used to sniff for food.

The sharp, pointed teeth are perfect for ripping food.

Great white shark

This shark eats other animals. It mainly hunts and feeds on fish, but also eats turtles, dolphins, and seals.

The front fins can be used to slow down the shark.

The tail moves from side to side to power the shark forward.

Sharks have been around for more than **400 million years!**

Whale shark

This is the largest fish in the world. It can travel thousands of miles every year.

Each whale shark has its own pattern of spots.

Hammerhead shark

Hammerhead sharks use their wide heads to pin down stingrays on the ocean floor.

The eyes are a long way apart, letting the shark see far to spot its prey.

Sharks under threat

The number of sharks is getting smaller because they are hunted for their fins, teeth, and fish oil. Scientists try to learn about sharks, so that they can help them.



Ships

Sea transportation comes in all shapes and sizes, from tiny sailing boats to large cruise liners and gigantic container ships. They can be used for sports, to take people on vacation, and to move things from one country to another.

SEE ALSO

- ▶ Oceans and seas p.187
- ▶ Sports p.239
- ▶ Trade p.257
- ▶ Transportation pp.258-259
- ▶ Work p.274



Containers are large metal boxes. Each container can hold around 6,000 shoe boxes.

The captain steers the ship from the bridge.

Cranes load containers on and off the ship.

Container ship

Some of the largest boats on the oceans are container ships. These giants of the seas can transport more than 15,000 containers full of items such as clothes, toys, and televisions to different countries.

Sailing boat

This tiny boat doesn't have an engine. It is powered by wind, which is caught in the sail to push the boat across the water.



The hold carries more containers. Some store hundreds of cars.



Cruise ship

A cruise ship is a floating hotel that takes tourists to different countries. On board are swimming pools, theaters, and water slides.

Submersible

This craft doesn't travel on the ocean, but underneath it. It can take scientists deep underwater to look at sea life and study the ocean floor.



Sickness

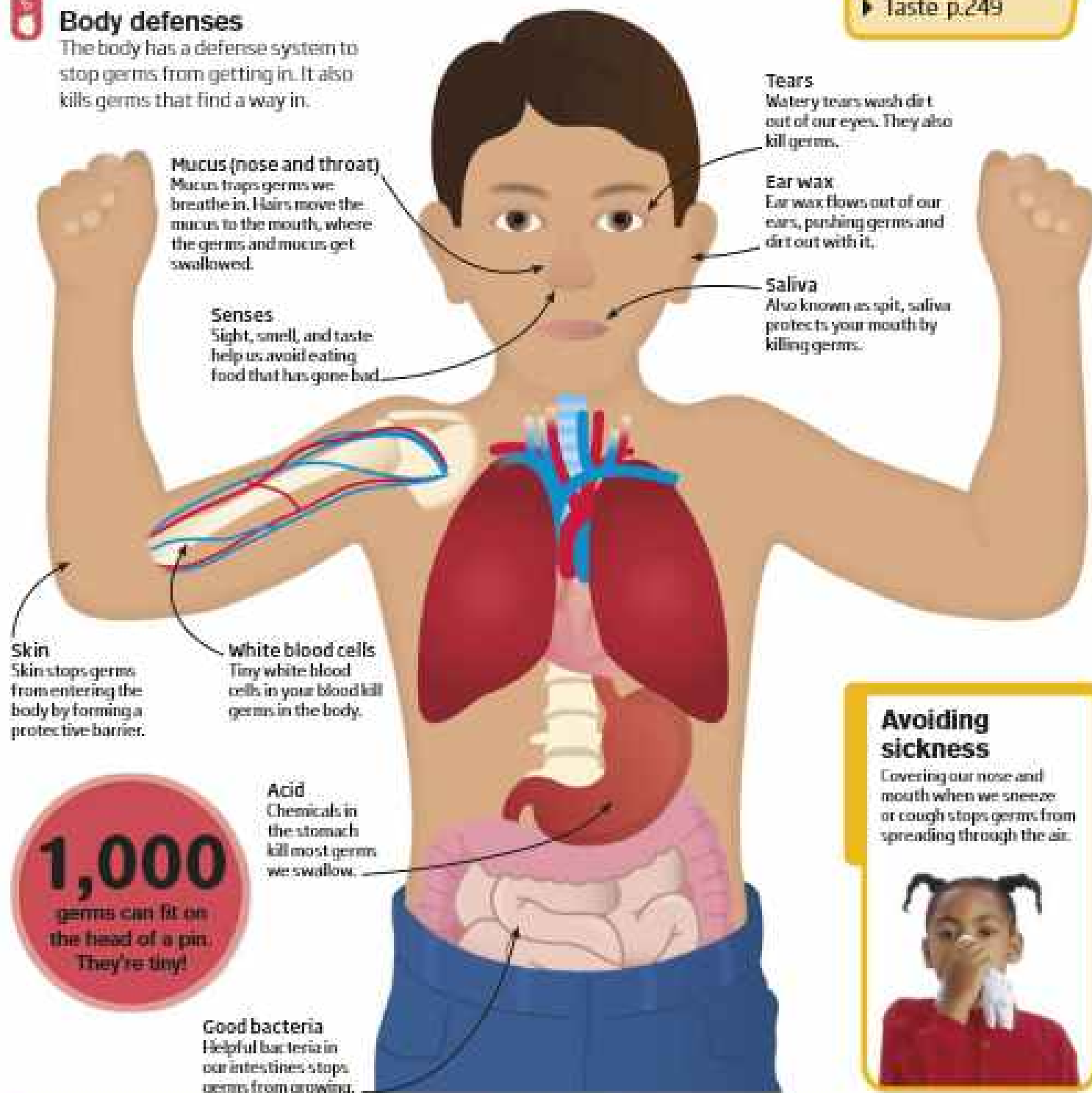
Tiny living things called germs can make us sick if they get inside our bodies. Germs are all around us—in the air, in our food, and on the things we touch. Our bodies have many different ways to stop them from harming us.

Body defenses

The body has a defense system to stop germs from getting in. It also kills germs that find a way in.

SEE ALSO

- ▶ Body cells p.41
- ▶ Human body p.130
- ▶ Medicine p.160
- ▶ Microscopic life p.165
- ▶ Skin p.229
- ▶ Taste p.249



Avoiding sickness

Covering our nose and mouth when we sneeze or cough stops germs from spreading through the air.



Sight

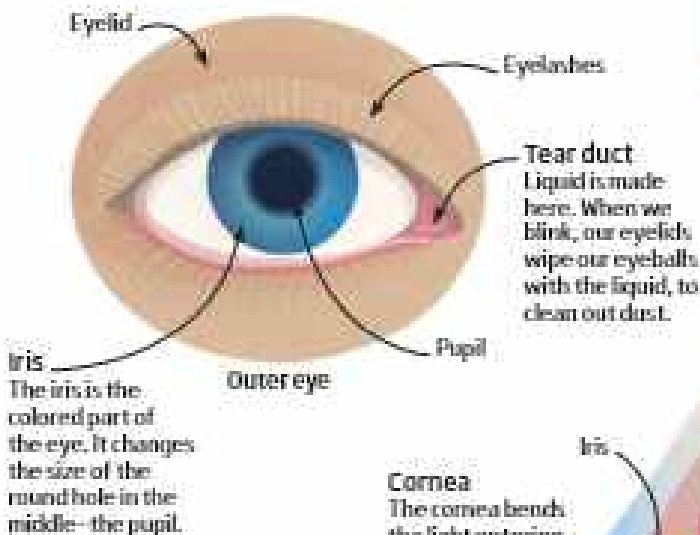
Sight is seeing the shapes, sizes, and colors of nearby and faraway objects within our surroundings. When we see, colored light is bouncing off objects and into our eyes.

SEE ALSO

- ▶ Body cells p.41
- ▶ Brain p.45
- ▶ Cells p.56
- ▶ Hearing p.127
- ▶ Light p.147
- ▶ Muscles p.173
- ▶ Smell p.232

How we see

Tiny sensors at the back of the eye receive light and send signals to the brain to make an image from the light we see.

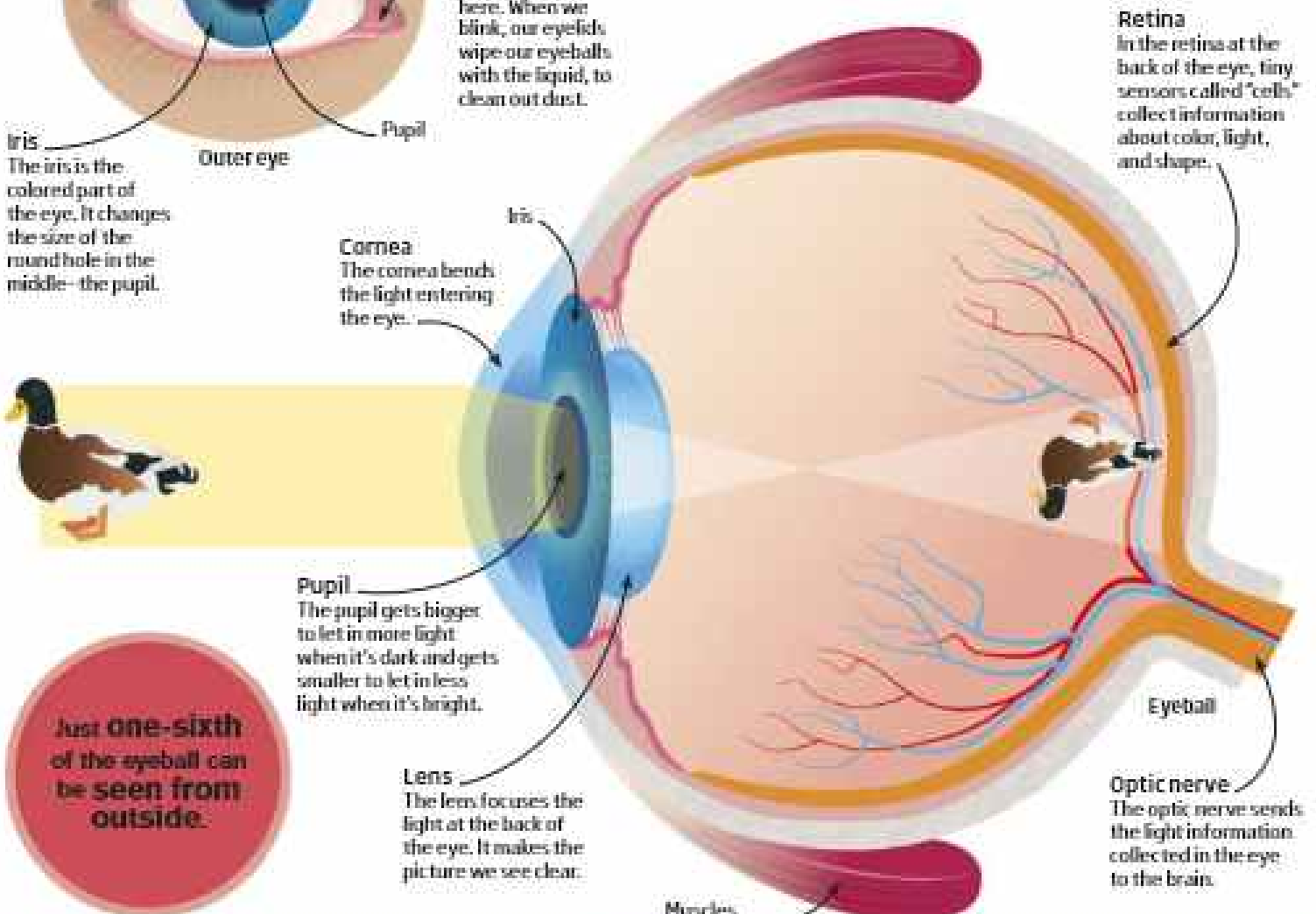


Glasses

If an eye lens doesn't focus the light in the right place, the image is blurred. Glasses contain lenses that change where the light focuses in the eye, so the image becomes clear.



Some people use glasses to help them read



Just one-sixth of the eyeball can be seen from outside.

Sinking and floating

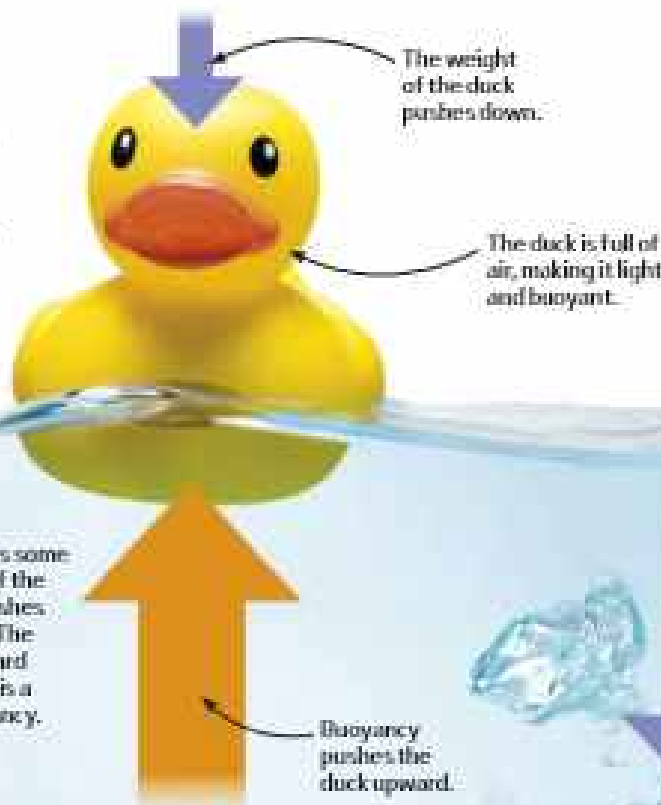
When we place objects in water they can float on the surface or sink below it. The heavier and more packed together a material is, the more likely it is to sink. Materials like stone and metal usually sink, while wood and plastics usually float.

SEE ALSO

- Forces p.108
- Gases p.117
- Gravity p.125
- Materials p.157
- Metals p.162
- Ships p.224

Floating

If the downward force of an object's weight is less than the force of the water pushing up on it, it will float. The more air an object contains, the more likely it is to float.



Salt water has more **buoyancy** than fresh water, so we can **float** more easily in the ocean than in a lake.

Massive ships

It seems amazing that huge metal ships float while small metal coins sink. Ships float because they are full of air and because they have a large surface area. The force of buoyancy pushing upward is spread out and is greater than the ship's overall weight.



Skeleton

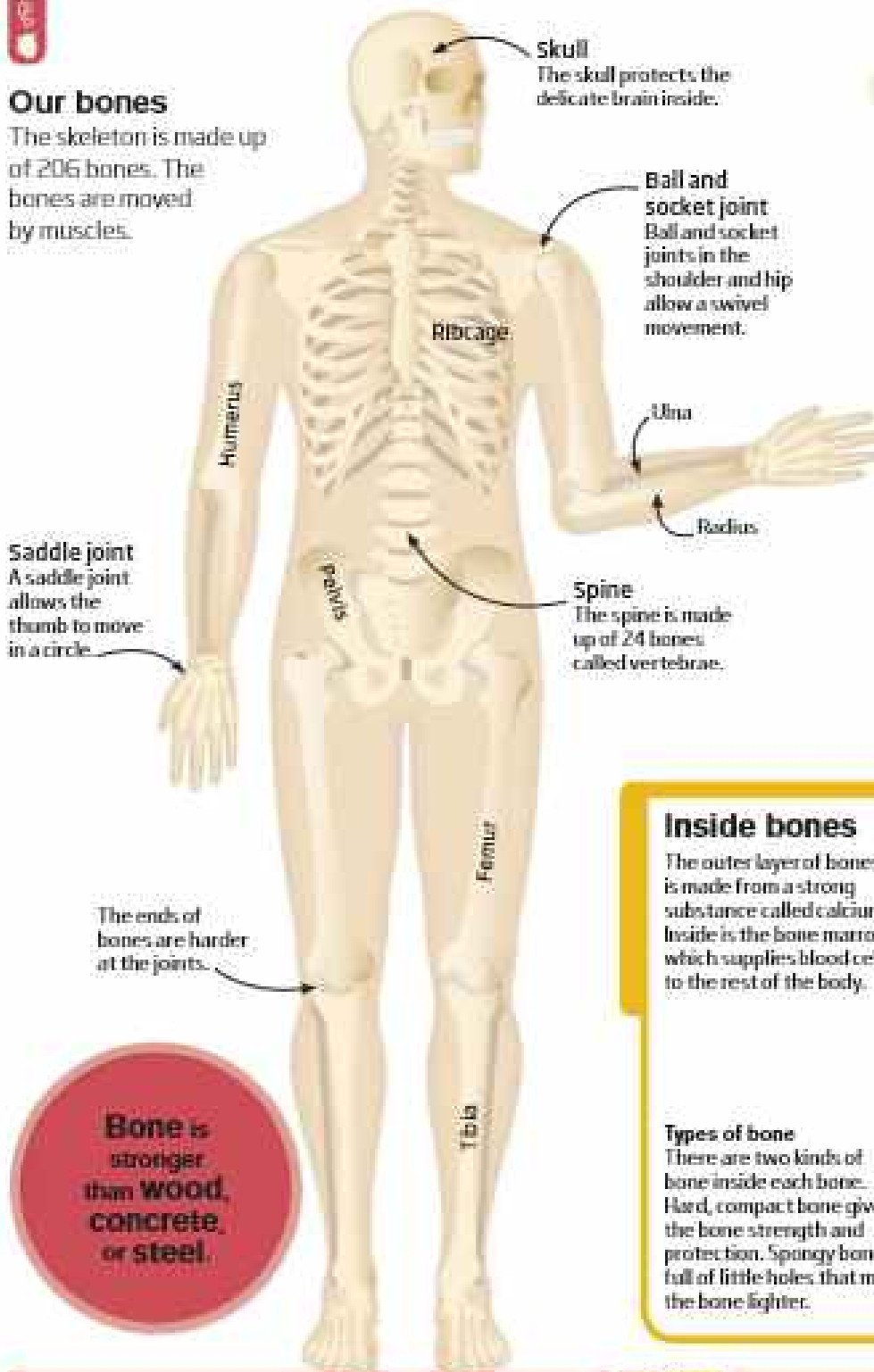
All the bones in the human body fit together in the skeleton. They make up the shape of the human body. The skeleton forms a protective cage around the soft, inner organs like the lungs and the heart.

SEE ALSO

- ▶ Brain p.45
- ▶ Heart p.128
- ▶ Human body p.130
- ▶ Lungs p.149
- ▶ Muscles p.173

Our bones

The skeleton is made up of 206 bones. The bones are moved by muscles.



Skull
The skull protects the delicate brain inside.

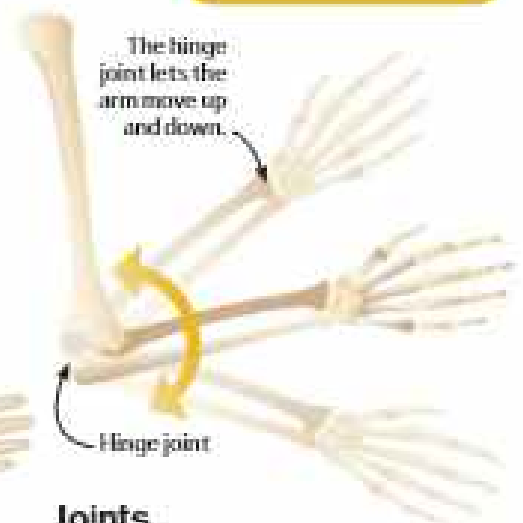
Ball and socket joint
Ball and socket joints in the shoulder and hip allow a swivel movement.

Saddle joint
A saddle joint allows the thumb to move in a circle.

Spine
The spine is made up of 24 bones called vertebrae.

The ends of bones are harder at the joints.

Bone is stronger than wood, concrete, or steel.



Joints

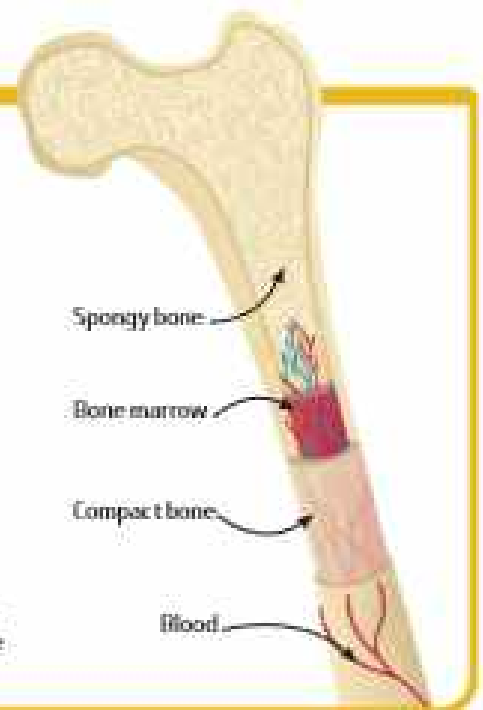
Joints are the places where one bone joins another to allow movement. They let our bones move side to side, up and down, or in circles. Joints have fluid in them to help make movement smooth.

Inside bones

The outer layer of bones is made from a strong substance called calcium. Inside is the bone marrow, which supplies blood cells to the rest of the body.

Types of bone

There are two kinds of bone inside each bone. Hard, compact bone gives the bone strength and protection. Spongy bone is full of little holes that make the bone lighter.



Skin

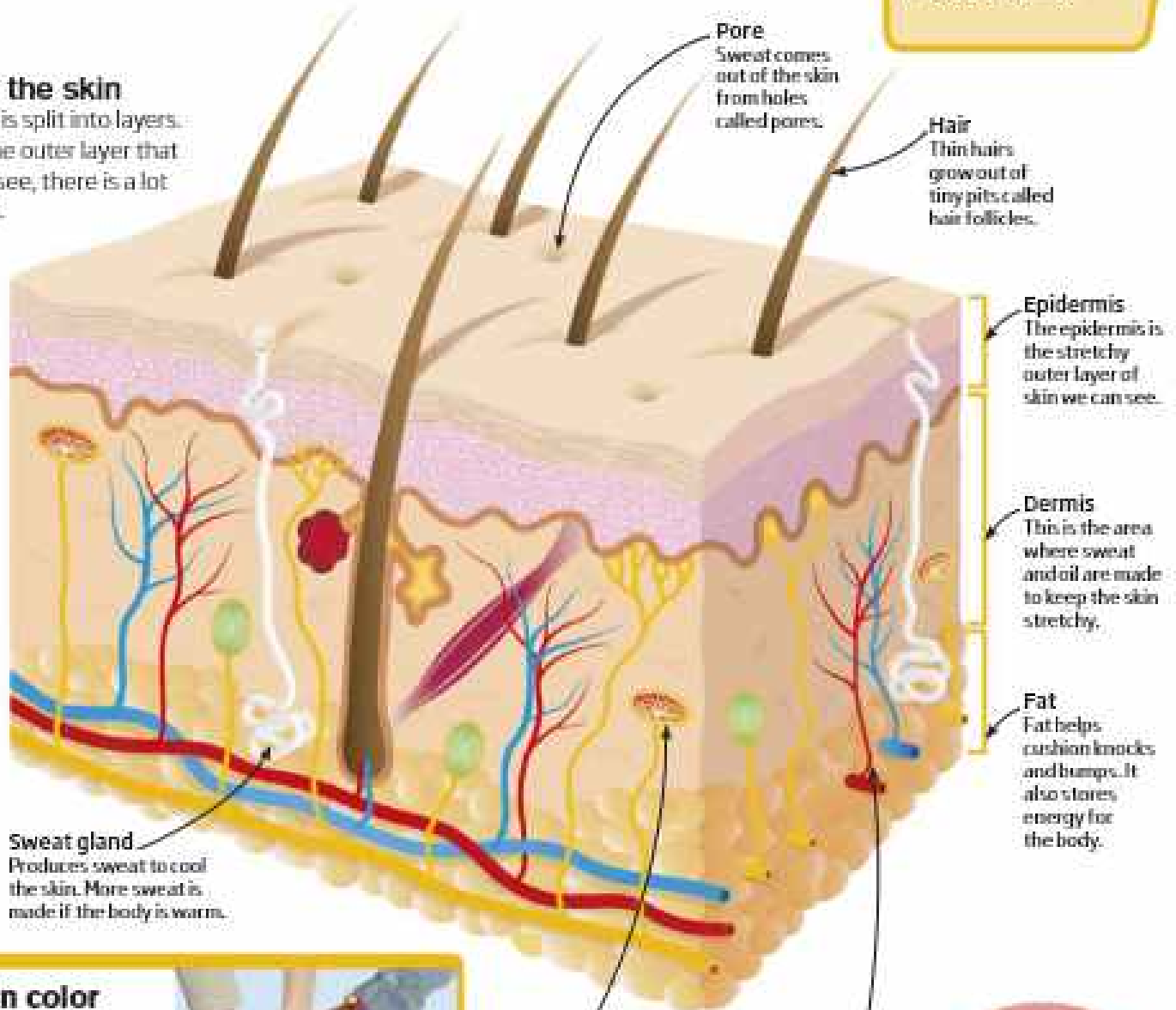
Skin is the stretchy outer layer of the body. It keeps our inside parts inside and germs outside. It protects us from water and sunshine, and keeps us at the right temperature. The outer layer continually flakes off as new skin is made.

SEE ALSO

- ▶ Body cells p.41
- ▶ Cells p.56
- ▶ Genes p.119
- ▶ Heart p.128
- ▶ Human body p.130
- ▶ Touch p.256

Inside the skin

The skin is split into layers. Below the outer layer that you can see, there is a lot going on.



Sweat gland
Produces sweat to cool the skin. More sweat is made if the body is warm.

Skin color

A chemical in our skin called melanin controls skin color. The more melanin you have in your skin, the darker it is. It is made in the epidermis, the top layer of skin.



Nerves
Nerves send signals to the brain that tell us the texture, temperature, and pressure of the things we touch.

Blood vessels
These tubes carry blood around the body. They get wider to allow more blood to flow and to help the body to cool down.

Skin is the body's biggest organ.
It weighs **8.8 lbs (4 kg)**

Slavery

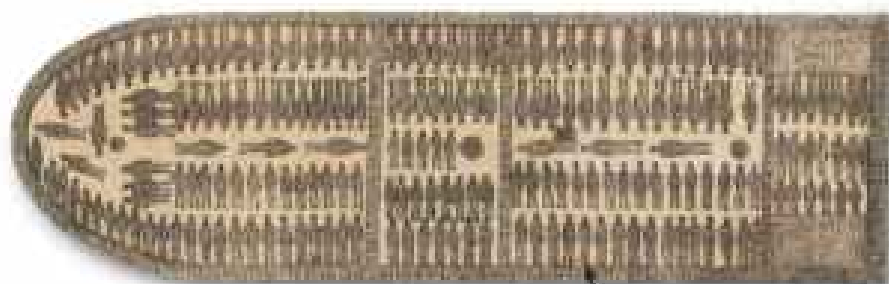
Slaves are people who have had their rights taken away and are treated like property. They may have been captured in war, owe more money than they can pay, or have parents who are slaves. Slavery has been used throughout history, but today it is against the law in every country in the world.

SEE ALSO

- ▶ Africa p.12
- ▶ Ancient Rome p.20
- ▶ Law p.145
- ▶ North America p.184
- ▶ Ships p.224
- ▶ Trade p.257

Slave labor

In the early 1800s, big farms called plantations in the US used slave labor to pick cotton, cut sugar cane, and perform other tasks. Slaves worked long hours in very bad conditions.



The slave trade

To keep American plantations supplied with workers, ships carried slaves from Africa to America. Between 1450 and 1850, 12 million people crossed the Atlantic Ocean.

People were often chained and packed tightly into slave ships.

Slavery today

Though slavery is now illegal, more than 20 million people are still kept as slaves, mostly in Asia and Africa. Groups around the world are working to solve this terrible problem.



Sleep

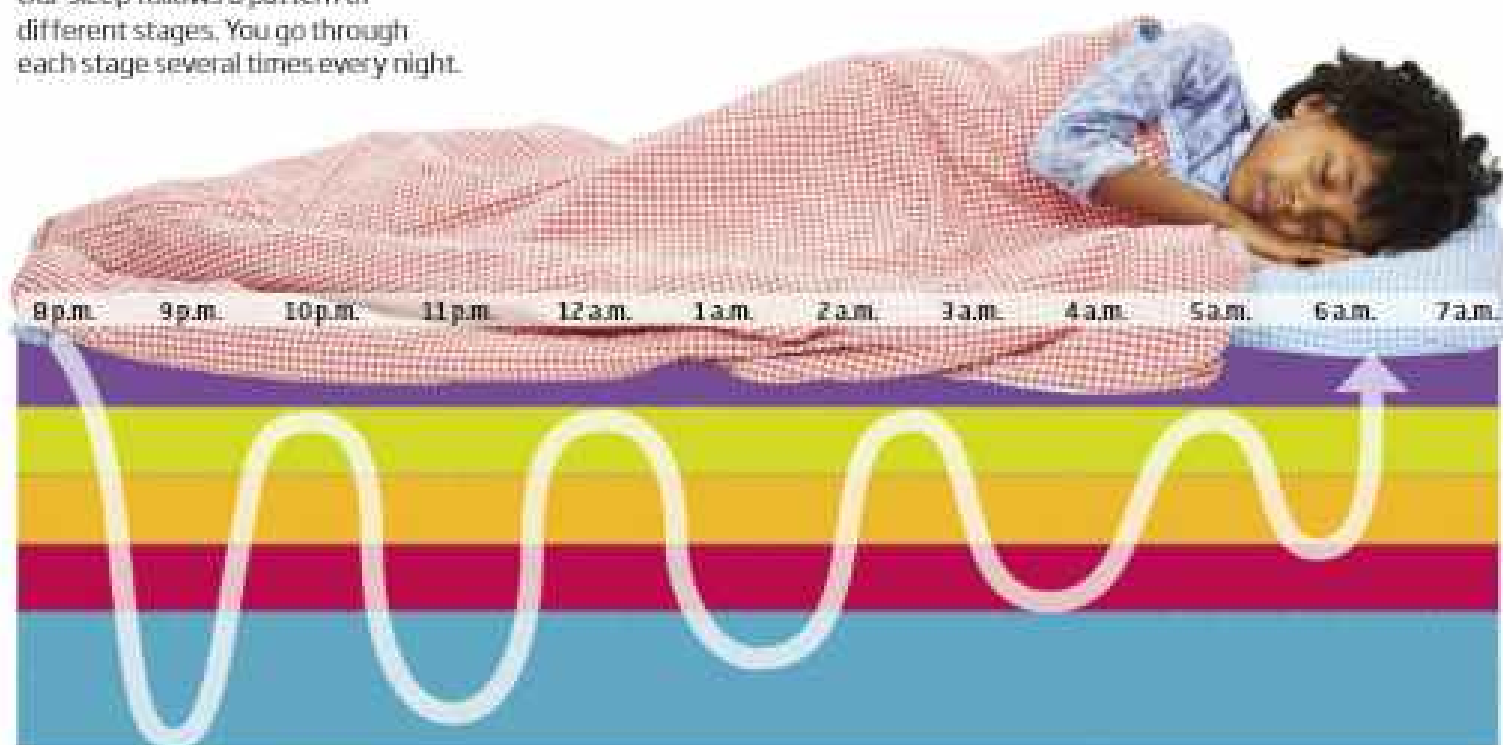
Every night we sleep and our bodies rest, repair, and grow. While we sleep our brains sort through the information gathered from our senses and some of it is deleted and some is stored as memories. We need sleep to stay fit and healthy.

SEE ALSO

- ▶ Brain p.45
- ▶ Eating pp.104-105
- ▶ Hibernation p.129
- ▶ Human body p.130
- ▶ Sickness p.225

Sleep patterns

Our sleep follows a pattern of different stages. You go through each stage several times every night.



■ Awake

When we wake up, we become alert and aware of our surroundings.

■ Dreaming

We dream when we're close to waking. We have three to seven dreams a night.

■ Light sleep

Breathing slows down but the brain is still active and you can wake up easily.

■ Shallow sleep

We move around more during shallow sleep than during deep sleep.

■ Deep sleep

This is when our bodies are growing and repairing muscle, tissue, and bones.

Why do we sleep?

Sleep is important because without it our brain and body slowly stop working.



Memory

Our brain deletes useless information and stores more important things while we sleep.



Energy

If we don't get enough sleep, we lack energy and want sugary food.



Healing

Our bodies heal better and faster when we get plenty of solid sleep.



Growth

Our bodies grow and repair muscles and bones while we are in a deep sleep.

You will spend **one-third** of your life asleep. This is around **30 years** in total!

Smell

Smell is one of our senses. When tiny parts of something float through the air and into our nose we identify a smell. The brain tells us what it smells like compared to other things we've smelled before.

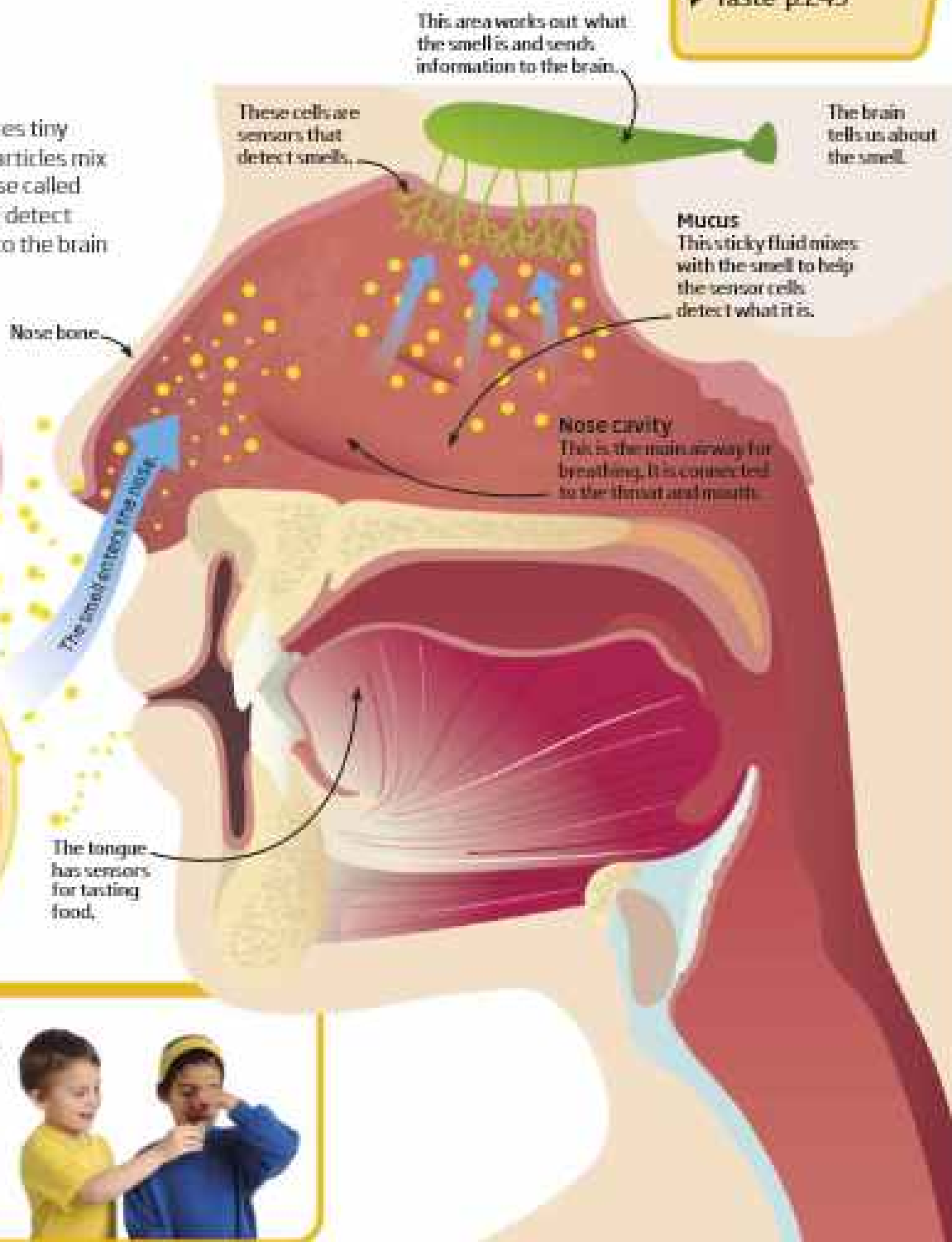
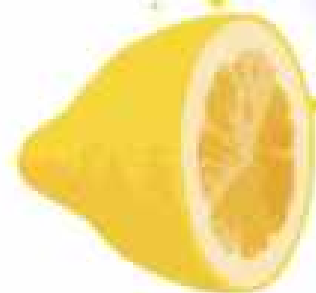
SEE ALSO

- ▶ Brain p.45
- ▶ Hearing p.127
- ▶ Human body p.130
- ▶ Muscles p.173
- ▶ Skeleton p.228
- ▶ Taste p.249

How we smell

Anything that smells releases tiny particles into the air. The particles mix with a sticky fluid in our nose called mucus. Sensors in the nose detect the smell and send signals to the brain to identify it.

We can detect more than **10,000** different smells with our nose!



Taste and smell

Smell and taste are closely linked. If you hold your nose, you will find it harder to figure out what something tastes like.

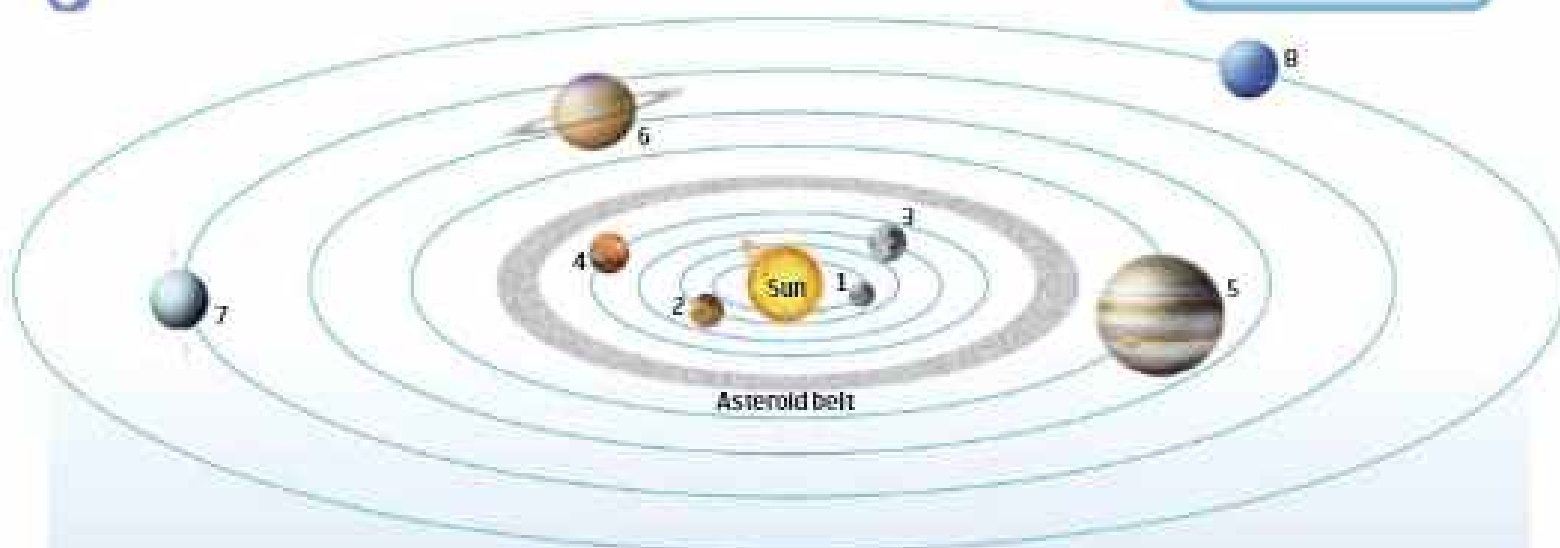


Solar system

The solar system is made up of our star, the sun, and everything that travels around it. This includes eight planets and their moons, asteroids, and comets. Scientists think the solar system formed 4.6 billion years ago, from a massive spinning cloud of gas and dust.

SEE ALSO

- ▶ Asteroids p.30
- ▶ Comets p.68
- ▶ Earth p.83
- ▶ Jupiter p.141
- ▶ Neptune p.183
- ▶ Sun p.247
- ▶ Universe p.263



Orbiting planets

The sun is the center of the solar system. Everything in the solar system travels around, or orbits, the sun.

1. Mercury
Mercury is the smallest planet.

2. Venus
The surface of Venus is super-hot.

3. Earth
Earth has lots of liquid water.

4. Mars
Mars has a red, dusty surface.

5. Jupiter
Jupiter is the largest planet.

6. Saturn
Saturn is famous for the rings around it.

7. Uranus
Uranus is thought to be the coldest planet.

8. Neptune
Neptune has strong winds and giant storms.

Types of planet

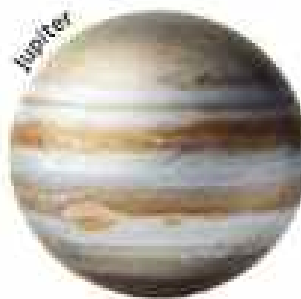
There are three types of planet in the solar system. The rocky planets orbit close to the sun, and the gas and ice giants orbit farther away.



Rocky planets
Mercury, Venus, Earth, and Mars are small and rocky with solid surfaces.



Ice giants
Uranus and Neptune are made of a mixture of gas and icy materials.



Gas giants
Jupiter and Saturn are huge planets made of gas.

Kepler-16b

There are many other solar systems in the universe. Scientists have even found a planet, called Kepler-16b, that orbits two stars.



Solids

Solids are materials that keep their shape. They don't flow like water—they stay where you put them. Solids are useful for making many things, from mobile phones to houses. They are usually hard materials, but soft materials that keep their shape are solid, too.

SEE ALSO

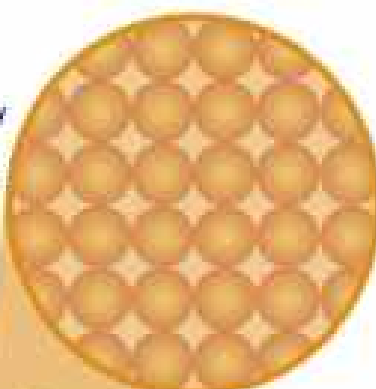
- ▶ Changing states p.57
- ▶ Gases p.117
- ▶ Liquids p.148
- ▶ Materials p.157
- ▶ Metals p.162
- ▶ Plastic p.195

What solids do

Solids can be hard, bendy, strong, squishy, see-through, or magnetic. The behaviors of a solid are called its properties.

Particles

Solids are made from tiny particles that are close together. If enough heat is added, they start to move past each other to become liquid.



Solids make a pile, not a pool or puddle.



Solid rocks become liquid lava when heated to very high temperatures.

Solids keep their shape.

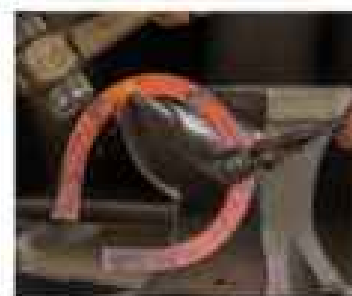
Cutting solids

We can cut solids into different shapes. Wood is a solid material that comes from trees. We cut and shape it to make furniture.



Shaping solids

Metal is a hard solid we can find underground. We heat metal to make it soft, then bend and hammer it into the shape we want. When it cools, metal keeps its shape.



A hot metal horseshoe

Making new solids

We can make new solid materials by mixing other materials together. For example, gelatin is made by adding hot water to gelatin powder to make a liquid. The liquid then cools into a solid.

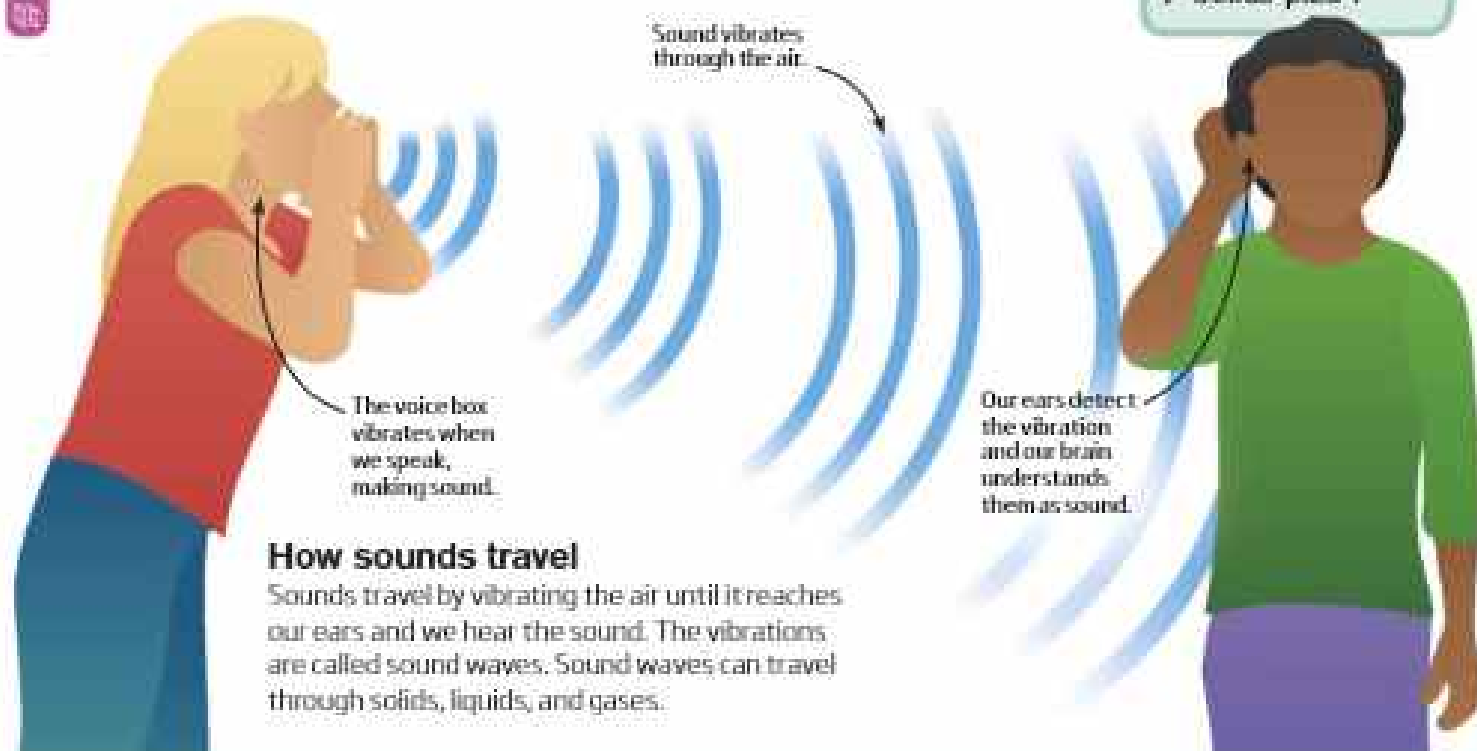


Sound

A sound is made when something vibrates. The bigger the vibration, the louder the sound is. The faster something vibrates, the higher pitched the sound is. Sound vibrations travel through things to get to our ears.

SEE ALSO

- ▶ Gases p.117
- ▶ Hearing p.127
- ▶ Liquids p.148
- ▶ Musical instruments p.175
- ▶ Music pp.176-177
- ▶ Solids p.234

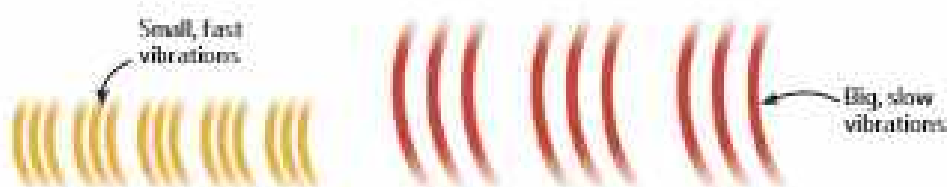


How sounds travel

Sounds travel by vibrating the air until it reaches our ears and we hear the sound. The vibrations are called sound waves. Sound waves can travel through solids, liquids, and gases.

Volume

Volume changes depending on how big a vibration is. The bigger the vibration, the louder the sound. How high or low a sound is depends on how fast something vibrates. The faster the vibration, the higher the sound.



Things sound louder when they are close to us.

Small drum
We can hit a small drum harder to make it louder, but it vibrates at the same speed so it has a constant note.



Big drum
The big drum vibrates more slowly, so it has a lower note compared to the small drum. It's bigger, so it makes bigger vibrations that are louder.



South America

The continent of South America is surrounded by ocean, except at the top where it joins with North America. One-third of the continent is covered by a huge jungle, called the Amazon rain forest. The Andes mountain range stretches all the way down one side of South America.

SEE ALSO

- ▶ Amphibians p.15
- ▶ Explorers p.96
- ▶ Incas p.132
- ▶ North America p.184
- ▶ Rain forests p.204
- ▶ World p.275

The giant tortoises of the Galápagos Islands can live for more than 150 years.



Giant tortoise

About South America



Population: 422.5 million



Highest point: Aconcagua



Lowest point: Laguna del Carbón



Biggest desert: Atacama Desert



Longest river: Amazon

Amazon rain forest

The world's biggest forest, the Amazon, is home to thousands of species of plants and animals. Many native, or indigenous, tribes live in the forest, too.

The skin of the golden poison dart frog is covered in a deadly poison.



Andean condor



Virgin de la Paz



Pitcher plant



Bogota Cathedral



Anaconda



Amazon theater



Canoes



Piranha



Kapok tree



Capybara



Qualea grandiflora



Flute



Llama



Reed boat



Salt lakes



Pampas grass



Soccer



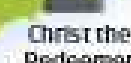
Polo



Andean goose



Leatherback turtle



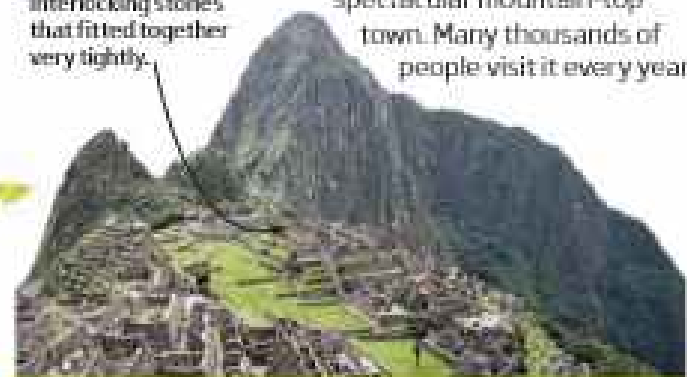
Christ the Redeemer

This soapstone statue is 128 ft (39 m) high. It overlooks the Brazilian city of Rio de Janeiro.

Machu Picchu

Built by Inca emperor Pachacuti in 15th-century Peru, Machu Picchu is a spectacular mountain-top town. Many thousands of people visit it every year.

Inca builders used interlocking stones that fitted together very tightly.



Perito Moreno Glacier



Magellanic penguin

Space travel

Space travel is how we explore the solar system and learn about our place in the universe. Most space travel is done using robot spacecraft called probes. The farthest in space that humans have traveled themselves is to the moon.

SEE ALSO

- ▶ Astronauts p.31
- ▶ Moon p.171
- ▶ Exploration pp.180-181
- ▶ Radio p.203
- ▶ Robots p.212
- ▶ Solar system p.233

Humans in space

To get into space, people use super-powered spaceships. Space shuttles like *Atlantis* were used to take people into space for 30 years. Now people use the Russian *Soyuz* spacecraft.

It takes a robot spacecraft about **six months** to travel to **Mars**.

The outside fuel tank is filled with liquid hydrogen and oxygen to power the shuttle's engines.

Astronauts sit in the cockpit.

Solar cells use the sun's power to make the probe work.

Juno probe

Magnetometer measures magnetic fields.

Robots in space

Probes gather data using cameras, magnetometers, and radars, then send the data back to Earth.

This booster rocket gives extra power.

Launch of the space shuttle *Atlantis*

Extreme environment

Space is not an easy place for people to be in. It can be both extremely hot and cold. There are dangerous rays from the sun and there is no air to breathe. Spaceships and stations are carefully designed to keep astronauts safe.



Astronaut Karen Nyberg washes her hair on board the International Space Station.

Spiders

Spiders are animals with two body segments and eight legs. They are predators, hunting other small animals to eat. They do not chew their food, but turn it into liquid before sucking it up.

SEE ALSO

- ▶ Animal groups p.22
- ▶ Animal homes p.23
- ▶ Food chains p.107
- ▶ Insects p.134
- ▶ Invertebrates p.139
- ▶ Sight p.226

Tarantula

Some of the world's largest spiders are tarantulas. As they get bigger, their old skin comes off and they grow a new one.



Tarantulas bite with their venomous fangs, but their bite is less harmful to humans than a bee's sting.

Leg hairs help tarantulas feel the vibrations of other animals nearby.

Spiders can grow back their legs if they break off.

The row of eyes help the jumping spider see all around it.



Jumping spider

Jumping spiders can jump 30 times their body size. They use their good eyesight to spot other animals.



The body is designed for speed.

Huntsman spider

These spiders don't build webs, but hunt and forage for insects. Females can go for three weeks without eating.

Spider webs

Many spiders build webs, using silk that they make in their bodies. Some spiders use their web to trap and store the insects they eat.





Sports

Sports are physical activities performed by individuals or teams of players. There are usually rules to sports, and team sports often take place in a set amount of time. Individuals and teams compete against one another to achieve the best result or the highest score.

SEE ALSO

- ▶ Ancient Greece p.18
- ▶ Bicycles p.36
- ▶ Dance p.76
- ▶ Games pp.240-241
- ▶ School pp.272-273

Athletics

Athletics is a group of sports that take place around a running track or on a sports field. Big athletic competitions such as the Olympics have lots of events.

Sprinting
A sprint is a race that is shorter than 400 m.

Spectators
Spectators watch sports and cheer on their favorite athletes.

Archery
Archers use a bow to fire arrows at a circular target.

Javelin
A javelin is a long spear thrown as far as possible.

High jump
High jumpers compete to jump the highest over a bar.

Shotput
Athletes compete to throw a small, heavy ball the farthest distance.

Judges

Long jump
Long jumpers try to launch themselves as far as possible along a sandpit.

Hurdles
Athletes jump over a series of obstacles called hurdles on a running track.

Team sports

Sports such as football, rugby, and cricket are played between two opposing teams. The team that scores the most goals or gets the highest number of runs or points wins the game.

Soccer teams pass a round ball by foot.

Basketball teams pass a round ball by hand.

Football players pass an unusually shaped ball by hand.

Gymnastics

Strong gymnasts use their bendy bodies for moves including flips and handstands. They use the floor as well as props such as balancing beams.

Gymnasts can do impressive moves while standing on their hands.

Winter sports

In cold weather, competitors use snow and ice to ice-skate, ski, snowboard, and bobsled. Many winter sports involve performing tricky moves, or racing.

Skier

Games

A game is an activity or sport played using a set of rules. Games are played by individuals or by teams of players, who compete against each other. Some games use balls and bats or rackets and are played on special courts; others use boards with specially designed game pieces.

Board games

Tabletop games, such as chess and backgammon, are played on specially built boards. Each player has a number of pieces, such as chessmen, that they move according to a set of rules. Board games were first played in ancient Egypt 5,500 years ago.

A chessboard has a grid of black and white squares.



The start of a game of chess

Stone ring to hit the ball through.



Ball and wall games

The ancient Mayans of Central America played a ball game on long, narrow, stone-walled courts. Players used a solid rubber ball, which they had to keep in play by using only their hips and arms—they weren't allowed to touch it with their hands or feet.

Stone-walled Mayan ball court

Racket sports

Tennis, badminton, and squash are sports played using a racket on a special court. In tennis, opponents hit the ball across a central net, and in badminton they hit a shuttlecock. In squash, the players hit the ball against a wall.



The tennis racket frame has strings pulled tightly across it.

Ping pong ball



Australian rules football



Handball



Baseball



The first Olympic Games were held in ancient Greece in 776 BCE.

The Olympics

Every four years, athletes compete for their country in the Olympic Games. The Olympics includes lots of different sporting events, including athletics, gymnastics, and team sports.



Olympic sprint relay race

Lacrosse ball



Football



Soccer ball



Golf ball

The first computer game was developed in 1947. Players fired a dot at a target.

Computer games

Computer games are played on a computer or through a console on a TV screen. Many games include special effects and music. They can be one-player or multi-player games.



Volleyball



Basketball



Rugby ball

Ball sports

Ball sports are among the most popular games of all. Teams of up to 15 players per side play in matches of soccer, rugby, basketball, football, Australian rules football, cricket, and other games. Huge crowds watch these matches, many of which are shown on television.

Stars

Stars are balls of very hot gas, deep in space. From Earth, they look like tiny dots but are actually huge. The smallest ordinary star is about the same size as Jupiter. Stars shine because the gases inside them constantly crash together in a process called fusion.

SEE ALSO

- Color pp.26-27
- Galaxies p.116
- Light p.147
- Solar system p.233
- Sun p.247
- Temperature p.252

Sizes and colors

Stars come in different sizes and colors. A star's color depends on how hot its surface is. The hottest stars are blue, and the coolest ones are red.

Blue supergiant

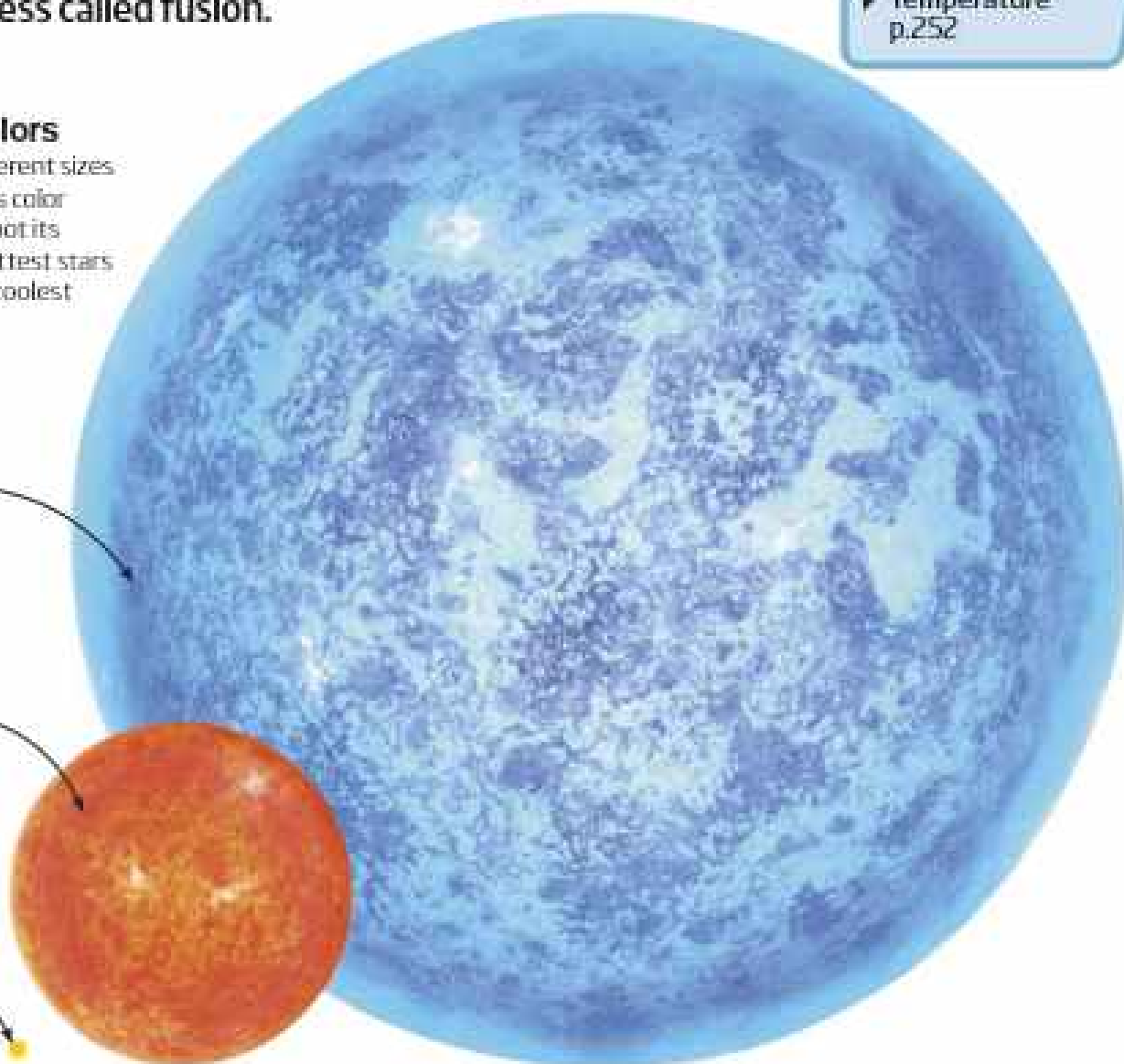
These stars are very young and extremely hot.

Red giant star

Red giants are older stars with cool surface temperatures.

Sun

Our sun is a middle-aged, medium-sized star with a medium surface temperature.



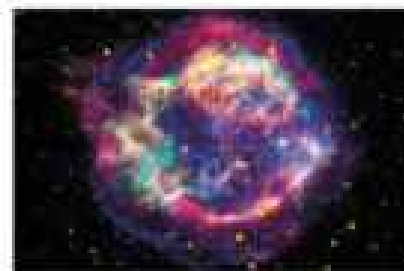
The nearest star

The star closest to the Earth after our sun is Proxima Centauri, a red dwarf with at least one planet. This star is 9,000 times farther away than Neptune, the planet farthest from the sun.



Star deaths

Some stars end their lives in a spectacular explosion called a supernova. Others slowly fade as they run out of energy.



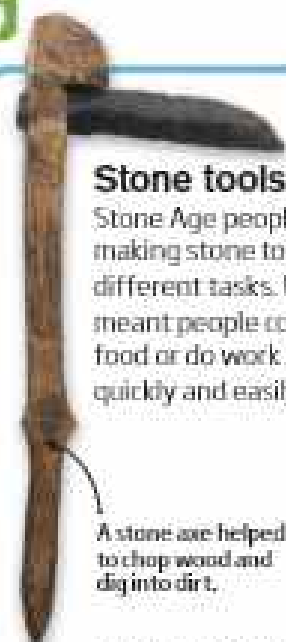
Cloud of material from a supernova explosion

Stone Age

The Stone Age was a period of time that began about 3.3 million years ago and lasted until about 4,000 years ago. In the Stone Age, people made tools out of stone for the first time. They used them to cut meat and plants, to build shelters, and, in the late Stone Age, for farming.

SEE ALSO

- ▶ Art p.28
- ▶ Buildings p.48
- ▶ Caves p.55
- ▶ Early humans p.82
- ▶ Farming p.98
- ▶ Food p.106



Stone tools

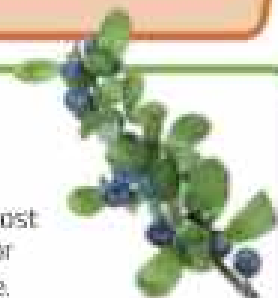
Stone Age people began making stone tools to do different tasks. Using tools meant people could get food or do work more quickly and easily.

A stone axe helped to chop wood and dig into dirt.

People made handaxes to help them cut meat and chop hard plants.

Finding food

Finding food was the most important part of life for people in the Stone Age. They ate wild plants, and animals from land and sea.



Blueberries



Salmon

Bison
Hunting for big animals, such as bison, could be dangerous.



Cave painting

Some Stone Age people made beautiful artwork on cave walls, often showing the animals they would hunt. These paintings are still being found today.



Lascaux cave paintings in France

Paint could be made from powdered fats and minerals.



Buildings

The first Stone Age buildings were made of wood and animal skins. By the end of the Stone Age, people had begun building large stone structures called megaliths.

Stonehenge in England is a famous Stone Age megalith that is still standing today.



Homes

A home is a house or building in which you live. That home might be built of canvas, stone, bricks, wood, or ice. It could even be carved out of solid rock. Homes can be built separately as single, detached houses or built together as row houses or terraces. Some homes are built on top of each other together in blocks called apartments.

Early homes

Some early people made their homes in caves or holes dug into sides of mountains and hills. Others cut down trees to build simple wooden huts, which they draped with animal skins.

Some early people built simple grass shelters. The layering of grasses is called **thatching**.

Caves were easy to turn into homes, as they didn't have to be built.

Caves in Cappadocia, Turkey

Grand palaces

The kings and queens of Europe were very rich and lived in magnificent palaces and castles. They feasted in great halls and received visitors in grand throne rooms.



Neuschwanstein Castle in Germany

Eco homes

Today, some homes are specially built to work with the natural world, or environment. They don't use up as much energy as ordinary homes.

Solar panels on the roof make energy from sunshine.



Rainwater is collected for reuse.

Walls have added layers to save heat (this is called **insulation**).



Plan for
3D-printed
houses on the
planet Mars

Future homes

What will homes of the future look like? Some might be built using a method called "3D printing." With this method, robots add material layer upon layer to create a three-dimensional shape.

Igloo



Extreme homes

In the cold Arctic, some Inuit people build shelters called igloos out of blocks of ice. Igloos keep out the wind and are warm inside.

Houseboat
in India



Moving homes

Some people live in homes that can move from place to place. They float in houseboats on water or live in wheeled caravans towed by cars or horses.

Architecture

Architecture is the art of designing and making buildings. In Barcelona in Spain, architect Antoni Gaudí was inspired by nature and decorated his buildings with different materials and colorful patterns.

House
designed by
Antoni
Gaudí



Storms

Storms are powerful winds that often bring rain, thunder and lightning, snow, hail, dust, or sand. Storms can cause serious damage if they have very high wind speeds or heavy rain that leads to flooding. Tornadoes, hurricanes, and thunderstorms are all types of storm.

SEE ALSO

- ▶ Climate change p.60
- ▶ Clouds p.64
- ▶ Electricity p.87
- ▶ Erosion p.93
- ▶ Water cycle p.270
- ▶ Weather p.271

Tropical giants

The biggest and most destructive storms, called hurricanes or typhoons, happen in tropical areas, above warm water. They begin when groups of smaller storms spin together into a spiral shape.



Hurricanes are given **people's names**, such as Alex, Matthew, and Patricia.



Thunderstorms

Storms with thunder and lightning are common in summer. They often have heavy rain or hail that can break things and cause floods.



Tornadoes

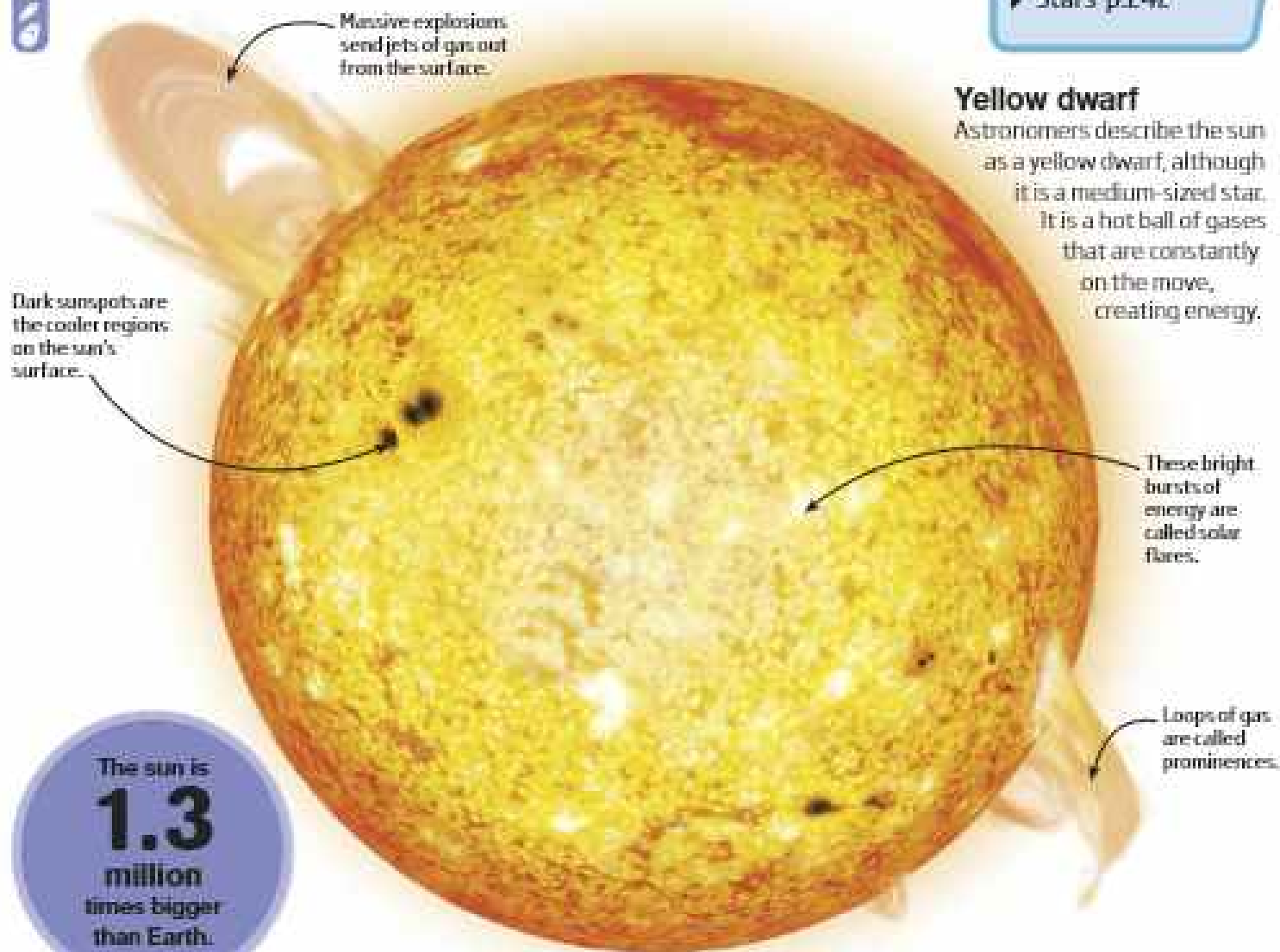
Tornadoes are fast-spinning columns of air that form during massive thunderstorms. A tornado can destroy everything in its path.

Sun

The sun is the star at the center of our solar system. It is one of at least 100 billion stars in our Milky Way galaxy. The sun gives off light and heat, making it possible for life to exist on Earth.

SEE ALSO

- Atmosphere p.33
- Atoms p.34
- Gases p.117
- Magnets p.151
- Milky Way p.167
- Solar system p.233
- Stars p.242



Yellow dwarf

Astronomers describe the sun as a yellow dwarf, although it is a medium-sized star. It is a hot ball of gases that are constantly on the move, creating energy.

Auroras

Toward the top and bottom of the Earth (at its poles), particles from the sun meet Earth's magnetic field. This colorful glow is called an aurora.



Final stages of the sun

In about 5 billion years time, the sun will have used up most of the gases that keep it shining. It will collapse into a small, very hot white dwarf. It will then slowly cool down and eventually fade away.



Symmetry

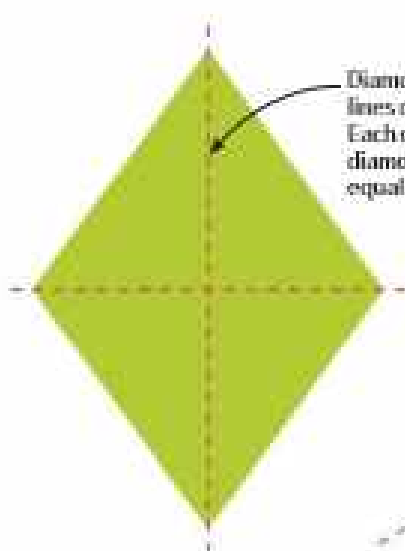
There are two types of symmetry. Reflective symmetry is when lines can be drawn to divide a shape into identical parts. Rotational symmetry is when a shape can be turned around (rotated) and still look the same.

SEE ALSO

- ▶ Art p.28
- ▶ Flowers p.103
- ▶ Human body p.130
- ▶ Shapes p.222
- ▶ Sight p.226
- ▶ Games pp.240-241

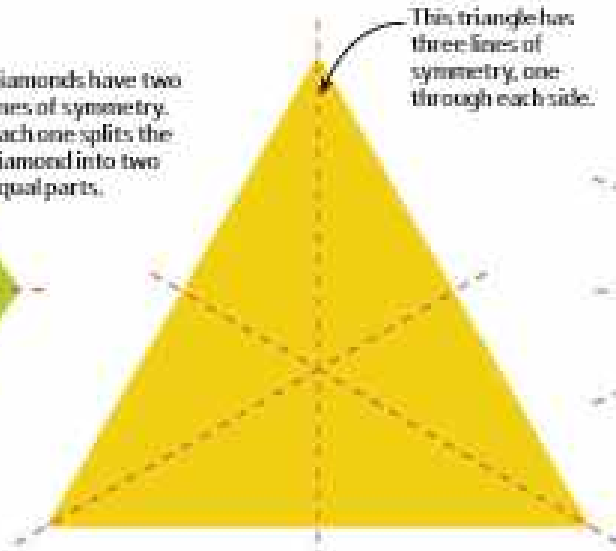
Reflective symmetry

Lines of symmetry are like folding a shape in half. If a shape is symmetrical, both sides of the fold will look the same. Shapes can have more than one line of symmetry.



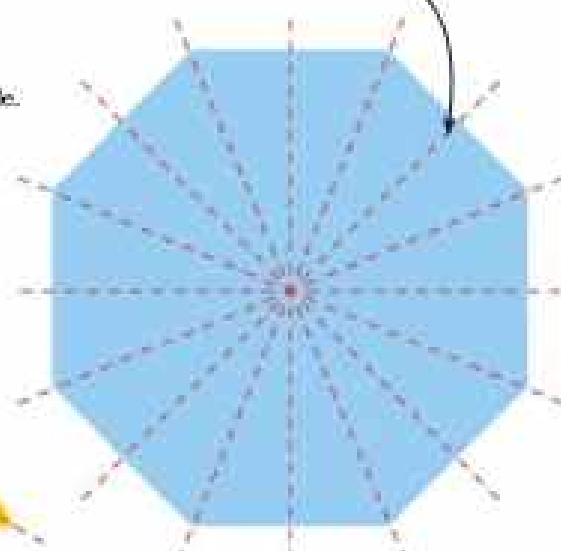
Diamond

Diamonds have two lines of symmetry. Each one splits the diamond into two equal parts.



Equilateral triangle

This triangle has three lines of symmetry, one through each side.



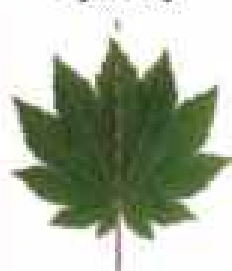
Octagon

A regular octagon has eight lines of symmetry, through its sides and angles.

Symmetry in nature

Nature has many examples of both types of symmetry. Most animals, including humans, are roughly the same on both sides - they have a line of symmetry down the middle.

Reflective symmetry



Leaf

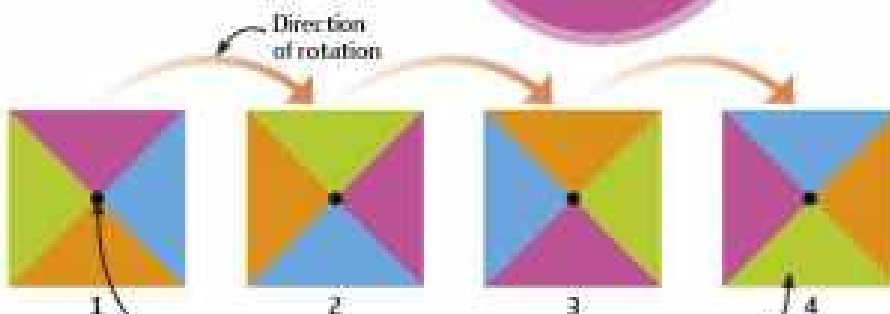
Rotational symmetry



Starfish

Rotational symmetry

If a shape can be turned and its outline looks the same, it has rotational symmetry. This sequence shows how a square has rotational symmetry.



1

2

3

4

Point of rotation

A square can turn within its shape four times.

A circle has a never-ending number of lines of rotational symmetry.

Taste

When we eat food, tiny bumps in our mouth sense if it tastes sweet, sour, salty, savory, or bitter. The mouth sends information to our brain, which works out the flavors of what we are eating and drinking.

SEE ALSO

- ▶ Color pp.26-27
- ▶ Brain p.45
- ▶ Digestion p.79
- ▶ Food p.106
- ▶ Fruit and seeds p.115
- ▶ Smell p.232

Taste and smell

The sense of taste from the tongue and the sense of smell from the nose work together to tell us how our food tastes.

Bitter

Bitter-tasting foods include olives, coffee beans, and cocoa beans.



Salt

Salt is added to dishes to help them to taste better. Our bodies need a small amount of salt to stay healthy, but too much salt is bad for us.



Taste buds

The little bumps in our tongue and mouth have tiny taste sensors in them called taste buds. We have around 10,000 taste buds.



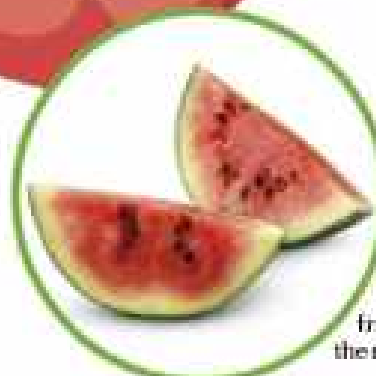
Sour

Lemons, limes, and grapefruit taste sour. A sour taste can also be a warning that food has gone bad.



Savory

Savory flavors include soy sauce and parmesan cheese.



Sweet

Foods such as honey and fruit taste sweet because of the natural sugar they contain.

The body regrows all of our taste buds every two weeks.

Telephones

Telephones allow people to speak to each other from anywhere in the world. They turn the sound of our voices into signals, which are sent through radio waves or cables to another phone. That phone then changes the signals back into sound.

SEE ALSO

- Codes pp.66–67
- Communication p.69
- Computers p.71
- Electricity p.87
- Hearing p.127
- Internet p.138

Where does your voice go?

When you speak into a phone, the sound of your voice is turned into electrical signals. A network of telephone lines and cell towers let us speak to people over long distances.



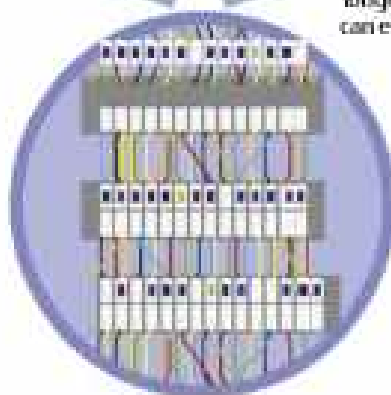
Cell tower
These towers send and receive signals between mobile phones and a telephone exchange.



Telephone lines
Phone lines carry phone signals long distances by holding the wires up above the ground. For longer distances, cables can even go underwater.



Mobile phones
Mobile phones send and receive signals as radio waves. They don't work if they are too far away from a cell tower.



Telephone exchange
This place connects telephone calls using computers. It sends incoming signals to the right phone.



Wired phones
Landline phones have a cable that plugs into a wall. They send a signal through a network of wires.

Past and present

The way that phones send and receive signals has changed since they were first invented. Early telephones sent sounds short distances through pipes or tubes. From the 1800s, telephones sent electrical signals through wires. Mobile phones use radio waves to pick up signals.



First telephone
The telephone was invented in 1876 by Alexander Graham Bell, a Scottish music teacher.



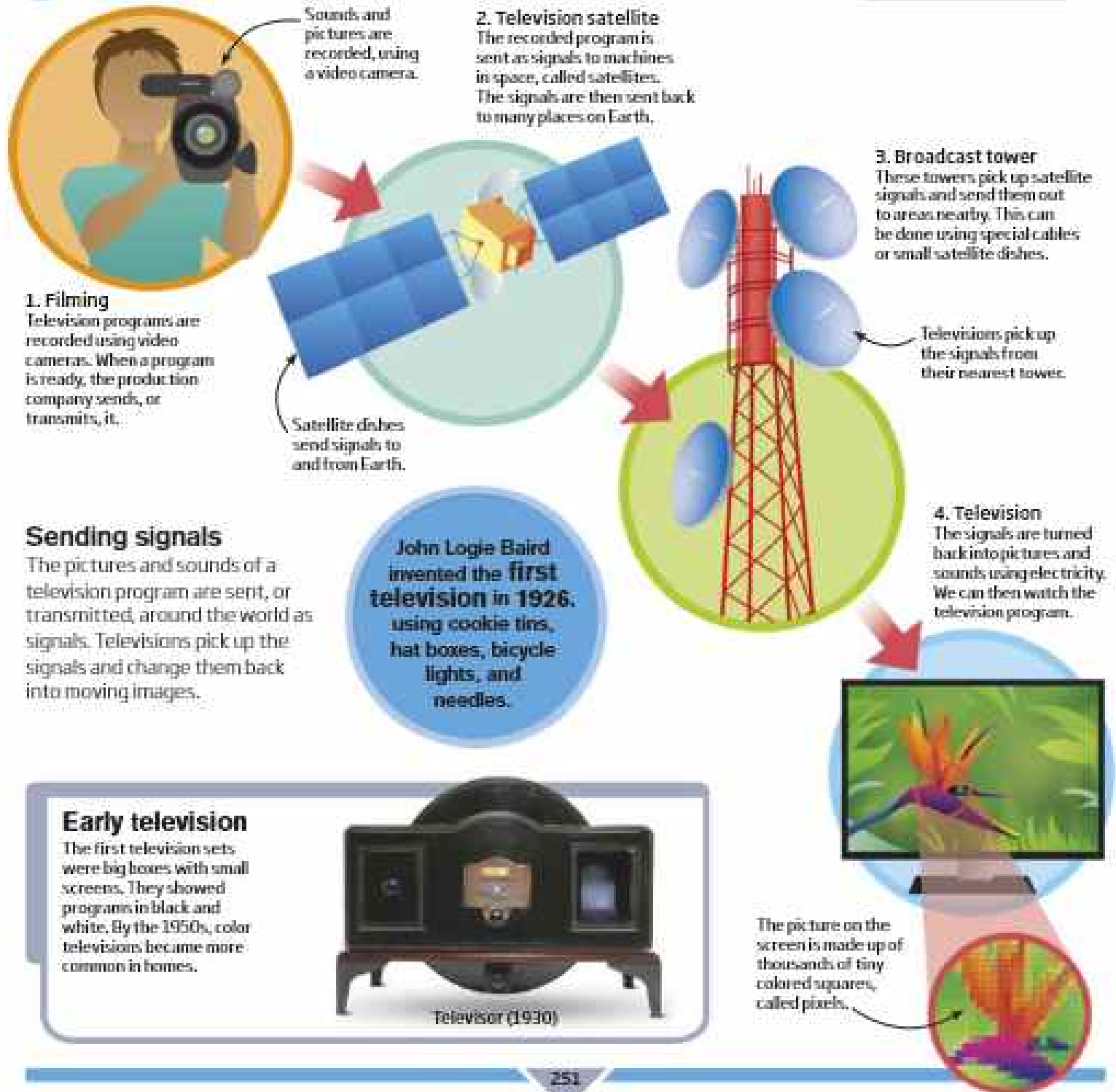
Smartphone
A smartphone is a pocket computer that can be used to make phone calls, record videos, and play games.

Television

Television lets people watch the news, documentaries, films, cartoons, and game shows without leaving the house. It is one of the world's most popular forms of entertainment.

SEE ALSO

- Communication p.69
- Factories p.97
- Hearing p.127
- Radio p.203
- Satellites p.215
- Sight p.226

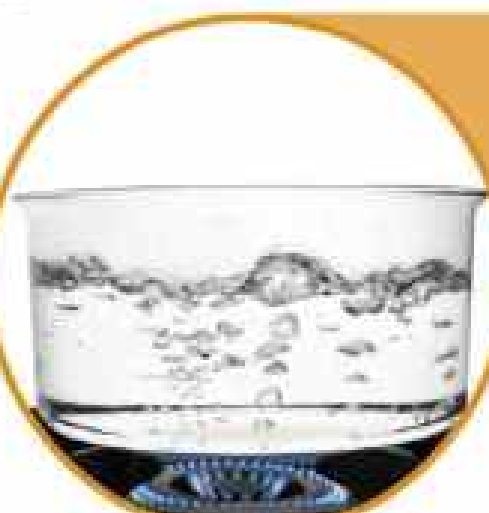


Temperature

Temperature is a measurement of how hot or cold something is. We measure temperature in degrees of Fahrenheit ($^{\circ}\text{F}$) or Celsius ($^{\circ}\text{C}$). We can use a thermometer to measure the temperature of the air, liquids, or the human body.

SEE ALSO

- Changing states p.57
- Gases p.117
- Human body p.130
- Liquids p.148
- Measuring p.159
- Solids p.234



Water boils

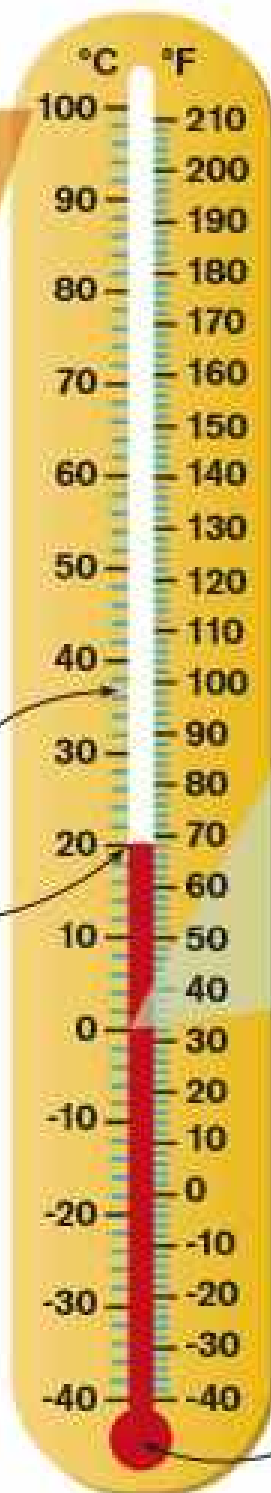
When liquid water reaches a temperature of 212°F (100°C), it boils and changes from a liquid into a gas called water vapor.

Body temperature

Room temperature
We use the term "room temperature" to describe normal conditions. The average temperature of a room is 68°F (20°C).

Body temperature

A healthy body temperature is about 98.6°F (37°C). Doctors can check body temperature by putting an electric thermometer in our mouth or ear.



Thermometer

Thermometer

A thermometer is an instrument that measures temperature. We read the temperature off it in degrees.

Digital thermometer



Water freezes

When the temperature of the air around us gets to 32°F (0°C), liquid water freezes and becomes a solid called ice.

Lightning is $53,540^{\circ}\text{F}$ ($29,727^{\circ}\text{C}$), which makes it the hottest natural thing on Earth.

The red liquid goes up or down to show the temperature.

Theater

People have acted out stories for thousands of years. These stories are plays, and the theater is the exciting place in which they are performed. Theater performers try to make you believe that the characters in a play are real, and that the events they show are actually happening.

SEE ALSO

- ▶ Ancient Greece p.18
- ▶ Books p.44
- ▶ Buildings p.48
- ▶ Clothing pp.52-63
- ▶ Film p.100
- ▶ Music pp.176-177

On stage

The area where plays are performed in a theater is called the stage. Many actors can be on stage at the same time. Music, sounds, and lighting on stage make the play more exciting.

British author Agatha Christie's play *The Mousetrap* has been performed more than 25,000 times!

Actor

The people in plays who pretend to be characters are called actors.

Costume

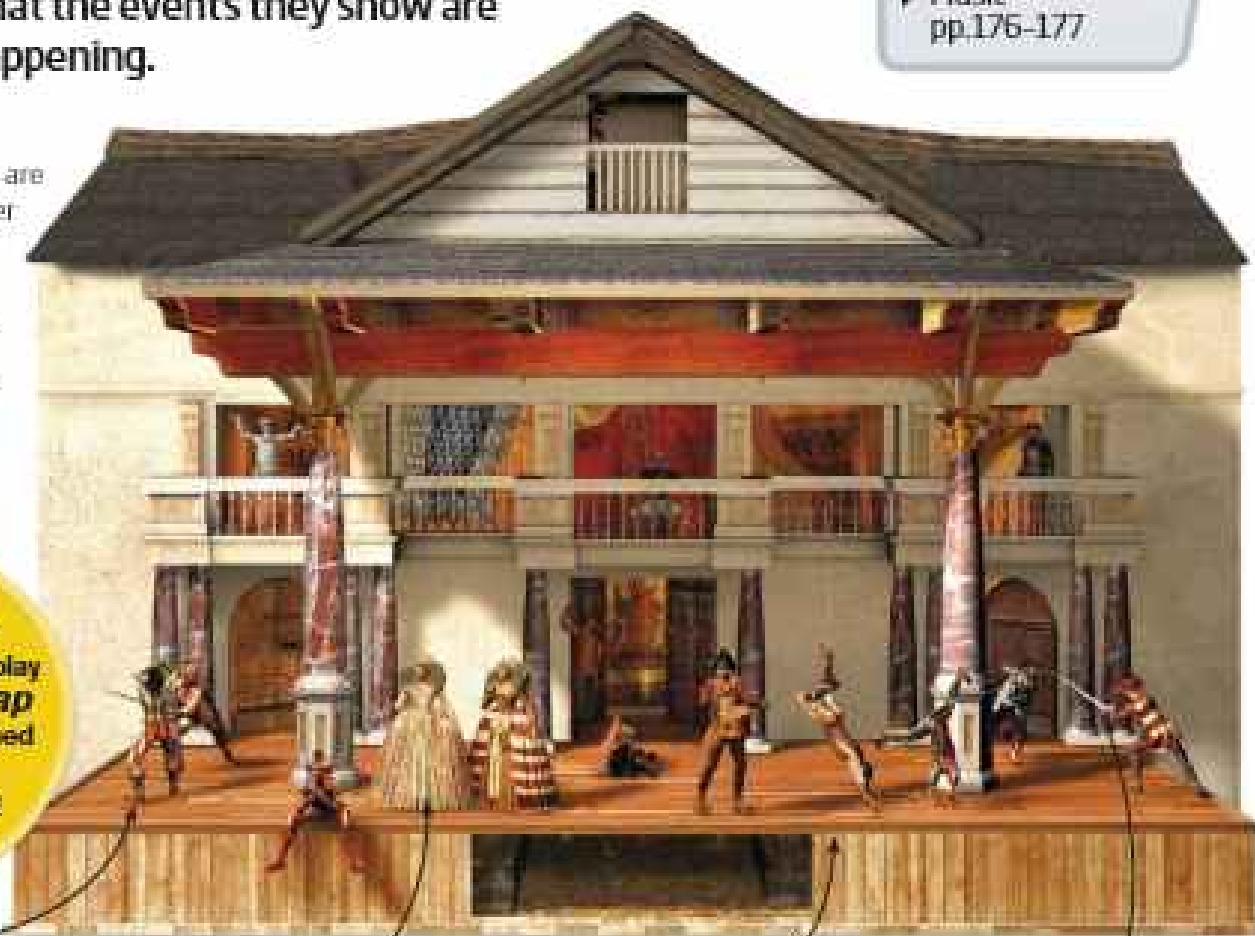
The clothing an actor wears is called their costume.

Stage

The stage is usually in front of the audience.

Prop

Props are things used in plays to make them more lifelike, such as weapons.



Ancient plays

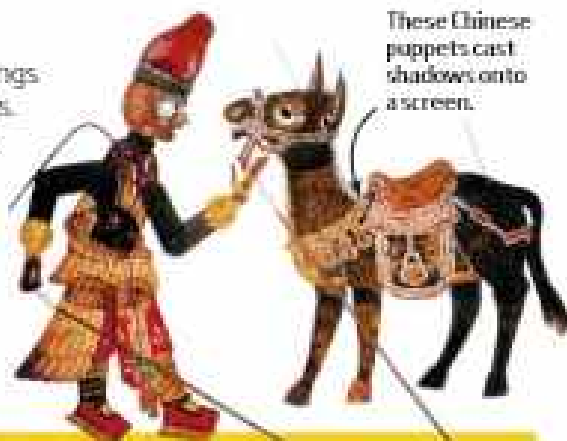
The first plays were written in ancient Greece, around 700 BCE. Greek playwrights mainly wrote sad plays, called tragedies, and funny plays, called comedies.



Greek theater masks

Puppets

Models controlled by strings or rods are called puppets. They are given voices by performers and tell stories on small stages. Puppet shows have been performed for at least 3,000 years.



Tides

Tides are daily changes in the level of the sea on the coast. They are mainly caused by the moon's gravity, which is an invisible force that pulls on the Earth. When the water is high up the coast, it is called high tide, and when it falls it is low tide.

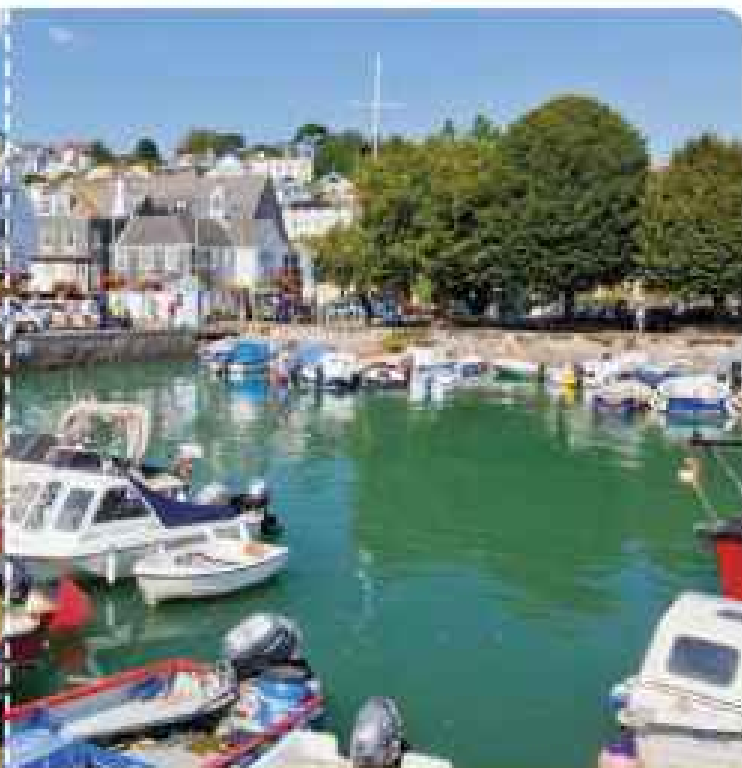
SEE ALSO

- ▶ Day and night p.77
- ▶ Gravity p.125
- ▶ Moon p.171
- ▶ Oceans and seas p.187
- ▶ Seashore p.220
- ▶ Sun p.247



Low tide

When the pull of the moon is weak, water levels fall and the tide goes out.

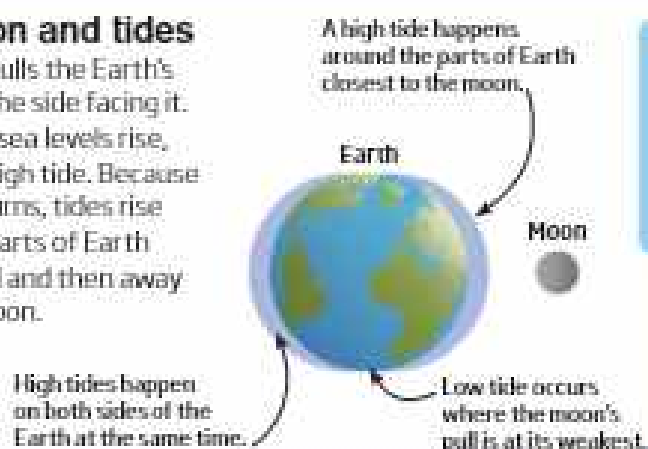


High tide

When the pull of the moon is strong, water levels rise and the tide comes in.

The moon and tides

The moon pulls the Earth's oceans on the side facing it. This makes sea levels rise, creating a high tide. Because the Earth turns, tides rise and fall as parts of Earth turn toward and then away from the moon.



Living between tides

The part of the coast that is covered up and then uncovered between the tides is called the intertidal zone. Many living things are found here. They have to be tough to cope with battering waves at high tide, and air and sunlight at low tide.

Mussels live on rocks. They shut their shells when the tide is out.



Time zones

Clocks do not show the same time all over the world. If they did, it would be dark at noon and light at midnight in some places. To avoid this, the world is divided into 24 areas called time zones. There is a difference of one hour between time zones that are next to each other.

SEE ALSO

- ▶ Aircraft p.13
- ▶ Clocks p.61
- ▶ Day and night p.77
- ▶ Maps p.155
- ▶ Sun p.247
- ▶ World p.275

Times around the world

Time zones are based on the time in Greenwich, in London, which is known as Greenwich Mean Time. The time in the zones to the west of Greenwich are earlier, and the times in the zones to the east are later.



Sundials

Before time zones were invented, people worked out the local time from the position of the sun in the sky using a sundial. The shadow cast by a pointer on the dial showed the time.



The pointer's shadow shows the time.

Jet lag

Traveling quickly across several time zones can confuse our bodies, which still think they are in the original time zone. This state, called jet lag, can cause tiredness, headaches, and problems sleeping.



Touch

Touching is how we feel the world around us. When we touch something, sensors in our skin send information to our brain. We can tell if things are rough or smooth, hot or cold, and how much something pushes against us.

SEE ALSO

- ▶ Body cells p.41
- ▶ Brain p.45
- ▶ Human body p.130
- ▶ Muscles p.173
- ▶ Skin p.229
- ▶ Temperature p.252

Feeling things

There are tiny sensors in our skin called cells or neurons. These neurons collect information about what we touch and send electrical signals to the brain.



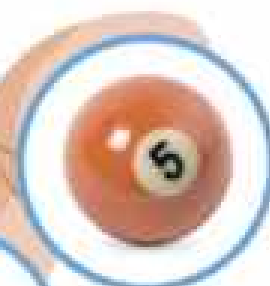
Hard and soft

We can feel how hard things are by how much they push back against our touch.



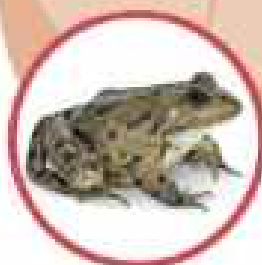
Hot and cold

We can detect if things are hot or cold. If something is too hot, our skin tells us to move away quickly.



Smooth and rough

We are able to feel very small bumps and differences in texture.



Wet and dry

We can tell the difference between wet, sticky, and dry things just by touching them.

Blind people can read by touching a series of tiny bumps on a page called **braille**.

Pain

The neurons in your skin can also detect damage. If we cut or burn ourselves, the neurons send a message to our brain that we feel as pain.

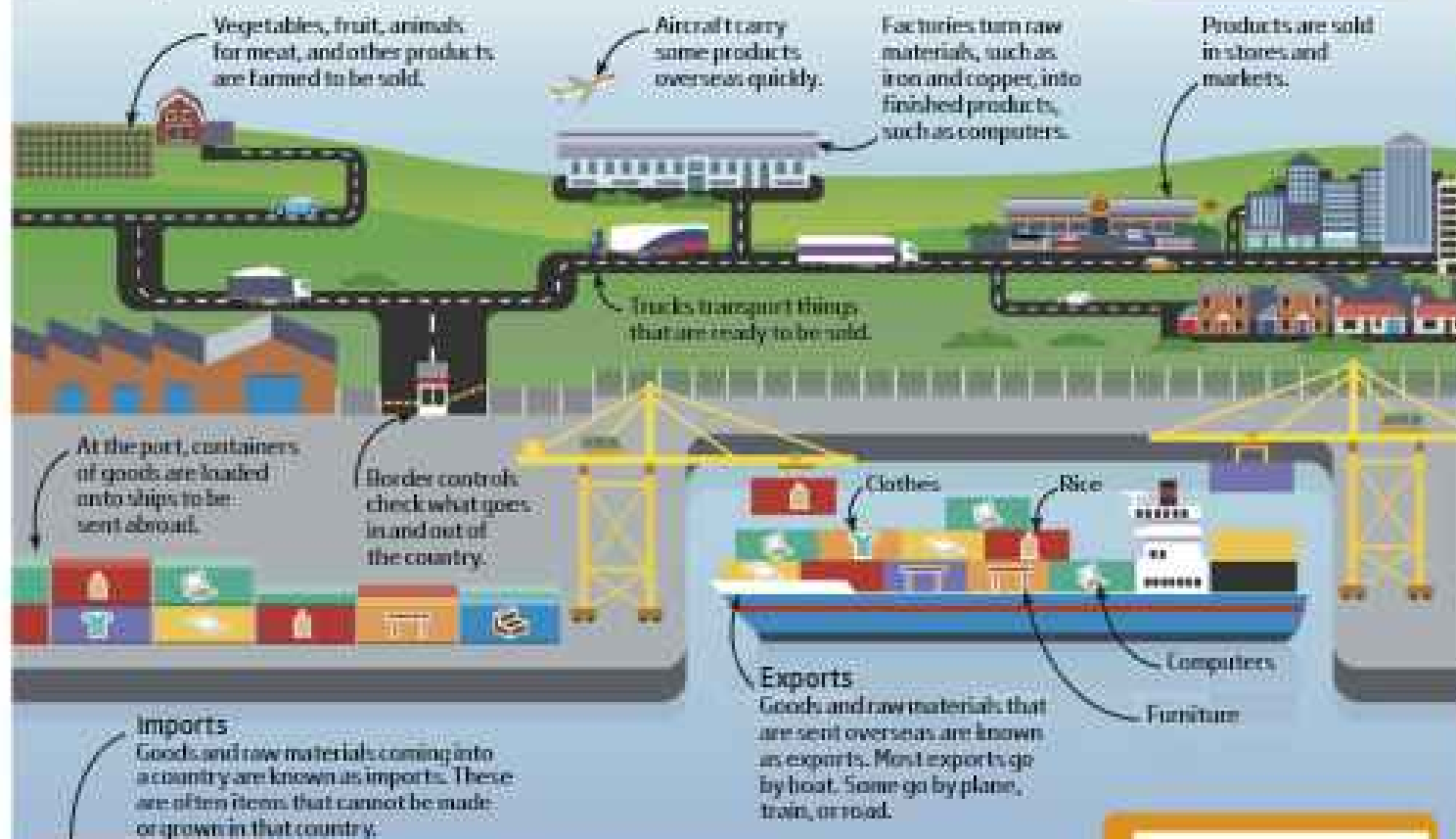


Trade

Trade is buying and selling. We trade raw materials, like metal, to make things, as well as trading the things they are made into, like phones. Everything we eat, wear, and use is the result of trade. You can also buy and sell services, which are jobs people do, such as computer coding.

SEE ALSO

- Farming p.98
- Governments p.123
- Materials p.157
- Money p.169
- Transportation pp.258-259
- Work p.274



Spice trade

One of the oldest trades in the world is the spice trade. Cinnamon, turmeric, and other spices are grown in Asia and used around the world to flavor food. When the trade began, spices were carried by land across Asia.



Cinnamon

International trade

Countries all over the world send goods to one another. They can make money by charging the other country tax (money) for the right to sell its goods within their borders.

Coastguards make sure that ships make it to land safely.



The **story** of...

Transportation

Humans have been inventing new ways of moving from one place to another for thousands of years. At first, people used animals for transportation on land. Later, the wheel was invented, then engines. People started crossing water using rafts and simple dugout canoes, while air travel began with hot-air balloons. We have even traveled into space!

Green travel

The bicycle is one of the most environmentally friendly (green) forms of transportation, because it has no engine to release harmful gases into the air. Other forms of green travel include electric cars and buses that run on clean hydrogen gas.



Horse and cart

Animals

Animals were our first type of transportation other than walking. At first, people rode on them. In 3500 BCE the wheel was invented, and carts and carriages were pulled by horses, oxen, and other animals.



Model of a boat, from ancient Egypt

Crossing water

The first boats were log boats, carved out of tree trunks, and basic rafts made from reeds and sticks. People used them to travel around and also for fishing.

Cyclists and their passengers wear a helmet to keep their head protected if they fall.

Ancient boats were powered by people using oars.

Turning the pedals makes a bicycle's wheels go around.





Model T Ford

Cars for all

In 1908, the Model T Ford became the first car that was cheap enough for many people to buy. More than 15 million were built. Most cars at the time cost nearly \$3,000, but the Model T Ford was only \$850.

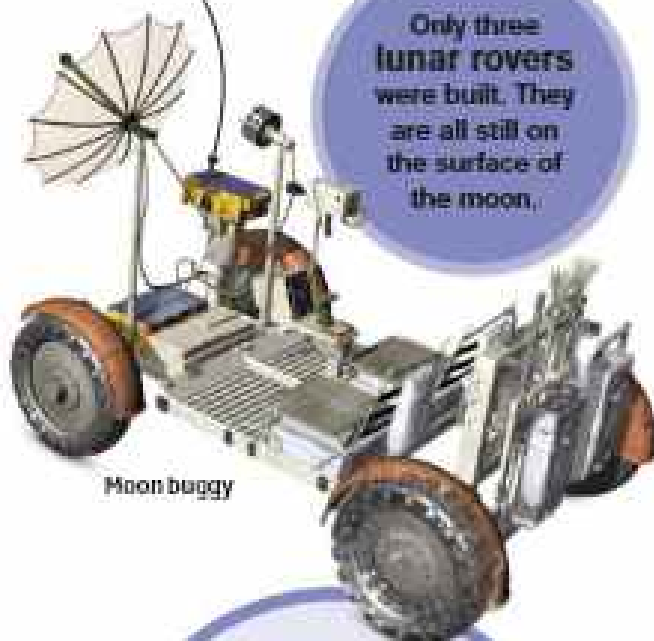
Air travel

A few decades after the first powered flight in 1903, aircraft were developed to take people around the world faster than ever before. Today, the longest nonstop flight takes 17 hours and 27 minutes from New Zealand to Qatar.



Poster for Korean Air Lines

Video camera records moving color images of the moon.



Moon buggy

Only three lunar rovers were built. They are all still on the surface of the moon.

There are more than one billion bikes on the planet.

Lunar Rover

The Lunar Roving Vehicle was designed to transport astronauts on the moon's surface. Three of these battery-powered craft drove on the moon. They could transport two astronauts at speeds of up to 8 mph (13 kph).

Trains

Trains are vehicles that move along tracks. The first trains were powered by steam, but modern trains use diesel, electricity, or even magnets. They are a fast way for passengers to travel and for goods to be transported.

SEE ALSO

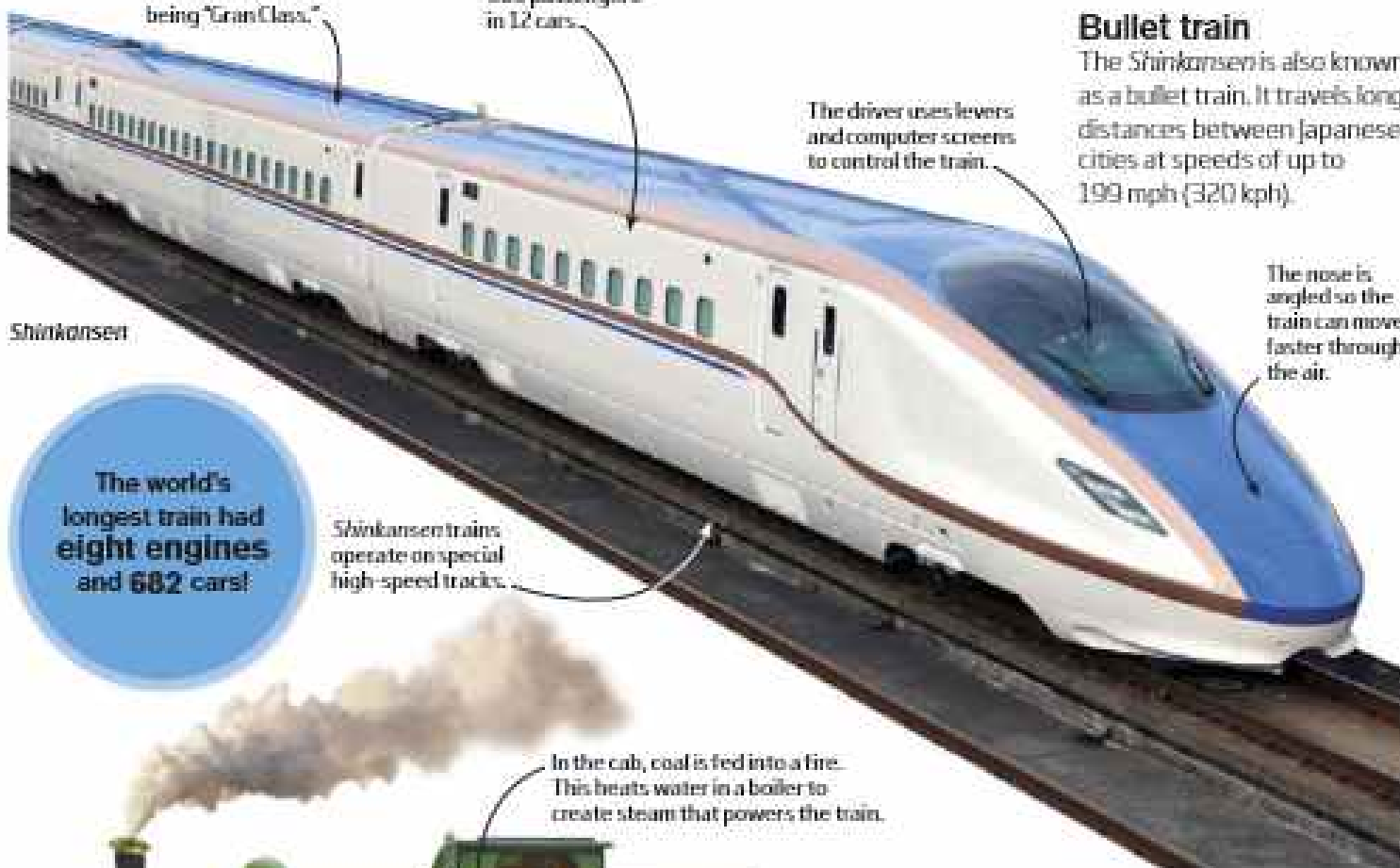
- ▶ Asia p.29
- ▶ Engines p.92
- ▶ Inventions pp.136-137
- ▶ Magnets p.151
- ▶ Trade p.257
- ▶ Transportation pp.258-259

There are three types of seating, the most luxurious being "Gran Class."

Each train can carry more than 900 passengers in 12 cars.

The driver uses levers and computer screens to control the train.

The nose is angled so the train can move faster through the air.



Shinkansen

The world's longest train had eight engines and 682 cars!

Shinkansen trains operate on special high-speed tracks.

In the cab, coal is fed into a fire. This heats water in a boiler to create steam that powers the train.



Aerolite, 1902

The large, powered wheel is called the driving wheel.

Steam engine

The earliest trains used steam for power. Water was heated by a coal fire in the engine. The first working train, Stephenson's *Rocket*, was built in 1829.

Bullet train

The Shinkansen is also known as a bullet train. It travels long distances between Japanese cities at speeds of up to 199 mph (320 kph).

Underground trains

Many cities have underground trains known as subway, or metro, systems. Avoiding the busy traffic above, these trains can quickly move people around a city.



The Paris metro

Trees

A tree is a plant with a woody stem called a trunk. Trees are found all over the world, except in Antarctica. The two main types of tree are deciduous and evergreen.

SEE ALSO

- Forests p.109
- Fruit and seeds p.115
- Habitats p.126
- Materials p.157
- Photosynthesis p.191
- Plants p.194

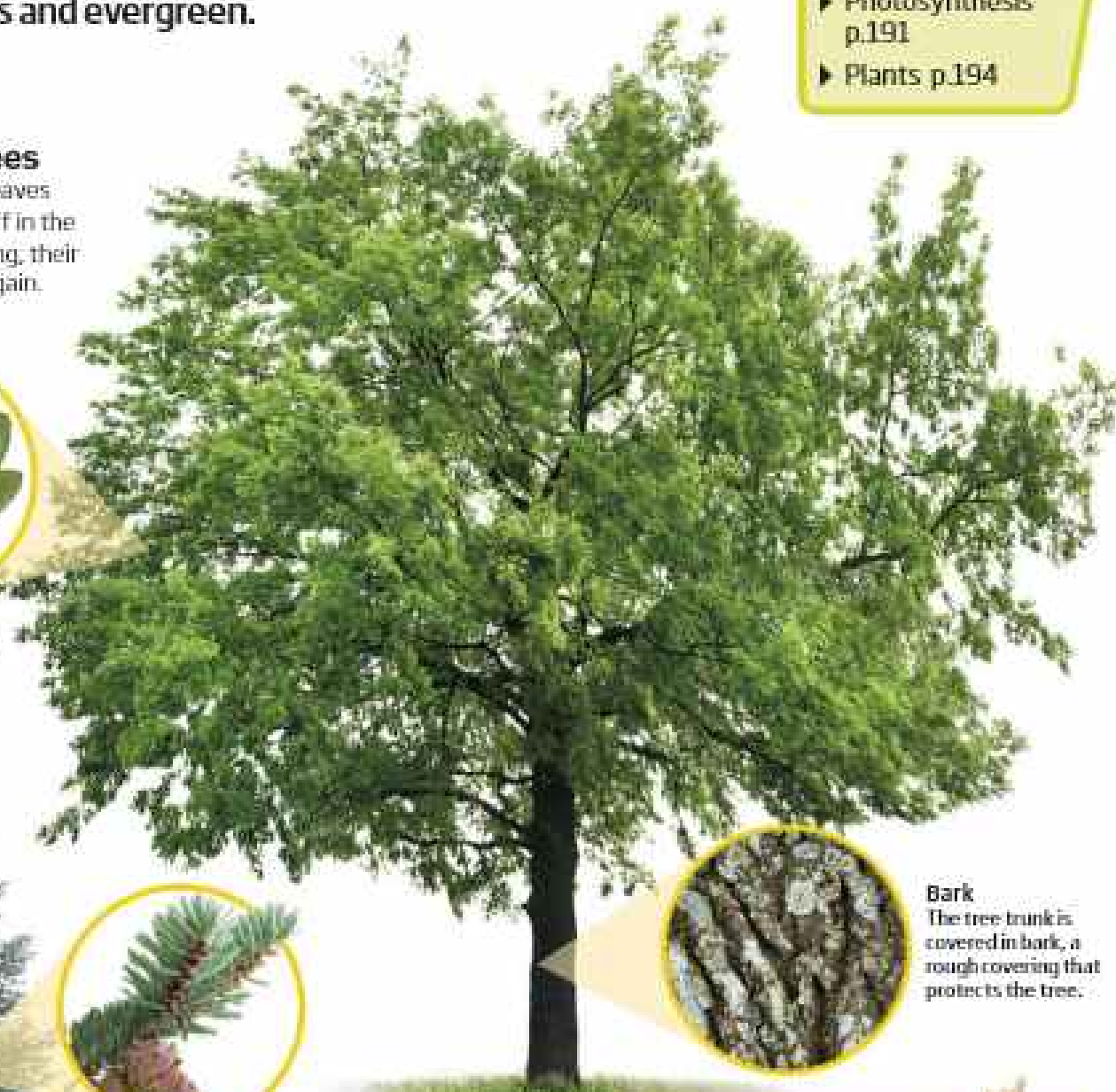
Deciduous trees

These trees have leaves that die and drop off in the autumn. In the spring, their leaves grow back again.



Leaves

Leaves make the food a tree needs to grow. They come in all shapes and sizes depending on the type of tree.



Oak tree



Bark

The tree trunk is covered in bark, a rough covering that protects the tree.



Sicilian fir



Needles

Needles are leaves that are curled up into a tough pointed shape.

Evergreen trees

These trees keep their leaves all year round. They have flat, hard leaves called needles or scales.

Tree rings

You can tell how old a tree is by counting the rings in its trunk. Each ring shows a year in the tree's life.



Turkish Empire

For hundreds of years, the Ottoman Turks ruled one of the largest empires the world has ever seen. It stretched from North Africa across the Middle East to the Indian Ocean. The Ottomans were Muslims, but they ruled over many different people.

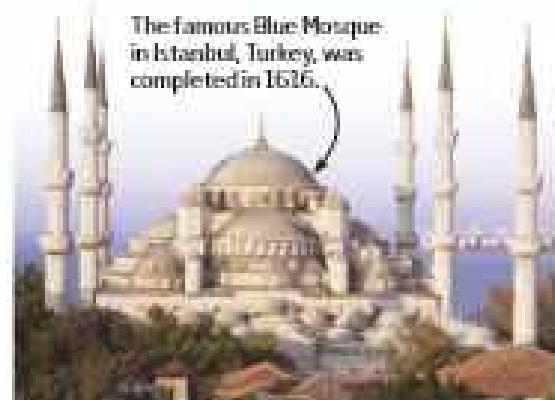
SEE ALSO

- ▶ Africa p.12
- ▶ Asia p.29
- ▶ Buildings p.48
- ▶ Crafts p.75
- ▶ Europe p.94
- ▶ Flags p.102
- ▶ Religion p.208

Ottoman leader

In 1299, a Turkish leader, Osman I, founded what was to become a new Turkish empire – the Ottoman Empire. The same family of sultans ruled this empire for 600 years.

Osman I led the Turks from 1299 to 1323.



The famous Blue Mosque in Istanbul, Turkey, was completed in 1616.

Religious empire

The Ottomans were Muslims, which means they followed the religion of Islam. They built grand buildings called mosques to pray in. Many of their mosques are still in use today.

The capital of the Ottoman Empire was **Constantinople**, which is now known as **Istanbul**.

Flower patterns were often used to decorate Iznik pottery.

Turkish art

The Turks made beautiful pottery in the town of Iznik in northwest Turkey. They also wove wool carpets and tapestries.

The republic

The Ottoman Empire ended in 1922, and the sultans were no longer in charge. The next year, Turkey became a republic, with its people voting to choose the leaders.



Flag of Turkey

Universe

The universe is everything around us: matter, energy, and space. This means the Earth, the solar system, the Milky Way, and other galaxies are part of the universe. The universe is very big and is always changing.

SEE ALSO

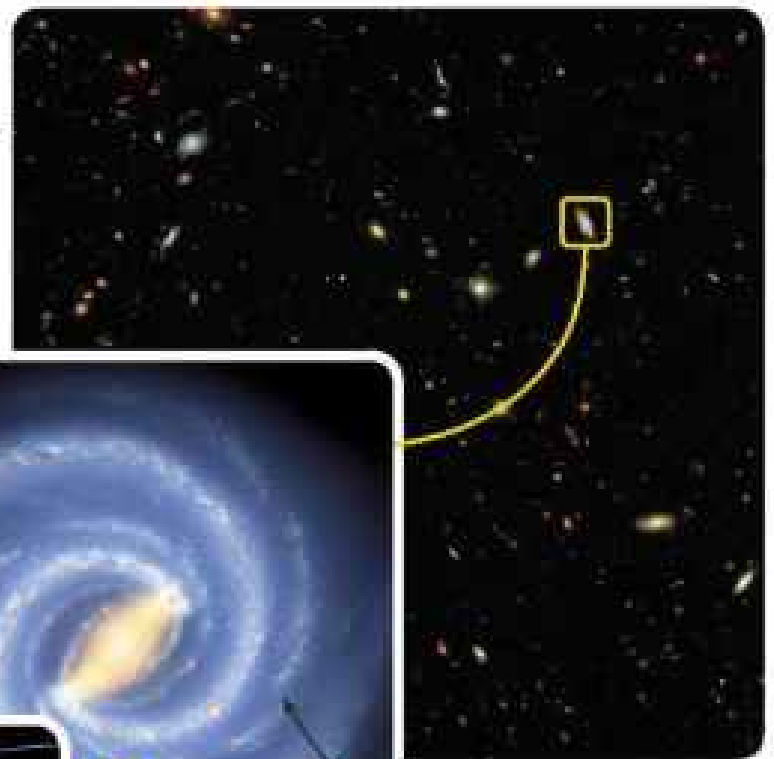
- ▶ Atoms p.34
- ▶ Big Bang p.37
- ▶ Earth p.83
- ▶ Galaxies p.116
- ▶ Milky Way p.167
- ▶ Solar system p.233

Where are we?

The universe is so huge that it is hard to understand. This diagram shows how Earth fits with the rest of the universe.

Universe

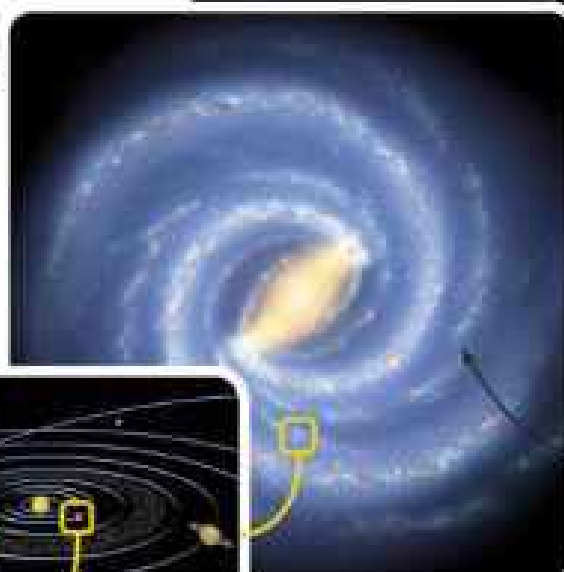
The universe is made up of billions of galaxies that cluster together, with huge empty spaces between them.



The universe has **no center**, and is filled with galaxies in all directions—it goes on forever.

The Milky Way

The solar system orbits the center of our home galaxy, which is known as the Milky Way.



Scientists thought the Milky Way was the only galaxy until the early 20th century.

The solar system

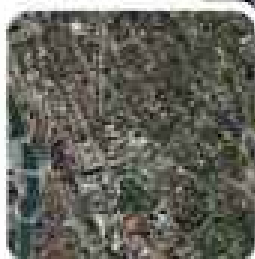
The sun and its family of planets are known as the solar system.



The Earth
Our planet is one of eight planets that move around the sun.



Earth was thought to be the center of the universe until the 16th century.



City landscape

Our planet is home to 7 billion people living in towns, cities, and the countryside.

Dark matter

Scientists think that dark matter is made of particles smaller than atoms. Dark matter is invisible to us, but we know it exists because its force of gravity pulls nearby space objects toward it.

Visible matter
20 percent



Dark matter
80 percent

Uranus

Uranus is the third largest planet in the solar system, after Jupiter and Saturn. It is the second farthest planet from the sun. From Earth, Uranus looks like a very faint star.

SEE ALSO

- Atmosphere p.33
- Elements p.90
- Gases p.117
- Mixtures p.168
- Neptune p.183
- Solar system p.233

Ice giant

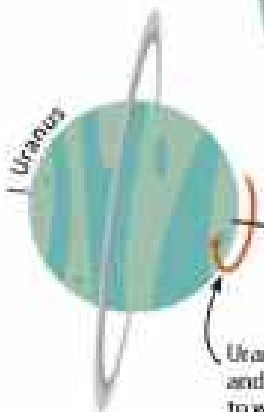
Uranus is an "ice giant"—it has a rocky core that is surrounded by a mixture of liquid ices. Uranus has no solid surface.

The atmosphere of Uranus is mostly made of hydrogen and helium gas. It is very cold.

Rolling planet

Most planets spin like tops on their axis, but Uranus spins on its side like a rolling ball. Its tilt was probably caused by a giant crash with another planet-sized body.

Earth is slightly tilted and spins from west to east.



Uranus is very tilted and spins from east to west.

Few features

The spacecraft *Voyager 2* visited the Uranus system in 1986. Images sent back to Earth revealed 10 new moons and two new rings but few other features.



Voyager 2

Uranus has thin, dark rings that are hard to see.

Uranus is the **coldest planet** in the solar system, with temperatures as low as **-371°F (-224°C)**.

Venus

Venus is a rocky planet that is only slightly smaller than Earth. It is the second planet away from the sun, and sits between Mercury and Earth. Venus spins very slowly and has the longest day of all the planets in the solar system.

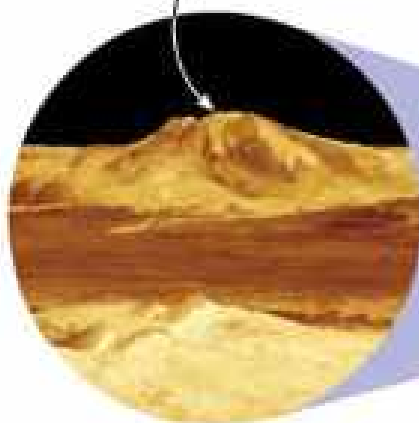
SEE ALSO

- ▶ Atmosphere p.33
- ▶ Earth p.83
- ▶ Gases p.117
- ▶ Mercury p.161
- ▶ Solar system p.233
- ▶ Temperature p.252
- ▶ Volcanoes p.268

Harsh planet

Venus's rocky surface is extremely hot. Temperatures can reach more than 878°F (470°C), which is hot enough to melt metal.

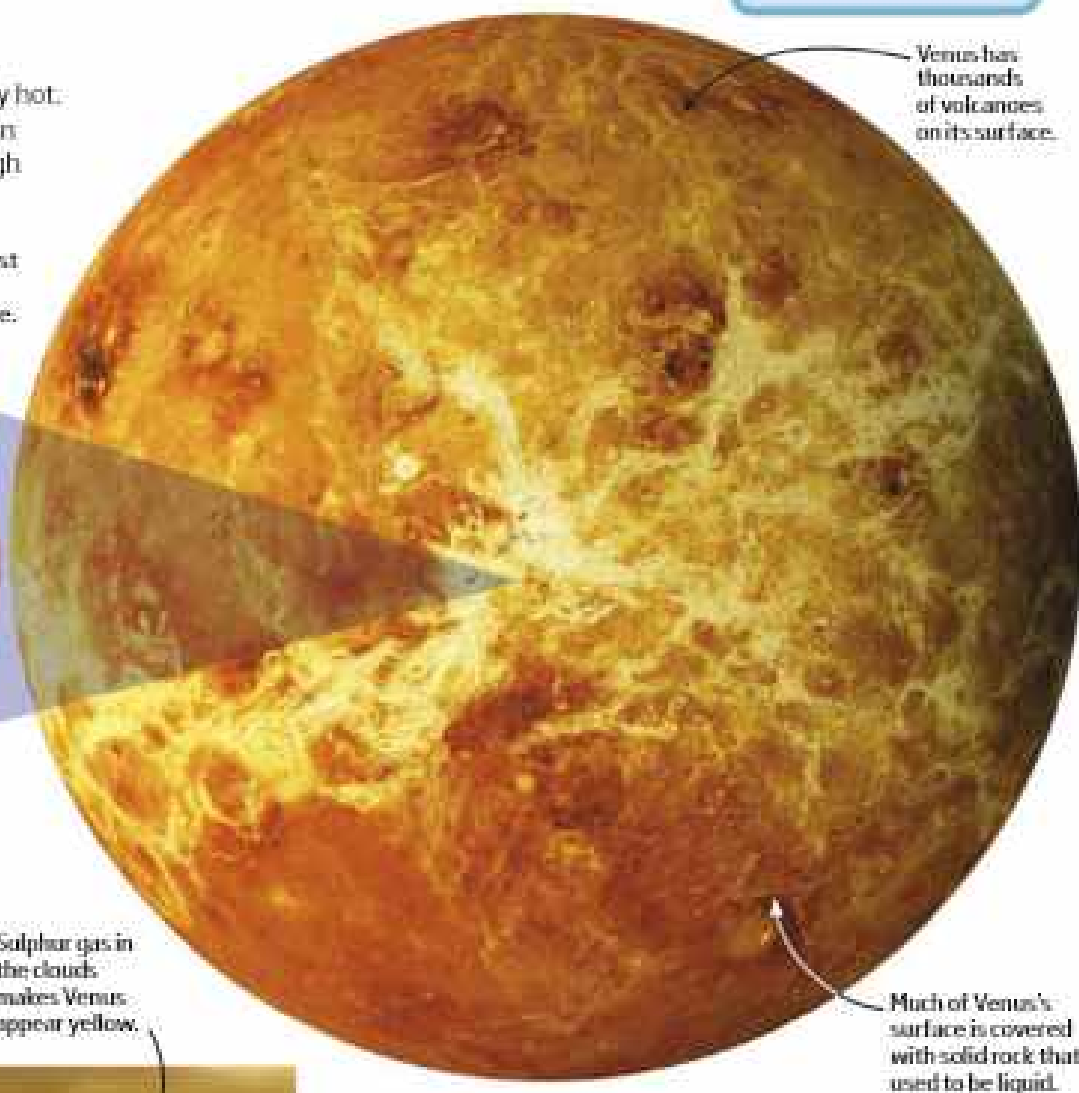
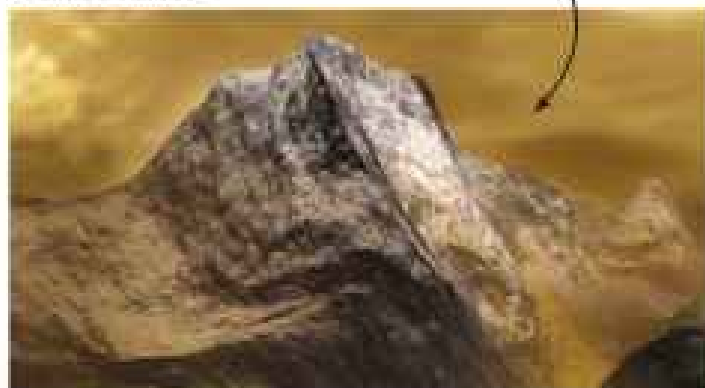
Maat Mons is the biggest volcano on Venus. It is 245 miles (395 km) wide.



Atmosphere

Venus is surrounded by a thick layer of poisonous gases. This atmosphere makes it hard for scientists to see Venus's surface.

Sulphur gas in the clouds makes Venus appear yellow.



Venus has thousands of volcanoes on its surface.

Much of Venus's surface is covered with solid rock that used to be liquid.

Transit of Venus

Venus is closer to the sun than the Earth is. We sometimes see Venus moving in front of the sun. It looks like a small, dark disk moving across the bright Sun. This is called the transit of Venus.



Vertebrates

Vertebrates are animals that have a backbone. The skeleton is a frame that helps the body move around. Mammals, amphibians, reptiles, fish, and birds are all vertebrates.

SEE ALSO

- ▶ Amphibians p.15
- ▶ Birds p.39
- ▶ Fish p.101
- ▶ Invertebrates p.139
- ▶ Mammals p.154
- ▶ Reptiles p.210
- ▶ Skeleton p.228

Mammals

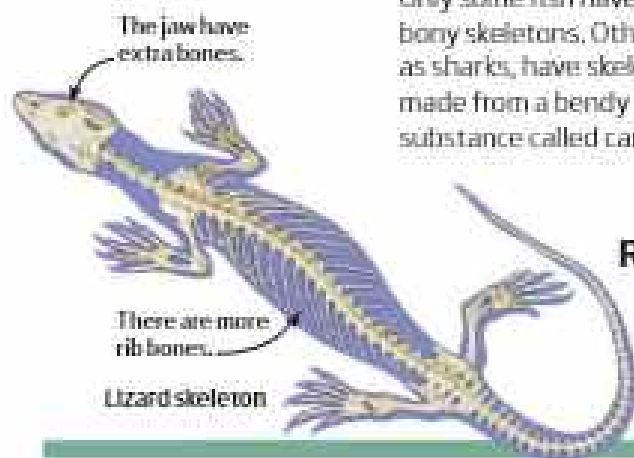
All mammals have a similar skeleton. Only mammals have a lower jawbone that is joined to the skull by a hinge.



Frog skeleton

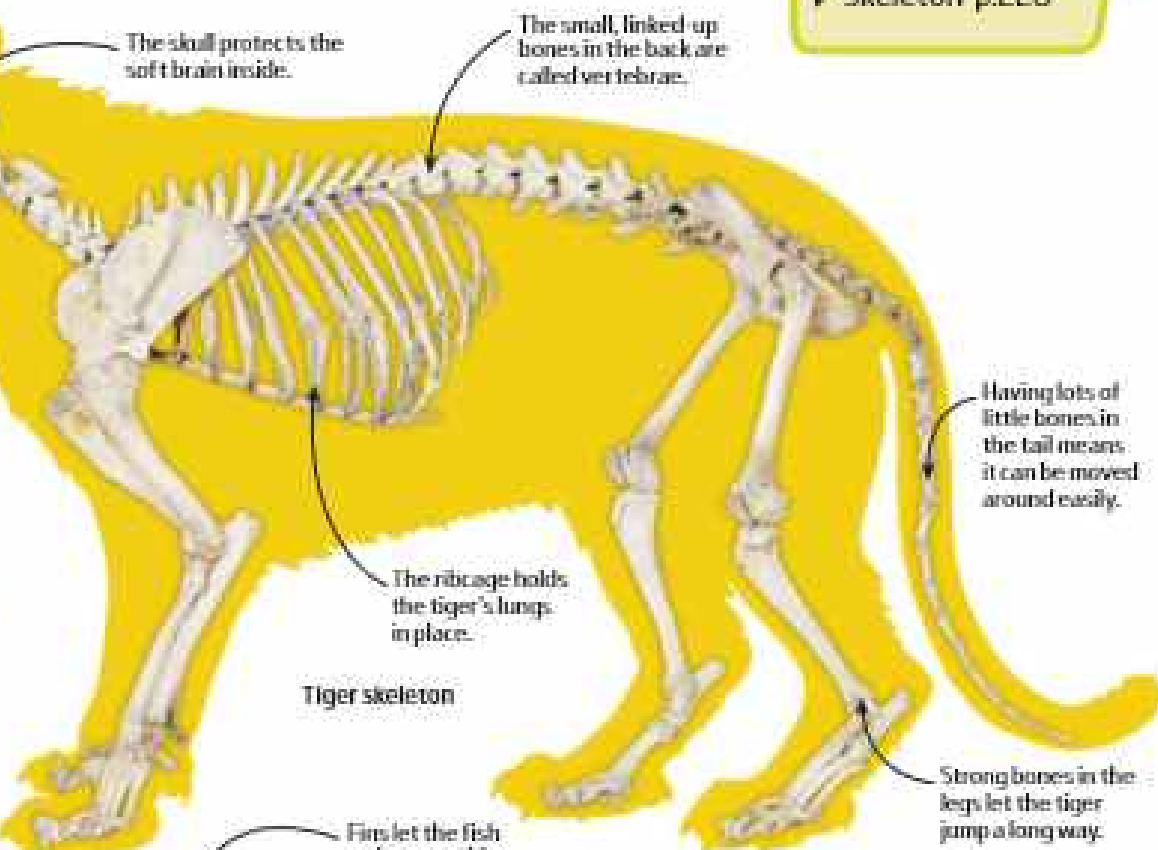
Amphibians

Frogs and toads don't have ribs. They have strong leg bones for jumping.



There are more rib bones.

Lizard skeleton



Tiger skeleton



Fish skeleton

Fish

Only some fish have bony skeletons. Others, such as sharks, have skeletons made from a bendy substance called cartilage.

Reptiles

Reptiles have more bones in their skeletons than other animals. This makes them very bendy.

Bird skeletons are full of holes. This helps them to be as light as possible.

Birds

Most birds have light bones, to let them fly. Penguins have heavier bones so they can dive deep in the water.



Penguin skeleton

Vikings

Starting in the year 800, people from Norway, Sweden, and Denmark set out to travel long distances and explore the world. We call these people Vikings. At home they had been farmers and craftspeople. On their travels, Vikings traded with or sometimes stole from others.

SEE ALSO

- Crafts p.75
- Europe p.94
- Explorers p.96
- Myths and legends p.178
- Oceans and seas p.187
- Ships p.224

Viking longship

Longships were fast ships Vikings used to travel across the Atlantic Ocean and up rivers in Europe. They were powered by oars and a sail.

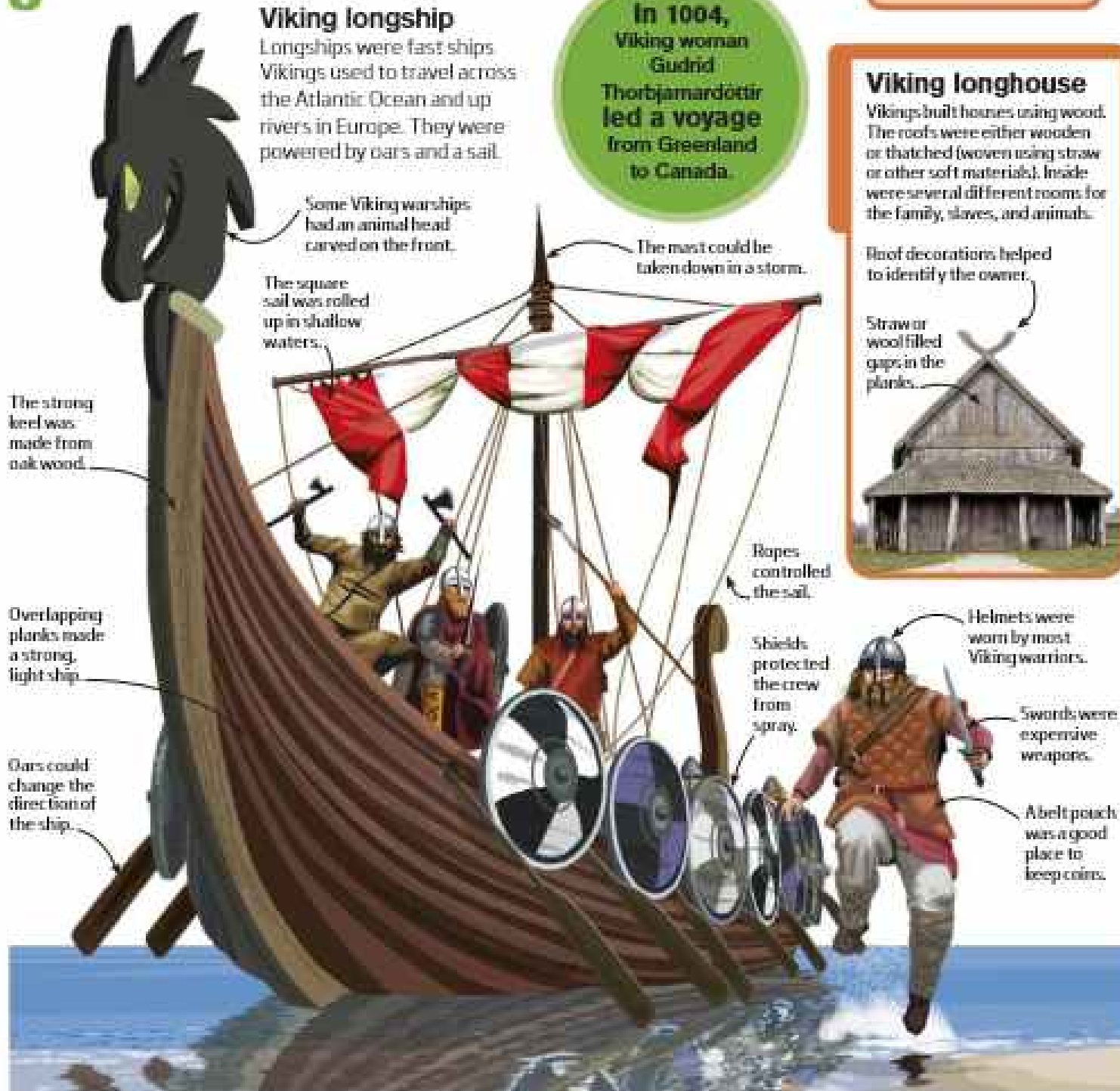
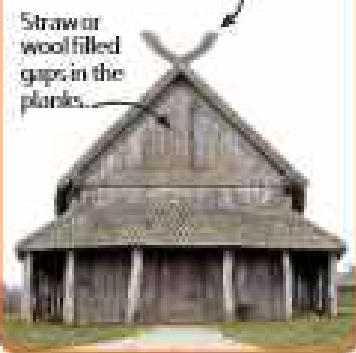
In 1004,
Viking woman
Gudrid
Thorbjarnardóttir
led a voyage
from Greenland
to Canada.

Viking longhouse

Vikings built houses using wood. The roofs were either wooden or thatched (woven using straw or other soft materials). Inside were several different rooms for the family, slaves, and animals.

Roof decorations helped to identify the owner.

Straw or wool filled gaps in the planks.



Some Viking warships had an animal head carved on the front.

The square sail was rolled up in shallow waters.

The mast could be taken down in a storm.

The strong keel was made from oak wood.

Overlapping planks made a strong, light ship.

Oars could change the direction of the ship.

Ropes controlled the sail.

Shields protected the crew from spray.

Helmets were worn by most Viking warriors.

Swords were expensive weapons.

A belt pouch was a good place to keep coins.

Volcanoes

A volcano is a mountain or crater that forms when melted rock, called magma, breaks through the Earth's surface. As soon as the magma breaks through, or erupts, from, a volcano, it is known as lava. Every year around the world, between 50 and 70 volcanoes erupt.

SEE ALSO

- ▶ Earth's surface p.84
- ▶ Earthquakes p.85
- ▶ Inside Earth p.135
- ▶ Rock cycle p.213
- ▶ Rocks and minerals p.214

Volcanic eruption

Volcanoes erupt in different ways. In some eruptions, lava gently flows out or spurts like a fountain. In other eruptions, gas, ash, and rocks explode out of the volcano.

Small pieces of lava fall around the crater, forming a cone-shaped mountain.

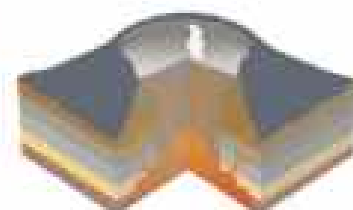
Gas in the magma can make it blast high into the sky, forming a lava fountain.

About 80 percent of volcanic eruptions take place under the sea.

Lava flows are slow-moving rivers of melted rock that can bury or destroy everything in their path.

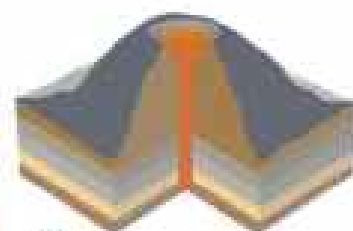
Types of volcano

Volcanoes come in all shapes and sizes. Some are small, cone-shaped hills formed in a single eruption. Others are giant mountains built up by many eruptions.



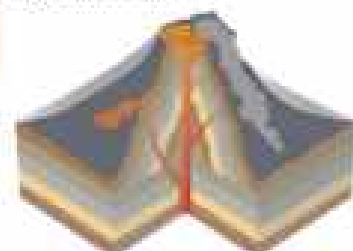
Caldera

The biggest volcanic eruptions leave behind an enormous crater known as a caldera. Some craters fill with water and become lakes.



Cinder cone

Built of fragments of cooled lava called cinders, cinder cones are the smallest and most common type of volcano.



Stratovolcano

This type of volcano is made up of layers of ash and lava from many eruptions. Stratovolcanoes are steep-sided and cone-shaped.

Volume

In math, volume is the amount of space inside a shape. The volume of a shape is measured in "cubed" units, such as cubic inches (in^3) or cubic centimeters (cm^3).

SEE ALSO

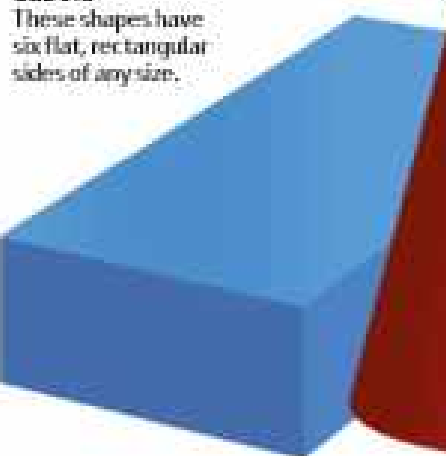
- ▶ Ancient Greece p.18
- ▶ Human body p.130
- ▶ Measuring p.159
- ▶ Numbers p.185
- ▶ Science p.217
- ▶ Shapes p.222

3-D shapes

Three-dimensional (3-D) shapes have length, width, and height. While 2-D shapes such as squares are flat, 3-D shapes have volume.

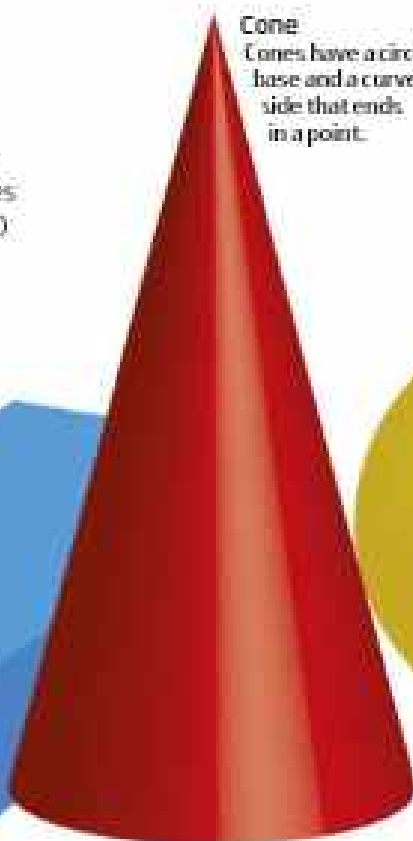
Cuboid

These shapes have six flat, rectangular sides of any size.



Cone

Cones have a circular base and a curved side that ends in a point.



Cylinder

A cylinder has circular ends and a long middle section.



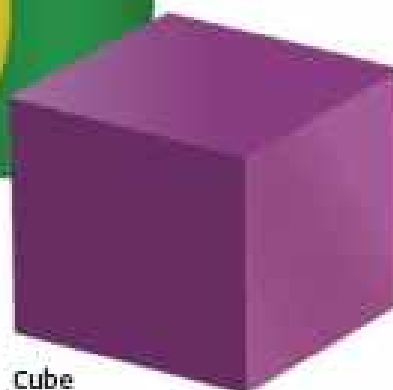
Sphere

A sphere is shaped like a ball. If you cut it in half, the sliced face would be circular.



Cube

Cubes have six equal-sized, square sides.



Finding volume

The volume of any object can be found by putting it in water. The volume of the water is measured first. The object is added, then the water is measured again.

Carefully measure the volume of the water. Then add your object.

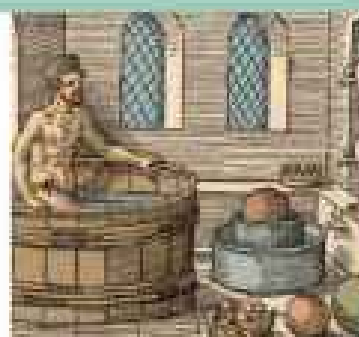


The amount the water level changes is the volume of the object.



Eureka!

An ancient Greek mathematician called Archimedes realized that the amount of water he pushed out when he got in the bath was the same volume as his body. He shouted "Eureka!" which means "I have found it!"



Archimedes in the bath

Water cycle

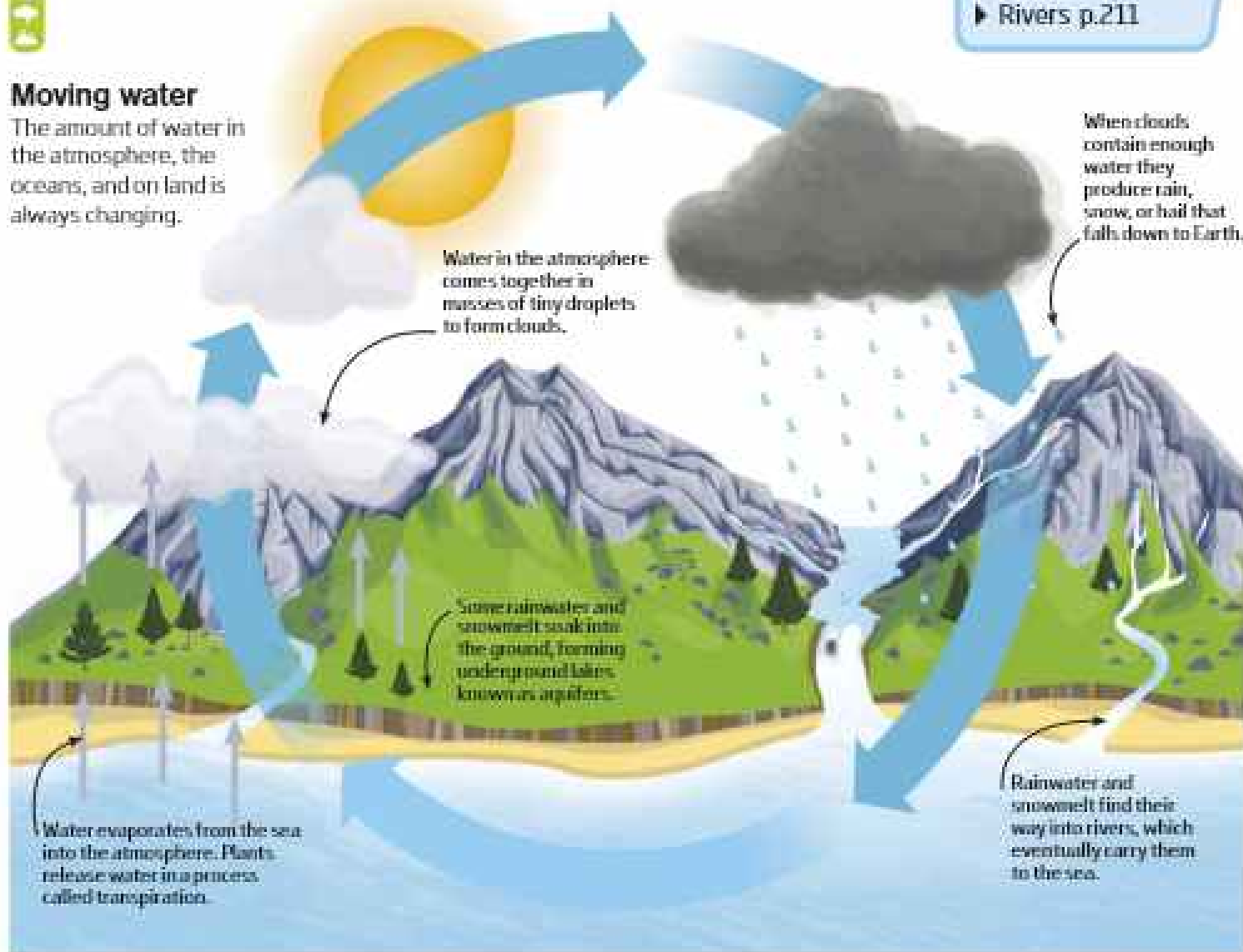
The Earth always has the same amount of water, but it is constantly moving between the oceans, rivers, underground reservoirs, ice caps, and the atmosphere. This continuous movement is called the water cycle.

SEE ALSO

- ▶ Clouds p.64
- ▶ Water pp.120-121
- ▶ Glaciers p.122
- ▶ Lakes p.143
- ▶ Oceans and seas p.187
- ▶ Rivers p.211

Moving water

The amount of water in the atmosphere, the oceans, and on land is always changing.



Water of life

Without water there would be no life on Earth. Even plants and animals that live in very dry places, such as deserts, need some water to stay alive.



Breaking the cycle

Humans break the water cycle in several ways. We dam rivers, suck up water from underground, and use water for washing and drinking.



Weather

The weather is what is happening in the atmosphere, or air and sky, outside. It could be sunny or cloudy, windy or calm, rainy or dry, or foggy or clear. In tropical parts of the world, it is hot and sunny most of the time. Further north or south, the weather can be different every day.

SEE ALSO

- ▶ Atmosphere p.33
- ▶ Changing world pp.50-51
- ▶ Clouds p.64
- ▶ Seasons p.221
- ▶ Storms p.246
- ▶ Water cycle p.270

Sunny

When there is bright sunshine, it is often warm with clear blue skies. Plants grow well in this kind of weather. If it is too hot and dry, however, they might die.



Windy

The wind is the movement of the air. Winds may be warm or cold, depending on the direction they blow in from. Very strong winds can damage buildings and blow down trees.



Rainy

Water droplets that fall from clouds are called rain. Plants need rain to grow, but too much rain can cause floods. When it is very cold, rain falls as snow.



Foggy

Fog and mist are made up of water droplets. They are clouds at ground level. Fogs are thicker than mists. People driving in fog need to be very careful as it is hard to see ahead.



The **story** of...

School

A school is where children go to learn subjects, such as reading and writing, that help them to understand the world. Going to school gives us the knowledge and skills that help us to get a job.

First schools

Boys first started to go to school in ancient Greece, Rome, China, and India. Later, in Europe, church schools were set up. Girls were not always sent to school.

A school in ancient Rome



Madrasa

In parts of the Islamic world, children go to a school known as a madrasa. Here they learn more about their religion by studying the Quran.



Students take **ten billion trips** every year on **school buses** in the United States and Canada.

Extra flashing lights help children see that the school bus is coming.

School buses have extra mirrors to help the driver spot children.



Education for all

Today, both boys and girls go to school from around the age of five. They learn math, reading, and writing. Older children study other subjects, too.



College

A college or university is where people over 18 can study a subject in great detail for three or four years. They are awarded a degree when they graduate.



Graduation cap



Degree certificate

Almost **two million** children are home-schooled in the US.



Getting to school

Many students walk to school, others are taken there in special school buses or by car or train. In the US, school buses are painted bright yellow.



Home school

Some children stay at home during the day and are given lessons by their parents. They get to study all the subjects they would learn at school. Children that live a long way from the nearest school join in lessons over the internet.



Victorian desktop blackboard

Tools for school

In the past, children wrote out their lessons using chalk on a small blackboard. Today, some schools use computers and tablets, although tests and homework usually have to be written out on paper.

Work

Work provides people with money to pay for the things they need, such as food and somewhere to live. There are lots of different types of work you can do. Many jobs need special skills that you must train for. Some people stay doing the same work all their lives. Others learn new skills and change jobs.

SEE ALSO

- ▶ Farming p.98
- ▶ Inventions pp.136-137
- ▶ Law p.145
- ▶ Medicine p.160
- ▶ Money p.169
- ▶ School pp.272-273

Jobs

The type of work a person does is called their job. Different jobs require different skills. People might work on a building site, in a hospital, or at a school.

Factory workers

Workers in factories produce everything from washing machines to cars and phones. Machines and computers help them to do this work.

Pilot flying a plane

Farm workers

Farmers use land to grow and sell crops like wheat. They raise animals such as cows, sheep, and chickens.

Truck driver

Office workers

Some people work in buildings called offices. They do different jobs and often use computers, phones, and books.

Teacher

Teachers help young people learn new skills and information at school.

Construction workers

Construction workers build new buildings and repair old ones. They build roads, dig ditches for pipes and cables, and lay train tracks.

Computer coder

Lawyer

Help center

Streetsweeper

Bus driver

Scientists

Scientists invent new products and medicines. They test everything from bridges to how healthy your blood is.

Police

Shop assistants

Shop workers sell things to people in stores, such as food, clothes, shoes, books, and music.

Creative work

Some people use their imaginations to create websites, design books and posters, and make music and films.

Medical staff

Doctors, nurses, and surgeons in clinics and hospitals try to cure you if you're ill. Carers look after people who need extra help.

Market traders

Many people sell fruit, vegetables, plants, flowers, and household goods in market stalls.



World

The Earth and everything that lives on it make up the world. We often show the world as a map. Just over one-quarter of it is land, which is divided into seven huge areas, called continents. The rest of the world is covered in water, or oceans.

SEE ALSO

- Changing world pp.50-51
- Climate change p.60
- Earth p.83
- Earth's surface p.84
- Maps p.155

Where we live

We live on all of the world's seven continents. Apart from Antarctica, the continents are divided into areas called countries.

The world's smallest country is Vatican City. It is contained within the city of Rome, in Italy.

The world's biggest country is Russia. It stretches from Eastern Europe across Asia to the Pacific Ocean.

The world's biggest city is Tokyo, in Japan. More than 35 million people live there.

The world's richest country is Qatar in Western Asia.

The world's poorest country is the Central African Republic, in Africa.

About the world



Population:
7.5 billion



Highest point:
Mount Everest



Lowest point:
Mariana Trench



Biggest desert:
Sahara Desert (hot)
or Antarctica (cold)



Longest river:
Nile

There are more
than **7.5 billion**
people in the world,
living in about
200 countries.

City life

Half of all the people on Earth live in cities. Many live in one of the world's 35 megacities, each of which has more than ten million people.



This nighttime photograph of the megacity Paris, in France, was taken from space.

World War I

In 1914, war broke out in Europe and spread across the world. In this war, planes and tanks were used for the first time. The fighting lasted for four years and millions of soldiers were killed. Peace was declared in 1918.

Trench warfare

In Western Europe, the opposing armies defended the land they held by digging lines of deep trenches. Trenches protected the soldiers from enemy fire but were dangerous and very dirty.

Soldiers went over the top of the trench to fight the enemy.

Sand bags protected against rifle fire.

Barbed wire kept enemy soldiers out.

A bayonet on the end of a gun was used to stab the enemy.

Heavy duty boots were worn.

Rats ran everywhere in the trenches, spreading disease.

Trenches were very muddy and often filled with water.

Gas masks could be worn if the enemy attacked using poisonous gas.

SEE ALSO

- ▶ Europe p.94
- ▶ Factories p.97
- ▶ Flags p.102
- ▶ Work p.274
- ▶ World p.275
- ▶ World War II p.277
- ▶ War pp.278-279



Women at war

With the men away fighting, women worked in factories to make weapons and ammunition (bullets and shells). They also worked on farms.

Allies and Central Powers

The warring countries formed two groups, with the Allies fighting the Central Powers.

Allies:



Britain



France



Italy



Russia



USA

Central Powers:



Germany



Austria-Hungary



Ottoman Empire

World War II

In 1939, Germany invaded Poland and a war broke out in Europe. Fighting spread across the world, with massive battles on land, at sea, and in the air. The war lasted six years and was the most violent conflict in history, with more than 60 million people killed. Peace was restored in 1945.

SEE ALSO

- ▶ Aircraft p.13
- ▶ Europe p.94
- ▶ Religion p.208
- ▶ Ships p.224
- ▶ World p.275
- ▶ World War I p.276
- ▶ War pp.278-279



Kindertransport

Almost 10,000 mainly Jewish children who were threatened with persecution were brought to safety in Britain from mainland Europe. This was called the *Kindertransport*, or "children's transport."

A single pilot sat in the cockpit. He also fired the guns.



British fighter plane

British Spitfire fighter aircraft brought down many German planes during the Battle of Britain in 1940.

The turret rotated to point the main gun at the enemy.



German tank

Germany made thousands of powerful, well-armed tanks to attack both Western Europe and the Soviet Union (Russia).

Aircraft were kept on the deck ready to take off.



The ship was painted in different colors to confuse the enemy.

US ship

The US Navy fought a series of fierce battles in the Pacific Ocean against the Japanese Navy.

Allies and Axis

The four main Allies faced the three Axis nations of Germany, Italy, and Japan. Fighting between them took place in Europe, Africa, Asia, and Oceania.

Main Allies:



Britain



France



USA



Soviet Union

Axis:



Germany



Italy



Japan

War

Throughout history, people have fought each other for land, money, and power. They have fought to defend a religion or overthrow a ruler or a government. Wars are expensive and kill thousands of people. They can last for years. Many don't agree with war, believing it is always wrong to kill people.

Early warfare

Men fought the first wars armed with battle-axes, wooden clubs, knives, spears, and shields. They didn't have uniforms, so it was sometimes difficult for them to know if they were fighting their enemies or their friends.

A full suit of armor weighed up to 90 lb (40 kg), which is about the same as 40 dictionaries.

Horses wore armor to protect their head, neck, and sides.

Knight on horseback

The Trojan War ended when Greek soldiers tricked their way into the city of Troy, by hiding inside a wooden horse.

Long wars

War can last a long time. The Trojan War lasted for ten years, while the Greeks and Persians fought each other for 50 years in the 400s BCE. The Hundred Years' War between England and France lasted for 116 years, from 1337 to 1453, although fighting did not take place every day.

Knights in armor

In medieval Europe, knights rode into battle on horseback, wearing suits of metal armor. The armor protected them from arrows and spears, but was heavy to wear and meant they couldn't see very well.

The Spanish Armada was made up of 130 ships that carried 2,500 guns and 30,000 soldiers and sailors.



A painting of the Spanish Armada in 1588.

War at sea

Many battles take place at sea. Sea battles can be very dangerous as ships can quickly sink, killing everyone on board. If rival boats get close together, soldiers can scramble across onto an enemy ship.

Civil wars

Most wars are fought between countries, but wars can also break out between groups within a country—these are known as civil wars. The US fought a war against Britain to win its independence in 1783, and then fought a bitter civil war from 1861 to 1865.



This flag was used by the rebel states during the Civil War.



The first official flag of the USA had 13 stars and 13 stripes to represent each of the colonies in 1777.

A modern submachine gun can fire

1,200 bullets

a minute, which is 20 every second.

Gunpowder

Gunpowder is an explosive material that was first invented in China in the 800s. It can be used in guns to propel bullets and shells very fast and over long distances against an enemy army.

Prussian soldier



War graves

Cost of war

Soldiers are killed or die of their wounds, and civilians caught up in the fighting can be killed by accident. If a person refuses to fight in a war because they don't believe in killing people, they are known as a conscientious objector.

Writing

Writing is putting the words we speak down on paper or on a screen. We do this using sets of characters, such as the letters in an alphabet. Characters represent the different letters or words of a language. The many different languages in the world have different systems of writing.

SEE ALSO

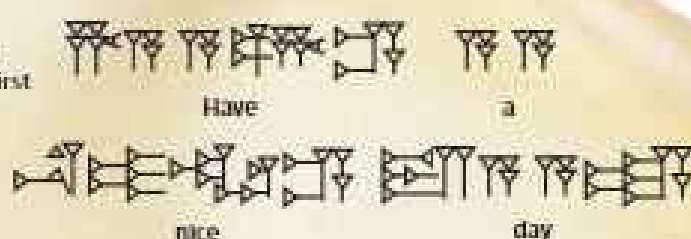
- ▶ Storytelling pp.42-43
- ▶ Books p.44
- ▶ Bronze Age p.47
- ▶ Codes pp.66-67
- ▶ Language p.144
- ▶ Philosophy p.189

Writing systems

Written characters can be joined together to form words, or sometimes form words on their own. Different systems are written left to right, right to left, or downward.

Cuneiform

Used in ancient Iraq, this was one of the first writing systems. Cuneiform means "wedge-shaped."



English

The English alphabet has 26 letters. These letters are used by many languages across the world.

Have a nice day

Chinese

These characters are formed of pictures that often show an object. A character forms one word or part of a word.

祝你过一个好天

Have a nice day

Cyrillic

This alphabet is used to write Russian and other Eastern European and Central Asian languages.

Хорошего дня!

Have a nice day

Hindi

The Hindi language of India is written in the beautiful Devanagari alphabet. It has 47 different letters.

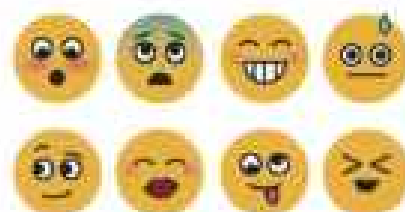
आप का दिन अच्छा बीते!

Have a nice day

Some Chinese dictionaries list more than **40,000** characters!

Emojis

Emoji means "picture character" in Japanese. Emojis are used on mobile phones and computers as a quick way to show feelings or words.



Calligraphy brush

Fountain pen

Pencil

Zoo

Zoos are home to animals from all over the world. Scientists work in zoos to learn about the animals, and how animals live in the wild. The oldest zoos have been around for hundreds of years. Millions of people visit zoos to see animals and learn more about them.

Natural spaces

Zoos try to keep animals in spaces that are like where they would be in the wild. This is good for the animals and helps people learn about these places.



Zoo people

Lots of different people work in zoos. Zookeepers look after the animals every day; zoologists are scientists that study the animals; and vets keep the animals healthy.



Conservation

Zoos protect animal numbers in the wild and work to stop animals from becoming extinct. For example, California zoos have set up breeding programs to save the endangered California condor.



Bad zoos

Not all zoos are good. Some don't look after their animals properly or keep them in the right spaces. Good zoos are part of zoo organizations that make sure they keep animals safe and healthy.



SEE ALSO

- ▶ Animal families p.21
- ▶ Conservation p.72
- ▶ Farming p.98
- ▶ Pets pp.152-153
- ▶ Mammals p.154
- ▶ Work p.274

Zoo visit

Here are some things to bear in mind if you visit a zoo.



Don't feed the animals. Feeding animals food that is not part of their diet can make them sick.



Don't make loud noises. Loud noises scare the animals, so try not to shout.



Listen to the zookeepers. The keepers know a lot. You can learn about animals by listening to them and reading the signs.



Reference

In this section you'll find a useful collection of lists and diagrams packed with helpful information.

Artists

An artist is someone who creates art by painting, sculpting, or making. Even early humans made cave paintings. Many artists are famous for inventing new styles of art and ways of painting.

Giotto (around 1266-1337)

Italian painter who started painting in a more lifelike way. His pictures mark the start of the Renaissance style of painting, which was more realistic than what had been before.

Jan van Eyck (around 1390-1441)

The first great painter to develop the use of oil paints. He came from what is now Belgium.

Leonardo da Vinci (1452-1519)

Italian painter, inventor, and thinker who painted people with natural expressions. His most famous works are the *Mona Lisa* and the wall painting *The Last Supper*.

Michelangelo Buonarroti (1475-1564)

Italian painter, sculptor, architect, and poet, often called simply "Michelangelo." His large religious paintings on the ceiling and walls of the Sistine Chapel in Rome are among the most famous of all artworks.

Raphael (Raffaello Sanzio, 1483-1520)

Italian painter of religious works and portraits. He used Leonardo's and Michelangelo's techniques to make paintings that influenced art for hundreds of years.

Titian (around 1488-1576)

Painter from Venice, Italy, whose works include mythological scenes and realistic portraits of people, and are well known for their bright colors.

Peter Paul Rubens (1577-1640)

Artist and diplomat who lived in what is now Belgium. He was the most famous painter of the Baroque style, which came after the Renaissance and features dramatic situations and emotions.

Claude Lorrain (around 1600-1682)

French landscape painter who mainly worked in Italy. His landscapes often include ancient ruins, and inspired the fashion for landscape gardening, where people tried to make their land look like his paintings.

Rembrandt (Rembrandt van Rijn, 1606-1669)

Dutch painter whose great skill as an artist helped him paint people's emotions. Many of his best paintings are self-portraits.

Francisco Goya (1746-1828)

Spanish artist who became official painter to the King of Spain, but whose works also include nightmare scenes and paintings of the horrors of war.

Katsushika Hokusai (1760-1849)

Japanese artist who excelled in painting scenes from everyday life and landscapes. Many of his works feature the snow-capped Japanese volcano, Mount Fuji.

J. M. W. Turner (1775-1851)

English landscape painter whose works show his interests in travel, the sea, history, and literature. In his later paintings the scenes are sometimes almost completely hidden by mist, rain, or snow.

John Constable (1776-1837)

English landscape painter known for his everyday countryside scenes. His famous paintings include *The Hay Wain* and *The Cornfield*.

Eugène Delacroix (1798-1863)

French painter of the Romantic period, when art, writing, and music focused on emotions. He chose dramatic subjects, deliberately painting so that his individual brushstrokes could be seen.

Paul Cézanne (1839-1906)

French painter, sometimes called the father of modern art. He mainly painted landscapes and still lifes (objects like flowers and fruits), building up his pictures with large blocks of color.

Claude Monet (1840-1926)

French landscape painter who invented the Impressionist style of art, which tried to paint the overall effect of a moment in time.

Vincent van Gogh (1853-1890)

Dutch painter who developed a unique style featuring bright colors and dramatic brushstrokes. He was not well known until after he died.

Edvard Munch (1863-1944)

Norwegian painter who had a tragic childhood and painted many works expressing fear and anxiety. His most famous painting is *The Scream*.

Qi Baishi (1864-1957)

Popular Chinese artist whose many works include a variety of subjects such as paintings of individual animals and plants.

Henri Matisse (1869-1954)

French painter. His brightly colored, usually cheerful works are sometimes abstract, but usually he painted recognizable objects in a simplified style.

Abanindranath Tagore (1871-1951)

Indian painter and author who helped develop Indian art that was less dependent on British influence (Britain ruled India at the time). His uncle was the poet Rabindranath Tagore (see "Writers" list).

Pablo Picasso (1881-1973)

Spanish artist, probably the most famous painter of the 20th century. He painted in a variety of modern art styles and helped invent Cubism, which includes lots of geometric shapes such as squares and triangles.

Edward Hopper (1882-1967)

American painter of realistic scenes, often city streets or buildings, either deserted or with lonely-looking people in them.

Diego Rivera (1886-1957)

Mexican painter best known for his colorful, action-packed wall paintings which often have a political message. Husband of Frida Kahlo.

Mark Rothko (1903-1970)

American abstract artist whose work features rectangular blocks of color painted without sharp edges.

Salvador Dalí (1904-1989)

Spanish painter and sculptor who belonged to the art movement called Surrealism, which created made-up subjects. His work features dreamlike impossible scenes, painted in a highly realistic way.

Frida Kahlo (1907-1954)

Mexican painter known for her self-portraits. She had a complex life, affected by an accident and illness as a child. Wife of Diego Rivera.

Jackson Pollock (1912-1956)

American painter best known for his "action paintings" - abstract works created by dribbling swirls of paint on a canvas.

Andy Warhol (1928-1987)

American founder of Pop Art, which takes everyday images such as soup cans or celebrities' faces and uses them as the basis for artworks.

Antony Gormley (born 1950)

British sculptor whose works include the huge outdoor winged figure the *Angel of the North* near Newcastle upon Tyne, UK.



Writers

People have written things down for thousands of years. Writing can include books, poems, or plays. It can tell a story or record facts.

Homer (around 800 BCE)

Legendary blind author of the Greek epic poems the *Iliad* and the *Odyssey*, set at the time of the Trojan War.

Sappho (around 630 BCE)

Greek poet famous for her passionate love poetry. Only a small amount of her work now survives.

Qu Yuan (around 340–278 BCE)

Ancient Chinese poet and public servant. His most famous poem is called *The Lament*.

Virgil (70–19 BCE)

Roman author of the epic poem the *Aeneid*, which tells the legendary story of the creation of the city of Rome.

Imru' al-Qais (around 500)

Arabian poet whose works are full of passionate feeling. He is sometimes called the father of Arabic poetry.

Dante Alighieri (1265–1321)

Italian author of the *Divine Comedy*, a three-part epic poem describing hell, heaven, and purgatory (a place in between heaven and hell).

Geoffrey Chaucer (around 1343–1400)

English author of *The Canterbury Tales*, entertaining stories told in the voice of different pilgrims (people travelling to a sacred place).

Miguel de Cervantes (1547–1616)

Spanish writer whose comic book *Don Quixote*, about the adventures of a well-meaning but silly knight, is often described as Europe's first novel.

William Shakespeare (1564–1616)

English playwright and poet whose many famous plays include *Hamlet*, *Macbeth*, *Romeo and Juliet*, and *A Midsummer Night's Dream*.

Molière (1622–1673)

Famous French actor and author of funny comic plays. Molière was his stage name; his real name was Jean-Baptiste Poquelin.

Matsuo Bashō (1644–1694)

Japanese poet, a master of the short type of Japanese poem called a haiku, which contains just 17 syllables (single sounds in words).

Voltaire (1694–1778)

French writer and thinker, whose real name was François-Marie Arouet. He attacked old-fashioned ideas in his funny and controversial writings.

Johann Wolfgang von Goethe (1749–1832)

German writer, poet, and thinker, whose wide-ranging works include *Faust*, a long drama finished just before his death.

Robert Burns (1759–1796)

Scotland's national poet. He wrote or revised the words for hundreds of Scottish songs including *Auld Lang Syne*.

William Wordsworth (1770–1850)

English poet who used nature as a source of inspiration.

Sir Walter Scott (1771–1832)

Scottish writer and poet. He was the first great historical novelist, with works including *Rob Roy*, *Old Mortality*, and *The Heart of Midlothian*.

Jane Austen (1775–1817)

English author whose funny and clever novels, including *Emma* and *Pride and Prejudice*, are still popular today.

Hans Christian Andersen (1805–1875)

Danish writer best known for his children's stories including *The Ugly Duckling*, *The Little Mermaid*, and *The Snow Queen*.

Charles Dickens (1812–1870)

English author of many famous novels including *Oliver Twist*, *David Copperfield*, and *A Tale of Two Cities*.

Charlotte Brontë (1816–1855)

English author of *Jane Eyre* and other novels. Her sisters Emily (1818–1848, author of *Wuthering Heights*) and Anne (1820–1849) are also well-known writers.

Charles Baudelaire (1821–1867)

French poet whose subjects include city life and the unhappy side of emotions. He was a big influence on later poets.

Leo Tolstoy (1828–1910)

Russian author of the famous novels *Anna Karenina* and *War and Peace*.

Emily Dickinson (1830–1886)

American poet whose deeply felt, personal poems only became well known after her death.

Lewis Carroll (1832–1898)

English author and mathematician. His real name was Charles Lutwidge Dodgson. He wrote the stories *Alice's Adventures in Wonderland* and *Through the Looking Glass*.

Mark Twain (1835–1910)

American author, whose real name was Samuel Langhorne Clemens. His many works include the novels *The Adventures of Tom Sawyer* and *The Adventures of Huckleberry Finn*.

Oscar Wilde (1854–1900)

Irish author whose works include the play *The Importance of Being Earnest* and his only novel, *The Picture of Dorian Gray*.

Rabindranath Tagore (1861–1941)

Indian poet, novelist, composer, and thinker who wrote mainly in the Bengali language. He won the Nobel Prize for Literature in 1913.

H. G. Wells (1866–1946)

English author and thinker. He wrote works of science fiction such as *The Time Machine* and *The War of the Worlds*.

James Joyce (1882–1941)

Irish author of famous novels including *Ulysses* and *Finnegans Wake*.

Virginia Woolf (1882–1941)

English novelist whose works feature a style of writing called stream-of-consciousness where you read a person's thoughts as they think them.

T. S. Eliot (1898–1965)

American-English poet whose works include *The Waste Land*. His humorous poems about cats became the inspiration for the musical *Cats*. He won the Nobel Prize for Literature in 1948.

Ernest Hemingway (1899–1961)

American author whose books include *A Farewell to Arms* and *For Whom the Bell Tolls*, both set in wartime. He won the Nobel Prize for Literature in 1954.

George Orwell (1903–1950)

English novelist and essay writer. He wrote the famous political novels *Animal Farm* and *Nineteen Eighty-Four*.

Gabriel García Márquez (1927–2014)

Colombian author whose novels, originally written in Spanish, include *One Hundred Years of Solitude* and *Love in the Time of Cholera*. He won the Nobel Prize for Literature in 1982.

Wole Soyinka (born 1934)

Nigerian playwright, poet, and novelist whose works often deal with African political and social issues. He won the Nobel Prize for Literature in 1986.

J. K. Rowling (born 1965)

British author of the hugely successful *Harry Potter* series of books about a young wizard.



Alphabets and writing systems

An alphabet is a set of marks that each means a sound. It is used to write down the words of a language.

Ancient Greek letters

The ancient Greeks used an alphabet with 24 letters. The Latin alphabet is based on it.

Ancient Greek

Αα alpha	Ββ beta	Γγ gamma	Δδ delta	Εε epsilon	Ζζ zeta	Ηη eta	Θθ theta
Ιι iota	Κκ kappa	Λλ lambda	Μμ mu	Νν nu	Ξξ ksi	Οο omicron	Ππ pi
Ρρ rho	Σσς sigma	Ττ tau	Υυ upsilon	Φφ phi	Χχ chi	Ψψ psi	Ωω omega

Latin letters

The Latin alphabet is still used today in many European languages. Three letters have been added since ancient times: J, U, and W.

Latin (Roman)

Aa	Bb	Cc	Dd	Ee	Ff	Gg	Hh	Ii	Jj	Kk	Ll	Mm
Nn	Oo	Pp	Qq	Rr	Ss	Tt	Uu	Vv	Ww	Xx	Yy	Zz

Arabic letters

The Arabic alphabet has 28 letters. It reads from right to left and does not have separate capital letters. Some vowels have their own letters, but some are added to the consonants.

Arabic

ص	ش	س	ز	ذ	د	ر	ز	ز	ز	ا	ا	ا	ا
ي	ا	و	و	و	و	و	و	و	و	و	و	و	و

Chinese characters

Chinese writing does not have an alphabet; instead, symbols called characters represent whole words. More than one language is spoken in China. Mandarin is the most common.

Chinese (Mandarin)

女 woman	男 man	子 child	头 head	手 hand	脚 foot	日 sun	月 moon
土 earth	水 water	火 fire	金 metal	木 tree	山 mountain	云 cloud	龙 dragon
狗 dog	猫 cat	马 horse	鸟 bird	北 north	南 south	大 big	小 small
刀 knife	叉 fork	辣 hot	冷 cold	春天 spring	夏天 summer	秋天 autumn	冬天 winter



Scientists

For thousands of years, scientists have made all sorts of important inventions and discoveries. Today, they are still answering important questions about the universe.

Aristotle (384–322 BCE)

Ancient Greek philosopher and scientist. His ideas on physics are out of date but he was a good biologist, pointing out many facts about animals for the first time.

Aristarchus of Samos (around 310–230 BCE)

Greek astronomer who first suggested that the Earth goes around the sun, instead of the other way round, as was thought before. Copernicus came up with the same idea much later on.

Zhang Heng (78–139 CE)

Chinese scientist and mathematician who invented a device that would detect earthquakes up to 310 miles (500 km) away.

Galen (around 129–200)

Greek doctor who studied the parts of the human body. Although many of his ideas were later proved to be wrong, people treated his writings on medicine very seriously for more than 1,300 years.

Alhazen (around 965–1039)

Arab mathematician, astronomer, and physicist. He was probably the best scientist of medieval times, writing a major work on the theory of light and vision.

Nicolaus Copernicus (1473–1543)

Polish astronomer who showed both that the Earth was not standing still, but instead spinning on its axis once a day, and that it orbits the sun once a year, instead of the sun orbiting the Earth.

Galileo Galilei (1564–1642)

Italian physicist and astronomer. He was the first person to use a telescope in astronomy, discovering among other things that Jupiter had moons.

Johannes Kepler (1571–1630)

German astronomer who improved Copernicus's theory that the Earth and other planets moved around the sun, by showing that their orbits are ellipses (oval shapes), not circles.

William Harvey (1578–1657)

English doctor who discovered that the heart pumps blood around the body, pushing it outward through arteries and back through veins.

Isaac Newton (1642–1727)

English physicist and mathematician who explained gravity for the first time. In physics, he introduced his famous "three laws of motion," which explain how objects move and interact with each other.

Carl Linnaeus (1707–1778)

Swedish biologist who introduced the idea of naming living things by giving them a name in Latin, for example *Homo sapiens* for humans.

James Hutton (1726–1797)

Scottish geologist whose work showed that the Earth's rocks formed over a huge time period as a result of very slow changes.

Antoine Lavoisier (1743–1794)

French chemist, often called the father of modern chemistry. He introduced the idea of a chemical element, and named the gas oxygen.

Alessandro Volta (1745–1827)

Italian physicist who in 1800 invented the electric battery, which first allowed a steady electric current to be produced. The unit of electricity, the volt, is named in his honor.

Michael Faraday (1791–1867)

English physicist and chemist. He showed that a moving magnet creates an electric current in a wire and invented the theory of electric and magnetic fields to explain his discoveries.

Charles Darwin (1809–1882)

English biologist whose 1859 book *On the Origin of Species* argued that new species can evolve from existing ones by natural selection.

Ada Lovelace (1815–1852)

English mathematician who put together the world's first computer program. She wrote it for a never-completed mechanical computer built by the inventor Charles Babbage.

Gregor Mendel (1822–1884)

Austrian science teacher and monk. He carried out careful experiments on plants to show how features such as flower color and seed shape are passed on to the next generation.

Louis Pasteur (1822–1895)

French chemist who proved that tiny living things cause rotting and decay. He also showed how people could be protected from diseases by immunizing them.

Dmitri Mendeleev (1834–1907)

Russian chemist who created the first periodic table of elements. He arranged the elements in increasing size of their atoms and whether they have similar properties.

Marie Curie (1867–1934)

Polish–French physicist. Along with her husband Pierre, she was one of the first people to research radioactivity, and discovered the radioactive elements radium and polonium. She won Nobel prizes in 1903 and 1911.

Ernest Rutherford (1871–1937)

New Zealand physicist who discovered that all atoms have a tiny central nucleus containing most of their mass (weight). He won the Nobel Prize for Chemistry in 1908.

Albert Einstein (1879–1955)

German-born physicist best known for his theories of relativity, including that matter and energy can be turned into each other (described by his famous equation $E = MC^2$). He won the Nobel Prize for Physics in 1921.

Alfred Wegener (1880–1930)

German weather scientist who suggested that the Earth's continents slowly move over time (continental drift).

Niels Bohr (1885–1962)

Danish physicist who added to Ernest Rutherford's ideas to suggest that electrons move around an atom in fixed orbits. He won the Nobel Prize for Physics in 1922.

Dorothy Hodgkin (1910–1994)

English chemist who worked out how to discover the shapes of complicated molecules in the body, such as penicillin and insulin. She won the Nobel Prize for Chemistry in 1964.

Alan Turing (1912–1954)

English mathematician and founder of computer science. During World War II he helped crack German military codes, and later he was involved with some of the first practical computers designed for general use.

Francis Crick (1916–2004) and James Watson (born 1928)

Crick (an English physicist) and Watson (an American biologist) co-discovered the spiral (double helix) shape of DNA in 1953. With a third scientist, they won the Nobel Prize for Medicine in 1962.

Rosalind Franklin (1920–1958)

English chemist who provided much of the evidence that Francis Crick and James Watson used in discovering the spiral shape of DNA.

Lynn Margulis (1930–2011)

American biologist who developed the theory that the complicated cells of animals and plants came from smaller bacteria-sized cells that started to live inside each other.

Stephen Hawking (born 1942)

English physicist who has helped us to understand black holes, the origin of the Universe, and the nature of time.

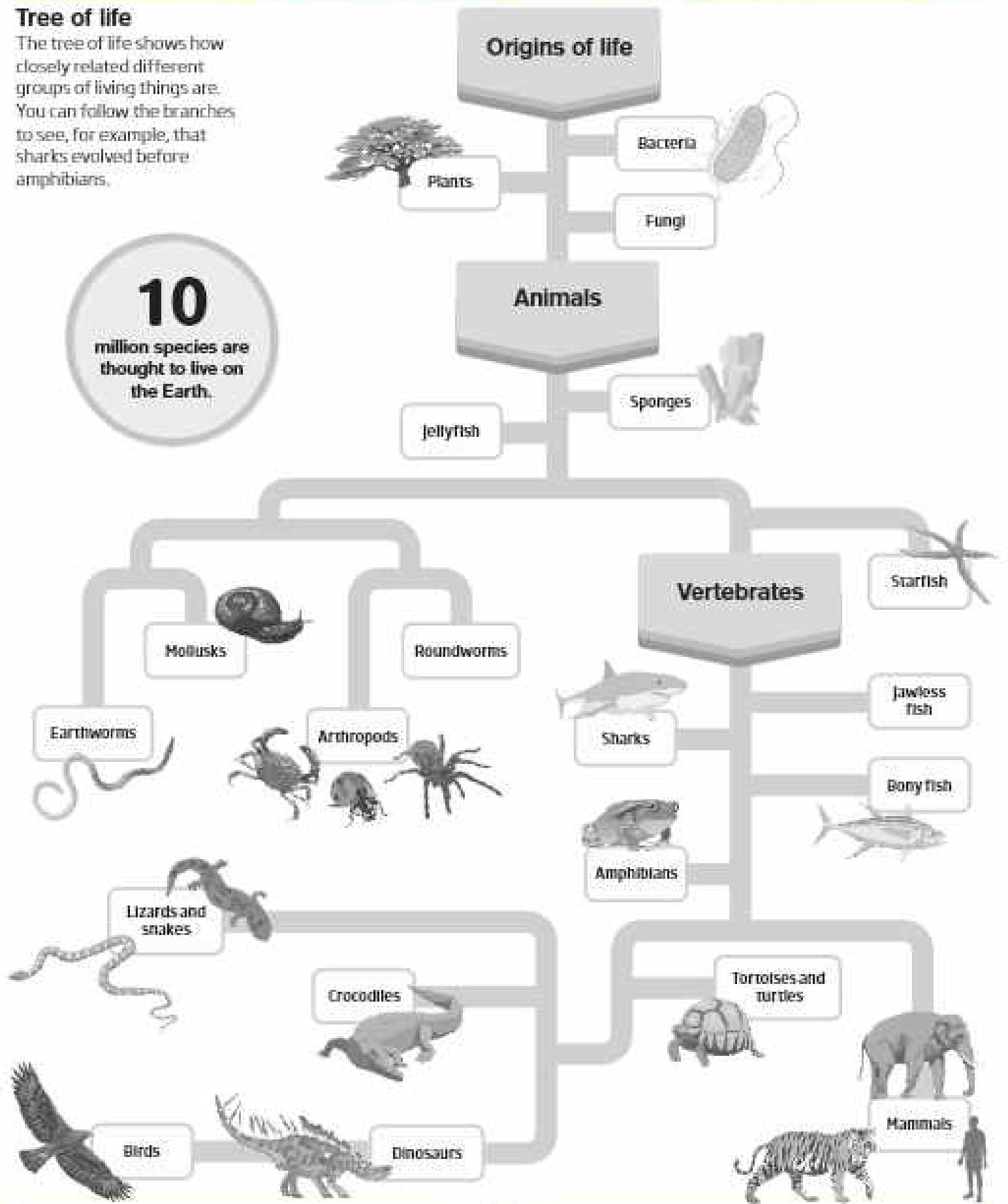


Tree of life

The tree of life shows how closely related different groups of living things are. You can follow the branches to see, for example, that sharks evolved before amphibians.

10

million species are thought to live on the Earth.





Multiplication

If you times two numbers together you are multiplying them. You can use this table to quickly work out the answer to multiplying any two numbers between 1 and 20.

When we multiply a number by itself, we say it has been "squared."

To work out what 3×2 equals, find the first number (3) on the left hand column.

Find the second number (2) on the top row and follow the column down to where it meets the row for the first number.

By following down the column across from 3 and down from 2, you can see that $3 \times 2 = 6$.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
2	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
3	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60
4	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60	64	68	72	76	80
5	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100
6	6	12	18	24	30	36	42	48	54	60	66	72	78	84	90	96	102	108	114	120
7	7	14	21	28	35	42	49	56	63	70	77	84	91	98	105	112	119	126	133	140
8	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120	128	136	144	152	160
9	9	18	27	36	45	54	63	72	81	90	99	108	117	126	135	144	153	162	171	180
10	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200
11	11	22	33	44	55	66	77	88	99	110	121	132	143	154	165	176	187	198	209	220
12	12	24	36	48	60	72	84	96	108	120	132	144	156	168	180	192	204	216	228	240
13	13	26	39	52	65	78	91	104	117	130	143	156	169	182	195	208	221	234	247	260
14	14	28	42	56	70	84	98	112	126	140	154	168	182	196	210	224	238	252	266	280
15	15	30	45	60	75	90	105	120	135	150	165	180	195	210	225	240	255	270	285	300
16	16	32	48	64	80	96	112	128	144	160	176	192	208	224	240	256	272	288	304	320
17	17	34	51	68	85	102	119	136	153	170	187	204	221	238	255	272	289	306	323	340
18	18	36	54	72	90	108	126	144	162	180	198	216	234	252	270	288	306	324	342	360
19	19	38	57	76	95	114	133	152	171	190	209	228	247	266	285	304	323	342	361	380
20	20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340	360	380	400

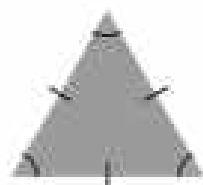


Flat shapes

Flat, or 2-D, shapes have length and width but no depth. Triangles have three straight sides, and quadrilaterals have four straight sides.

KEY

- Equal angles
- Right angle
- Equal sides
- Equal sides



Equilateral triangle
All three sides and all three angles in an equilateral triangle are equal.



Scalene triangle
All three sides and all three angles in a scalene triangle are different.

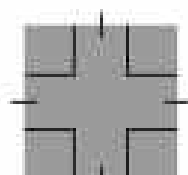


Isosceles triangle
An isosceles triangle has two sides of equal length and two angles of equal size.

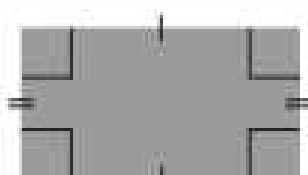


The right angle is always opposite the longest side.

Right-angled triangle
A right-angled triangle has one right angle, but the other angles and sides vary.



Square
All four sides in a square are of equal length and all four angles are right angles.



Rectangle
A rectangle has two pairs of sides that are the same length and four right angles.



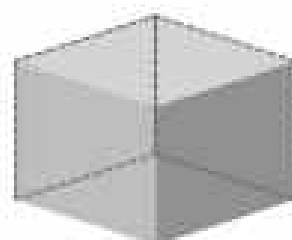
Parallelogram
A parallelogram has two pairs of sides that are equal length and two pairs of angles that are the same size.



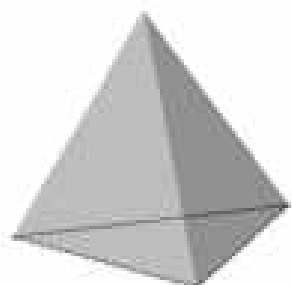
Rhombus
All the sides in a rhombus are equal length and it has two pairs of angles of equal size.

Solid shapes

Solid, or 3-D, shapes have depth as well as length and width.



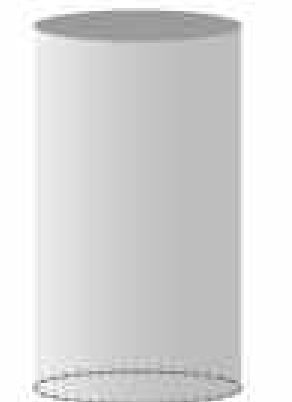
Cube
A cube has 12 edges of equal length and six faces of equal size.



Pyramid
A pyramid can have a triangular base, as seen here, or a square base. A triangular-based pyramid has four faces and six edges.



Cone
Cones have two faces and only one edge. The point is above the center of the circular base.



Cylinder
A cylinder has three faces and two edges. The two circular faces are opposite each other.



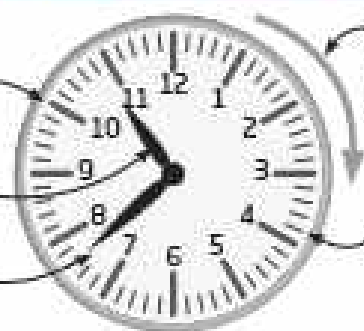
Time

We divide time into hours, minutes, and seconds. A clock or watch is used to tell what time of day it is.

The clock face is divided into 12 hours. There are 24 hours in a full day.

The hour hand shows what hour of the day it is.

The minute hand shows how many minutes in an hour have passed. It is longer than the hour hand.



The hands move down on the right and up on the left. We call this direction "clockwise."

Each hour is divided into 60 minutes.



O'clock

When the minute hand points to 12, the time is exactly the hour shown by the hour hand.



Half past

When the minute hand points to 6, we know it is half past the hour.



Minutes past

When the minute hand is on the right of the clock, we say how many minutes are past the hour.



Quarter past

When the minute hand points to 3, we know it is quarter past the hour.



Quarter to

When the minute hand points to 9 we know it is quarter to the hour.



Minutes to

When the minute hand is on the left of the clock, we say how many minutes are to the hour.

The planets

There are eight planets in our solar system, but they are all very different. Here you can compare them—the days and hours given are equal to the time they take on Earth.

243

Earth days in a single day on Venus.

Planet name	Distance from the sun	Width	Orbit time around the sun	Day length	Number of moons
Mercury	36 million miles (58 million km)	3,032 miles (4,879 km)	88 days	59 days	0
Venus	67 million miles (108 million km)	7,521 miles (12,104 km)	225 days	243 days	0
Earth	93 million miles (150 million km)	7,926 miles (12,756 km)	1 year	24 hours	1
Mars	142 million miles (228 million km)	4,220 miles (6,792 km)	1 year 322 days	25 hours	2
Jupiter	494 million miles (778 million km)	88,846 miles (142,984 km)	11 years 315 days	10 hours	At least 67
Saturn	887 million miles (1,427 million km)	74,898 miles (120,536 km)	29 years 163 days	11 hours	At least 62
Uranus	1,784 million miles (2,871 million km)	31,763 miles (51,118 km)	84 years 6 days	17 hours	At least 27
Neptune	2,795 million miles (4,498 million km)	30,775 miles (49,528 km)	163 years 289 days	16 hours	At least 14



US presidents

The president is the head of state and head of government for the United States of America. Alongside the name of each president is the party (political group) that they belong to.

George Washington	Federalist	1789–1797
John Adams	Federalist	1797–1801
Thomas Jefferson	Democratic-Republican	1801–1809
James Madison	Democratic-Republican	1809–1817
James Monroe	Democratic-Republican	1817–1825
John Quincy Adams	Independent	1825–1829
Andrew Jackson	Democrat	1829–1837
Martin Van Buren	Democrat	1837–1841
William H. Harrison	Whig	1841 •
John Tyler	Democrat	1841–1845
James K. Polk	Democrat	1845–1849
Zachary Taylor	Whig	1849–1850 •
Millard Fillmore	Whig	1850–1853
Franklin Pierce	Democrat	1853–1857
James Buchanan	Democrat	1857–1861
Abraham Lincoln	Republican	1861–1865 □
Andrew Johnson	Democrat	1865–1869
Ulysses S. Grant	Republican	1869–1877
Rutherford B. Hayes	Republican	1877–1881
James A. Garfield	Republican	1881 □
Chester A. Arthur	Republican	1881–1885
Grover Cleveland	Democrat	1885–1889
Benjamin Harrison	Republican	1889–1893
Grover Cleveland	Democrat	1893–1897
William McKinley	Republican	1897–1901 □
Theodore Roosevelt	Republican	1901–1909
William H. Taft	Republican	1909–1913
Woodrow Wilson	Democrat	1913–1921
Warren G. Harding	Republican	1921–1923 •
Calvin Coolidge	Republican	1923–1929
Herbert Hoover	Republican	1929–1933
Franklin D. Roosevelt	Democrat	1933–1945 •
Harry S. Truman	Democrat	1945–1953
Dwight D. Eisenhower	Republican	1953–1961
John F. Kennedy	Democrat	1961–1963 □
Lyndon B. Johnson	Democrat	1963–1969
Richard Nixon	Republican	1969–1974
Gerald Ford	Republican	1974–1977
Jimmy Carter	Democrat	1977–1981
Ronald Reagan	Republican	1981–1989
George H. W. Bush	Republican	1989–1993
Bill Clinton	Democrat	1993–2001
George W. Bush	Republican	2001–2009
Barack Obama	Democrat	2009–2017
Donald Trump	Republican	2017–

Assassinated □

Died in office •

Timeline of the Civil War

The Civil War was a major conflict in American history. It began when 11 southern states, unhappy that Abraham Lincoln had been elected president and concerned he would outlaw slavery, seceded (broke away) from the Union. They created their own Confederate States of America, also known as the Confederacy. The war between the North and the South that followed lasted four years and caused the deaths of more than 600,000 soldiers. When it ended in April 1865, the Union was restored, and more than four million slaves were freed.

1850

Compromise of 1850: New states are allowed to decide for themselves whether to be slave states or free states.

1857

The Supreme Court rules that slaves are not US citizens and are not protected by the Constitution.

October 1859

John Brown and his men raid a military arsenal at Harpers Ferry, Virginia, hoping to arm a slave uprising. Brown is caught and hanged.

November 1860

Republican candidate Abraham Lincoln wins the US presidential election, but without any support from the southern states.

December 20, 1860

South Carolina secedes (breaks away) from the Union.

February 1861

Representatives from Alabama, Florida, Georgia, Louisiana, Mississippi, and South Carolina meet to form the Confederate States of America, with Jefferson Davis as president. Later they are joined by Virginia, Texas, North Carolina, Tennessee, and Arkansas.

April 12, 1861

Confederate forces attack and capture Union-held Fort Sumter in South Carolina. The Civil War begins.

July 21, 1861

First Battle of Bull Run (Virginia): Confederate victory.

April 6–7, 1862

Battle of Shiloh (Tennessee): Union victory under Ulysses S. Grant, but with heavy losses.

February 6, 1862

Fall of Fort Henry (Tennessee): Union victory.

February 16, 1862

Battle of Fort Donelson (Tennessee): Union victory.

March 9, 1862

Battle of the Monitor and the Virginia: Draw.

April 25, 1862

Union forces take New Orleans.

May 31–June 1, 1862

Battle of Seven Pines (Virginia): No clear winner.

June 26–July 2, 1862

The Seven Days Battles (Virginia): Confederate victory. This is a hard-fought victory for the Confederates.

August 29–30, 1862

Second Battle of Bull Run (Virginia): Confederate victory.

September 17, 1862

Battle of Antietam/Sharpsburg (Maryland): No clear winner.



September 22, 1862

The Union's Emancipation Proclamation declares all slaves in the rebelling Confederate areas to be free.

December 13, 1862

Battle of Fredericksburg (Virginia): Confederate victory.

December 31, 1862-January 2, 1863

Battle of Stones River (Tennessee): No clear winner.

May 1-6, 1863

Battle of Chancellorsville (Virginia): Confederate victory.

May 18-July 4, 1863

Siege of Vicksburg (Mississippi): Union victory.

July 1-3, 1863

Battle of Gettysburg (Pennsylvania): Union victory. This battle is a turning point in the war.

September 19-20, 1863

Battle of Chickamauga (Georgia): Confederate victory.

November 19, 1863

Lincoln delivers the Gettysburg Address. This famous speech was given at the dedication of a national cemetery for Union soldiers, close to the site of the Battle of Gettysburg.

November 23-25, 1863

Battle of Chattanooga (Tennessee): Union victory.

May 5-6, 1864

Battle of the Wilderness (Virginia): No clear winner.

May 8-12, 1864

Battle of Spotsylvania (Virginia): No clear winner.

May 11, 1864

Battle of Yellow Tavern (Virginia): Union victory.

June 3, 1864

Battle of Cold Harbor (Virginia): Confederate victory. This is the last major Confederate victory.

June 18, 1864

Siege of Petersburg (Virginia) begins: Union victory.

August 5, 1864

Battle of Mobile Bay (Alabama): Union victory.

September 2, 1864

Fall of Atlanta (Georgia): Union victory.

November 8, 1864

Lincoln is reelected as US president.

November 15, 1864

The Union's Savannah Campaign, or the "March to the Sea," begins.

November 30, 1864

Battle of Franklin (Tennessee): Union victory.

December 15-16, 1864

Battle of Nashville (Tennessee): Union victory.

February 1, 1864

The Thirteenth Amendment, outlawing slavery, is proposed.

April 2, 1864

Fall of Petersburg and Richmond (Virginia): Union victory.

April 9, 1864

Robert E. Lee surrenders to Grant at Appomattox, Virginia. This marks the effective end of the Civil War, although fighting continues for a few weeks afterward.

December 1865

The Thirteenth Amendment is ratified (made law). Slavery is outlawed.

Major religions

Religion is a set of beliefs and ideas about a god or many gods. There are lots of different religions and they are practiced all over the world.

Baha'i

Founded in Iran in the 19th century, the Baha'i religion seeks to achieve peace and togetherness for all humankind.

Buddhism

Buddhism was developed around 500 BC by an Indian prince who later became called the Buddha (the "Enlightened One"). It teaches the need for a spiritual journey to free people from wants and pain.

Cao Dai

A modern religion founded in Vietnam in 1926 which believes in peace and non-violence.

Christianity

A faith centered on the belief that Jesus Christ, who preached in Palestine around 2,000 years ago, is the son of God and came down to Earth to rescue people from sin (doing wrong).

Confucianism

An ancient Chinese religious philosophy that began with the teachings of the philosopher Confucius, who lived around 500 BC.

Hinduism

An ancient Indian religion that includes a belief that every person goes through a cycle of life, death, and then rebirth in a future life. Hinduism features many different gods and goddesses.

Islam

Followers of the Islamic faith are called Muslims. They believe that the true word of Allah (God) was revealed to the Prophet Muhammad around 607 AD and written down in the sacred text called the Quran.

Jainism

Jainism is an ancient Indian religion that emphasizes non-violence toward people and animals.

Judaism

The religion of the Jewish people, Judaism also influenced the development of Christianity and Islam. Jews worship one God and their sacred text is called the Torah.

Shamanism

A system of belief common in small traditional societies. Shamans are individuals who believe they have special access to the spirit world, often by going into a dream, which they use to help their community.

Shinto

The traditional religion of Japan, followers of Shinto believe that there are spirits called *kami* everywhere in the world.

Sikhism

A faith that developed in northern India around 1500 and encourages tolerance between religions. Sikhs worship one God and Sikh men traditionally wear a turban to cover their hair, which they leave uncut.

Taoism

An ancient Chinese religion and philosophy that involves accepting, and following, the natural power of the Universe.

Zoroastrianism

Zoroastrianism is an ancient religion of Persia (modern-day Iraq) that features the idea of an unending struggle between good and evil. It is only a small religion today.



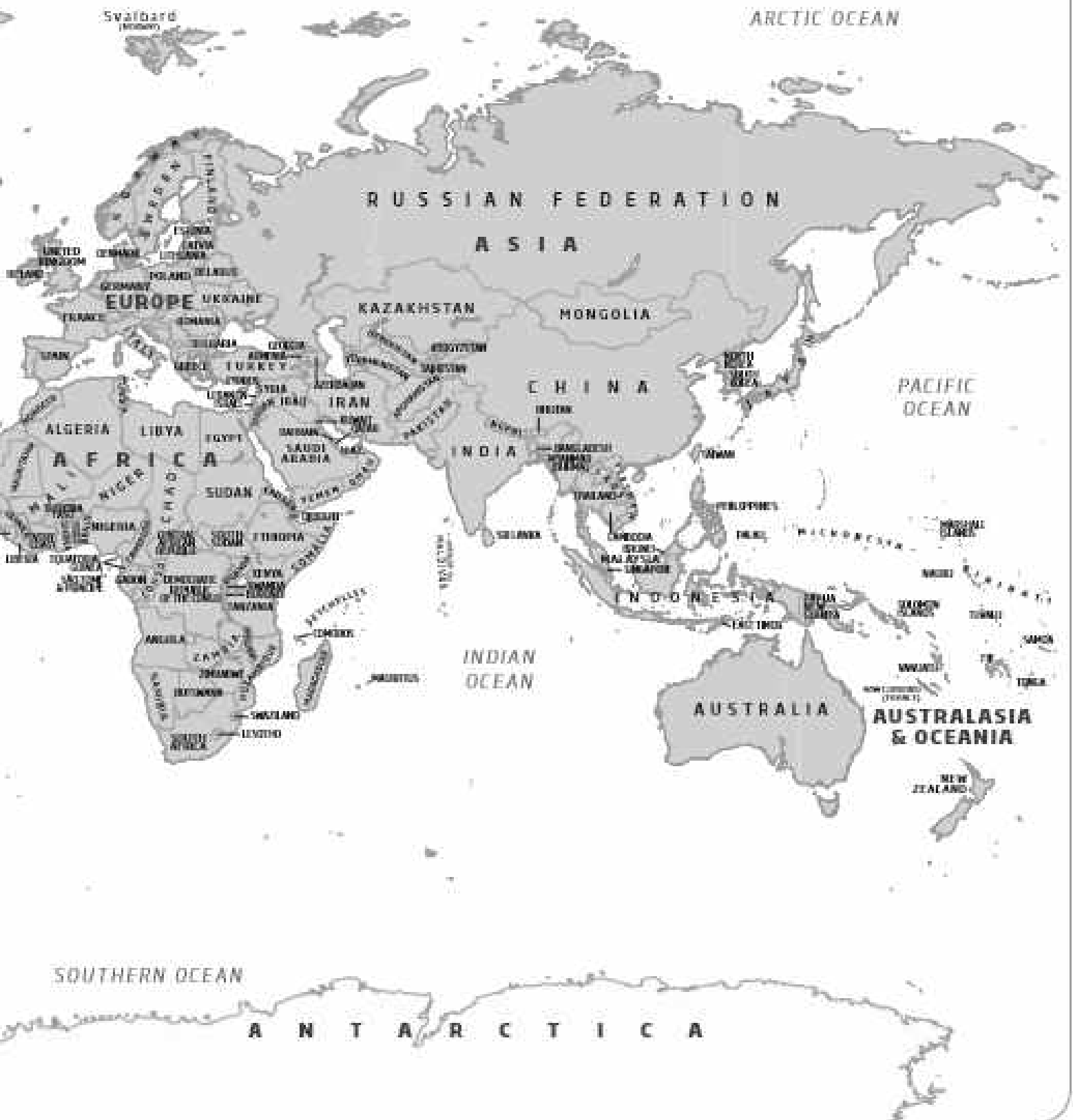
The world

Land covers about a third of the Earth's surface. The land is broken up into seven large blocks called continents. The continents are divided into smaller areas called countries.

Europe

This map shows Europe in more detail, as there is not enough space to show all the European countries on the main map.







Glossary

abstract

Art that does not copy real life exactly, but may only look a bit like an object, or represent a feeling

adaptation

Way in which an animal or a plant becomes better-suited to its habitat

algebra

Type of math that uses letters to stand for numbers or amounts

amphibians

Cold-blooded vertebrates that start life in water before moving between land and water when fully grown

ancestor

Ancient relative

ancient

Very old

antenna

One of two feelers found on an insect's head with which they can sense their surroundings

appliance

Machine used for a certain job that is usually electrical, such as a toaster

aqueduct

Channel or bridge built to carry water

architect

Person who plans and designs buildings

armor

Hard covering that provides protection

army

Organized group of soldiers

artificial

Object that is made by humans and is not found naturally

asteroid

Small, rocky object that orbits the sun

asteroid belt

Area of the solar system between Mars and Jupiter containing a large number of asteroids

astronaut

Someone who is trained to travel and work in a spacecraft

astronomy

Study of space

atmosphere

Layer of gas that surrounds a planet

atom

Smallest part of an element that can take part in a chemical reaction

attract

When two things pull toward each other

avalanche

Sudden movement of a large amount of snow or rock down a mountainside

axis

Imaginary line that passes through the center of a planet or star, around which it rotates

bacteria

Tiny organisms that live everywhere on Earth, such as inside food, soil, or even in the human body

BCE

Before Common Era, or all the years before year 0

beliefs

Set of views that people hold about the world, life, and the afterlife

biology

Study of living things and their relationship with their habitat. Someone who studies biology is called a biologist

birds

Warm-blooded vertebrates with a beak and feathers that can usually fly. They lay hard-shelled eggs to produce young

black hole

Object in space with such a strong force of gravity that nothing can escape it, not even light

boil

When a liquid is heated to a temperature at which it bubbles and turns into a gas or vapor

boundary

The point where one area ends and another begins

breed

Variety of a pet or farmed animal; for example, a pug is a breed of dog

burrow

Hole or tunnel dug in the ground by an animal, to live in

calendar

Breakdown of the year into days and months; used to work out the date

camouflage

Colors or patterns that help something appear to blend in with its environment

carnivore

Animal that eats only meat

cartilage

Tough but flexible material found in animals that, among other things, makes up the human nose and ears, and the skeletons of sharks

catapult

Ancient war machine used to hurl rocks over long distances

CE

Common Era, or all the years after year 0

chemical

Substance used in, or made by, a reaction between particles such as atoms

chemistry

Study of chemicals and their reactions. Someone who studies chemistry is called a chemist

chrysalis

Hard casing, often camouflaged, that a caterpillar wraps itself in during metamorphosis

circuit

Loop that an electric current travels around

circulation

Path that blood travels around the body, out from the heart through arteries and back to the heart through veins

citizen

Someone who lives in a certain city or country is a citizen of that place

civil war

War between people who belong to the same country

civilization

Society where people have built a complex city or country

climate

Weather that is usual for an area over a long period of time

code

Written commands, or language, used in a computer program

cold-blooded

Animal with a body temperature that goes up and down to match the surrounding air or water temperature

colony

Large group of animals that live together

comet

Object made of dust and ice that orbits around the sun, developing a tail as it gets close to the sun

competition

Contest between two or more people, groups, or living things, where one is trying to win a prize or resource

computers

Machines that can perform difficult tasks by following programs

condensation

When a gas cools and becomes liquid. Often seen as droplets of water that form on cold surfaces, such as windows

conductor

Substance that allows heat or electricity to pass through it easily

coniferous tree

Type of evergreen tree, usually with needlelike leaves

conquer

Act of one country taking over another country

conservation

Trying to stop a plant or animal from becoming extinct

consumer

Animal that eats a producer or other consumers

continent

One of seven large areas of land into which the world is divided: Africa, Antarctica, Asia, Europe, North America, Oceania, and South America

coral

Hard outer skeleton of tiny sea animals, which can build up into large coral reefs

core

Center part of a planet, star, or moon

country

Area of Earth that is governed by the same leaders and has the same flag

court

Place where it is decided if someone has broken the law

**crater**

Bowl-shaped dent on the surface of a planet or other body in space, caused by a collision with a rock

crime

Activity that is against the law

crop

Group of plants that are grown as food

crust

Outer layer of a planet

culture

Way of life and beliefs of the people of a region or country

deciduous tree

Type of tree that loses all its leaves at the same time during the winter or the dry season

decomposer

Living thing, often a fungus, that breaks down dead matter to create nutrients

deforestation

Destruction of forests

democracy

System of government where people outside the government have a say in how the country is run, usually by voting

desert

Dry region that gets 10 in (25 cm) or less of rainfall in a year. Deserts can be hot or cold

dictator

Ruler with total power

dinosaur

Group of reptiles, often very big, that lived millions of years ago

diplomat

Person from one country who travels to another to make sure the two countries have a good relationship

direction

Way in which an object is traveling, such as up or down, or right or left

disability

Something that makes certain activities difficult or impossible for a person to do

disease

Condition that makes a person ill, often caused by germs

drought

Period when there is little or no rainfall

earthquake

Shaking of the Earth's surface caused by moving tectonic plates or volcanic activity

eclipse

When an object in space passes into the shadow of another object

election

Event where people vote to decide who will be part of the government

electricity

Type of energy that can be used to power appliances such as lights. It is also found naturally as lightning

element

One of 118 chemical substances that are made of the same type of atom, such as gold, oxygen, and helium. The elements are arranged on the periodic table

emperor

Ruler of an empire

empire

Large area with different peoples, ruled by a single government or person

endangered

When an animal or plant species is in danger of becoming extinct

energy

Source of power such as electrical energy or heat energy

environment

Surroundings in which something lives

equality

Equal rights for all people

equator

Imaginary line around the center of the Earth that is an equal distance from the North and South poles

erosion

Gradual wearing away of rocks due to water and weather

eruption

When lava, ash, rock, or gas shoots or flows out of a volcano

ethical

Something that is done the right way, with thought as to how it will affect others

evaporation

When a liquid is heated and turns into a gas or vapor

evolution

Process where living things change, over many generations, to become new species

exoplanet

Planet that orbits a star other than the sun

exoskeleton

Hard outer casing of animals such as arthropods that do not have an inside skeleton

experiment

Test to see how something works

explorer

Someone who travels to unmapped places to find out what is there

extinction

When all of a particular animal or plant species dies out and there are none left in the world

factory

Building where products are made

fertilize

Process by which cells from a male and female join to create offspring, such as male plant pollen and a female plant ovum joining to create a seed

fins

Flattened limbs found on animals that live in water that help them swim

fish

Cold-blooded vertebrates that live underwater and have scales

flexible

Bendy

float

Stay at the surface of a liquid, rather than sinking

forage

Searching for food in the wild

force

Push or pull that causes things to start moving, move faster, change direction, slow down, or stop moving

foreign

Something or someone from a different country or place

fossil

Remains of a dead dinosaur, other animal, or plant, which has been preserved in rock over time

fossil fuels

Fuels made from animals and plants that died millions of years ago, such as coal and oil

friction

Force created when two surfaces rub or slide against each other

fuel

Substance that is burned to create heat or power

fungi

Group of living things, including mushrooms and molds, that break down dead plants and animals to make their food

galaxy

Huge group of stars, gas, and dust held together by gravity

gas

State of matter with no fixed shape, such as air, that fills any space it is in

generation

Group of living things that are of a similar age, and usually related; for example, brothers and sisters are one generation and their parents are another

genetics

Study of the genes in DNA that cause characteristics like hair color to be passed from one generation to the next. Someone who studies genetics is called a geneticist

geometry

Type of math that deals with solids, surfaces, lines, angles, and space

germs

Tiny life forms, such as bacteria or viruses, that cause disease

gills

Organs of fish and some amphibians that allow them to breathe underwater

glacier

Huge, thick sheet of ice moving very slowly, either down the side of a mountain or over land. Glaciers help to shape and form the landscape

government

Group of people who run a country

grasslands

Open land covered in grass and sometimes a few small bushes

gravity

Invisible force that pulls objects toward each other

habitat

Natural home environment of an animal or plant

hardware

Physical parts of a computer, such as the keyboard and screen

hatch

Process by which an animal breaks out of an egg

hemisphere

Top or bottom half of the Earth

**herbivore**

Animal that eats only plants.

herd

Group of animals, particularly hoofed mammals.

hibernation

Period of inactivity that some animals go through in the winter.

holy

Something or somewhere sacred to a religion.

hurricane

Violent storm with extremely strong winds that can cause a great deal of damage.

identical

Two or more people or things that look exactly the same.

imports

Goods or services bought from another country.

incubation

Keeping an egg warm until it hatches.

instruction

Command that tells something or someone what to do.

insulator

Substance that does not allow heat or electricity to pass easily through it.

interact

When two or more things communicate or do something that affects the other.

internet

Network that links computers across the world.

invertebrate

Animal that does not have a backbone.

jewel

A precious gemstone that has been cut and polished.

joint

Place in the body where two bones meet, such as the knee or elbow.

king

Man who rules a country.

knowledge

Understanding of a topic.

laboratory

Place where scientific experiments are done.

lake

Large body of water surrounded by land.

landfill

Place where garbage is buried in the ground.

landslide

Sudden movement of a large amount of earth down a hill or mountainside.

latitude

Horizontal line around the Earth that tells you how far up or down the globe a place is.

lava

Hot, melted rock on the Earth's surface.

light

Type of energy that allows humans and other animals to see, and plants to make food.

light year

Distance traveled by light in a year, equal to about 5.9 trillion miles (9.5 trillion km).

liquid

State of matter that flows and takes the shape of any container it is in, such as water.

longitude

Vertical line around the Earth that tells you how far east or west around the globe a place is.

lungs

Breathing organs found inside the body of vertebrates.

luxury

Expensive activity or item that is not necessary but wanted.

machine

Something that is powered by energy and carries out a task.

magma

Hot, melted rock below the Earth's surface.

magnetic field

Area of magnetism surrounding a magnet or a planet, star, or galaxy.

magnetism

Invisible force that is created by magnets, which pull certain metals toward them.

magnify

Make something appear larger than it is.

mammals

Warm-blooded vertebrates that have skin covered in hair and feed their young milk.

mantle

Thick layer of hot rock between the core and the crust of a planet or moon.

manuscript

Handwritten book, poem, or other document.

mate

When a male and female animal produce young together.

material

Substance that can be used to make or build things. It can be natural or made by humans.

mathematics

Study of numbers and equations. Someone who studies mathematics is called a mathematician.

matter

Stuff that all things are made of.

melt

When a solid is heated and becomes a liquid.

memory

Ability to remember things that have happened, or where computers store their information.

merchant

Person whose job is buying and selling things, often from a foreign country.

metamorphosis

Process by which some animals transform themselves into a different form from youth to adulthood.

meteor

Streak of light caused by a meteoroid burning up as it enters Earth's atmosphere, sometimes called a "falling star."

meteorite

Rock from space that lands on a planet or moon's surface.

microscope

Instrument that magnifies things and is used to look at tiny objects.

microscopic

Very small and only able to be seen with a microscope.

migration

Regular movement of animals over long distances, often to feed or breed.

Milky Way

Galaxy we live in.

mine

Place where naturally occurring resources such as coal, iron, copper, or gold, and gemstones such as diamonds and rubies are dug out of the ground.

mineral

Natural substance that grows in crystals, such as salt. Minerals can be polished to make gemstones. All rocks are made from minerals.

mixture

Combination of more than one type of thing.

monument

Statue put up to remember a person or event.

moon

Object made of rock, or rock and ice that orbits a planet or an asteroid.

navigation

Way of finding a path from one place to another.

Nobel Prize

Special prize given to people for different subjects in both science and the arts, once a year.

novel

Story book.

nucleus

Central part of an atom or cell.

nutrients

Food or substance that gives a living thing the energy or chemicals that it needs to live, grow, and move.

omnivore

Animal that eats both plants and meat.

orbit

Path an object takes when traveling around another object when pulled by its gravity.

orchestra

Group of musicians and their instruments playing together.

organ

Body part that has a certain job; for example, the heart, which pumps blood.

organism

Living thing.

particle

Extremely small part of a solid, liquid, or gas.

periodic table

Set arrangement of elements into a grid.

persecution

Bad treatment of people because of their beliefs.

philosophy

Study of how we live, such as whether things are wrong or right. Someone who studies philosophy is called a philosopher.

**photosynthesis**

Process that green plants use to make food from sunlight

physics

Study of the universe and forces. Someone who studies physics is called a physicist

planet

Large, round object that orbits a star

poisonous

Substance that may be deadly if touched or eaten

pollen

Powder that comes from flowering plants and is used in pollination

pollination

Transfer of pollen from one plant to another so those plants can reproduce

pollution

Harmful substances in the air, soil, or water

power source

Energy that is used to make a machine work, such as electricity

predator

Animal that hunts other living animals for food

prehistoric

Time before written history

prey

Animal that is hunted for food

primate

Type of mammal, which includes monkeys and humans

prison

Building where people who have broken the law are locked up as a punishment

probe

Unmanned spacecraft designed to study objects in space and send information back to Earth

producer

Living thing such as a plant that makes its own food and is eaten by animals

program

Set of instructions a computer follows to complete a task

queen

Woman who rules a country

recycle

Use something old to make something new

renewable

Type of energy that will not run out, such as solar power

reptiles

Cold-blooded vertebrates with scaly skin that usually reproduce by laying eggs

republic

State ruled by elected officials instead of a royal family or emperor

robot

Machine that is programmed by a computer to do different tasks

rock

Naturally occurring solid made from different minerals, such as granite. Rocks make up the surface of planets and moons

rover

Robot used to explore the surface of a rocky planet or moon

reaction

Effect when two chemicals cause a change in each other

reflect

When light or sound bounces off a surface

repel

When two objects push away from each other

reproduce

Have young

reservoir

Large store of something, usually water

satellite

Any object that goes around a planet, usually a moon or a human-made machine

scavenger

Animal that feeds on the leftover meat of another animal that has already died

shadow

Area of darkness formed when light rays are blocked by a solid object

shield

Object that protects something from damage or attack

society

Organized group of people with a shared culture

software

Programs and instructions that are used by a computer

solar system

The sun and all the objects that orbit it, including the planets

solid

State of matter that holds its shape

solidify

When a liquid cools and becomes a solid

solution

Mixture that is created when a solid dissolves in a liquid and disappears

sound

Form of energy that is produced when objects vibrate, or shake

space

Place beyond Earth's atmosphere

spacecraft

Vehicle that travels in space

species

Specific type of an animal or plant that can mate and produce young together

spectrum

Range of something; for example, the range of colors in a rainbow

stalactite

Piece of rock that hangs down from the roof of a cave and looks like an icicle

stalagmite

Piece of rock that points upward, slowly growing from the floor of a cave

star

Huge, hot sphere of gas in space that releases energy from its core and gives off heat and light

sustainable

Able to be supported for a long time

tame

Animal that is used to people, such as a pet dog or cat

technology

Using scientific knowledge to create machinery and devices, such as computers

tectonic plate

Large, slow-moving piece of the Earth's crust

telescope

Instrument used to look at distant objects

temperature

Measure of how hot or cold things are

temple

Home for a god or gods and a place for worshipping them

traditional

When something has been done in the same way for a long time

transmit

Pass something, such as information, between two places

tropical

Area or climate with hot temperatures

tsunami

Giant wave created by an earthquake or a volcanic eruption

turbine

Wheel or rotor that is turned to make power

universe

All space and everything in it

venomous

Substance that may be deadly if injected by an animal or plant, through a sting or fangs

vertebrate

Animal that has a backbone

vibrate

Moving back and forth small amounts very quickly

volcano

Opening in the Earth's crust, usually in the shape of a mountain, out of which lava, ash, rock, and gas erupt, sometimes explosively

voyage

A journey, often over water

warm-blooded

Animal that keeps a constant body temperature

weight

Amount of the force of gravity that acts on an object, making it feel heavy. The more mass something has, the larger the force of gravity on the object, and the heavier it feels

wildlife

Animals found in a certain area

womb

Organ in which baby animals develop

worship

Praying to a god or gods

X-ray

Radiation used to create shadows of bones and other organs in the body. Images on an X-ray photo can reveal internal damage and disease

young

Babies, or not very old

zoo

Place where wild animals are kept, so people can see and study them



Index

Pages numbers in **bold** refer to main entries.

3-D printing 91, 245
3-D shapes 222, 269, 288

Aa

aboriginal people 186
acid 225
actors 253
adults 146
aerials 203
Africa **12**, 180
Age of Exploration 96, 180–181
air 117
air pollution 198
aircraft **13**, 33
 air transportation 259
 engines 92
 solar-powered 181
 World War II 277
algebra 105
allergies 105
alligators 210
alphabets 280, **284**
aluminum 90
Amazon rain forest 204, 236
American West **14**
ammonites 95, 202
amphibians **15**, 22, 86, 266
Amundsen, Roald 24
ancient civilizations
 ancient China **16**
 ancient Egypt **19**
 ancient Greece **18**
 ancient India **19**
 ancient Rome **20**
 Aztecs **35**
 Incas **132**
 Maya **158**
 Native Americans **179**
Andes 51, 172, 236
Andromeda galaxy 167
angles 222
animals
 amphibians **15**
 animal families **21**
 animal groups **22**
 birds **39**
 cows 56
 colors 26
 conservation **72**, 281
 evolution **95**
 farming **98**
 fish **101**
 food chains **107**

animals continued
 habitats **126**
 hibernation **129**
 homes **23**, 55
 insects **134**
 invertebrates **139**
 mammals **154**
 metamorphosis **163**
 migration **166**
 mythical creatures 178
 pets **54**, **152–153**
 reptiles **210**
 stories 42
 tree of life **206**
 vertebrates **266**
 zoology 38
 zoos **281**
animation 100
Anning, Mary 219
Antarctica **24**, 197
antibiotics 137, 218
ants 21, 78, 134
apes 170
apples 115
aqueducts 20
Arabic 144, 284
arachnids 139
Archimedes 269
architecture 245
Arctic **25**, 197, 245
Aristotle 218
art **28**, 131, 179, 209, 243, 262
arteries 128
artists **282**
Asia **29**
assembly lines 97
asteroids **30**, 164, 200
astronauts **31**, 83, 171, 237
astronomy 19, **32**, 73, 167,
 218–219
athletics 239
atmosphere
 Earth **33**, 49, 50, 83, 164, 270
 planets 183, 264, 265
atoms **34**, 37, 58, 90, 192
auroras 33, 247
autumn 221
avalanches 93
axes 136, 243
axolotls 15
Aztecs **35**
Bb
babies
 animals 21, 154
 human 146

Babylonians 145, 155
backbones 266
bacteria 38, 56, 165, 218, 225
Baird, John Logie 251
balanced forces 108
ball game (Maya) 154, 240
ball sports 241
ballet 76
balloons 117
banks 169
bark 261
the Bastille 113
bats 55, 129
batteries 59, 71
bathing 75
bears 129
beavers 23
beetles 134
beryllium 199
bicycles **36**, 162, 258–259
Big Bang **37**
big cats 54
biology **38**, 217
birds **22**, **39**, 78, 266
 display 26
 eggs 86
 flightless 186
 migration 166
 nests 23
birds of prey 39
black holes **40**
Blackbeard 193
blind people 256
blood 128, 149
blood cells 41, 225
blood vessels 229
Blue Mosque 262
blue supergiant stars 242
board games 240
boats see ships
body systems 130
Bollywood 76
bones 82, 127, 228, 266
books **43**, **44**, 201
botany 38
Brachiosaurus 80
braille 256
brain **45**, 99, 127, 148, 231,
 232, 256
brass instruments 188
breathing 130, **149**
bridges **46**
broadcasting 251
bronchi 149
bronchioles 149
Bronze Age **47**
bubbles 117
Buddhism 208
budgies 153
buildings **48**
bullet train 260

buoyancy 227
Buj Khalifa 48
buses, school 272–273
butterflies 134, 163, 166

Cc

cactuses 78
cavaliars 15
calcium 90
calderas 268
Callisto 141
camels 78
cameras 190
camouflage 54
carbohydrates 106
carbon 195
carbon atoms 34
carbon cycle **49**
carbon dioxide 49, 60, 117, 191
carnivores 107, 154
carriages 258
cars **52**, 92, 97, 259
cartilage 266
carts 258
cash machines 150
Cassini-Huygens spacecraft 216
castles **53**, 244
caterpillars 163
cats **54**, 152, 153
cave dwellings 244
cave paintings 28, 243
caves **55**
cells **56**
 human **41**
cell phones see mobile phones
Central Powers 276
ceratopians 80
cereal crops 98
Chiron 196
cheetahs 154
chemical engineering 91
chemistry **58**, 217
chess 240
children 133, **146**, 277
chimpanzees 170
China 29, 42, 102, 188,
 279, 280
 ancient **16**, 178, 272
Chinese New Year 206–207
chloroplast 56
Christianity 208
Christmas 206
chrysalises 163
cinder cones 268
circles 222
circuits **59**
circulation 128, 130
cities 275
civil engineering 91
civil wars 279
classical music 176



classification 22
cliffs 93
climate change 25, 51, **60**, 122,
143, 198
clocks **61**, 159
clones 119
clothing 27, **62-63**
the cloud 138
clouds **64**, 270, 271
coal 110
codes **66-67**
coding **65**, 66
coins 169, 201
college 273
colonies 21
color **26-27**
Columbus, Christopher 96, 180
comets **68**, 164
communication 66, **69**, 138,
215, 250
compasses **70**, 182, 218
compounds 58, 168
computer games 69, 71, 241
computers **71**, 137, 192, 212, 280
circuit boards 59
coding 65, 66
Internet 69, 138
condensation 57
conduction 157
conductors 189
conifers 109, 126, 194
consciousness 279
conservation 72, 281
Constantinople 262
constellations **73**
constitutions 123
construction 48
container ships 224
continents 50-51, 275, 293
Cook, James 96
cooking 104-105
copper 47
coral reefs **74**
core, Earth's 135
cornua 226
costumes, traditional 63
countries 275, 292
courts 145
crafts **75**, 132
cranes 48
craters 164
crayons 280
creation 37
creation stories 42, 178
crocodiles 124, 210
cruise ships 224
crust, Earth's 85, 135
crustaceans 139
cubed units 269
cuneiform 42, 280
currency 169

cycling 36
Cyrillic 280
cytoplasm 41, 56

Dd
da Gama, Vasco 96, 180
daimyo 131
dairy products 106
dams 211, 270
dance **76**
dark matter 116, 263
dark zone 187
Darwin, Charles 95
day 77
Day of the Dead 206
deciduous trees 109, 126, 261
decimals 112
decomposers 107
deep-sea zone 187
defense system, body 225
Degas, Edgar 28
Deimos 156
democracy 123
Denmark 102
denominators 112
dermis 229
deserts **78**, 126, 270
diamonds 118
diaphragm 149
dictatorships 123
digestion **79**, 130
digital clocks 61
digital money 169
digital radio 203
dinosaurs **80**, 111, 184, 202
directions 70
Diwali 207
DNA 67, 119
doctors 160, 274
documentaries 100
dogs **81**, 95, 152, 153
dolphins 154
dormice 129
dragons 178, 206-207
drawing 28, 219
dreams 231
drinks 97, 106, 148
droughts 60, 149
dustmites 165
dwarf planets 196

Ee
ears 127, 225, 235
Earth **83**, 233, 263, 289
atmosphere **33**
changing **50-51**
gravity **125**
inside **135**
magnetic field 70, 135, 151
moon's orbit 32, 171
orbits the sun 32

rotation 77
seasons **221**
surface **84**
weather **271**
earthquakes 84, **85**
eating 79, 104-105, 249
echinoderms 139
eclipses 77
eco homes 244
ecology 38
Edison, Thomas 137, 217
eels 101
eggs 39, **86**, 146, 163
Egyptians, ancient 12, **17**, 66,
152, 162, 218, 258
Eid 207
Eiffel Tower 94
electorate 123
electric cars 52
electrical engineering 91
electricity **87**, 120, 192
circuits 59
conducting 157
insulators 157, 195
power stations 110
electronics, recycling 205
electrons 34, 58
elements 58, **90**
elephants 12, 19, 154
elliptical galaxies 116
email 69
emojis 280
energy **60**, **88-89**
electricity **87**
food 106, 107
fossil fuels **110**
engineering **91**
engines 52, 89, **92**, 260
English language 144, 280
Enigma codes 67
environment 121, 198,
205, 258
epidermis 229
equality 189
erosion **93**
erotics 122
eruptions, volcanic 268
esophagus 79
Europe 141
Europe **94**, 276, 277, 292
evaporation 52, 168
evergreen trees 261
evolution **82**, **95**, 286
exercise 173
exosphere 33
experiments 218
exploration **180-181**, 267
explorers **96**
exports 257
expressions, facial 99, 173
eyes 54, 119, 226

Ff
faces 99, 173
factories **97**, 133, 174, 212
fairy tales 43
fall 221
fandancing 76
farming **98**, 105, 140, 274
fashion 27, 62-63
fat, body 229
fat cells 41
fats 106
faults 85
feathers 39
feeding (touch) 256
feelings (emotions) **99**
fems 194
festivals **206-207**
fiction 44
filaments 103
film 43, **100**, 190
filtering 168
firm 101
fire 82, 105
fish 22, 74, **101**, 120, 181, 266
eggs 86
farming 98
sharks 223
fishing 72
flags **102**, 193
flammability 157
Flaming, Alexander 218
flight 137, 209, 258, 277
floating 227
floods 60
flowers **103**, 194
fog 271
food **104-105**
digestion 79
farming 98
nutrition **106**
plants 191
taste 249
food chains 89, **107**
football 239, 241
forces **108**
buoyancy **227**
friction **114**
gravity **125**
magnets **151**
forests 72, **109**, 126, 202, 204
Formula 1 52
fossil fuels 51, 60, 88, **110**
fossils 80, 95, **111**, 202, 219
foxes 81
fractions **112**
French Revolution **113**
friction 108, **114**
Frobisher, Martin 96
frogs 15, 78, 129, 163, 236, 266
fruit 98, 106, **115**
fungi 38, 174



Gg

Gagarin, Yuri 31, 33, 181
galaxies 37, **116**, 167, 263
Galilean moons 141
Galilei, Galileo 32
games 69, **240-241**
Ganymede 141
garbage 121, 198, 205
gases 57, 110, **117**
Gaudi, Antonio 245
gears 61
gemstones **118**
genes **119**
gerbik 152
Germany 102, 277
germs 160, 225
Giant's Causeway 94
gills 101, 120, 174
giraffes 12, 95, 124
glaciers 55, 93, **122**
glass 48, 75, 205
glasses 226
gods and goddesses 18, 132, 158,
179, 183, 208
gold 44, 90, 199, **200-201**
Golden Gate Bridge 46
goldfish 101
governments **123**
GPS 155, 182, 215
grasshoppers 134
grasslands 124, 126
gravity 100, **125**
Great Barrier Reef 74
Great Red Spot 141
Great Wall of China 16
great white sharks 223
Greece, ancient **18**, 67, 241, 269,
272, 278
 alphabet 284
 medicine 160
 philosophy 189
 theater 253
green transportation 258
Greenwich Mean Time 255
griffins 178
Gris, Juan 28
growing up 146
growth 231
guinea pigs 153
gunpowder 279
gymnastics 239

Hh

habitable zone 83
habitats **126**
 conservation 72
 coral reefs 74
 deserts 78
 forests 109
 grasslands 124
 intertidal zone 254

mountain 172
oceans and seas 187
polar regions 197
rain forests 204
seashore 220
hair 119, 130, 154
Halley's Comet 68
hammerhead sharks 223
Hammurabi, King 145
hamsters 152
Han Dynasty 16
Hang Son Doong cave 55
hard drive 71
Harry Potter 44
hearing **127**
heart **128**
heat
 conducting 157
 friction 114
helicopters 13
helium 90, 117
heraldry 142
herbivores 107, 154
herds 21
hibernation **129**
hieroglyphs 12, 66
hill forts 140
Himalayas 84
Hindi 144, 280
Hinduism 207, 208
Hippocrates 160
Hokusai 28
HolLens 69
home school 273
homes **244-245**
 animals 23
 caves 55
 hominins 82
 horse-drawn vehicles 258
 houseboats 245
 Hua Mulan 178
 Hubble, Edwin 167
 Hubble Space Telescope 33, 215
human biology 38
human body **130**
 brain 45
 breathing 149
 carbon 49
 cells 41
 defense system 225
 digestive system 79
 energy 88
 feelings 99
 genes 119
 heart 128
 ingredients 130
 life cycle 146
 muscles 173
 senses 127, 226, 232, 249, 256
 sickness 225
 skeleton 228

skin 229
temperature 252
water 121, 130, 148
humanoid robots 212
humans, early **82**, 188, 202
hunting 243
huntsman spiders 238
hurricanes 246
Huygens, Christian 61
hydroelectric power 120
hydrogen 58, 120

Ii

ice ages 51, 202
ice caves 55
igloos 245
igneous rocks 213, 214
imports 257
Incas **132**, 236
India 29, 63, 102
 ancient **19**, 42, 272
Industrial Revolution
 89, **133**
inheritance 119
insects 105, **134**, 139
 pollen 103
 spiders 238
instruments
 medical 160
 musical 175, 176-177, 188
insulators 157, 195
International Space Station 31,
33, 212
international trade 257
Internet 69, **138**
intertidal zone 254
intestines 41, 79
inventions 16, 20, **136-137**, 217
invertebrates 22, **139**, 238, 286
io 141
iris (eye) 226
iron 135, 140, 164
Iron Age **140**
irregular galaxies 116
Islam 76, 207, 208, 262, 272
Istanbul 262

Jj

Japan 63
 Imperial 53, **131**
jet engines 92
jet lag 255
jewelry 47, 75, 118, 200
jobs 274
joints 228
jousting 142
Judaism 207, 208, 277
judges 145
jumping spiders 238
Jupiter **141**, 233, 289
juries 145

Kk

kangaroos 154
Kanya 102
Kepler-16b 233
keyboards
 computer 71
 musical instruments 175
kimonos 63
Kindertransport 277
kiwis 186
knights **142**, 278

Ll

lakes 122, **143**, 211
land pollution 198
language 66, 67, 144
 computer 65
Latin 144, 284
Latin dance 76
latitude 182
lava 135, 234, 268
law **145**
leaves 27, 109, 115, 191, 194,
248, 261
leeches 168
legends 178
lemurs 12, 170
length 159
lens (eye) 226
Leonardo da Vinci 209, 218-19
leopards 54
levers 150
life
 on Earth 50, 83, 247, 270
 tree of life 286
life cycle **146**
light **147**
light bulbs 59, 217
lightning 87, 246, 252
lines of symmetry 248
lionfish 101
lions 54, 107, 124
liquids 57, **148**, 159
lizards 78, 210, 266
log boats 258
longhouses, Viking 267
longitude 182
longships, Viking 267
Louis XVI of France 113
lunar rovers 259
lungs **149**

Mm

machines 133, **150**
Machu Picchu 132, 236
madrasas 272
Magellan, Ferdinand 96
magma 135, 213, 268
magnetic field 70, 135, 151
magnetism 108
magnets **151**



magnetometers 70
mammals 22, 86, **154**, 266
mammoths 51, 202
Mandarin 144, 294
maps **155**, 182
 Europe 292
 the world 292-293
Marie Antoinette 113
markets 274
Mars **156**, 233, 237, 289
marsupials 154
masks 35, 200-201, 253
mass production 97
materials 58, **157**
 buildings 48
 changing states 57, 148, 234
 elements 90
 plastic 195
mathematics
 fractions 112
 measuring 159
 multiplication 287
 numbers 185
 shapes 222
 symmetry 249
 volume 159, 269
matter 192
Matterhorn 172
Maya **158**
measuring **159**
 temperature 252
 volume 269
meat-eating plants 194
mechanical engineering 91
mechanics 192
medicine 137, **160**, 218, 274
 equipment 192
 robots 212
medieval period 44, 142, 160, 278
meerkats 107, 124
melanin 119, 229
melting 57
membranes, cell 41, 56
memory 231
Mercury **161**, 233, 289
Mesopotamia 121
mesosphere 33
metals 30, **162**, 200, 205, 234
 precious **199**
metamorphic rocks 213, 214
metamorphosis **163**
meteorites 24, 162, **164**
meteors 200
metro systems 260
Mexico 206
microbiology 38
microorganisms 38, **165**
microscopes **165**
migration **166**
milk, mammals 154
Milky Way 40, 116, **167**, 247, 263

minerals 30, 118, 214
mines 110, 162, 200
minotaur 178
Mississippi River 184
mitochondria 41, 56
mixtures **168**
mobile phones 69, 182, 190,
 250, 280
Model T/Ford 259
molecules 34, 58
mollusks 139
monarchy 123, 290
money **169**, 201
monkeys 124, **170**
monotremes 86
moose 221
the moon **161**, 171, 219
 eclipses 77
 gravity 125
 landings 31, 171, 181
 lunar rovers 259
 tides 254
moons, planets 141, 156, 161,
 183, 196, 216
Morse Code 66
mosques 262
moss 194
motherboards 71
motor racing 52
motte and bailey castles 53
mountains 50, 51, 84, 122, **172**
mouth 79, 249
movement
 energy 88
 friction 114
 muscles 173
movies see film
mucus 225, 232
Mughal Empire 19
multiplication **287**
mummies 17
muscles 130, **173**
mushrooms **174**
music 131, **176-177**, 188
 instruments **175**
musicals 100
myths **178**

Nn

Native Americans 14, **179**
natural selection 95
navigation 70, 73, 155, **182**, 215
negative numbers 185
Neptune **183**, 233, 289
nerves 41, 45, 127, 229
nervous system 45, 130
nests 23
neurons 45, 256
Newton, Isaac 125
newts 15
night 77

Nile, River 12, 17
nitrogen 117
nonfiction 44
North America **184**
North Pole 180
nose 149, 225, 232
notation, musical 177
novels 43
nuclear physics 192
nucleus 34, 41, 56, 58
numbers 112, **185**
numerators 112
nurses 274
nutrition **186**

Oo

Oceania **186**
oceans and seas **187**
 coral reefs 74
 exploration 181
 garbage in 198
 habitats 126, 197
 prehistoric life 202
 sea battles 279
 sea levels 51
 seashore 220
 tides 254
 volcanoes 268
 water cycle 270
oil 110
Olympic Games 18, 241
optic nerve 226
oral history 42
orbit
 Earth 32
 moon 171
 Pluto 196
orchestras **188**
Osman I, Sultan 262
ostriches 86
otters 21
Ottomans 262
ovaries 103
oxbow lakes 143
oxygen 34, 49, 58, 117, 120,
 129, 191
 breathing 149

Pp

pain 256
paint 26
painting 28, 201, 243
palaces 244
Panama Canal 184
pandas 29
Pangaea 50
parchment 44
Paris 275
parrots 39
Parthenon 18
particles 34, 58, 90

gas 117
liquids 148
solids 234
Passover 207
pens 115
pencils 280
pendulums 61
penguins 21, 24, 39, 197, 266
pens 280
percussion instruments 175, 188
periodic table 90
perspective 209
Piercing, Pietro 209
petals 103
pets 54, 81, **152-153**
pharaohs 12, 17
philosophy **189**
Phobos 156
photomessaging 69
photography **190**
photosynthesis **191**
physics **192**, 217
pirates **193**
pitch 235
pixels 251
place value 185
planes 13, 92, 277
planets 181, 219, 289
 dwarf planets 196
 Earth **83**, 263
 Jupiter **141**
 Mars **156**
 Mercury **161**
 minor planets 183
 Neptune **183**
 Saturn **216**
 solar system **233**
 types of 233
 Uranus **264**
 Venus **265**
plankton 165
plants 38, **194**
 cells 56
 flowers 103
 food chains 89, 107
 fruit and seeds 115
 habitats 126
 photosynthesis 191
 trees 261
plastic 121, 137, **195**, 198, 205
plates, tectonic 85
platinum 199
plays 253
Pluto 183, **196**
poetry 43
Poland 277
polar bears 25, 197
polar habitats 126, **197**
poles 151
police 145
pollen 103



pollution 60, 72, **198**
 polygons 222
 pop music 176
 population 275
 pores 229
 pottery 18, 75, 262
 power stations 110
 prairies 124
 praying mantises 139
 precious metals **199**
 prehistoric life **202**
 prehumans 82
 presidents **290**
 primary colors 27
 primary consumers 107
 primates 170
 prime ministers **289-290**
 printing 29, 44
 primes 147
 processors 71
 producers 107
 programming, computer 65, 66
 prominences 247
 protein 105, 106
 protons 34, 58
 Proxima Centauri 242
 pulleys 150
 pulling 108
 pupils 226
 puppets 253
 pushing 108
 pyramids 12, 17, 35, 158

Qq
 questions and answers 189

Rr
 rabbits 153
 racket sports 240
 radio **203**
 radio waves 203, 250
 railroads *see* trains
 rain 211, 221, 246, 270, 271
 rainbows 26
 rain forest 126, **204**, 236
 RAM 71
 reactions, chemical 58
 recycling 72, 195
 red giant stars 242
 reflecting colors 27
 reflection 147
 reflective symmetry 248
 religion 206-207, **208**, 291
 remote control 203
 Renaissance **209**, 218
 renewable energy 89
 reptiles 22, 86, **210**, 266
 reservoirs 143
 retina 226
 rice 16, 29
 right and wrong 189

Ring of Fire 84
 rings, planetary 216, 264
 rivers 55, 93, 121, 143, **211**, 270
 robots 71, 97, 150, 181, **212**, 237
 rock cycle **213**
 rock music 176
Rocket (steam train) 136-137,
 260
 rockets 237
 rocks **214**, 234
 erosion 93
 metals 162
 space 164
 roller coasters 88
 Rome, ancient **20**, 28, 62, 144,
 230, 272
 Röntgen, Wilhelm Conrad 219
 room temperature 252
 roots 115, 174, 191, 194
 rotational symmetry 248
 royal courts 62-63
 Russia 275

Ss
 Sahara Desert 12, 78
 sailing boats 224
 salamanders 15
 saliva 225
 Samurai 131
 San Andreas Fault 85
 sandstone 213
 saris 63
 satellites 138, 182, **215**, 251
 Saturn **216**, 233, 289
 sauropods 80
 savannas 124
 scales 101, 159
 schools **272-273**
 science **217, 218-219**
 ancient Greece 269
 ancient India 19
 biology **38**
 chemistry **58**
 physics **192**
 Renaissance 209, 218
 science fiction 100
 scientists 217, 274, 285
 scorpions 139
 Scott, Robert 24
 Scratch 68
 screws 150
 sculpture 28
 sea *see* oceans and seas
 seahorses 101
 seashore **220**
 seasons **221**
 secondary colors 27
 secondary consumers 107
 sedimentary rocks 213, 214
 seeds 103, **115**
 sexes 225

hearing **127**
 sight **226**
 small **232**
series continued
 taste **249**
 touch **256**
 separation, mixtures 168
 servers 138
 sewing machines 150
 shadows 147
 shapes **222**
 flat and solid **288**
 symmetry **248**
 volume 268
 sharks **223**
Shinkansen trains 260
 ships **224**
 early 258
 explorers 96
 floating 227
 paddle steamers 184
 pirate 193
 slave trade 230
 Viking 267
 World War II 277
 shoguns 131
 shops 274
 sickness 160, **225**
 sifting 168
 sight **226**
 sign language 144
 signal flags 102
 Sikhism 208
 silver 199
 singing 176
 sinkholes 55
 sinking **227**
 skeleton 130, **228**, 266
 skiing 114
 skin 119, 130, 225, **229**, 256
 skydiving 33
 skyscrapers 48
 slate 213
 slavery **230**
 sleep 129, **231**
 smartphones 138, 250
 small **232**, 249
 smokers, ocean 187
 snails 139
 snakes 78, 124, 152, 210
 soccer 239, 241
 society
 Egyptian 17
 Imperial/Japanese 131
 Incas 132
 Roman 20
 solar energy 60, 87, 89, 181, 192
 solar flares 247
 solar system 32, 37, 68, 167,
 233, 247, 263
 see also planets

soldiers 20, 142, 278-279
 solidifying 57
 solids 57, **234**
 solubility 148, 157
 songbirds 39
 sound 127, **235**
 sound waves 235
 South America **236**
 South Pole 24, 180
 space
 asteroids **30**
 astronauts **31**
 astronomy **32**
 black holes **40**
 comets **68**
 constellations **73**
 eating in 105
 exploration 188, 191
 galaxies **116**
 meteorites **164**
 Milky Way **167**
 pets in 153
 robots 212
 stars **242**
 travel **237**
 universe **263**
 see also planets
 spacecraft 101, 201, **237**
 Apollo missions 171
 Cassini-Huygens 216
 International Space Station 31
 lunar rovers 259
 Mars rovers 156
 satellites **215**
 space probes 161, 237
 space shuttles 237
 Voyager 2 264
 spacesuits 31
 spaghettiification 40
 Spanish language 144
 sperm 146
 spice trade 96, 180, 257
 spiders 139, 153, **238**
 spiral galaxies 116, 167
 spores 174
 sports 36, 121, **239**, **240-241**
 spring 221
 squares 227, 288
 squirrels 142
 stage 253
 stalactites 55
 stalagmites 55
 starfish 139, 248
 stars 37, 73, 167, 219, **242**
 states, changing **57**, 148, 234
 static electricity 87
 Statue of Liberty 184
 steam engines 89, 92, 133,
 136-137, 260
 stems 115, 174, 191,
 194, 261



stopper 124
stigma 103
stingrays 101
stomach 79, 225
Stone Age 62, 202, **243**
stone buildings 48
Stonehenge 243
storms 60, 141, **246**
storytelling **42-43**
stratosphere 33
stratovolcanoes 268
streams 143, 211
street dance 76
string instruments 175, 188
submersibles 224
sugar 106
suits 63
sultans 262
summer 221
the sun 68, 89, 147, 233,
 242, 247
 Earth's orbit 32
 eclipses 77
 movement 77
 seasons 221
sundials 61, 255
sunlight 191
sunlit zone 187
sunshine 271
supernovas 242
surface, Earth 51, **84**, 135
suspension bridges 46
sweat glands 229
switches 59
symmetry **248**

Tt

tadpoles 163
Taj Mahal 19, 29
tanks 277
tarantulas 238
taste 232, **249**
teachers 274
team sports 239
tears 225, 226
tectonic plates 84
teenagers 146
teeth 79
telegraph 69
telephones 69, **250**
telescopes 32
television **251**
temperature **252**
 Earth 64
 measuring 159
 planets 161, 264, 265
temples 18, 35, 158
termite mounds 23, 124
tertiary consumers 107
texting 69
thatch 244

theater **253**
thermometers 252
thermosphere 33
theropods 80
thinking 45, 189
thunderstorms 246
tides **254**
tigers 266
time 61, 159, **289**
time zones **255**
tin 47
Titan 216
toads 15, 266
togas 62
Tokyo 275
tongue 54, 232, 249
tools
 animals 170
 first human 82
 Iron Age 140
 measuring 159
 school 271
 Stone Age 243
tornadoes 246
tortises 210
touch **256**
Touatou 30
trachea 149
tractors 150
trade 96, 180, 224, 257
trains 14, 92, 133, 136-137,
 260
Transit of Venus 265
transpiration 270
transportation **258-259**
 bicycles **36**
 cars **52**
 rivers 211
 ships **224**
 space travel **237**
 trains **260**
treasure 193
tree of life **206**
trees 78, 109, 172, **261**
trench warfare 276
triangles 222, 248, 288
Triceratops 80
Triton 183
Trojan War 278
troposphere 33
trunks (traces) 261
tsunamis 85
tundra 126
Turing, Alan 67
Turkey, Republic of 262
Turkish Empire **262**
turtles 74, 210
twilight zone 187
twins 119
two-dimensional shapes 222
Tyrannosaurus rex 80

Uu

U-shaped valleys 93, 122
UFOs (unidentified flying
objects) 64
ultraviolet light 214
Uluru 186
underground trains 260
United Kingdom 102
United States 102, 123, 184
 American West 14
 Civil War 279, **290-291**
 presidents **290**
 slavery 230
universe 37, 40, **263**
Uranus 183, 233, **264**, 289
urinary system 130
USB ports 71

Vv

vaccinations 160
vacuoles 56
value 169
valves, heart 128
van Gogh, Vincent 28
Vatican City 275
vegetables 98, 106
veins 128
Venus 233, **265**, 289
Venus fly trap 194
vertebrates 22, **266**, 286
vibrations, sound 235
video calls 69
video games 71
Vikings **267**
viruses 38, 160, 165
voice box 149
volcanic rock 213
volcanoes
 Earth 84, 135, 184, 213,
 268
 planets 156, 196, 265
volume (maths) 159, **269**
volume (sound) 235
Voyager 2 spacecraft 264

Ww

wading birds 39
wagon trains 14
war 67, 276, 277, **278-279**
water 58, 106, **120-121**,
 148, 211
 boiling and freezing 252
 changing states 57
 energy 60, 89
 life on Earth 83, 270
 pollution 198
 sinking and floating 227
water cycle 64, **270**
water sports 121
waves 85
weapons, see war

weather 64, 66, 192, 215, 221,
 246, 271
weaver birds 23
weaving 63, 75
webs, spider 238
websites 138
wedges 150
weighing 159
white sharks 223
wheels 91, 136, 258
 bicycles 36
 cars 52
 tractors 150
whiskers 54
white blood cells 225
white light 147
whole numbers 185
Wi-Fi 138
wild cats 54
wind
 Earth 271
 Neptune 183
wind instruments 175, 188
wind sculptures 93
wind turbines 60, 87, 89
winter 221
winter sports 239
wireless gadgets 203
wolves 81
women at war 276
wood carvings 75
woodwind instruments 188
work 169, **274**
working dogs 81
the world **275**
 map 292-293
World War I **276**
World War II 63, **277**
worms 139
Wright brothers 137
writers **283**
writing **280**
 ancient China 16
 ancient Greece 67
 Bronze Age 47
 systems 284

Xx

X-rays 160, 192, 239

Yy

yellow dwarf stars 247

Zz

zebras 21, 124
Zheng He 96
zoology 38
zoos **281**

